

HOW EXCESS LATENT DIMENSIONALITY DELAYS MEMORIZATION IN DIFFUSION MODELS

Anonymous authors

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ABSTRACT

We study how latent width affects memorization in latent diffusion models, whose latent dimension d_{latent} typically exceeds the effective signal dimension of the data. For MLP score networks of fixed hidden width trained on synthetic mixtures with known signal dimension, the fraction of memorized samples after 5M steps falls from 30% at $d_{\text{latent}} = 5$ to near zero at $d_{\text{latent}} = 40$, and the learned score departs from the population score progressively later as the latent widens. The decline survives a dimension-controlled nearest-neighbor test restricted to the true signal subspace. On CelebA and CIFAR-10 VAE latents, comparisons above the VAE identity-preservation threshold show that the mean time to reach 5% memorization increases by $3.6\times$ and $1.4\times$, respectively, while FID follows a separate quality tradeoff. To investigate a possible mechanism, we evaluate the random-feature spectral theory of Bonnaire et al. (2025) at a two-atom covariance. In the separated regime, the feature spectrum has signal, noise-dimension, sample-specific and residual-floor blocks. Under gradient flow, modes learn in parallel, and their relative absorption times are set by spectral edges. Widening lowers the pre-activation variance and shrinks the nonlinear feature mass; holding that variance fixed largely removes the endpoint growth of the measured spectral separation. Frozen-feature results support this mechanism; its causal role in trained networks remains unestablished. A one-parameter onset clock derived from frozen latent representations provides a diagnostic whose predictive advantage over dimension-only baselines is dataset-dependent.

1 INTRODUCTION

Score-based diffusion models learn a score network to reverse a noising process (Ho et al., 2020; Song et al., 2021; Karras et al., 2022). Latent diffusion and related flow models generate in learned latent spaces (Rombach et al., 2022; Esser et al., 2024). Motivated by work on intrinsic dimension (Stanczuk et al., 2024; Kamkari et al., 2024), we study the regime $d_{\text{latent}} > d_{\text{intrinsic}}$. Fitting the empirical score can lead to training-set replication (Yi et al., 2023); collapse during reverse sampling (Biroli et al., 2024) is distinct from the late-training memorization observed in some settings (Bonnaire et al., 2025). Memorization raises privacy and copyright concerns (Carlini et al., 2019; Somepalli et al., 2023; Carlini et al., 2023) and a basic question about what the model learns. We investigate how latent dimension affects its onset.

We ask whether widening the latent space, with everything else fixed, delays memorization, and when a spectral model can explain the delay. On the first part the evidence is clear. For MLP score networks of fixed hidden width trained on synthetic mixtures with $d_{\text{intrinsic}} = 5$, the fraction of memorized samples after 5M steps falls from 30% at $d_{\text{latent}} = 5$ to near zero (0.1%) at $d_{\text{latent}} = 40$ over five seeds, and the learned score departs from the population score progressively later as the latent widens (Section 3). The decline is not the nearest-neighbor test losing sensitivity in high dimension: restricted to the true five-dimensional signal subspace, where the signal-copy control retains its sensitivity, memorization still falls from 29.9% to 1.2% against a chance level of 0.4% (Table 1). On CelebA and CIFAR-10 VAE latents, once the VAE is wide enough to preserve sample identity, the mean time to reach 5% memorization increases by $3.6\times$ and $1.4\times$, respectively, over post-threshold width comparisons, while FID follows a separate quality tradeoff (Section 5). On the second part the answer is partial: a solvable frozen-feature model identifies one mechanism for the delay (Section 4), but its transfer to learned representations is not established (Sections 5 and 6).

Our contributions are:

- **The phenomenon, with a dimension-controlled test.** We characterize the decline in memorization from 30% to near zero in a fixed-width d_{latent} sweep at known $d_{\text{intrinsic}}$, and test it with a projected variant of the nearest-neighbor ratio criterion. The variant is necessary: a sampler that copies signal coordinates and redraws null coordinates is invisible to the full-space test for $d_{\text{latent}} \geq 10$, so that test alone cannot separate a real delay from a loss of sensitivity (Sections 2 and 3).
- **A spectral explanation in frozen features.** Motivated by the random-feature analysis of Bonnaire et al. (2025), we interpret measured spectra using signal, noise-dimension, sample and tail index groups (Remark 4.1). Their separation and approximate edge scales are empirical and conditional, rather than a direct consequence of that theorem. Under gradient flow, modes learn in parallel and the measured ratio of signal to sample-block edges grows with d_{latent} as the pre-activation variance q falls and nonlinear feature mass shrinks. Interpreting this ratio as a memorization clock requires spectral-alignment assumptions. A control that holds q fixed as d_{latent} grows largely removes the endpoint growth ($1.09 \times$ against $3.17 \times$ over $d_{\text{latent}} = 40 \rightarrow 240$), supporting the operating-point explanation (Section 4).
- **Real-data evidence and the transfer boundary.** CelebA and CIFAR-10 sweeps quantify memorization timing and image quality after identity preservation. Measurements in trained MLPs and a spatial-DiT pilot delimit the transfer of the frozen-feature explanation (Section 5).

Relation to prior work. Bonnaire et al. (2025) relate random-feature diffusion training to generalization and memorization timescales, and derive a limiting spectral description for Gaussian data with general clean-data covariance. George et al. (2026) study Gaussian data supported on a subspace. We use these results as motivation for an anisotropic-mixture spectral approximation; finite-rank signal modes and noise averaging require additional arguments (Appendix I.1.1). The contribution is the operating-point interpretation and its empirical control, not a new general spectral theorem. Generative memorization is also studied through membership inference, reconstruction attacks and data-copying tests (Hayes et al., 2017; Hilprecht et al., 2019; Meehan et al., 2020), and diffusion replication and extraction (Somepalli et al., 2023; Carlini et al., 2023). Our nearest-neighbor ratio follows Bonnaire et al. (2025).

2 SETUP AND MEASUREMENT

Synthetic construction. Training data live in a signal subspace of dimension $d_{\text{intrinsic}} = 5$ embedded in a latent space of dimension d_{latent} . In the signal coordinates we sample a Gaussian mixture with $k = 10$ centers of norm 3 and unit variance around each center ($\sigma_{\text{sig}} = 1$); the remaining $d_{\text{latent}} - d_{\text{intrinsic}}$ coordinates receive independent null noise with $\sigma_{\perp} = 0.5$; the full vector is then rotated by a random orthogonal matrix Q so that no coordinate is privileged. The training set has $n = 500$ points. We sweep $d_{\text{latent}} \in \{5, 8, 10, 12, 15, 20, 25, 30, 35, 40\}$ with everything else fixed, so $d_{\text{latent}} = 5$ has no excess directions and the population score is known exactly at every d_{latent} .

Frozen-feature model. The random-feature network (RFNN) of Bonnaire et al. (2025) is

$$s_A(x_t) = \frac{1}{\sqrt{p}} A \tanh\left(\frac{1}{\sqrt{d_{\text{latent}}}} W x_t\right), \quad (1)$$

with $W \in \mathbb{R}^{p \times d_{\text{latent}}}$ drawn i.i.d. from $\mathcal{N}(0, 1)$ and frozen, and a readout $A \in \mathbb{R}^{d_{\text{latent}} \times p}$ trained by score matching at a single time $t = 0.01$. With $\phi(x_t) = p^{-1/2} \tanh(d_{\text{latent}}^{-1/2} W x_t)$, the empirical feature correlation matrix is

$$U = \frac{1}{n} \sum_{\mu=1}^n \mathbb{E}_{\xi} [\phi(x_t^{\mu}) \phi(x_t^{\mu})^{\top}], \quad (2)$$

where x_t^{μ} is the diffused training point μ and ξ the diffusion noise. Full-batch gradient flow on A is linear, and in the eigenbasis of U each mode decays independently,

$$a_i(T) - a_i^* = (a_i(0) - a_i^*) e^{-\lambda_i T}, \quad (3)$$

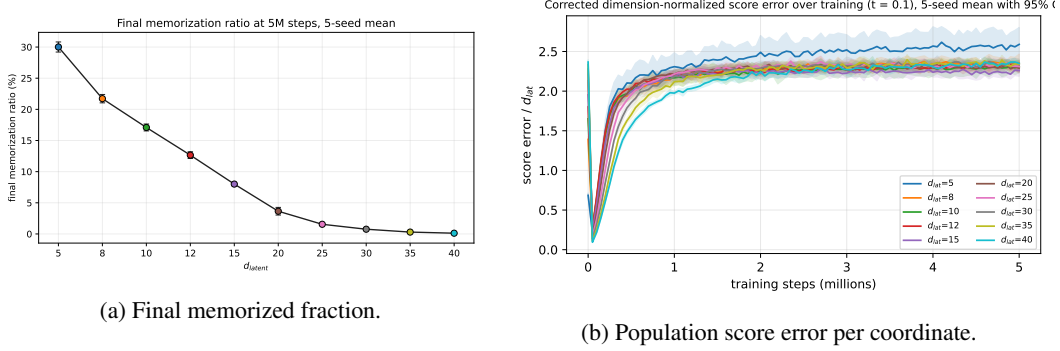


Figure 1: **Widening the latent space delays memorization.** Fixed-width MLP, $d_{\text{intrinsic}} = 5$, five seeds, 5M steps. (a) Final memorized fraction falls from 30% at $d_{\text{latent}} = 5$ to 0.1% at $d_{\text{latent}} = 40$. (b) Population score error at $t = 0.1$ per coordinate against the exact GMM score on held-out data. Every width reaches its minimum by 50k steps and then rises as the network fits its training set; the rise starts later at larger d_{latent} , and by 5M steps all widths have reached a similar level. The panel shows a delay, not a preserved score.

We estimate U by averaging K noise draws, denoted $U^{(K)}$; only $K = 1$ guarantees rank at most n (Appendix I.1.1). Thus eigenvalue λ_i is the learning rate of feature direction i and all modes are absorbed in parallel: no mode waits for another.

Trainable model. The trainable score network is an MLP with three hidden layers of width 256, GELU activations, and sinusoidal time features concatenated to x_t , trained by score matching over diffusion times with Adam at learning rate 10^{-4} , batch size 256, for 5M steps and five seeds. The hidden width is 256 at every d_{latent} , so the sweep changes only the input and output dimension and the parameter count stays roughly constant; a width-scaled sweep would be a capacity sweep.

Memorization metric. A generated sample $\hat{x}$ is counted as memorized when

$$\frac{\|\hat{x} - \text{NN}_1\|}{\|\hat{x} - \text{NN}_2\|} < \frac{1}{3},$$

where NN_1 and NN_2 are its nearest and second-nearest *training* points, the criterion of Bonnaire et al. (2025). We report the memorized fraction of generated samples, in latent space for synthetic data and in pixel space after decoding for images, as a mean over seeds.

Dimension-controlled variant. The same ratio is also computed after projecting generated and training points onto the true $d_{\text{intrinsic}} = 5$ signal subspace. This is needed because a sampler that copies a training point’s signal coordinates and redraws its null coordinates is invisible to the full-space test for $d_{\text{latent}} \geq 10$: the redrawn null coordinates dominate both distances and hide the copy (Table 1). The projected test cannot lose sensitivity as d_{latent} grows, so a decline that survives it is a real delay.

3 WIDENING THE LATENT SPACE DELAYS MEMORIZATION

Figure 1(a) shows the final memorized fraction after 5M steps as the latent dimension is swept from $d_{\text{latent}} = 5$, equal to the intrinsic dimension, to $d_{\text{latent}} = 40$ at a fixed hidden width of 256, five seeds per width (setup in Section 2). Once d_{latent} exceeds $d_{\text{intrinsic}}$ the fraction falls monotonically: 30.0, 21.7, 17.1, 12.7, 8.0, 3.7, 1.6, 0.8, 0.3 and 0.1% at $d_{\text{latent}} = 5, 8, 10, 12, 15, 20, 25, 30, 35, 40$. The added coordinates carry only isotropic noise, so the signal the network has to learn is the same at every width; what changes is how quickly it fits its training set. Holding the width fixed matters: scaling the hidden width with d_{latent} turns the sweep into a capacity sweep and no longer isolates the latent dimension.

The same delay is visible in score space, where the synthetic setting provides an exact target. Figure 1(b) reports the population score error at $t = 0.1$ per coordinate, against the exact GMM

Table 1: Dimension-controlled memorization test (5M steps, five seeds, 5000 samples each). The ratio test of Appendix B is applied in the full latent space and after projecting generated and training points onto the $d_{\text{intrinsic}} = 5$ signal coordinates. Signal copies reproduce a training point’s signal coordinates and redraw its null coordinates. α is the alignment of the learned score with the empirical score of the training set (signal coordinates, $t = 0.1$). Full version: Table 2.

	$d_{\text{latent}} = 5$	10	20	40
model samples, full-space test	29.9%	16.9%	3.6%	0.1%
model samples, signal-projected test	29.9%	16.8%	5.8%	1.2%
fresh draws, signal-projected test	0.4%	0.4%	0.4%	0.4%
signal copies, full-space test	57%	0.8%	0.0%	0.0%
signal copies, signal-projected test	57%	57%	58%	58%
α at 5M steps	0.75	0.54	0.39	0.31

score on held-out data. Every width reaches its minimum by the first evaluation at 50k steps and then rises as the network fits its training set. The rise starts later at larger d_{latent} : the error crosses 2.0 per coordinate at 0.35M steps for $d_{\text{latent}} = 5$ and at 1.05M for $d_{\text{latent}} = 40$, the same ordering as the memorization onset in panel (a). At 5M steps the error is 2.59 per coordinate at $d_{\text{latent}} = 5$ and 2.26–2.35 for $d_{\text{latent}} = 8$ –40: final errors are comparable while the timing differs. Extending the same fixed-width sweep to $d_{\text{latent}} = 240$ (Figure 17) makes the delay monotone over the full range. Measured as the time for the error to double from its minimum, it is at most 100k steps for $d_{\text{latent}} \leq 40$, 200k at $d_{\text{latent}} = 60$, 500k at 80, 800k at 100, 1.55M at 120, 2.5M at 140, and is not reached within 5M steps for $d_{\text{latent}} \geq 160$; beyond $d_{\text{latent}} \approx 60$ the per-coordinate error has not returned to its small- d_{latent} plateau by the horizon, ending at 2.59 for $d_{\text{latent}} = 5$ but 0.53 for $d_{\text{latent}} = 240$. The delay exceeds $25\times$ across the sweep.

The decline is not a metric artifact. A nearest-neighbour ratio test can lose sensitivity as the dimension grows, so the decline in panel (a) could in principle reflect the test rather than the model. Table 1 rules this out. A sampler that reproduced a training point’s signal coordinates while redrawing its null coordinates would be invisible to the full-space test for $d_{\text{latent}} \geq 10$: such signal copies are detected at 57% at $d_{\text{latent}} = 5$ but at 0.8% and 0.0% beyond. Repeating the test after projecting generated and training points onto the true $d_{\text{intrinsic}}$ -dimensional signal subspace, where signal copies are detected at every width, gives 29.9%, 16.8%, 5.8% and 1.2% for the model samples at $d_{\text{latent}} = 5, 10, 20, 40$, against a 0.4% chance level from fresh population draws. The models do not make signal-only copies: the decline is $25\times$ in the signal subspace, and the full-space test overstates it only mildly (3.6 against 5.8% at $d_{\text{latent}} = 20$, 0.1 against 1.2% at 40). The full table, including the exact-copy control, is Table 2.

Alignment with the empirical score. Empirical-score alignment provides a threshold-free second readout. Let s^* be the population score, s_{emp} the exact score of the empirical distribution of the n training points, and s_{θ} the learned score, all in the signal coordinates at $t = 0.1$. The alignment coefficient

$$\alpha = \frac{\langle s_{\theta} - s^*, s_{\text{emp}} - s^* \rangle}{\|s_{\text{emp}} - s^*\|^2} \quad (4)$$

is 0 for the population score and 1 for an exact fit of the training set. It reaches 0.75, 0.54, 0.39 and 0.31 at 5M steps for $d_{\text{latent}} = 5, 10, 20, 40$, and it passes 0.3 more than $30\times$ later at $d_{\text{latent}} = 40$ than at $d_{\text{latent}} = 5$. This is consistent with comparable final score errors in Figure 1(b) while memorization differs $25\times$: the per-coordinate gap $\|s_{\text{emp}} - s^*\|^2$ grows $3.6\times$ from $d_{\text{latent}} = 5$ to 40, so a smaller α can still incur substantial population error.

Two independent readouts, one of them threshold-free, agree that widening the latent space delays memorization; Section 4 asks what a solvable model says about why, and Section 5 tests the phenomenon on real data.

4 A SPECTRAL EXPLANATION IN FROZEN FEATURES

The frozen-feature score model of Section 2 has exactly solvable gradient-flow dynamics given U : each mode has an inverse-eigenvalue timescale. We use measured spectra and a conditional edge approximation to investigate the delay in Section 3. The dynamics are exact; the four-group interpretation and its connection to memorization require additional assumptions (Appendix I). Transfer to trained networks is tested separately in Section 5.

Parallel mode dynamics. By (3), the readout coefficient along the i -th eigenvector of U relaxes to its target as $1 - e^{-\lambda_i T}$, independently of every other direction (Lemma I.3). Each eigenvalue is a learning rate for one direction, and sorting the spectrum sorts the order of absorption. Writing S for the $d_{\text{intrinsic}}$ signal modes and $\mathcal{M}$ for the $n - d_{\text{latent}}$ index-defined sample modes (Remark 4.1 below), define characteristic edge-absorption times at a common relative parameter tolerance ε by

$$\tau_{\text{gen}} := \log(1/\varepsilon)/\lambda_{\min}^S, \quad (5)$$

$$\tau_{\text{mem}}^{\text{on}} := \log(1/\varepsilon)/\lambda_{\max}^{\mathcal{M}}. \quad (6)$$

These are spectral proxies: τ_{gen} tracks the slowest signal mode and $\tau_{\text{mem}}^{\text{on}}$ the fastest index-defined sample-block mode. That block can contain population-aligned spikes, so its edge need not mark observed memorization onset. Relating it to the train–test gap requires the alignment assumptions and spike/bulk distinction of Proposition I.10 and Definition I.11. Their ratio is independent of the common tolerance and clock normalization.

Four blocks and their edges. Let $M_t = \frac{1}{n} \sum_{\mu} \mathbb{E}[x_t^{\mu} x_t^{\mu \top}]$ be the second moment of the noised training set, with population signal and null eigenvalues $\alpha_t^2 = e^{-2t}(s^2/d_{\text{intrinsic}} + \sigma_{\text{sig}}^2) + \Delta_t$ and $\beta_t^2 = e^{-2t}\sigma_{\perp}^2 + \Delta_t$, and define the operating point

$$q(d_{\text{latent}}) := \frac{\mathbb{E}_{\text{data}} \text{tr } M_t}{d_{\text{latent}}} = \beta_t^2 + \frac{d_{\text{intrinsic}}(\alpha_t^2 - \beta_t^2)}{d_{\text{latent}}}, \quad (7)$$

the input variance per coordinate seen by $\tanh$ (Lemma I.6). Our homogeneous Gaussian surrogate replaces $\tanh(\sqrt{q}z)$ by its linear part plus an independent residual, with coefficients $a_1(q) = \mathbb{E}[\tanh'(\sqrt{q}z)]$ and $a_*(q)^2 = \mathbb{E}[\tanh(\sqrt{q}z)^2] - a_1(q)^2 q$; both are evaluated at $q(d_{\text{latent}})$, so nothing below is d_{latent} -independent.

Remark 4.1 (Index groups and approximate scales). For $d_{\text{intrinsic}} < d_{\text{latent}} < n < p$, partition the measured spectrum of $U^{(K)}$ at indices $d_{\text{intrinsic}}, d_{\text{latent}}, n$. In separated configurations we interpret these groups as:

- (i) $d_{\text{intrinsic}}$ signal modes, with scale $a_1(q)^2 \alpha_t^2 / d_{\text{latent}}$;
- (ii) $d_{\text{latent}} - d_{\text{intrinsic}}$ noise-dimension modes, with scale $a_1(q)^2 \beta_t^2 / d_{\text{latent}}$;
- (iii) $n - d_{\text{latent}}$ sample modes, with measured top edge $\kappa_{\text{top}} a_*(q)^2 / n$;
- (iv) $p - n$ tail modes, which are zero for a full-row-rank single-copy feature matrix but need not vanish after noise averaging.

These are index counts, not a theorem identifying four spectral components of U . The approximate scales require the Gaussian-surrogate and separation assumptions of Assumption I.14; signal and sample labels additionally require alignment. In the homogeneous surrogate, the Hermite coefficients satisfy

$$a_1(q)^2 q + a_*(q)^2 = \mathbb{E}[\tanh(\sqrt{q}z)^2]. \quad (8)$$

This identity does not establish individual block masses for fixed W .

The rank bound is exact only for $U^{(1)}$. Released spectra average $K = 50$ draws and can be full rank; their tail need not be separated from the sample group. The noise-dimension/sample gap is 0.6–1.3 decades in the reported $\sigma_{\perp} = 0.5$ configurations and nearly closes at $\sigma_{\perp} = 0.01$ (Figure 5). The approximate data-side band and its finite-width corrections are detailed in Appendix I.2.

Bonnaire’s noise-averaged asymptotic floor has weight $1 - (n + d_{\text{latent}})/p$ in its stated regime, unlike the single-copy rank deficit $1 - n/p$; its covariance argument is the clean covariance, not $\{\alpha_t^2, \beta_t^2\}$

after diffusion (Bonnaire et al., 2025). George’s subspace analysis uses a five-block pencil (George et al., 2026). Neither result directly establishes this index partition for a finite-rank Gaussian mixture; Appendix I.1.1 records the distinction.

Spectral separation as a ratio of two edges. Dividing the two absorption times, and inserting the edges of Remark 4.1 with $\ell_{\min}$ the smallest signal-subspace eigenvalue of the realised training set (Propositions I.26 and I.27),

$$R(d_{\text{latent}}) := \frac{\tau_{\text{mem}}^{\text{on}}}{\tau_{\text{gen}}} = \frac{\lambda_{\min}^S}{\lambda_{\max}^M} \simeq \frac{n}{d_{\text{latent}}} \cdot \frac{a_1(q)^2}{a_*(q)^2} \cdot \frac{e^{-2t\ell_{\min}} + \Delta_t}{\kappa_{\text{top}}}, \quad (9)$$

with $q = q(d_{\text{latent}})$ (Corollary I.28); the first equality is exact, whereas the last uses the edge approximation and a measured κ_{top} . R is *not* a count of modes: the $d_{\text{latent}} - d_{\text{intrinsic}}$ noise-dimension modes multiply no absorption time, because a bound of the form (number of buffer modes) $\times$ (time per mode) presumes that the flow traverses them sequentially, whereas under (3) they decay in parallel with the rest of the spectrum and enter R only through $q(d_{\text{latent}})$, κ_{top} and $\ell_{\min}$ (Remark I.31). d_{latent} enters through the fan-in $1/d_{\text{latent}}$ of the data-side edges and through $q(d_{\text{latent}})$, while $\ell_{\min}$ and κ_{top} also enter; growth of a_1^2/a_*^2 alone is not sufficient to guarantee increasing R . On the released spectra τ_{gen} grows $11\times$ in flow units over $d_{\text{latent}} = 5 \rightarrow 200$ (also $11\times$ in default repository steps, $788 \rightarrow 8663$), $\tau_{\text{mem}}^{\text{on}}$ grows $221\times$, and R grows monotonically from 21 to 428; predicted-versus-measured edges at every d_{latent} are in Appendix I.

The operating point, and a fixed- q control. What moves the edges is the de-saturation of $\tanh$. Over the sweep q falls from 2.76 to 0.33, so a_1^2 rises from 0.18 to 0.63 and a_*^2 falls $17\times$ ($0.079 \rightarrow 0.0047$): the ratio a_1^2/a_*^2 grows $58\times$ and the sample-block top κ_{top} collapses $13\times$, while n/d_{latent} shrinks $40\times$. Holding q fixed removes the variation in the Hermite coefficients; the remaining dimension dependence tests their contribution to the growth of R . We hold the input variance per coordinate $q_{\text{in}} = \mathbb{E}\|x\|^2/d_{\text{latent}}$ fixed by scaling $\sigma_{\perp}$ with d_{latent} ($\sigma_{\perp}^2 = (0.57d_{\text{latent}} - 14)/(d_{\text{latent}} - d_{\text{intrinsic}})$, coinciding with the $\sigma_{\perp} = 0.5$ sweep at $d_{\text{latent}} = 40$), and re-measure the spectrum at $n = 500$, $p = 4096$, $t = 0.01$ over three seeds (Remark I.34; Figure 2). The intervention largely removes the endpoint growth, although the fixed- q curve is non-monotone: R changes $1.09\times$ over $d_{\text{latent}} = 40 \rightarrow 240$ (92, 140, 158, 136, 119, 100) against $3.17\times$ in the baseline (92, 145, 201, 266, 287, 291), while the noise-dimension count grows $6.7\times$. The decomposition is consistent with (9): $\lambda_{\min}^S$ falls $4.3\text{--}4.4\times$ in both sweeps (the $1/d_{\text{latent}}$ of the signal edge), whereas the sample top falls $13.6\times$ in the baseline and only $4.8\times$ at fixed q , the missing factor being the shrinkage of $a_*(q)^2$. The isotropic control $\sigma_{\perp} = \sigma_{\text{sig}}$, where de-saturation is essentially absent, gives a non-monotone R that peaks near $d_{\text{latent}} \approx 40$ and then declines like n/d_{latent} (Remark I.33).

What the model says. Inside the frozen-feature model, lowering the operating point q contributes to separating signal and sample-block absorption times, as supported by the fixed- q control; the exactly solvable linear score model, the bridge from residual error to near-duplicate generation and the regime boundaries are in Appendix I. Trainable first layers and adaptive optimizers remain outside this model; Section 5 tests whether its explanation transfers to trained networks.

5 IMAGE EXPERIMENTS AND THE LIMITS OF TRANSFER

Setup. For each dataset (CelebA and CIFAR-10 at 32×32) and each d_{latent} we train a convolutional VAE on the full training split, freeze it, and train a depth-five MLP score model (hidden width 1024, sinusoidal time features, so parameter count is roughly constant across widths) on the encoder means of a farthest-first diverse subset of 1k training images, with SGD (learning rate 10^{-3} , momentum 0.8), batch size 512, 5M steps and five seeds (Appendix B). Memorization is the pixel-space nearest-neighbour ratio of 10k decoded samples against the 1k subset, evaluated every 100k steps; FID uses the same reference. A narrow VAE erases sample identity before the score model can copy it: the smallest width whose own reconstructions pass the memorization test on 95% of the subset is ≈ 70 on CelebA and ≈ 140 on CIFAR-10, and only widths above this identity threshold test the width effect.

Result. Post-identity-threshold width comparisons show longer times to memorization. The mean first evaluation step at which 5% of samples are memorized rises from 0.32M at $d_{\text{latent}} = 80$ to 1.16M

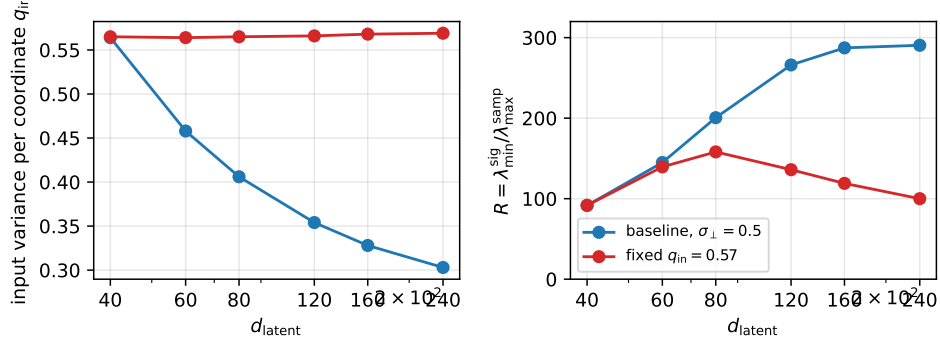


Figure 2: **Fixed- q control.** Left: the per-coordinate input variance $q_{in} = \mathbb{E}\|x\|^2/d_{latent}$ along the $\sigma_{\perp} = 0.5$ sweep (blue) and along a sweep in which $\sigma_{\perp}$ is scaled with d_{latent} to hold q_{in} at its $d_{latent} = 40$ value (red). Right: the edge ratio $R = \lambda_{min}^{sig} / \lambda_{max}^{smp}$ of the RFNN feature-correlation matrix ($n = 500, p = 4096, t = 0.01, K = 40$ noise draws, three seeds). Holding the operating point fixed largely removes the endpoint growth of R ($1.09\times$ against $3.17\times$ over $d_{latent} = 40 \rightarrow 240$) while the noise-dimension count grows $6.7\times$. Representative spectra with the four blocks are in Figure 5.

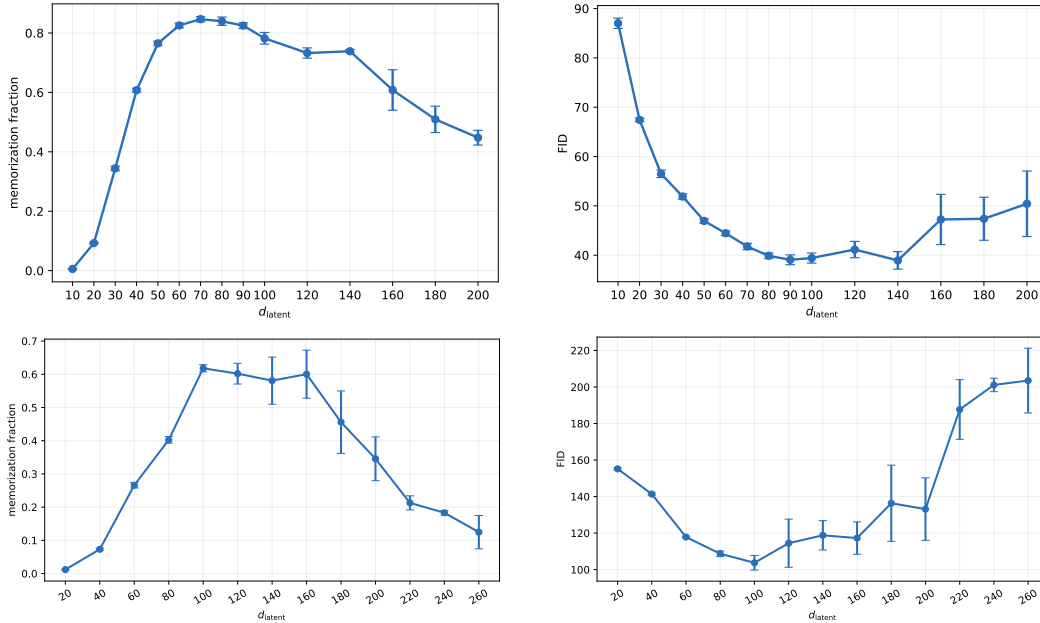


Figure 3: Final memorized fraction (left) and FID (right) after 5M steps against latent width, CelebA (top) and CIFAR-10 (bottom); five-seed means with 95% confidence intervals. Below the identity threshold the VAE hides memorization; above it, memorization falls with width while FID first improves and then worsens.

at $d_{latent} = 200$ on CelebA ($3.6\times$), and from 1.04M at $d_{latent} = 160$ to 1.50M at $d_{latent} = 240$ on CIFAR-10 ($1.4\times$; Table 7). Both comparisons start above the identity threshold; the CIFAR-10 trend is not strictly monotone. The final fraction after 5M steps peaks near the identity threshold ($d_{latent} = 70\text{--}80$ on CelebA) and falls by roughly half by $d_{latent} = 200$; on CIFAR-10 it stays high through $d_{latent} = 100\text{--}160$ and then declines. FID improves as the bottleneck is relieved, is best at intermediate widths, and worsens again at the widest latents (CelebA: 38.9 at $d_{latent} = 140$ to 47–50 at $d_{latent} = 160\text{--}200$), so width trades quality against memorization rather than buying both (Figure 3).

What carries over from the frozen-feature model. The mechanism of Section 4 acts through the operating point $q_d = \text{tr } M_t / d_{\text{latent}}$, the pre-activation variance of the isotropic frozen-feature surrogate. On real latents the nominal width is not the axis the model sees. Beyond $d_{\text{latent}} \sim 140$ on CelebA and ~ 160 on CIFAR-10 the coordinates the VAE adds are posterior-collapsed: their eigenvalues sit at the diffusion floor Δ_t (at least 15% of coordinates for CelebA $d_{\text{latent}} \geq 180$), the trace of the latent covariance is flat (252.5, 255.2, 260.4 at $d_{\text{latent}} = 160, 180, 200$), and the effective dimension saturates (89.5, 90.3, 87.9). The surrogate summarizes this change through q_d , which falls from 1.473 to 1.247 over CelebA $160 \rightarrow 200$. Reading the surrogate at these values gives the right sign, longer hitting times and smaller final fractions, and accounts for 0.55–0.82 of the log-slowdown of CelebA hitting times over $160 \rightarrow 200$ and 0.46 on CIFAR-10 over $160 \rightarrow 260$, but only 0.28 and 0.15 of the log-decline of the final fractions (Appendix I). The remainder is outside the surrogate.

What does not. Two boundaries. (i) The mechanism is verified in the random-feature network but not shown to be what delays memorization in trained networks. In the synthetic MLPs of Section 3, the first-layer mean-square pre-activation q_M is similar at 50k steps (0.71 at $d_{\text{latent}} = 5$, 0.68 at $d_{\text{latent}} = 40$); it separates later, growing with memorization at small width: $q_M(40)/q_M(5) = 0.96, 0.70, 0.47, 0.30$ at 0.05M, 0.25M, 1M and 5M steps (five seeds; protocol in Appendix D), while the memorized fraction at $d_{\text{latent}} = 5$ rises from 0.6% to 29.9%. These measurements do not establish a width-set difference in q_M preceding memorization. A candidate explanation consistent with the same data sits on the target side: the per-coordinate gap between the empirical and population scores grows $3.6\times$ (3.7 to 13.2) from $d_{\text{latent}} = 5$ to 40, so a memorizing network must fit a sharper target and grow its first-layer scale to do so. We state this as a hypothesis. (ii) The effect is not architecture-independent. In a CelebA pilot with a spatial $4 \times 4 \times d_{\text{latent}}$ VAE and a DiT-style score network, memorization rises with channel count, from about 22% at $d_{\text{latent}} = 5$ to roughly 48% by $d_{\text{latent}} = 8\text{--}10$, and the broad sweep shows no clean decline at large widths (Figures 28 and 30). Channel count there changes sixteen coordinates at once together with their spatial covariance, so it is not the same control; but the fixed-width MLP result cannot be assumed to transfer.

Predicting onset from the frozen latent spectrum. The appendix (Appendix I) turns the sample-mode eigenvalue of the frozen VAE latent spectrum into a one-constant clock for the memorization onset, $\tau_f(d_{\text{latent}}) = Cn/g(q_d, 1 - \Delta_t/q_d)$ with $q_d = \text{tr } M_t / d_{\text{latent}}$. Fit on all widths but one and evaluated on the held-out width, its log error is 0.41 on CelebA ($d_{\text{latent}} \geq 50$) and 0.15 on CIFAR-10 ($d_{\text{latent}} \geq 60$), comparable to the seven-parameter in-sample fit of the released predictor (0.34 and 0.15); but a clock that depends on d_{latent} alone reaches 0.13 on CelebA, the fitted constant differs by $2\times$ between datasets, and over the broader CelebA $50 \rightarrow 200$ sweep it captures only $1.78\times$ of the $3.9\times$ slowdown. These diagnostic fits include widths below the identity threshold. We therefore report it as a diagnostic, not a contribution.

6 DISCUSSION AND LIMITATIONS

What is established. At fixed hidden width, widening the latent space delays memorization: the synthetic memorized fraction falls from 30% to near zero over $d_{\text{latent}} = 5 \rightarrow 40$, the decline survives the signal-projected test (Table 2), and post-identity-threshold comparisons give $3.6\times$ and $1.4\times$ increases in the mean 5% hitting time on CelebA and CIFAR-10. In the frozen-feature model the measured spectral separation grows as q falls, and the fixed- q control (Remark I.34) supports an operating-point explanation rather than a mode-count explanation.

Theoretical scope. The mode dynamics are exact given the feature operator; the four-group edge approximation is not a direct specialization of prior spectral theorems. Noise averaging, finite-rank outliers and spectral alignment remain additional requirements. The empirical edge ratios and fixed- q comparison do not depend on proving those requirements.

The causal gap. Whether this mechanism operates in trained networks is the main open question. In the synthetic MLPs the first-layer mean-square pre-activation is similar at 50k steps ($q_M(40)/q_M(5) = 0.96$); it separates later, growing together with memorization at small d_{latent} (ratio 0.30 at 5M steps). Establishing whether a width-set operating point causes the MLP delay remains an open question after these observations. On real data the surrogate only partly captures the slowdown, and its fitted clock does not transfer across datasets with a single constant.

Boundaries. The spatial-DiT pilot on CelebA, with a $4 \times 4 \times d_{\text{latent}}$ latent grid, does not show the decline: memorization stays high across channel counts (Figure 28), so the effect is established only for vector latents with MLP denoisers. All sweeps use $n \leq 1000$ training points, and a sweep that scales width with d_{latent} is a capacity sweep. What would close the gap is a trained-network quantity that is set by width before memorization begins and predicts its onset; a candidate consistent with the data is the per-coordinate gap between the empirical and population scores, which grows $3.6\times$ from $d_{\text{latent}} = 5$ to 40.

AI DISCLOSURE

We used large language model (LLM) assistants, including agentic coding tools, throughout this project, and disclose those uses here.

Theoretical modeling and proof formulation. The statements and proofs of Section 4 and Appendix I were developed with substantial LLM assistance. An LLM-based audit of an earlier, human-written version of the theory identified errors in it (an incorrect scaling of the spectral edges and an invalid bound on the memorization delay); corrected statements and derivations were then drafted by LLM assistants, criticized by separate LLM reviewer passes, and checked against independent numerical simulations of the random-feature spectrum. Each statement in Appendix I is labeled *proved*, *sketched*, or *conjectured*. **AUTHORS: state what you verified by hand, e.g. “The authors verified every statement on which the main text relies.” Do not submit until this sentence is true.**

Code, experiments and data analysis. LLM assistants wrote analysis and plotting scripts, located and fixed an error in the analytic score used for the synthetic score-error metric, recovered the corrected metric from logged held-out losses, and designed and ran the control experiments of Remarks I.33 and I.34 and Table 2. **AUTHORS: state how the training code for the main synthetic and real-data sweeps was written.**

Writing and literature. LLM assistants drafted and edited text throughout the paper, checked the consistency of captions and reported numbers against the underlying data, and located bibliographic entries, each of which was checked against its primary source.

The authors reviewed the AI-assisted content and take full responsibility for the paper.

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A APPENDIX ORGANIZATION

The appendix follows the same experimental progression as the main paper. We begin with RFNN spectra, which isolate the kernel-level four-bulk mechanism under controlled synthetic data. We then turn to trainable synthetic MLPs, where the same latent-buffer idea is measured through memorization fraction and score error over optimization. Finally, we give the VAE and latent-diffusion diagnostics for CelebA and CIFAR-10, including reconstruction checks, FID, per-dimension memorization trajectories, a spatial-DiT pilot, and the frozen-latent spectrum memorization predictor.

B EXPERIMENTAL DETAILS

Synthetic latent data. The synthetic experiments draw a low-dimensional signal subspace with dimension $d_{\text{intrinsic}}$ and embed it in an observed latent space of dimension d_{latent} . Unless noted otherwise, the signal coordinates are sampled from an anisotropic Gaussian mixture and the remaining coordinates receive independent null noise. The nonlinear random-feature experiments use the same data design but study the spectrum of the feature kernel rather than generated samples. The trainable synthetic MLP experiments use the analytic Gaussian score as the target, which is why we report score error rather than image FID.

Synthetic experiment details. For Experiment 1, we fix $d_{\text{intrinsic}} = 5$ and sweep d_{latent} , so the exact null-buffer size is $d_{\text{latent}} - d_{\text{intrinsic}}$. For Experiment 2, we fix the observed latent space at $d_{\text{latent}} = 20$ and sweep $d_{\text{intrinsic}}$, so the exact null-buffer size decreases until it vanishes at $d_{\text{intrinsic}} = d_{\text{latent}}$. The default RFNN data are anisotropic GMM samples with $\sigma_{\text{signal}} = 1.0$, $\sigma_{\perp} = 0.5$, center scale 3.0, $n = 500$, diffusion time $t = 0.01$, and K Monte Carlo diffusion-noise samples used to estimate U ($K = 50$ for the $p = 64d_{\text{latent}}$ raw-data spectra, $K = 500$ for the $p = d_{\text{latent}} + n + 300$ set; see each run’s `config.json`). The K -draw average has full rank p with a soft shoulder at index n ; the single-draw matrix has exactly $p - n$ zero eigenvalues. The Gaussian robustness panels remove mixture centers and either sweep $\sigma_{\perp} \in \{0.1, 0.3, 0.5\}$ at fixed $\sigma_{\text{signal}} = 1.0$, or sweep $\sigma_{\text{signal}} \in \{0.5, 1.0, 1.5\}$ at fixed $\sigma_{\perp} = 0.3$. The synthetic MLP uses the same $d_{\text{intrinsic}} = 5$, $n = 500$, $\sigma_{\perp} = 0.5$ latent-distribution family as the main RFNN sweep, trains a fixed-hidden-width score MLP for 5M steps over seeds 42–46, and evaluates sample recall by the nearest-neighbor ratio on generated synthetic samples.

Width controls. The RFNN spectra are reported under three width rules: $p = 64d_{\text{latent}}$, $p = d_{\text{latent}} + n + 300$, and fixed $p = 1800$ with fixed total null energy. The first setting checks a high-feature-count regime. The second grows width just enough to cover the n data-supported directions plus a $d_{\text{latent}} + 300$ margin, so $\psi_p = p/d_{\text{latent}}$ approaches 1 at large d_{latent} and the noise-dimension block broadens accordingly. The fixed-width/fixed-energy control removes the two main confounds in the d_{latent} sweep: increasing model width and increasing total injected null variance.

Memorization metric. A generated sample $\hat{x}$ is counted as memorized when the ratio of its distances to its nearest and second-nearest training points, $\|\hat{x} - \text{NN}_1\|/\|\hat{x} - \text{NN}_2\|$, is below $1/3$, the nearest-neighbor ratio criterion of Bonnaire et al. (2025). Synthetic experiments apply it in latent space; image experiments apply it in pixel space to decoded samples. The relevant comparison set is the training subset used by the diffusion model, not the full VAE training set. Reported curves are the mean across five seeds with confidence bands unless the caption states that the figure is a deterministic VAE diagnostic.

VAE reconstruction memorization as an intrinsic-dimension proxy. Before training diffusion, we pass the same diverse 1k subset through each frozen VAE and apply the pixel-space nearest-neighbor ratio to the immediate reconstructions. This is not diffusion memorization. It measures whether the VAE bottleneck preserves enough sample identity for any downstream pixel-space memorization test to be meaningful. We use the rise and plateau of this reconstruction-memorization curve as a practical proxy for the dataset’s effective intrinsic-dimensionality threshold: below that point the latent code is capacity-limited, while above it increasing d_{latent} mostly adds excess buffer dimensions.

Real-data VAEs. CelebA and CIFAR-10 experiments first train one VAE per latent dimension on the full available training split, not on the 1k diffusion subset. The VAE is a five-stage convolutional encoder/decoder with channel widths (32, 64, 128, 256, 512). The final checkpoints use the ResNet-style variant: stride-two convolutional downsampling, GroupNorm/SiLU residual blocks, transpose-convolution upsampling, and a Tanh output. CelebA uses 21k training and 6k validation images from the local CelebA-HQ tar split at 32×32 grayscale resolution; CIFAR-10 uses the standard 50k/10k train/test split at 32×32 RGB resolution. The objective combines reconstruction with a KL penalty to the standard normal prior. In the capacity-loss form used by the final checkpoints, $\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \gamma|\text{KL} - C(t)|$, with $C(t)$ warmed up early in training; the logs also record β , reconstruction MSE/L1, KL, learning-rate reductions, and early stopping. This is why the appendix reports VAE reconstruction memorization before any diffusion result: if encode/decode alone cannot preserve sample identity, a pixel-space memorization test on generated samples is not yet meaningful.

Real-data latent diffusion. After selecting the best VAE checkpoint by validation objective, we encode the same farthest-first diverse 1k subset for each dataset and train a latent-space MLP score model on those codes. The final reported runs use a depth-five MLP with hidden width 1024, sinusoidal time features, SGD with learning rate 10^{-3} and momentum 0.80, batch size 512, and a 5M-step horizon over five seeds. The hidden width is fixed across d_{latent} , so increasing the latent dimension changes the input/output dimension and the amount of latent buffer, but not the dominant hidden-layer capacity. Training loss is logged every 10k steps; memorization and FID are evaluated every 100k steps using 10k generated samples, with the same 1k training subset as the FID reference.

Spatial-DiT pilot. The appendix also includes a CelebA pilot that replaces the vector VAE/MLP pipeline with a spatial 4×4 VAE and an unconditional DiT-style score model. In that setting the plotted d_{latent} is a channel count, so a single step in d_{latent} changes the latent state by 16 scalar coordinates. This makes the sweep harder to tune and interpret than the vector-latent MLP experiments: changing the channel count changes the scalar latent dimension, the VAE decoder, the first and last DiT projections, and the second-order cross-position/cross-channel covariance structure that the transformer sees. The DiT pilot therefore tests whether the buffer idea leaves a trace in a more realistic image architecture, not whether the clean vector-latent theory transfers without modification.

Optimization failures and clipped diagnostics. Nonfinite runs are treated as failed optimization runs, not as evidence of zero memorization. Gradient clipping was useful diagnostically because it prevented some high-dimensional CIFAR-10 runs from producing NaNs, but it also suppressed memorization almost entirely. The clean CIFAR-10 aggregate therefore uses unclipped replacement seeds with the same SGD hyperparameters rather than mixing clipped and unclipped regimes.

C RFNN SPECTRAL DIAGNOSTICS

The RFNN experiments are spectral diagnostics, not sampled diffusion runs. They therefore do not have an image-level memorization percentage. Their purpose is to check whether the kernel spectrum separates into the predicted signal, latent-null, sample, and rank-null regions under the same data choices used by the trainable experiments.

Experiment 2, the source-dimensionality sweep, uses the updated synthetic parameters from the final rerun: d_{latent} is fixed at 20, $d_{\text{intrinsic}}$ ranges over $\{2, 5, 8, 12, 16, 20, 25, 30, 35, 40\}$, and the whole sweep is repeated under $p = 64d_{\text{latent}}$, $p = d_{\text{latent}} + n + 300$, and fixed- p width controls. This experiment is the reverse of Experiment 1, the main d_{latent} sweep. It asks whether the noise-dimension block shrinks when more of the fixed observed latent space is occupied by true source variation, and whether over-intrinsic source structure merely compresses into the fixed observed space rather than creating additional observed signal modes.

Feature-scale caveat for RFNN spectra. The nonlinear feature map is kept as tanh, but the data scale is chosen to avoid saturation. If the preactivations saturate, the predicted four-bulk structure can fragment into extra internal sub-bulks. These saturation-induced sub-bulks reflect the feature nonlinearity being driven outside its approximately linear regime, not an additional latent-buffer effect.

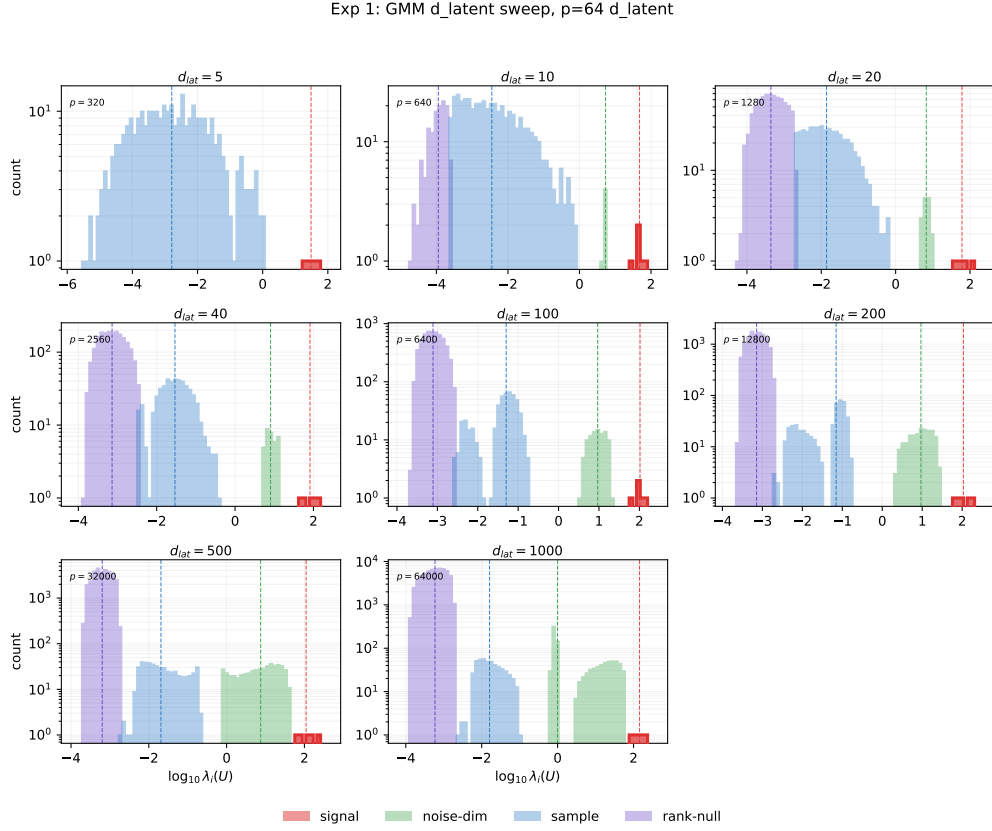


Figure 4: Experiment 1 RFNN four-bulk spectra for the latent-dimension sweep with $p = 64d_{\text{latent}}$. Here $d_{\text{intrinsic}} = 5$, $\sigma_{\text{signal}} = 1.0$, $\sigma_{\perp} = 0.5$, GMM center scale is 3.0, and the swept variable is $d_{\text{latent}} \in \{5, 10, 20, 40, 100, 200, 500, 1000\}$. Each panel plots the eigenvalue histogram after assigning eigenmodes to the signal, latent-null, sample, and rank-null index regions predicted by the theory. This highly overcomplete feature setting checks that the expected spectral ordering is visible when the RFNN has many more features as d_{latent} grows. Since the synthetic $d_{\text{intrinsic}} = 5$ is known exactly, the green block grows in the exact increasing-buffer regime $d_{\text{latent}} - d_{\text{intrinsic}}$, with little irregular variation beyond the deterministic change in block size. Colors are assigned by sorted eigenvalue-index blocks rather than fitted clusters: red denotes signal/source modes, green the observed latent-null/noise-dimension block, blue sample-specific modes, and purple rank-null modes. Dashed vertical lines mark the median log-eigenvalue of each colored block.

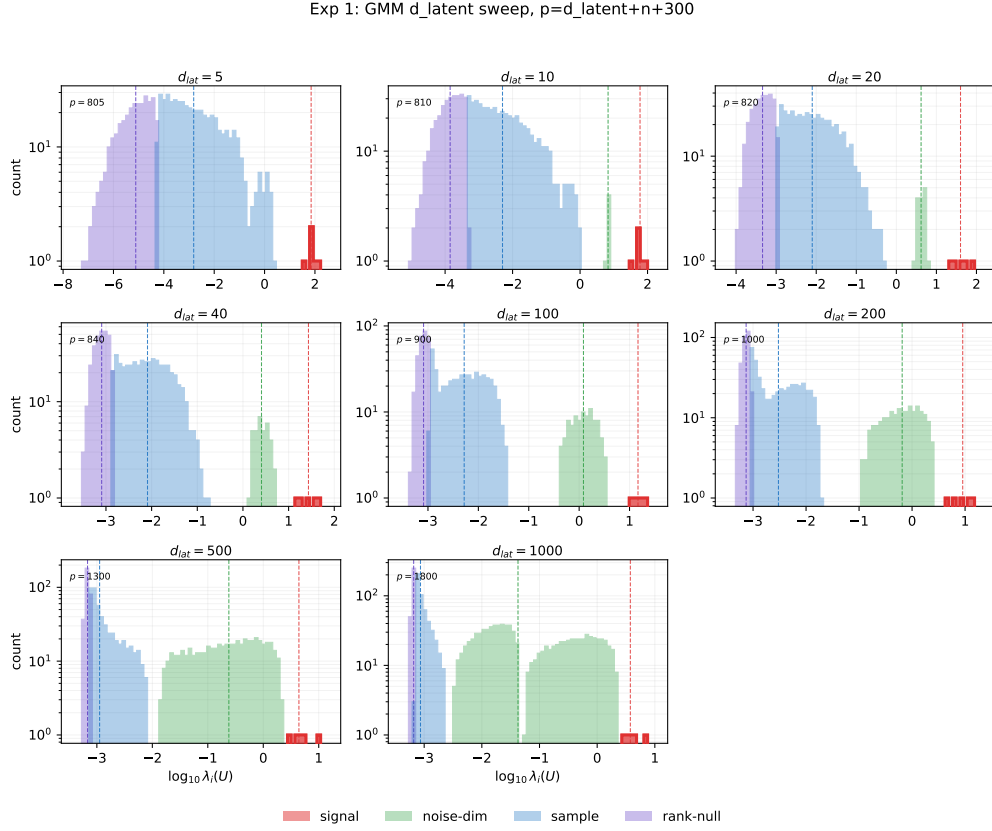


Figure 5: Experiment 1 RFNN four-bulk spectra for the latent-dimension sweep with $p = d_{\text{latent}} + n + 300$. The data sweep is the same as Figure 4: fixed $d_{\text{intrinsic}} = 5$, fixed $\sigma_{\perp} = 0.5$, and increasing d_{latent} . This width grows just enough to represent the latent, sample, and rank-null regions, so it checks that the four-bulk ordering does not require the much wider $p = 64d_{\text{latent}}$ feature map. The persistence of the same qualitative separation supports the interpretation that the effect comes from the latent-buffer geometry rather than from overcomplete RFNN width. The same near-monotone growth of the green block with the exact buffer size $d_{\text{latent}} - d_{\text{intrinsic}}$ appears even in this controlled-width setting. Colors are assigned by sorted eigenvalue-index blocks rather than fitted clusters: red denotes signal/source modes, green the observed latent-null/noise-dimension block, blue sample-specific modes, and purple rank-null modes. Dashed vertical lines mark the median log-eigenvalue of each colored block.

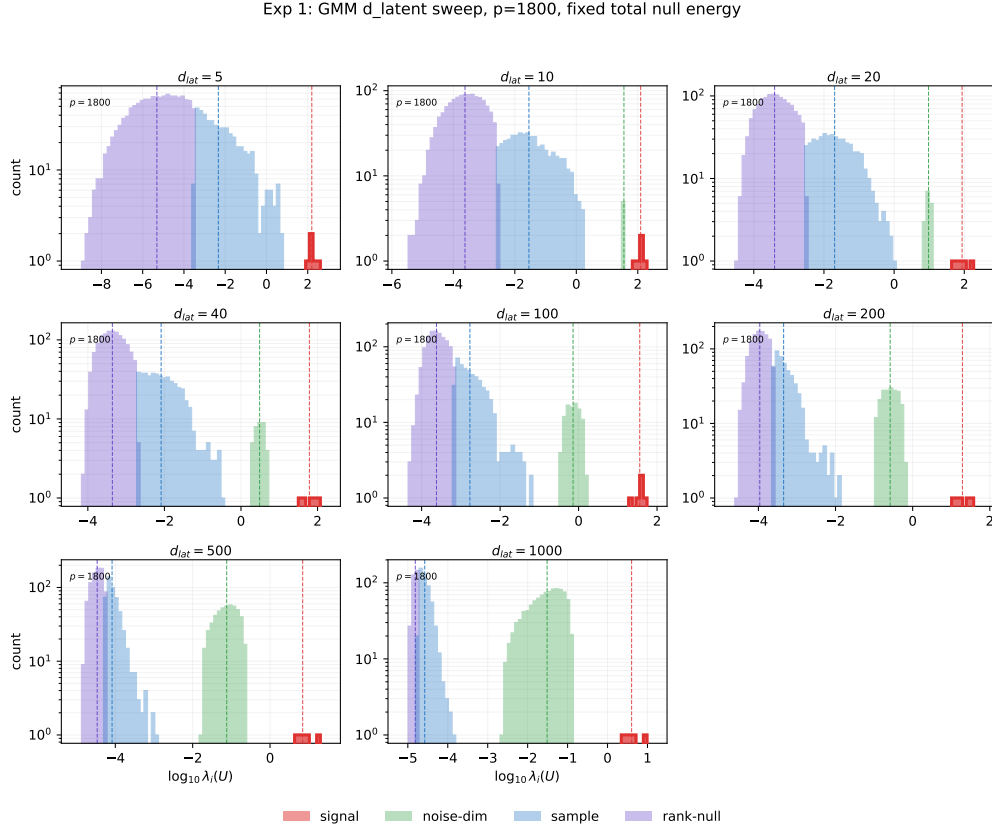


Figure 6: Experiment 1 RFNN four-bulk spectra for the latent-dimension sweep with fixed $p = 1800$ and fixed total null energy. The swept variable is still d_{latent} , but $\sigma_{\perp}^2$ is rescaled as $E_{\text{null}}/(d_{\text{latent}} - d_{\text{intrinsic}})$ so that the total null variance does not grow with latent dimension. This is the strictest control because increasing d_{latent} no longer increases either the number of random features or the total variance injected into null coordinates. The remaining spectral separation is therefore the cleanest evidence that the buffer arises from the distribution of modes across the latent space. The increasing regime is tied to the exact synthetic quantity $d_{\text{latent}} - d_{\text{intrinsic}}$, not to increasing RFNN width or total null energy. Colors are assigned by sorted eigenvalue-index blocks rather than fitted clusters: red denotes signal/source modes, green the observed latent-null/noise-dimension block, blue sample-specific modes, and purple rank-null modes. Dashed vertical lines mark the median log-eigenvalue of each colored block.

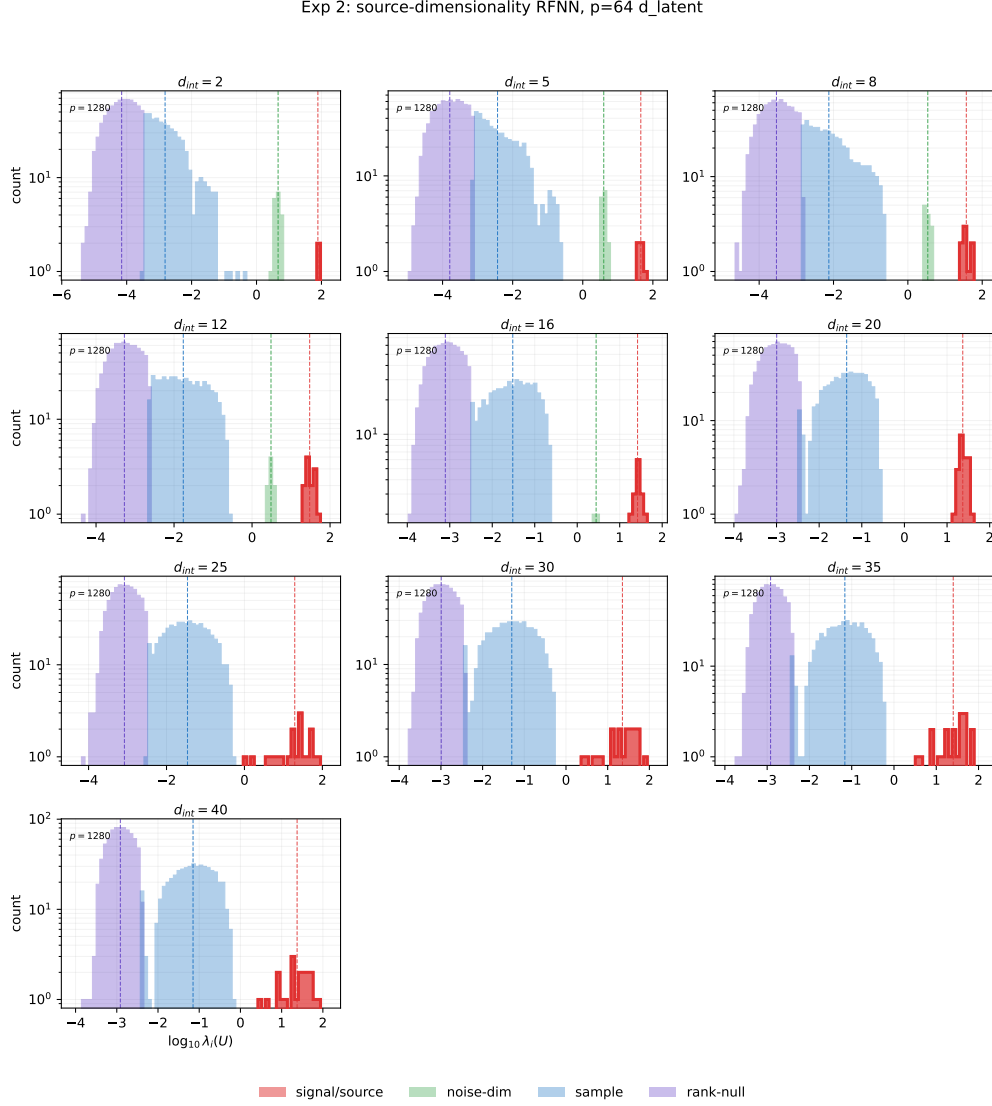


Figure 7: Experiment 2 RFNN source-dimensionality sweep with a fixed observed latent space under the $p = 64d_{\text{latent}}$ width rule. The learned latent dimension is held fixed at $d_{\text{latent}} = 20$, $\sigma_{\text{signal}} = 1.0$, $\sigma_{\perp} = 0.5$, and GMM center scale 3.0, while the intrinsic dimension of the source distribution sweeps $d_{\text{intrinsic}} \in \{2, 5, 8, 12, 16, 20, 25, 30, 35, 40\}$. When $d_{\text{intrinsic}} < d_{\text{latent}}$, the remaining observed coordinates form a genuine null buffer; as $d_{\text{intrinsic}}$ approaches d_{latent} , that buffer shrinks and the signal and sample regions become less separated. Because $d_{\text{latent}} = 20$ is fixed and known, this is the exact decreasing-buffer regime: the green block shrinks as $d_{\text{latent}} - d_{\text{intrinsic}}$ and disappears at $d_{\text{intrinsic}} = d_{\text{latent}}$. Colors are assigned by sorted eigenvalue-index blocks rather than fitted clusters: red denotes signal/source modes, green the observed latent-null/noise-dimension block, blue sample-specific modes, and purple rank-null modes. Dashed vertical lines mark the median log-eigenvalue of each colored block.

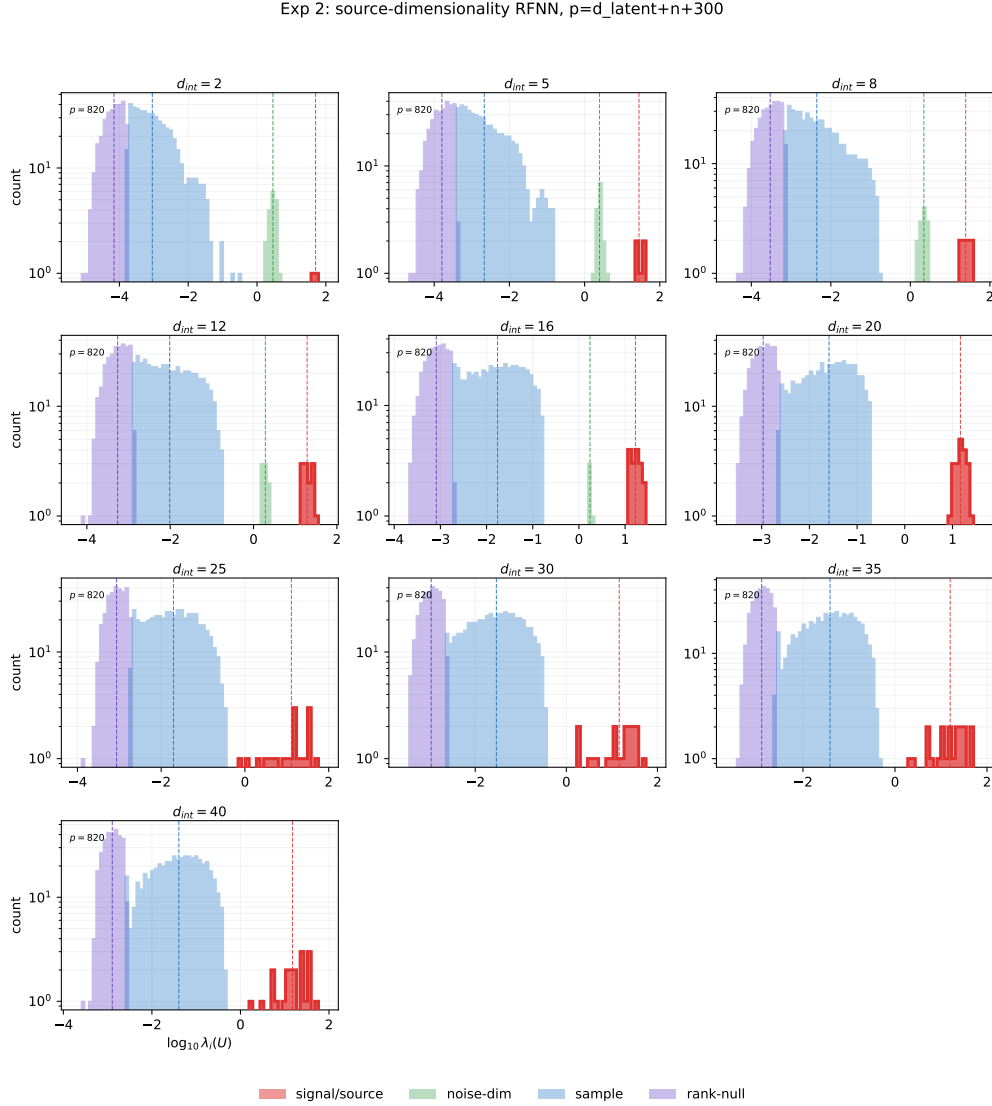


Figure 8: Experiment 2 RFNN source-dimensionality sweep with $p = d_{\text{latent}} + n + 300$. This panel uses the same fixed observed latent space as Figure 7 and the same $d_{\text{intrinsic}}$ sweep, but removes the highly overcomplete feature-count scaling. The same loss of separation as $d_{\text{intrinsic}}$ approaches d_{latent} shows that the effect is not specific to the widest RFNN parameterization. The smooth decrease of the null-buffer block with the exact gap $d_{\text{latent}} - d_{\text{intrinsic}}$ is the key diagnostic. Colors are assigned by sorted eigenvalue-index blocks rather than fitted clusters: red denotes signal/source modes, green the observed latent-null/noise-dimension block, blue sample-specific modes, and purple rank-null modes. Dashed vertical lines mark the median log-eigenvalue of each colored block.

Exp 2: source-dimensionality RFNN, fixed p

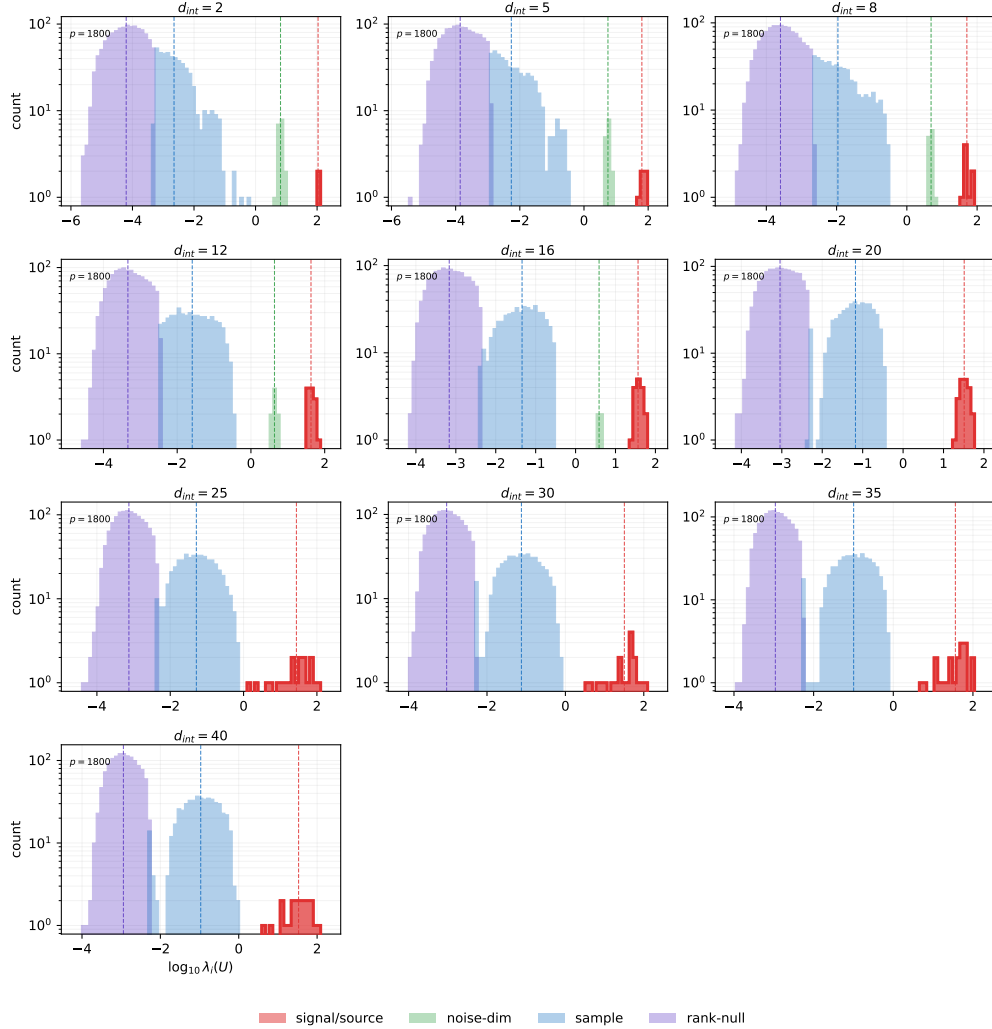


Figure 9: Experiment 2 RFNN source-dimensionality sweep with fixed width. For this control $d_{\text{latent}} = 20$ is fixed and $d_{\text{intrinsic}}$ is swept as in the other Experiment 2 panels, while RFNN width is held fixed instead of growing with d_{latent} or $d_{\text{intrinsic}}$. For $d_{\text{intrinsic}} > d_{\text{latent}}$, the source structure is compressed into the fixed observed latent space, so the high- $d_{\text{intrinsic}}$ regime should be read as an over-intrinsic stress test rather than a literal null-buffer regime. The fixed-width control makes the narrowing of the buffer visible without simultaneously changing RFNN capacity. The relevant regime is therefore decreasing until the exact boundary $d_{\text{intrinsic}} = d_{\text{latent}}$, followed by compressed over-intrinsic behavior rather than a new buffer. Colors are assigned by sorted eigenvalue-index blocks rather than fitted clusters: red denotes signal/source modes, green the observed latent-null/noise-dimension block, blue sample-specific modes, and purple rank-null modes. Dashed vertical lines mark the median log-eigenvalue of each colored block.

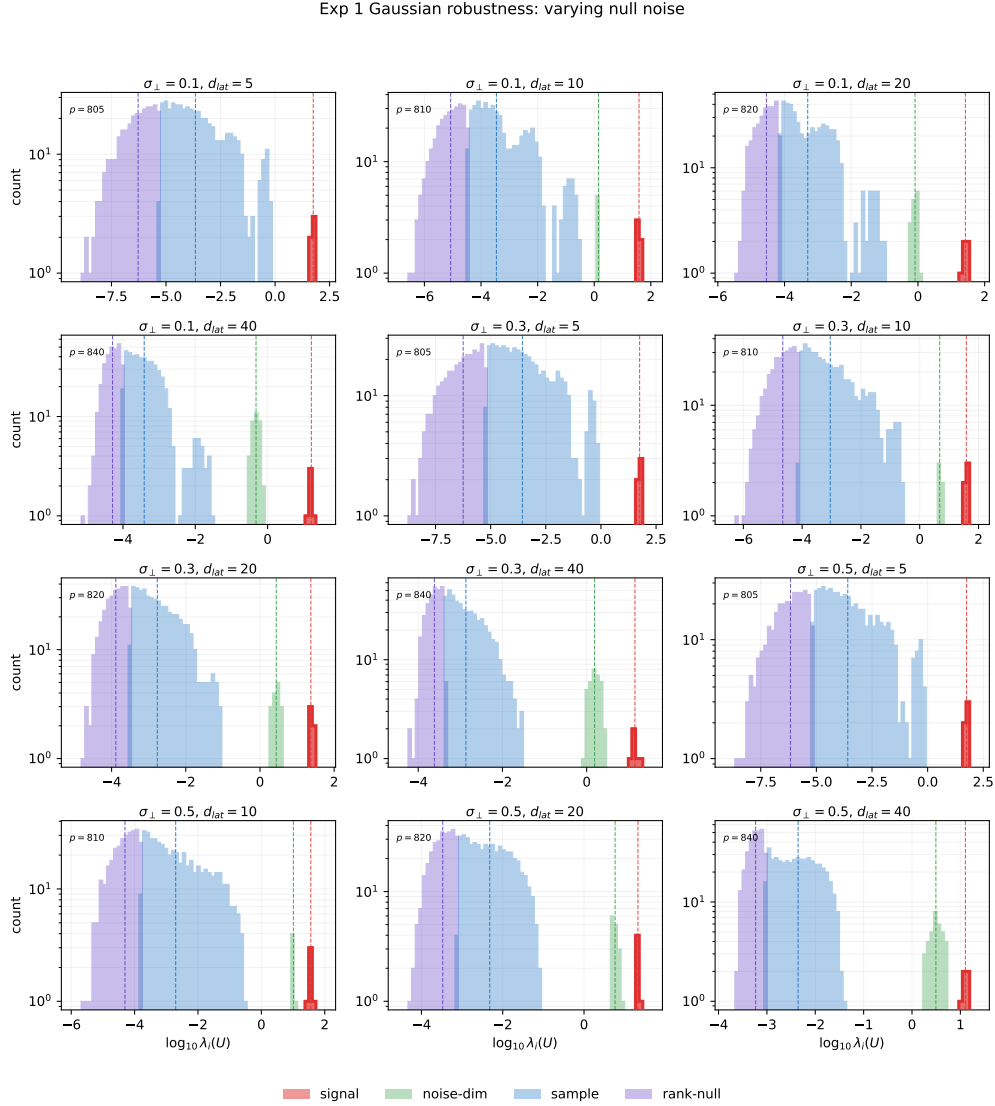


Figure 10: Experiment 1 RFNN latent-dimension sweep on pure anisotropic Gaussian data while varying the null-noise scale. Rows use $\sigma_{\perp} \in \{0.1, 0.3, 0.5\}$ at fixed $d_{\text{intrinsic}} = 5$, $\sigma_{\text{signal}} = 1.0$, and swept d_{latent} . This test removes mixture structure from the synthetic data and checks that the four-bulk decomposition is not an artifact of the default GMM. Increasing null noise shifts the latent-null region and changes the gap to the sample region, as expected from the buffer interpretation. Colors are assigned by sorted eigenvalue-index blocks rather than fitted clusters: red denotes signal/source modes, green the observed latent-null/noise-dimension block, blue sample-specific modes, and purple rank-null modes. Dashed vertical lines mark the median log-eigenvalue of each colored block.

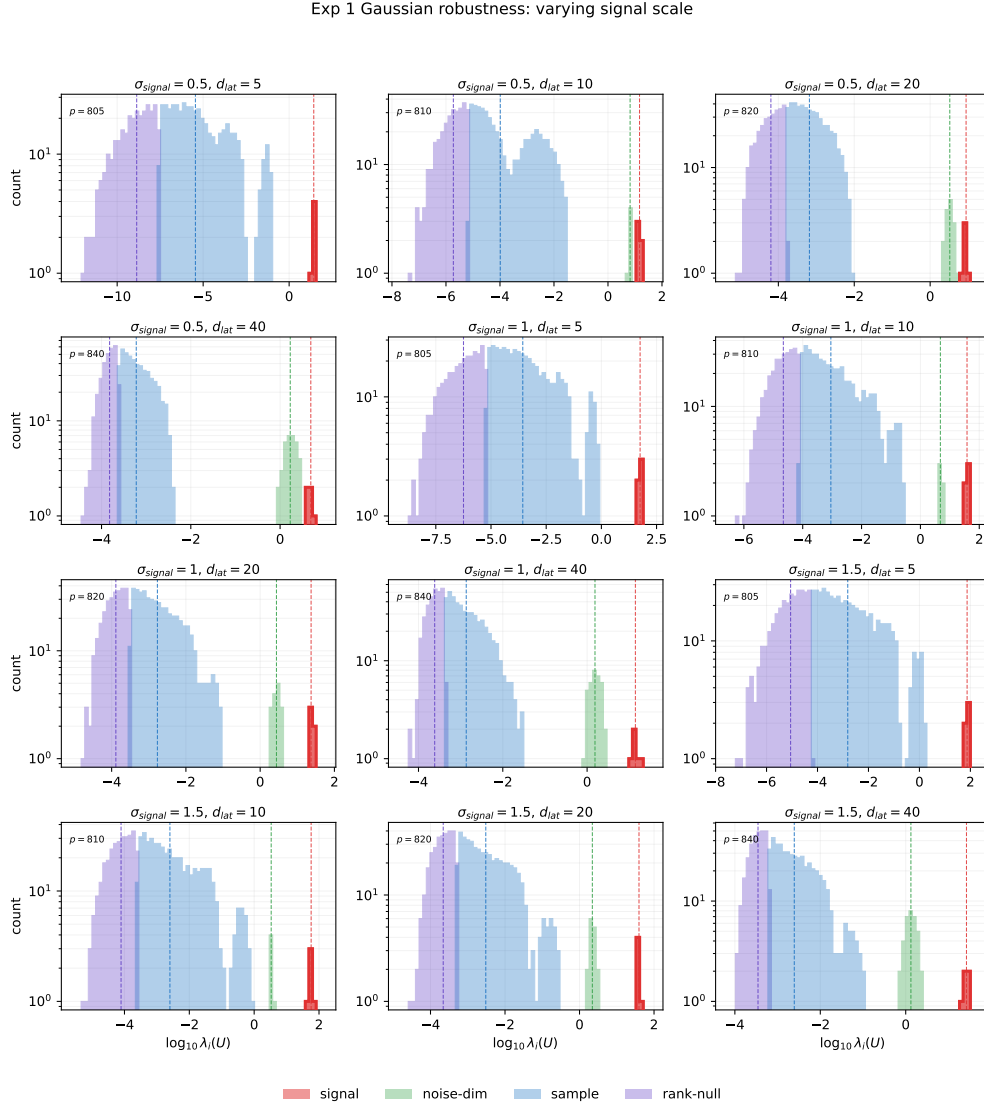


Figure 11: Experiment 1 RFNN latent-dimension sweep on pure anisotropic Gaussian data while varying the signal scale. Rows use $\sigma_{\text{signal}} \in \{0.5, 1.0, 1.5\}$ at fixed $d_{\text{intrinsic}} = 5$, $\sigma_{\perp} = 0.3$, and swept d_{latent} . The signal block moves with the strength of the structured coordinates, while the sample and null regions remain governed by the latent buffer and noise statistics. This confirms that the visible bulk ordering responds to the intended data-level parameters. Colors are assigned by sorted eigenvalue-index blocks rather than fitted clusters: red denotes signal/source modes, green the observed latent-null/noise-dimension block, blue sample-specific modes, and purple rank-null modes. Dashed vertical lines mark the median $\log$ -eigenvalue of each colored block.

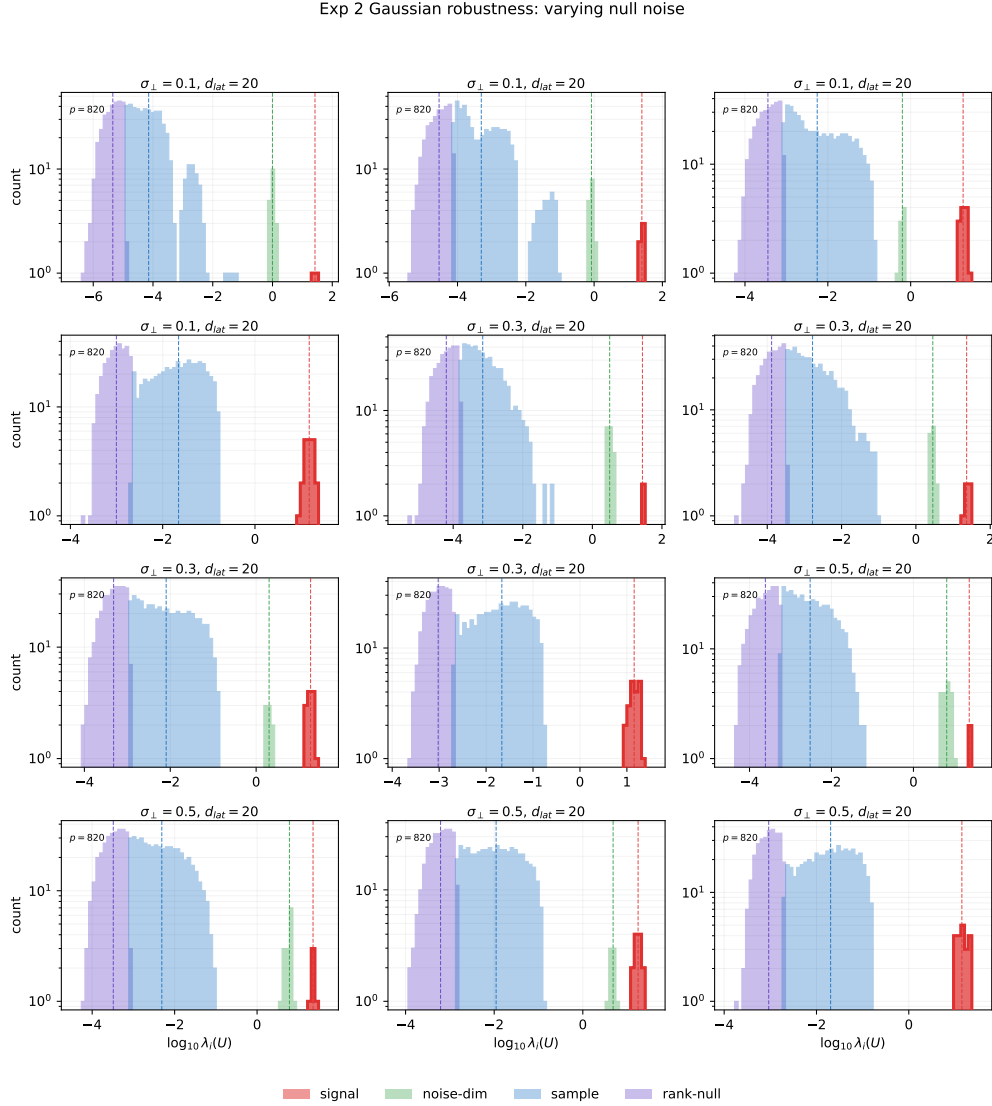


Figure 12: Experiment 2 RFNN source-dimensionality sweep on pure anisotropic Gaussian data while varying the null-noise scale. Rows use $\sigma_{\perp} \in \{0.1, 0.3, 0.5\}$ at fixed $d_{\text{latent}} = 20$ and swept $d_{\text{intrinsic}}$. The same buffer-collapse pattern appears when $d_{\text{intrinsic}}$ approaches the fixed observed latent dimension, and the amount of separation between null and sample regions changes with the null-noise scale. Colors are assigned by sorted eigenvalue-index blocks rather than fitted clusters: red denotes signal/source modes, green the observed latent-null/noise-dimension block, blue sample-specific modes, and purple rank-null modes. Dashed vertical lines mark the median log-eigenvalue of each colored block.

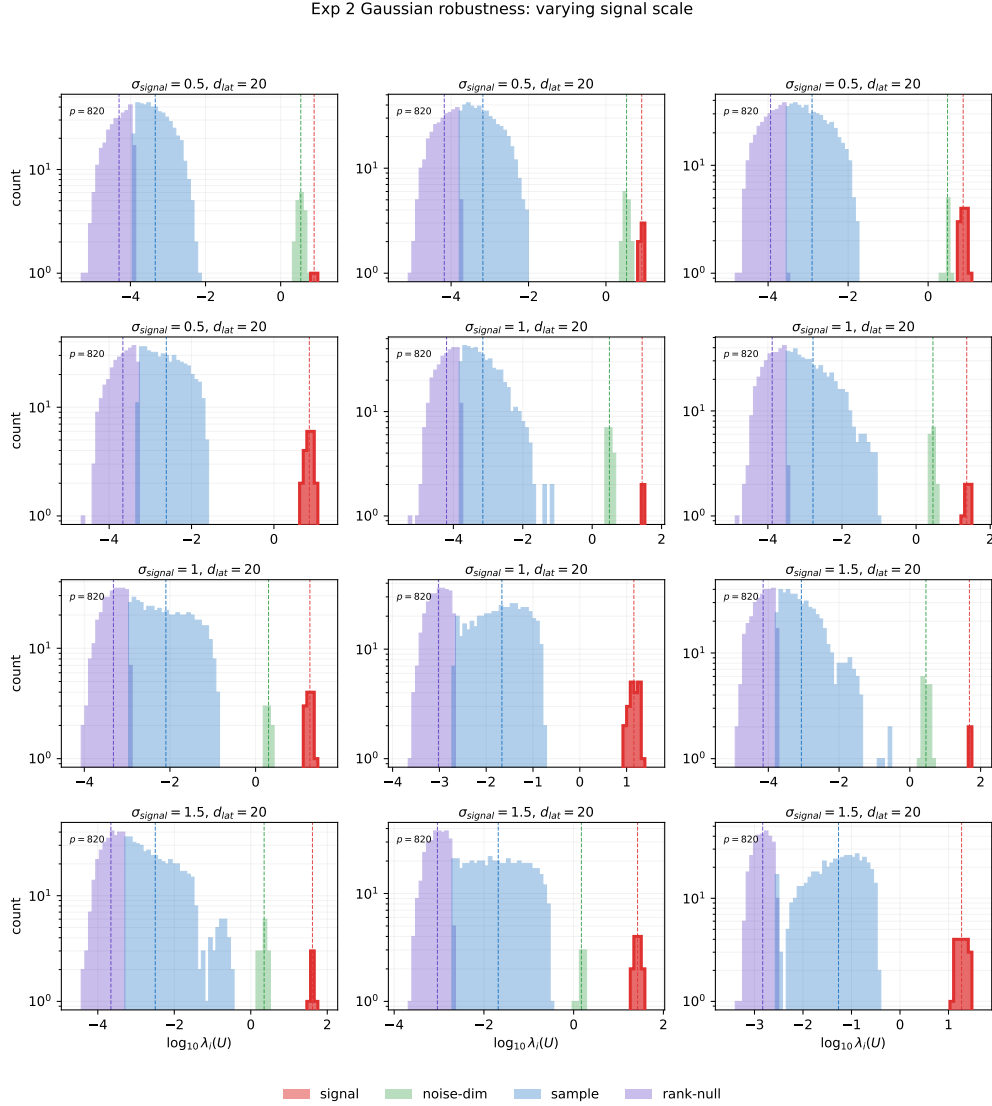


Figure 13: Experiment 2 RFNN source-dimensionality sweep on pure anisotropic Gaussian data while varying the signal scale. Rows use $\sigma_{\text{signal}} \in \{0.5, 1.0, 1.5\}$ at fixed $d_{\text{latent}} = 20$, $\sigma_{\perp} = 0.3$, and swept $d_{\text{intrinsic}}$. Larger signal scale sharpens the leading signal region, but the qualitative transition as $d_{\text{intrinsic}}$ approaches d_{latent} remains governed by the amount of available null-buffer dimension. Colors are assigned by sorted eigenvalue-index blocks rather than fitted clusters: red denotes signal/source modes, green the observed latent-null/noise-dimension block, blue sample-specific modes, and purple rank-null modes. Dashed vertical lines mark the median log-eigenvalue of each colored block.

D SYNTHETIC MLP DIAGNOSTICS

The synthetic MLP figures use five seeds, a 5M-step training horizon, and the nearest-neighbor memorization ratio described in Appendix B. This appendix repeats the main fixed-width sweep with the larger latent range $d_{\text{latent}} \in \{5, 8, 10, 12, 15, 20, 25, 30, 35, 40, 60, 80, 100, 120, 140, 160, 180, 200, 220, 240\}$. We report memorization percentage over training steps so that the onset, growth, and saturation of memorization remain visible.

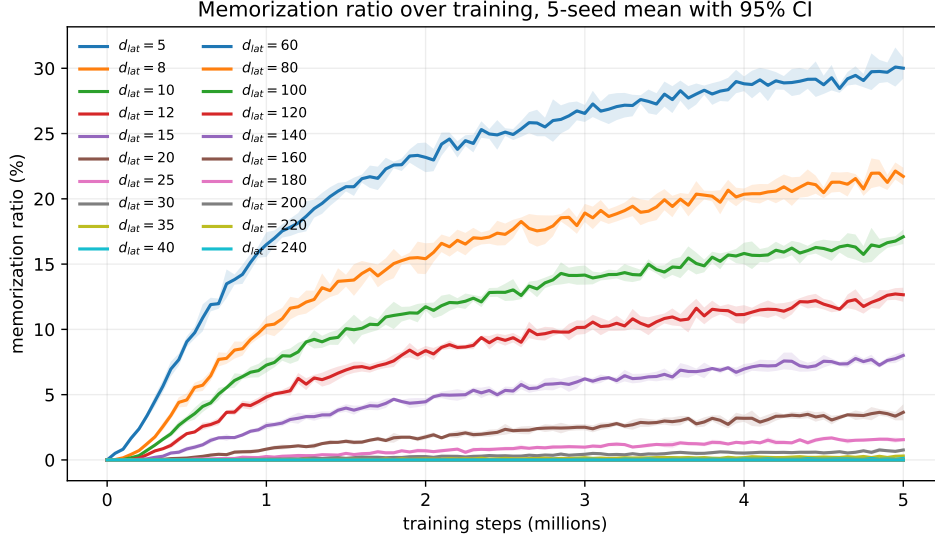
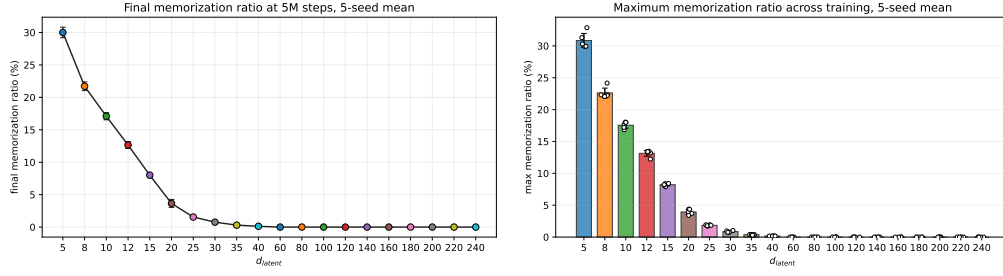


Figure 14: Synthetic trainable MLP memorization trajectories across latent dimension. The curves show when each model begins to reproduce individual training samples according to the nearest-neighbor ratio, while retaining the full shape of the rise rather than collapsing it to a single threshold. Smaller latent dimensions memorize quickly because the model reaches sample-specific directions after absorbing only a short null buffer; larger latent dimensions delay this rise, consistent with the RFNN picture that additional null directions must be learned before sample-specific modes dominate. The extended sweep shows that this continues through $d_{\text{latent}} = 240$ under the fixed-width MLP control. The sweep is almost ordered by the exact synthetic buffer size $d_{\text{latent}} - d_{\text{intrinsic}}$: after the no-buffer $d_{\text{latent}} = d_{\text{intrinsic}} = 5$ case, increasing d_{latent} produces a steadily lower and later memorization regime rather than large dimension-to-dimension variation.

Trained-MLP pre-activation measurement. We evaluate $q_M = \frac{1}{NH} \sum_{i,j} h_j(x_{t,i}, t)^2$, where h is the first linear layer’s output, including its bias and the Fourier time embedding, $H = 256$, $N = 4096$ held-out population draws, and $t = 0.1$. This is an uncentered second moment, not a centered variance; unlike the isotropic frozen-feature operating point q_d , it includes learned offsets. Test draws use seed 9999 and diffusion noise uses seed 123, fixed across checkpoints. Averaging over training seeds 42–46, q_M at 0.05M, 0.25M, 1M and 5M steps is (0.710, 1.118, 1.621, 2.813) for $d_{\text{latent}} = 5$ and (0.685, 0.780, 0.762, 0.836) for $d_{\text{latent}} = 40$; reported ratios divide these seed means. No initialization checkpoint is available, so this comparison starts at 50k steps.

Score-error recovery. The plotted population score error is estimated from saved held-out denoising losses using the DSM identity: per coordinate, $\hat{e} = L_{\text{test}}/\Delta_t - \hat{C}/d_{\text{latent}}$, where $\Delta_t = 1 - e^{-2t}$ and $\hat{C}$ estimates $\mathbb{E}\|s^* - s_{\text{cond}}\|^2$ using the corrected analytic GMM score on the same 2048 held-out noisy pairs. This corrects a rotation-frame error in the originally logged analytic-score diagnostic; training did not use that diagnostic. The identity holds in expectation; a finite-sample cross term remains, so this recovery is an estimate rather than an exact checkpoint evaluation. At 5M steps the five-seed mean is 2.59 for $d_{\text{latent}} = 5$, 2.26–2.35 for $d_{\text{latent}} = 8$ –40, and 0.53 for $d_{\text{latent}} = 240$.



(a) Final memorization at 5M steps.

(b) Peak memorization during training.

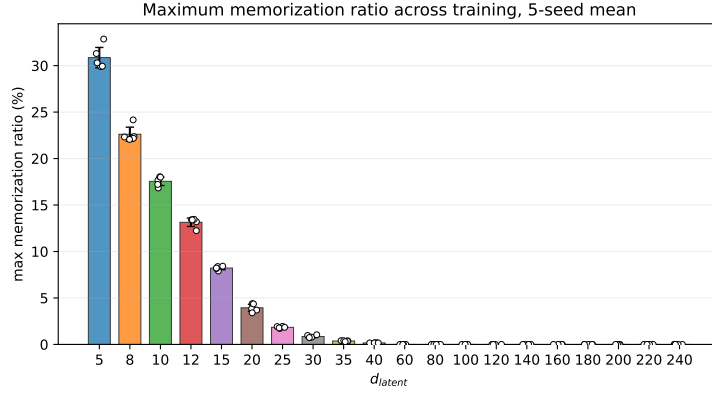


Figure 15: Synthetic MLP final and peak memorization summaries. The final memorization fraction measures the end-of-training state, while the peak memorization fraction captures transient sample recall that may not persist until 5M steps. The seed-level histogram exposes how much of the latent dimensional trend is stable across random seeds rather than being driven by a single unusually memorizing run. The final and peak summaries both support the same decreasing regime with increasing d_{latent} , including the high-dimensional extension beyond the main-text $d_{\text{latent}} \leq 40$ view.

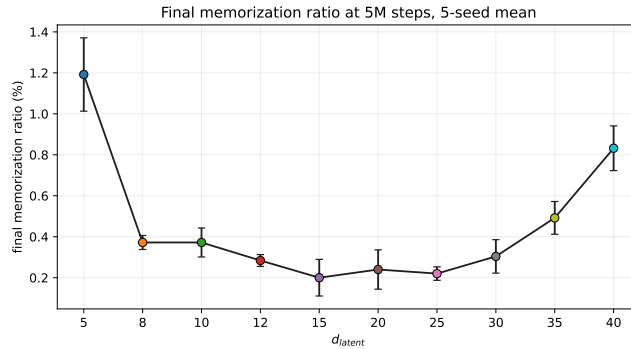
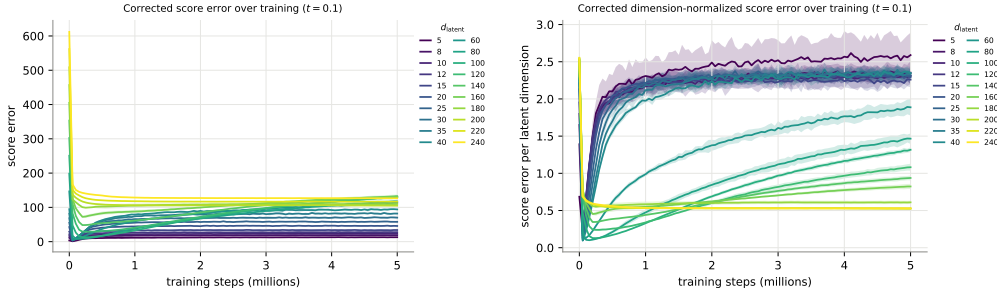


Figure 16: Width-scaled control for Figure 1: the same sweep with hidden width $8d_{\text{latent}}$ (40 to 320) instead of a fixed 256, five seeds, 5M steps. Memorization sits at the 1% level and rises again for $d_{\text{latent}} \geq 30$ as the network grows, so a width-scaled sweep is a capacity sweep and does not isolate the latent dimension. This is why the main text holds the width fixed.

Table 2: Dimension-controlled memorization test for the fixed-width synthetic MLP (5M steps, five seeds, 5000 samples each; same configuration as Figure 1). The nearest-neighbor ratio test of Appendix B is applied in the full latent space and after projecting generated and training points onto the true $d_{\text{intrinsic}} = 5$ signal coordinates. Controls: fresh draws from the population; exact copies of a training point plus the sampler’s terminal noise in every coordinate; and signal copies, which reproduce a training point’s signal coordinates and redraw its null coordinates. α is the projection coefficient of the learned score onto the exact empirical score of the training set in the signal coordinates at $t = 0.1$ (0: population score; 1: exact fit of the training set).

	$d_{\text{latent}} = 5$	10	20	40
model samples, full-space test	29.9%	16.9%	3.6%	0.1%
model samples, signal-projected test	29.9%	16.8%	5.8%	1.2%
fresh draws, signal-projected test	0.4%	0.4%	0.4%	0.4%
exact copies, full-space test	57%	69%	84%	97%
signal copies, full-space test	57%	0.8%	0.0%	0.0%
signal copies, signal-projected test	57%	57%	58%	58%
α (signal coordinates), 0.25M steps	0.54	0.37	0.22	0.09
α (signal coordinates), 5M steps	0.75	0.54	0.39	0.31



(a) Score error, all coordinates.

(b) Score error normalized per latent dimension.

Figure 17: Synthetic MLP score-error diagnostics: the population score error at $t = 0.1$ against the exact GMM score on a held-out set, five-seed mean with 95% CI, over the complete fixed-width sweep $d_{\text{latent}} = 5\text{--}240$ (the main-text panel of Figure 1 covers $d_{\text{latent}} \leq 40$). The left panel sums over all latent coordinates; the right panel divides by d_{latent} , which removes the mechanical scaling with d_{latent} . This distinction matters because a larger latent space has more coordinates to predict even if the average coordinate-wise error is comparable. Score error is the appropriate quality metric in this synthetic setting because the target distribution is specified analytically in latent space rather than through decoded images. Beyond $d_{\text{latent}} \approx 60$ the per-coordinate error has not returned to its small- d_{latent} plateau by the 5M-step horizon: it ends at 2.59 for $d_{\text{latent}} = 5$ but 0.53 for $d_{\text{latent}} = 240$, so the widest latents never begin the divergence from the population score that accompanies memorization at small d_{latent} .

Interpreting threshold-free memorization curves. Single hitting times such as τ_{mem} are convenient summaries, but they hide whether a run crosses a threshold sharply, gradually, or only transiently. The full memorization-fraction curves are therefore the primary diagnostic for the trainable MLP experiments. Thresholded hitting times are used later only in the spectral-predictor analysis, where multiple fixed memorization levels are compared against predicted iso-memorization times.

E REAL-DATA VAE DIAGNOSTICS

The real-data experiments separate two questions. First, the VAE must preserve enough image identity that nearest-neighbor memorization can be detected after encoding and decoding. Second, the diffusion model must have enough capacity and stable optimization to actually memorize the encoded subset. For this reason, the VAE figures appear before the diffusion figures: they determine which latent dimensions are still below the effective intrinsic dimensionality of the image manifold and therefore should not be expected to obey the excess-buffer prediction cleanly.

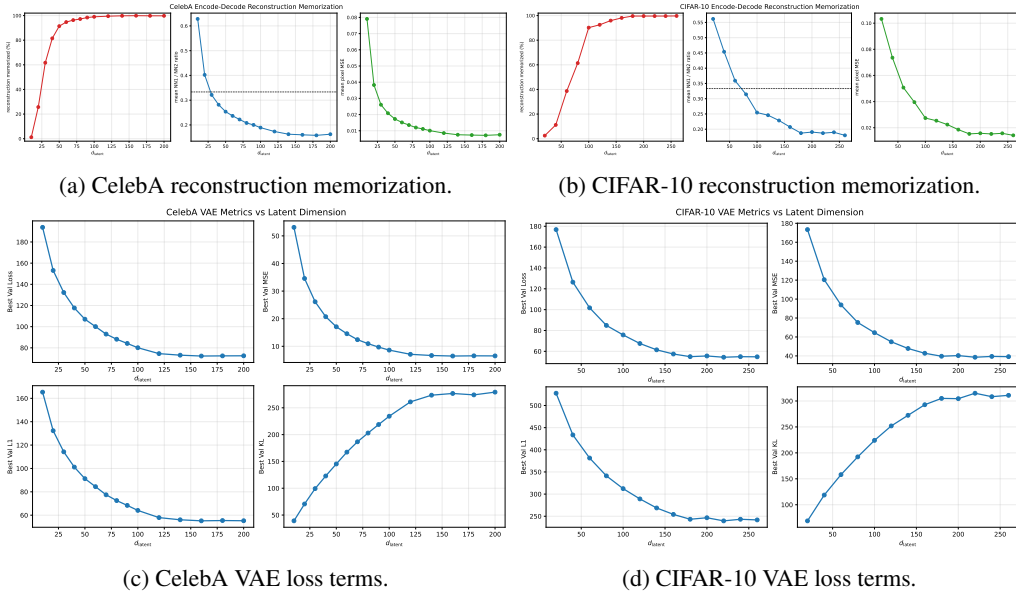


Figure 18: VAE diagnostics across latent dimension for CelebA and CIFAR-10. The reconstruction-memorization panels encode the 1k diffusion subset through the frozen VAE and decode it immediately, then apply the same pixel-space KNN ratio used for generated samples. This measures whether the VAE bottleneck itself preserves enough sample identity for memorization to be detectable. The loss panels show the corresponding validation reconstruction and KL behavior across dimensions for VAEs trained on the full training split. Low-dimensional VAEs blur or average away identity, so weak diffusion memorization at small d_{latent} can reflect a lossy representation rather than an intrinsically nonmemorizing diffusion model. In real data the intrinsic dimension is only approximate, but the reconstruction panels show the first regime cleanly: identity preservation increases with d_{latent} until the VAE is near the effective intrinsic dimensionality. Below that point, the model is capacity-limited and the spectral memorization predictor is not expected to follow the empirical curve closely.



Figure 19: CelebA VAE encode/decode contact sheet. The rows show original images and their final checkpoint reconstructions across latent dimensions, making visible the point at which identity-level facial features become stable under the VAE bottleneck. The visual trend is mostly increasing rather than noisy: larger d_{latent} steadily moves the VAE toward the approximate intrinsic-image regime where sample identity survives encode/decode. This qualitative check is necessary because the downstream KNN memorization metric can only detect sample recall when reconstructions preserve enough pixel-level identity.



Figure 20: CIFAR-10 VAE encode/decode contact sheet. The rows show original images and final checkpoint reconstructions. CIFAR-10 has much more class and texture variation than the CelebA subset, so visual reconstruction quality is an important check before interpreting downstream diffusion memorization. The higher-dimensional VAEs preserve more object identity and local structure, making pixel-space nearest-neighbor tests more meaningful. As with CelebA, reconstruction quality is best read as an increasing-capacity regime up to an approximate effective intrinsic dimension, not as arbitrary variation across latent widths.

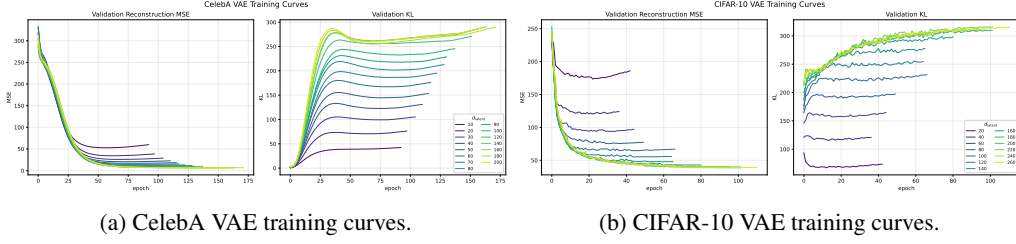


Figure 21: VAE training curves for CelebA and CIFAR-10. The reported VAE checkpoints are selected by validation objective after the reconstruction and KL objectives have largely stabilized, so the downstream latent-diffusion trends are not simply caused by comparing undertrained encoders at different latent dimensions. Together with Figures 18–20, these diagnostics justify treating high-dimensional VAE latents as a meaningful substrate for pixel-space memorization tests.

F REAL-DATA DIFFUSION DIAGNOSTICS

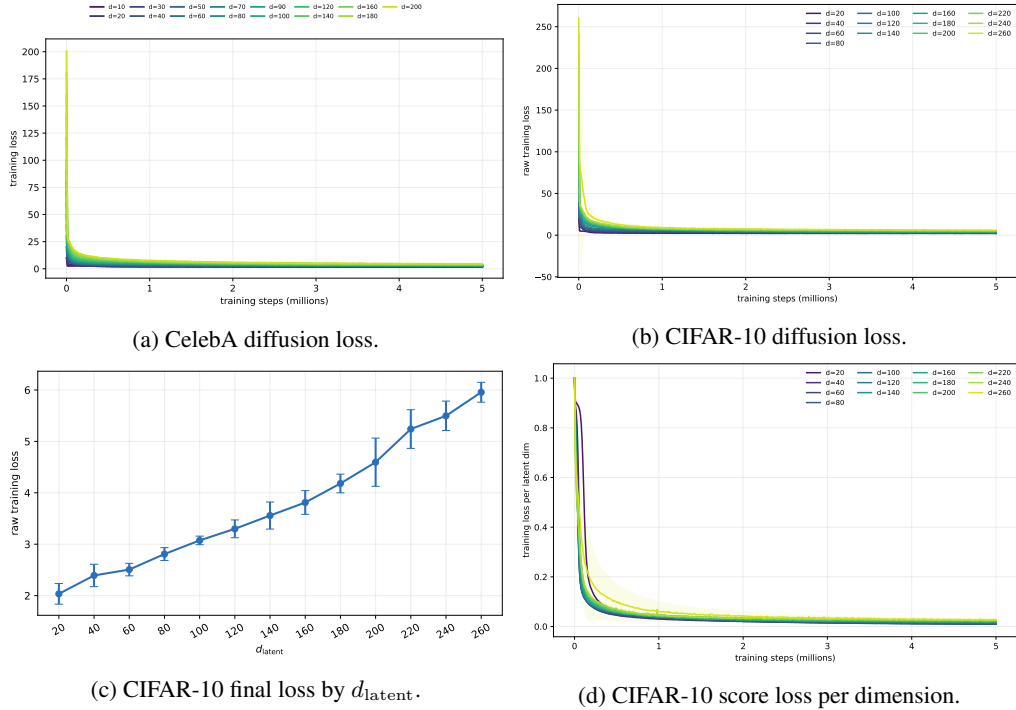


Figure 22: Optimization diagnostics for the real-data diffusion runs, shown before the memorization and FID panels because they establish that the models are actually training. The loss curves track whether differences in memorization and FID coincide with gross optimization failure, while the CIFAR-10 per-dimension score loss removes the trivial scaling of the objective with latent dimension. Gradient clipping was used only as a diagnostic intervention: it prevented some unstable runs from diverging, but it also strongly suppressed memorization and therefore changed the effective training regime. The aggregate CIFAR-10 results are reported using stable unclipped runs so that optimizer changes are not conflated with latent-dimensional effects.

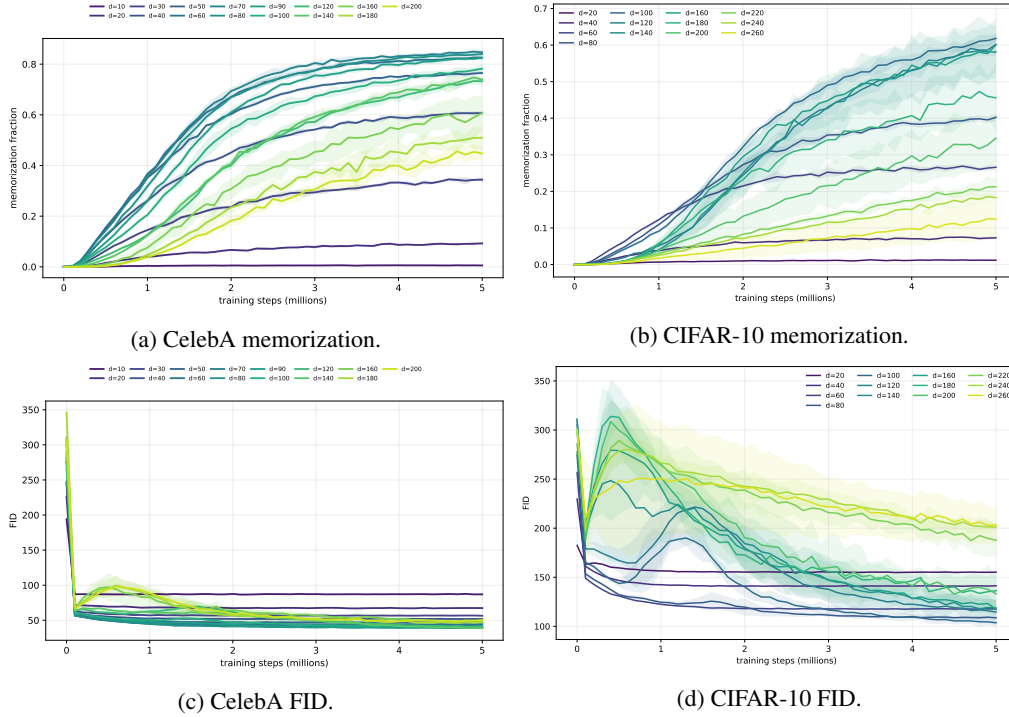


Figure 23: Real-data diffusion trajectories with five-seed confidence bands. The memorization panels report the fraction of generated samples that pass the pixel-space nearest-neighbor ratio against the diffusion training subset. The FID panels report image-quality evolution over the same optimization horizon. Reading the two rows together separates quality improvement from direct sample recall: a model can improve FID before it begins to reproduce exact training examples, and a high-dimensional latent can delay memorization even when the generated samples are already becoming visually plausible. Across d_{latent} , memorization follows a clean increasing-then-decreasing regime near the approximate effective intrinsic dimension, while FID follows the complementary decreasing-then-increasing regime: quality improves as the VAE bottleneck is relieved and worsens again when the latent score problem is too wide.

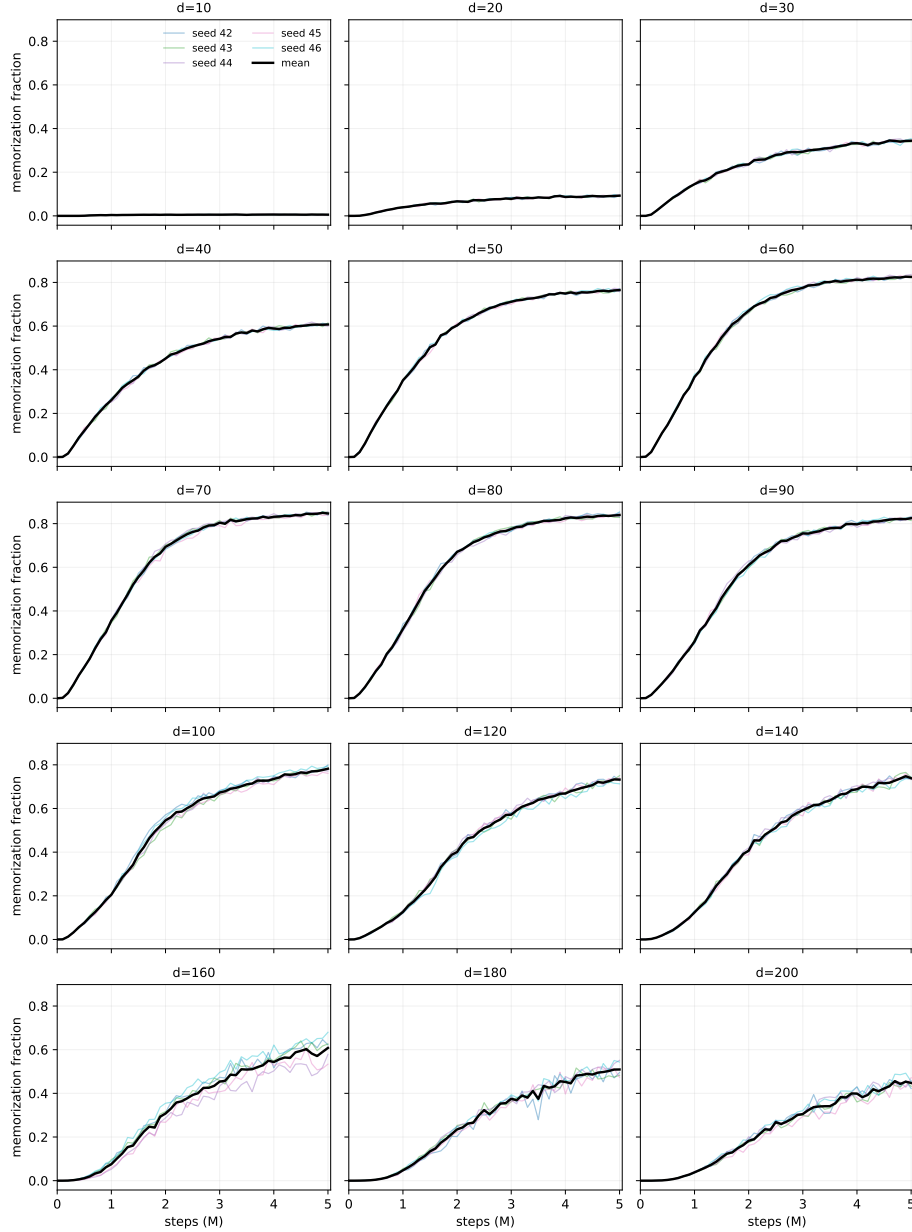


Figure 24: CelebA per-dimension memorization trajectories. Each panel isolates one latent dimension and shows the seed-level evolution of KNN memorization over the 5M-step run. This view is important because the aggregate curve can hide whether a dimension memorizes consistently across seeds or only through a few high-memorization runs. At very small d_{latent} , the simple latent-buffer prediction need not apply cleanly: the VAE may still be below the effective image dimension, so the decoder cannot reproduce enough sample-specific detail for pixel-space memorization to be detected. The same capacity bottleneck explains why spectral memorization predictions can miss the low- d_{latent} regime. Once past that bottleneck, the panels show a largely ordered increasing-then-decreasing memorization regime rather than unsystematic variation across adjacent latent dimensions.

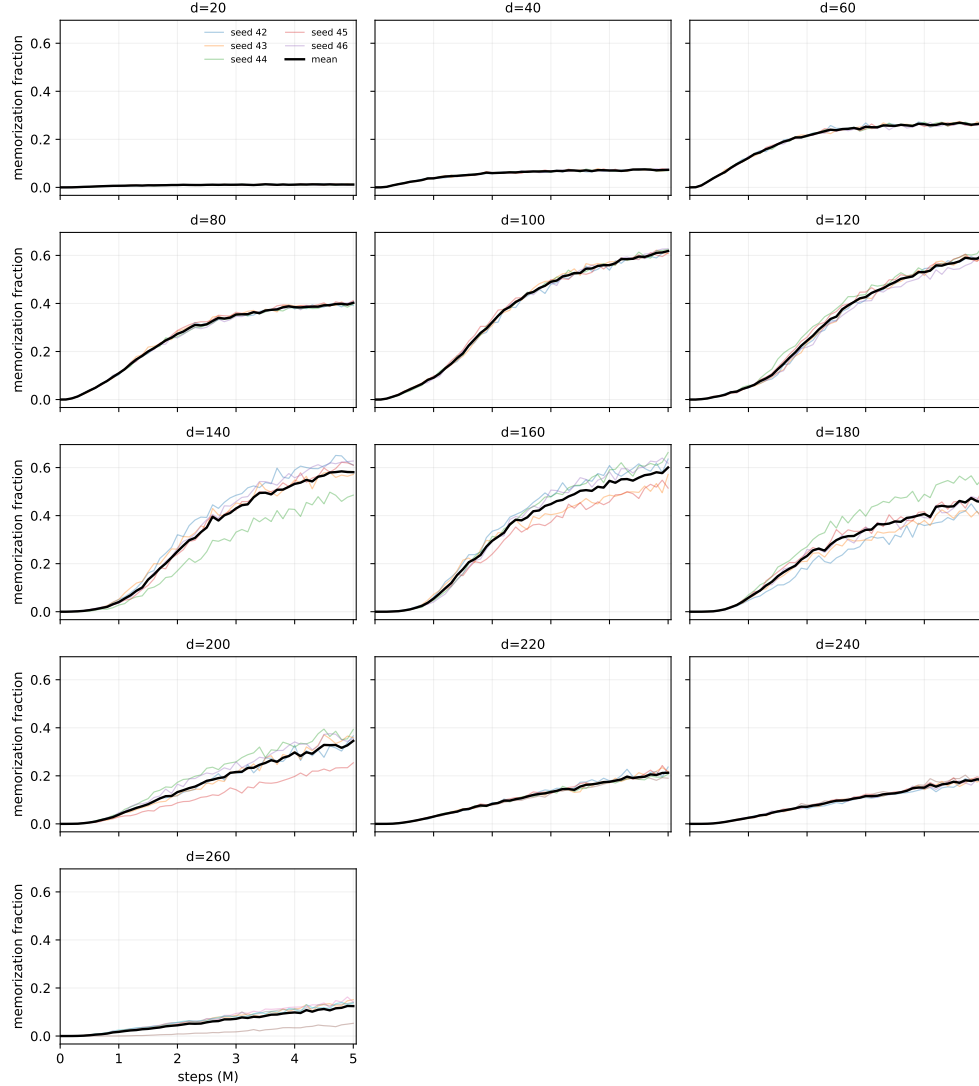


Figure 25: CIFAR-10 per-dimension memorization trajectories. The larger view exposes which latent dimensions memorize consistently across seeds and which remain limited by representation or optimization. As with CelebA, the low-dimensional regime mixes two effects: a smaller latent buffer and a VAE bottleneck that may be too lossy for pixel-space KNN memorization to detect exact sample identity. Because this is below the effective intrinsic dimensionality of the dataset, deviations from the spectral predictor are expected. The higher-dimensional panels then enter the decreasing excess-buffer regime, so the full sweep is best read as an approximate intrinsic-threshold transition followed by delayed memorization.

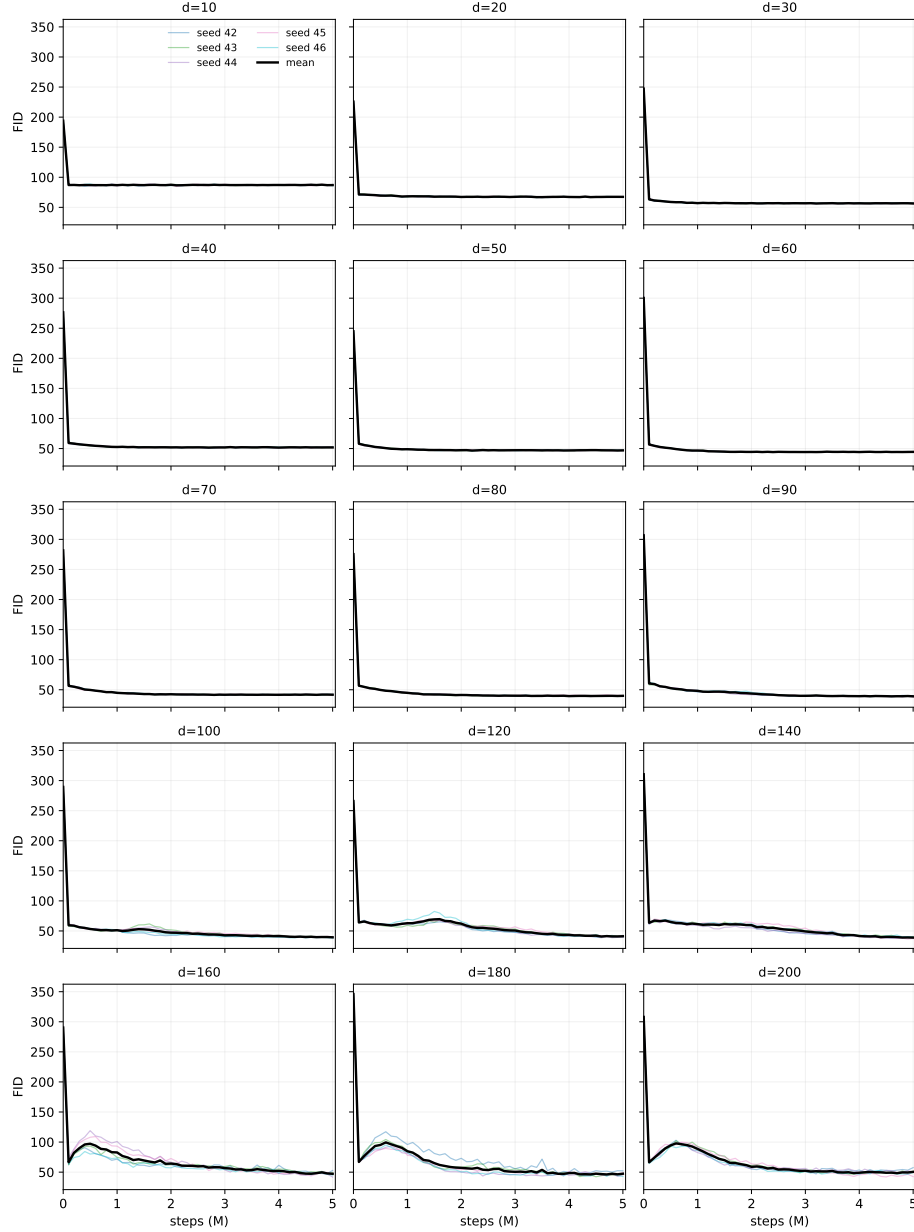


Figure 26: CelebA per-dimension FID trajectories. These panels use the same dimension-by-dimension organization as Figure 24 but track sample quality rather than memorization. The comparison shows where increasing d_{latent} improves representational fidelity and where it mainly changes the timing of sample recall. The FID trend follows the expected decreasing-then-increasing regime: it improves as d_{latent} approaches the approximate effective intrinsic dimension, then degrades when the latent score problem becomes overwide. In particular, a dimension with better FID is not automatically more memorizing; the two quantities are coupled by the VAE representation and diffusion optimization but measure distinct behaviors.

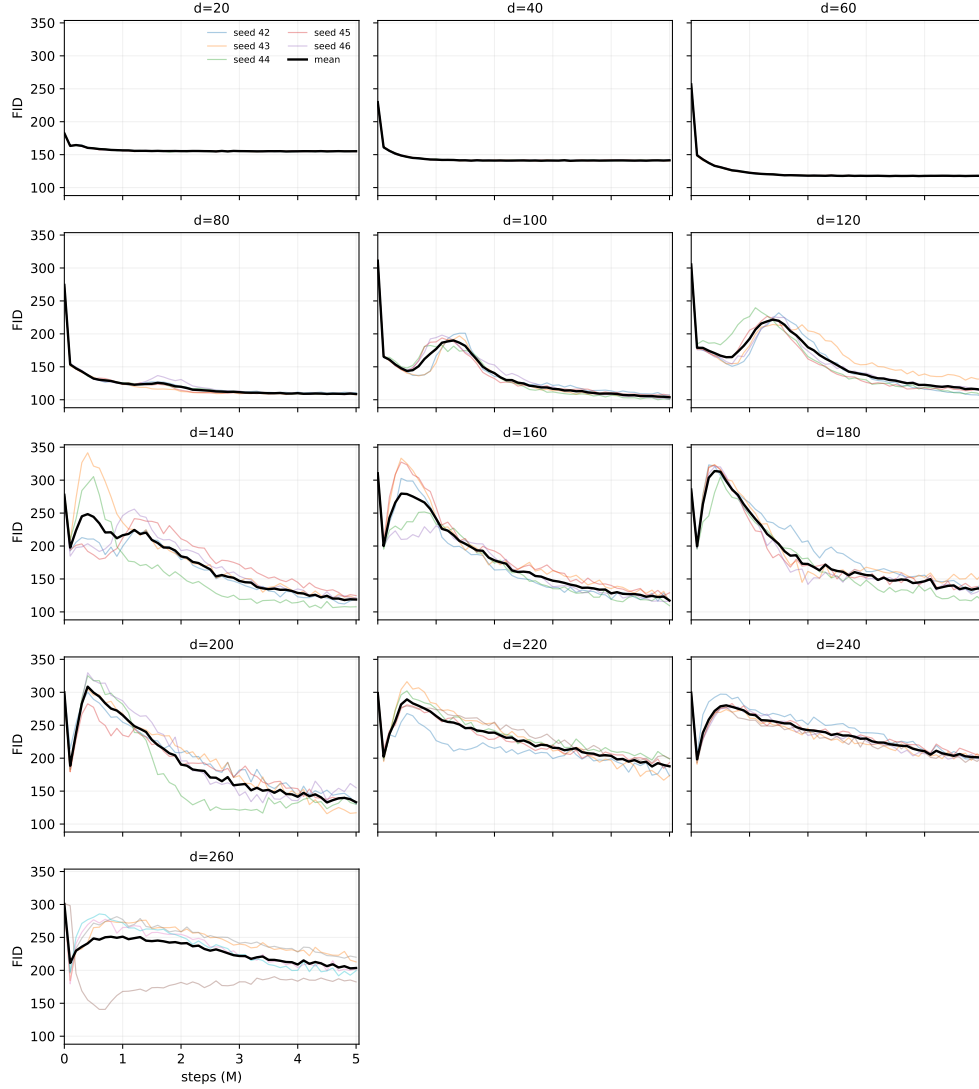


Figure 27: CIFAR-10 per-dimension FID trajectories. Plotted separately from the memorization trajectories, these curves show that sample quality and sample recall are not interchangeable metrics. Some latent dimensions improve FID before reaching high KNN memorization, while others memorize more strongly without proportionally improving image quality. The broad trend is still a smooth quality regime: FID decreases as the VAE bottleneck is relieved and increases again in the high- d_{latent} excess-latent regime.

G SPATIAL-DiT CELEBA PILOT

The main real-data experiments use vector latents and fixed-width MLP score models because that is the cleanest way to isolate the role of latent dimension. We also ran a CelebA pilot with a spatial VAE and a DiT-style score network to see what changes when the architecture looks more like a modern latent image generator. The VAE maps each 32×32 image to a $4 \times 4 \times d_{\text{latent}}$ latent grid. Thus the channel count shown on the horizontal axis is not the total latent dimension: $d_{\text{latent}} = 20$ already means 320 scalar latent coordinates, and $d_{\text{latent}} = 260$ means 4160 scalar latent coordinates. This is the main reason the spatial sweep is less clean than the vector-latent MLP sweep. A one-channel change in the plotted variable adds sixteen coordinates, and the covariance structure now includes both channel and spatial-position interactions.

For the broad pilot, we train a DiT-S-style unconditional trunk with hidden size 384, depth 12, 6 attention heads, MLP ratio 4, batch size 1024, learning rate 10^{-4} , EMA decay 0.9999, and gradient clipping at 1.0. The model is evaluated at 20k, 40k, 60k, 80k, 100k, 200k, 300k, 400k, and 500k steps using 1k generated samples, pixel-space nearest-neighbor memorization against the same diverse 1k subset, and FID against that subset. Because this pilot uses a different architecture, EMA, gradient clipping, and a spatial latent representation, we treat it as a diagnostic rather than as part of the main causal evidence. The useful takeaway is more modest: spatial latents appear to contain a lower-dimensional window or null area where added coordinates are not yet a clean monotone memorization control, and much finer tuning would be needed before using this architecture for the same sharp claim as the MLP.

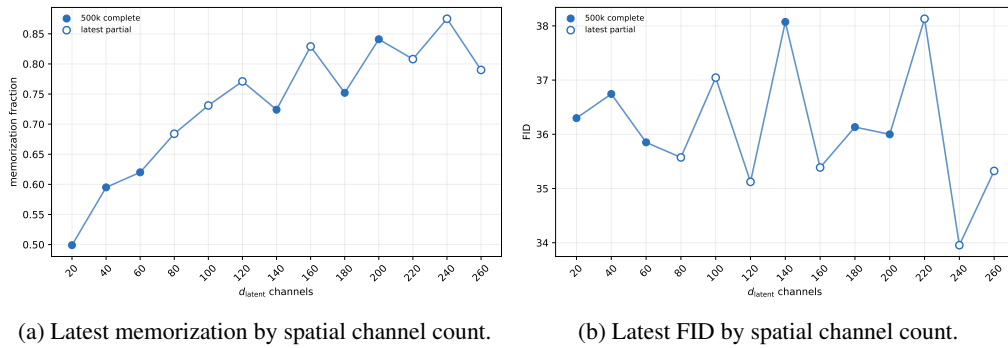


Figure 28: CelebA spatial-DiT pilot summary across channel counts. Each plotted d_{latent} corresponds to a $4 \times 4 \times d_{\text{latent}}$ latent grid, so the true scalar latent dimension is $16d_{\text{latent}}$. The broad seed-42 sweep therefore changes latent size in large jumps compared with the vector MLP experiments. FID is reasonably stable across many widths, while memorization is high and does not show the clean high-dimensional decline seen in the fixed-vector MLP. This does not rule out a buffer effect in spatial architectures; rather, it shows that channel count, spatial covariance, VAE quality, EMA, clipping, and DiT optimization all enter before the scalar buffer can be read off as a one-dimensional sweep.

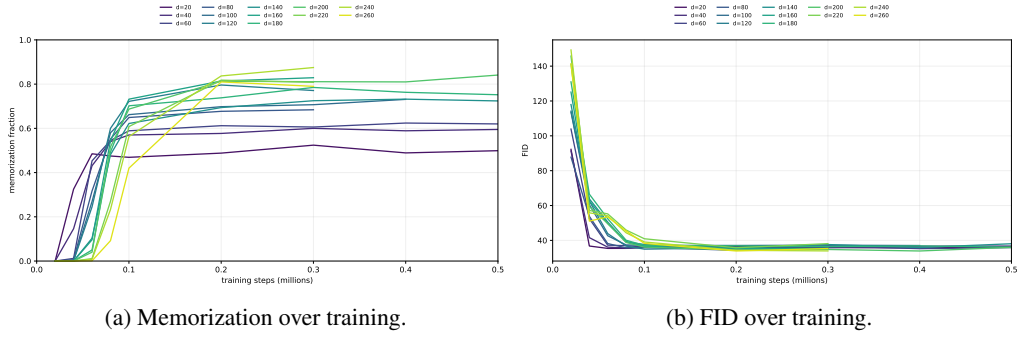


Figure 29: Spatial-DiT training trajectories for the broad CelebA pilot. The memorization panel uses the same decoded pixel-space nearest-neighbor ratio as the main real-data experiments. The FID panel uses the same 1k diverse subset as the reference. Unlike the vector MLP curves, these trajectories are not a clean function of the plotted channel count, because each width change also changes a 4×4 spatial representation and its second-order covariance structure.

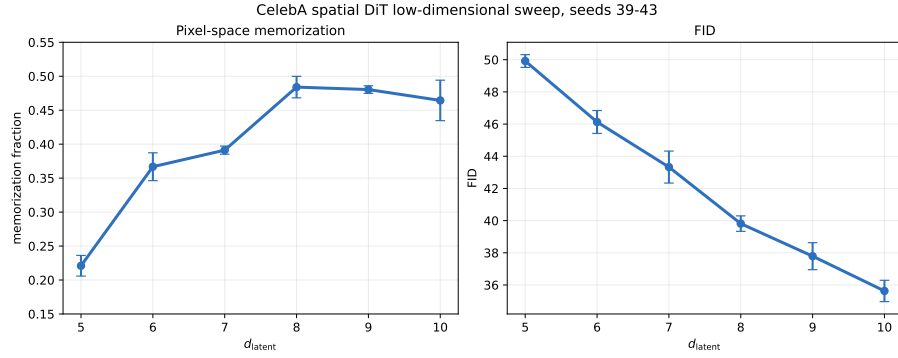


Figure 30: Low-dimensional spatial-DiT check on CelebA, averaged over seeds 39–43. The channel range $d_{\text{latent}} = 5\text{--}10$ corresponds to 80–160 scalar latent coordinates. Memorization rises from about 22% at $d_{\text{latent}} = 5$ to roughly 48% by $d_{\text{latent}} = 8\text{--}10$, while FID improves from about 50 to about 36. This lower-dimensional sweep suggests that there may be a representation-limited or partially null area at very small channel counts, but the transition is not the same clean increasing-then-decreasing curve as the vector-latent MLP.

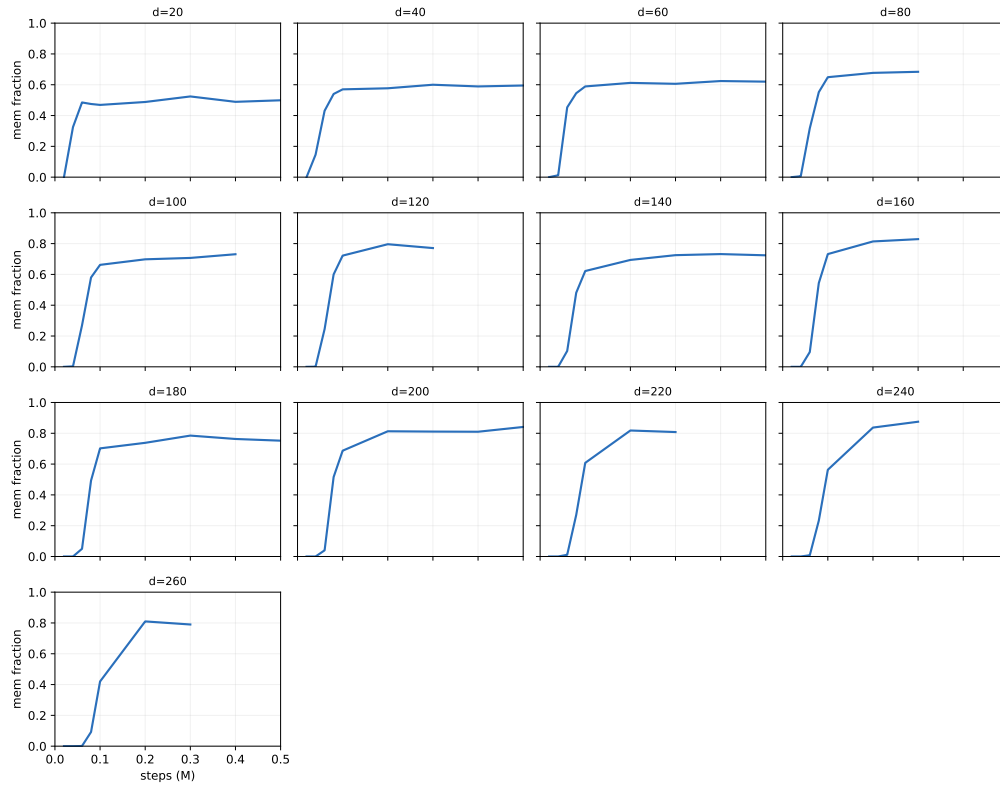


Figure 31: Per-channel spatial-DiT memorization trajectories. These panels are included to make clear that the broad pilot is not a direct replacement for the MLP evidence. Adjacent widths can behave differently because the architecture sees a structured grid rather than an unordered vector of latent coordinates.

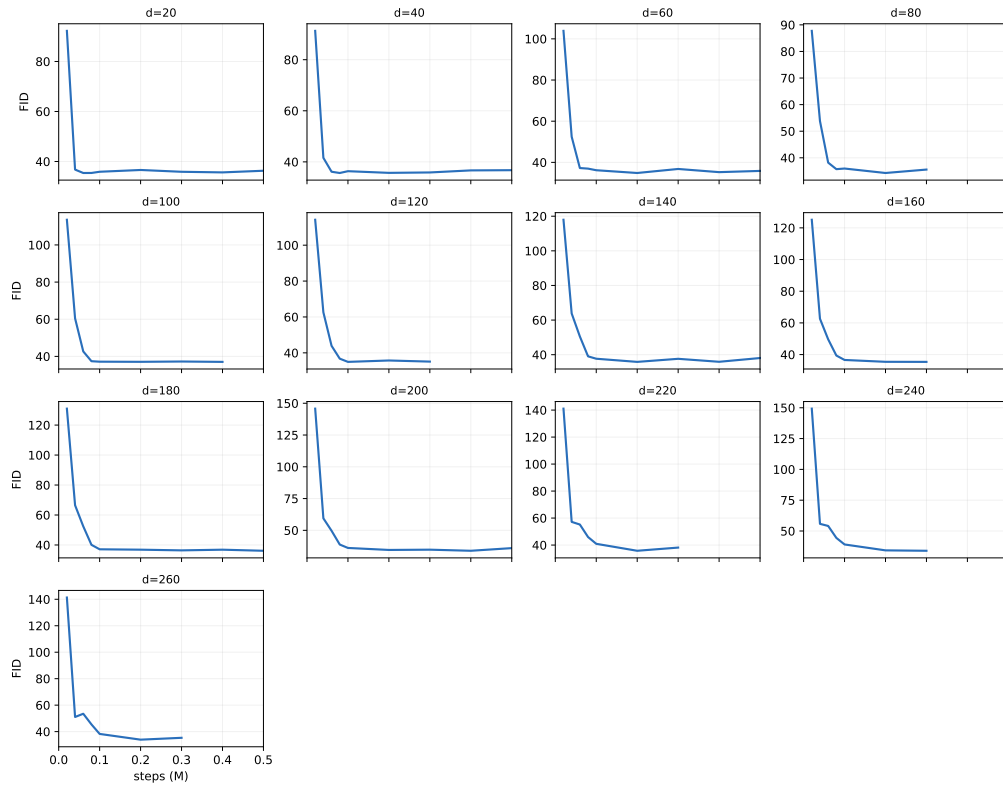


Figure 32: Per-channel spatial-DiT FID trajectories. Some spatial widths improve quality without reducing memorization, so sample quality and direct recall remain separate diagnostics in the DiT pilot as in the vector-latent MLP experiments.

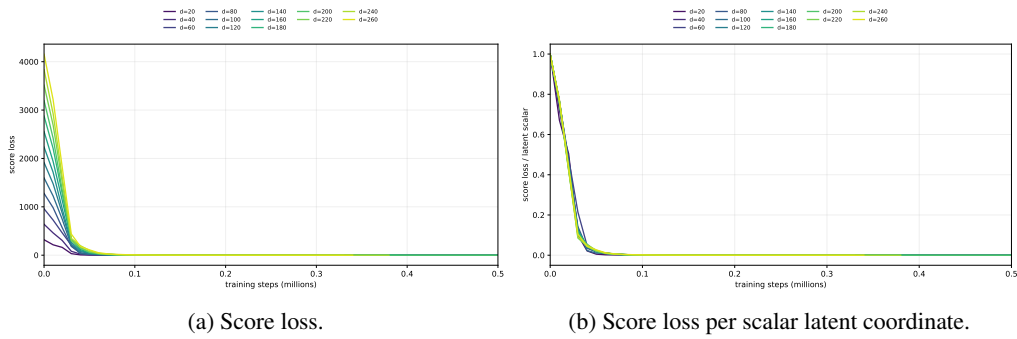


Figure 33: Spatial-DiT optimization diagnostics. The per-scalar loss divides by $16d_{\text{latent}}$ to remove the trivial increase in the number of latent coordinates. These curves help distinguish ordinary training instability from genuine changes in memorization or FID.

H DATA-SPECTRUM MEMORIZATION PREDICTOR DIAGNOSTICS

The spectral predictor is a data-only diagnostic inspired by the RFNN mode-timescale theory. It does not inspect diffusion-model weights. Instead, it asks whether the frozen VAE representation already contains enough geometry to predict when memorization should become measurable.

Terms in the predictor. For each dataset and latent width d , $Z_d \in \mathbb{R}^{n \times d}$ denotes the matrix of VAE encoder means for the same diverse training subset used by the diffusion model; in the main real-data experiments $n = 1000$. Rows are training images and columns are latent coordinates. We fix diffusion time $t = 0.1$ and form

$$M_t(d) = e^{-2t} Z_d^\top Z_d / n + (1 - e^{-2t}) I_d.$$

The first term is the empirical latent covariance after deterministic diffusion shrinkage, while the second term is the isotropic Gaussian noise injected by the forward diffusion process. The eigenvalues $\lambda_i(d)$ of $M_t(d)$ label the d_{latent} data-side modes of the feature correlation matrix (their RFNN rates are $\eta a_1(q_d)^2 \lambda_i / d_{\text{latent}}$); they do not label the $n - d_{\text{latent}}$ sample-specific modes that carry memorization (Appendix I.6). Training step is denoted by s . The scalar κ absorbs $\eta a_1(q_d)^2 / d_{\text{latent}}$ and is therefore only approximately d_{latent} -independent; it is fit once per dataset on all uncensored latent widths.

Given these eigenvalues, the spectral clock is

$$P_d(s) = \frac{\sum_i w_i(d) (1 - e^{-\kappa \lambda_i(d)s})^2}{\sum_i w_i(d)}.$$

The primary figures use $w_i = 1$, so $P_d(s)$ is the average absorbed-mode mass and lies between zero and one. The weighted diagnostic uses de-noised excess weights $w_i(d) = \max\{\lambda_i(d) - \beta_d, 0\}$, where β_d is an empirical isotropic-noise floor. To turn the clock into a memorization fraction, we use one anchored sigmoid response

$$\hat{m}_d(s) = \frac{\sigma(a(P_d(s) - b)) - \sigma(-ab)}{1 - \sigma(-ab)},$$

so that zero absorbed pressure gives zero predicted memorization. The constants (κ, a, b) are fit once per dataset using the dimensions where the VAE already preserves pixel-space identity: $d \geq 70$ for CelebA and $d \geq 140$ for CIFAR-10. For a memorization level q , the predicted hitting time is $\hat{\tau}_q(d) = \inf\{s : \hat{m}_d(s) \geq q\}$. Empirical hitting times are censored when a seed never reaches level q within 5M steps. The KNN memorization test itself is always measured in decoded pixel space using the nearest-neighbor ratio threshold $1/3$.

How n enters. The training-set size appears in two places. First, it changes the empirical covariance $Z_d^\top Z_d / n$ if the encoded subset is recomputed. Second, it changes the interpretation of a memorization fraction: the first absolute sample corresponds to level $q = 1/n$, while “first 1% memorized” is the fixed level $q = 0.01$. Figure 34 isolates the second effect using the fixed $n = 1000$ spectra from the main experiments. It is therefore a sensitivity diagnostic, not a replacement for a full empirical n -sweep in which VAEs and diffusion models are retrained or at least re-evaluated on newly encoded subsets.

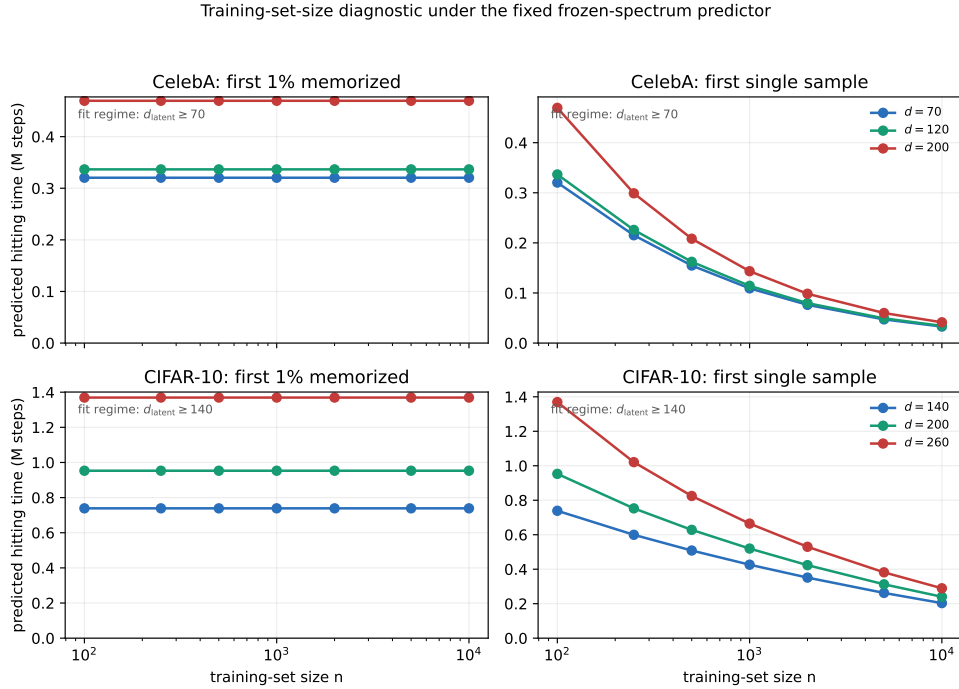


Figure 34: Training-set-size diagnostic for the frozen-spectrum predictor. The left column asks when the predicted memorization fraction first reaches 1%. Because this is a fixed fraction $q = 0.01$ and the plot deliberately holds the saved $n = 1000$ latent covariance spectrum fixed, the curves are flat: this is not evidence that true memorization dynamics are independent of n , but a diagnostic of what the current frozen-spectrum shortcut does and does not model. A real n -dependence for the 1% time would require recomputing $M_{t,n}(d)$ on different n -sized subsets, or adding an explicit sample-count/sample-bulk term to the predictor. The right column changes the question to the first absolute memorized training example, using level $q = 1/n$; this threshold moves earlier as n grows because one sample is a smaller fraction of a larger dataset.

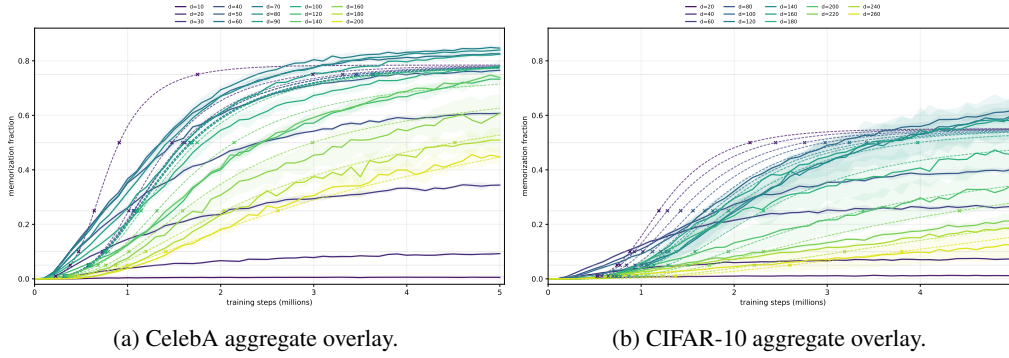


Figure 35: Spectral-predictor overlays on empirical memorization curves. For each latent dimension, the VAE-encoded training subset defines a diffused covariance spectrum; the spectral-pressure model then predicts when fixed memorization levels should be reached. The aggregate overlays show these predicted iso-memorization times on top of the mean empirical curves. For both datasets, the smallest latent dimensions are below the effective intrinsic dimensionality of the image manifold; in that capacity-limited regime, the VAE cannot faithfully encode sample identity, so the spectral predictor is not expected to match memorization onset as closely as it does after the representation becomes expressive enough. The point of the overlay is therefore not a single threshold hit, but the near-ordered increasing-then-decreasing memorization regime around the approximate intrinsic-dimension transition.

I THEORY: PROOFS AND DERIVATIONS

This appendix collects the full statements and proofs behind Section 4. Throughout, $\Delta_t = 1 - e^{-2t}$, the random-feature score model is $s_A(x_t) = p^{-1/2} A \tanh(Wx_t/\sqrt{d_{\text{latent}}}) = A\phi(x_t)$ with $W \in \mathbb{R}^{p \times d_{\text{latent}}}$ i.i.d. $\mathcal{N}(0, 1)$ frozen and $A \in \mathbb{R}^{d_{\text{latent}} \times p}$ trained from $A(0) = 0$, and $U = \frac{1}{n} \sum_{\mu} \mathbb{E}_{\eta}[\phi(x_t^{\mu})\phi(x_t^{\mu})^{\top}]$ is the feature second-moment operator. Every statement is labelled *proved*, *sketched* or *conjectured*; every displayed edge and timescale carries its d_{latent} -dependence explicitly. Sections are ordered as in the main text: setup and spectral timescales (Appendix I.1), the four spectral blocks of U (Appendix I.2), the buffer ratio (Appendix I.3), the exactly solvable linear score model (Appendix I.4), the bridge from residual sample-bulk error to near-duplicate generation (Appendix I.5), the fraction-level predictor (Appendix I.6), general latent spectra and posterior collapse (Appendix I.7), and regime boundaries (Appendix I.8, with a closing list of configurations outside the hypotheses).

I.1 SETUP, STANDING ASSUMPTIONS, AND SPECTRAL DEFINITIONS OF τ_{gen} AND τ_{mem}

Setting. Data: $x^{\mu} = Q(m_{c(\mu)} + \xi^{\mu})$, $\mu = 1, \dots, n$, with k centres m_c of norm s in $\text{span}(e_1, \dots, e_{d_{\text{intrinsic}}})$, $\xi^{\mu} \sim \mathcal{N}(0, \text{diag}(\sigma_{\text{sig}}^2 I_{d_{\text{intrinsic}}}, \sigma_{\perp}^2 I_{d_{\text{latent}} - d_{\text{intrinsic}}}))$, $Q \in O(d_{\text{latent}})$ Haar. Since W is Gaussian and the loss is rotation-equivariant we set $Q = I$. Forward process $x_t = e^{-t}x + \sqrt{\Delta_t}\eta$; $M_t = \frac{1}{n} \sum_{\mu} \mathbb{E}_{\eta}[x_t^{\mu}x_t^{\mu\top}] = e^{-2t}(\hat{C} + \hat{S}) + \Delta_t I$ with $\hat{C} = n^{-1} \sum_{\mu} m_{c(\mu)}m_{c(\mu)}^{\top}$ and $\hat{S} = n^{-1} \sum_{\mu} [\xi^{\mu}\xi^{\mu\top} + m_{c(\mu)}\xi^{\mu\top} + \xi^{\mu}m_{c(\mu)}^{\top}]$, including the finite-sample cross terms; population block eigenvalues $\alpha_t^2 = e^{-2t}(s^2/d_{\text{intrinsic}} + \sigma_{\text{sig}}^2) + \Delta_t$, $\beta_t^2 = e^{-2t}\sigma_{\perp}^2 + \Delta_t$. Features $\phi(x) = p^{-1/2} \tanh(Wx/\sqrt{d_{\text{latent}}})$; the code stores $W/\sqrt{d_{\text{latent}}}$ as $\mathcal{N}(0, 1/d_{\text{latent}})$ entries and omits $p^{-1/2}$ in its logged U , so *code eigenvalues are p times ours*. Two feature operators are distinguished throughout: the single-draw $U_k^{(1)}$ (one noise draw per point; rank $\leq \min(p, n)$) and its expectation $U = \mathbb{E}_{\eta} U_k^{(1)}$ (full rank; the operator governing the mean training dynamics). The released eigenvalue files contain $U^{(50)}$, an estimate of the expectation: they are full rank and show only a soft step at index n ($\lambda_n/\lambda_{n+1} = 1.00, 1.7, 1.7$ at $d_{\text{latent}} = 20, 60, 200$). Empirical objective L_n , population objective $L(A) = \frac{1}{2} \mathbb{E} \|A\phi(x_t) - y\|^2$, $\bar{U} = \mathbb{E}[\phi\phi^{\top}]$, $\bar{V} = \mathbb{E}[y\phi^{\top}] = \mathbb{E}[s^*(x_t)\phi(x_t)^{\top}]$ (tower property), $s^* = \nabla \log p_t$. Gaussian-equivalence coefficients $a_1(q) = \mathbb{E}[\tanh'(\sqrt{q}z)]$,

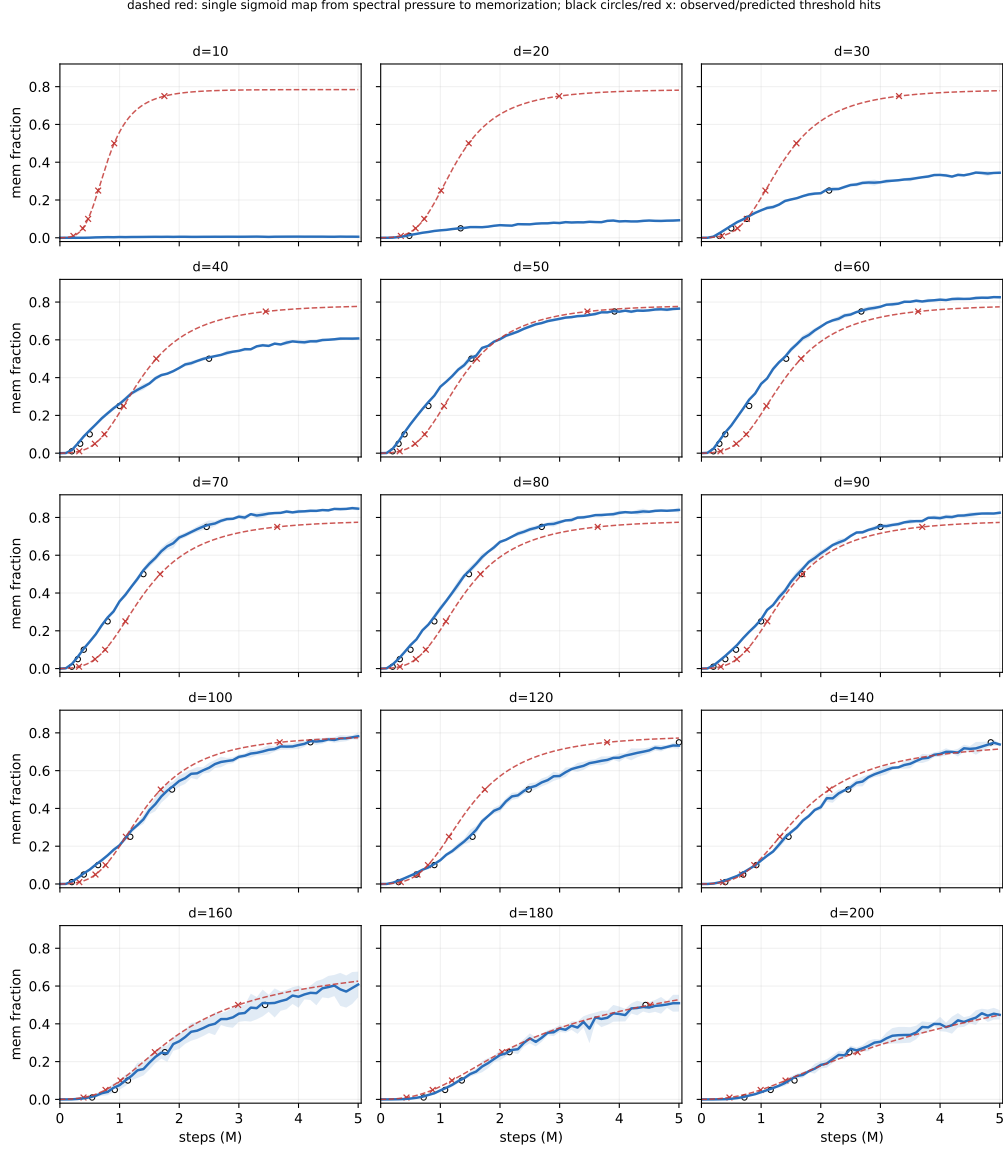


Figure 36: CelebA spectral-predictor small multiples. Each panel compares the predicted iso-memorization checkpoints against the empirical memorization curve for one latent dimension. This is the most direct visual check of the predictor because it preserves the full curve shape and makes censored or delayed thresholds visible without compressing them into a single cross-dimensional summary. Mismatches at the smallest d_{latent} values are expected because those VAEs are below the effective intrinsic dimension of CelebA and do not yet have capacity to preserve identity. At larger d_{latent} , the predicted checkpoints track the smooth slowdown and decline of memorization in the excess-buffer regime.

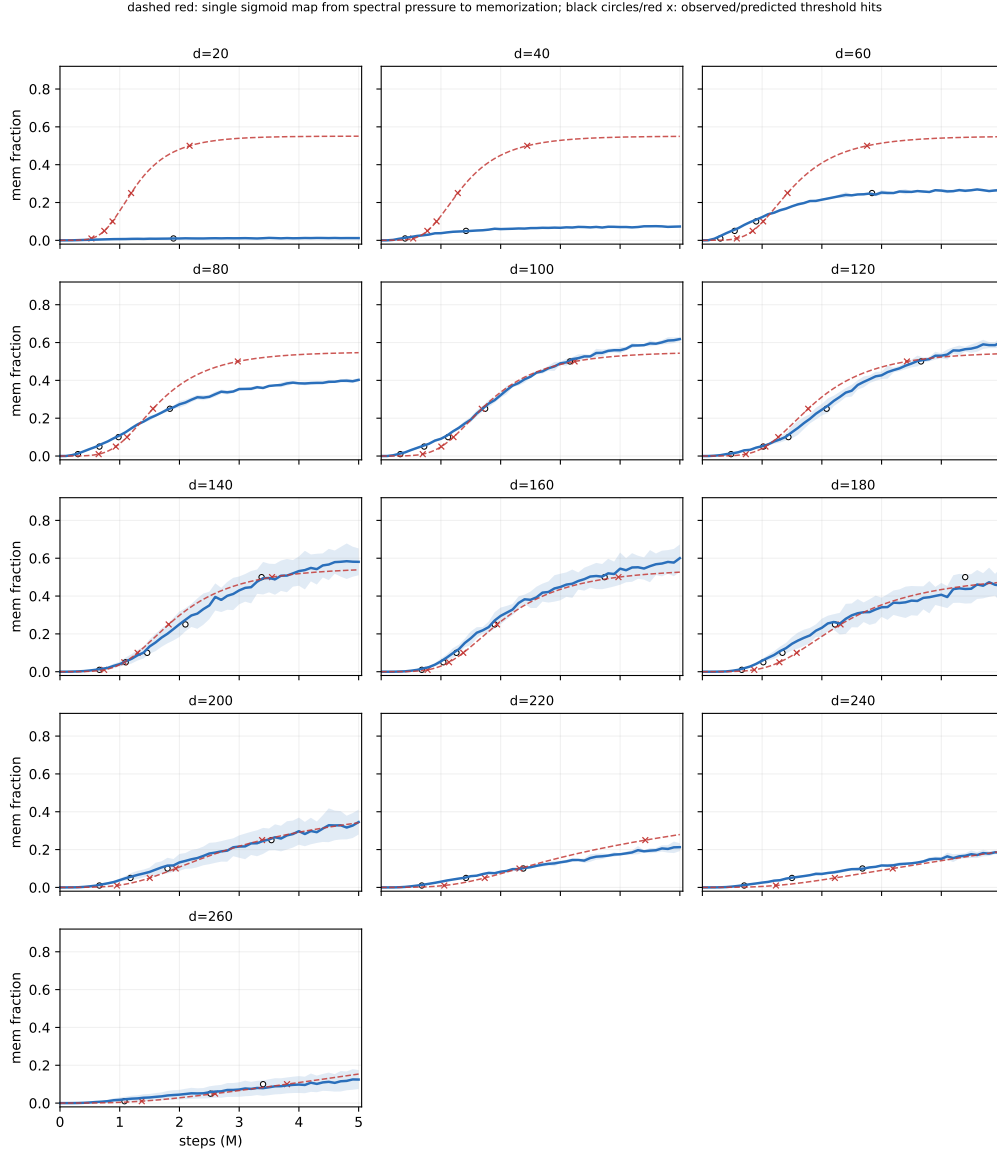


Figure 37: CIFAR-10 spectral-predictor small multiples. The same globally calibrated spectral-pressure rule is compared to each dimension’s empirical memorization curve. Deviations at low dimension are expected when the VAE bottleneck limits pixel-space identity: below the effective intrinsic dimension of CIFAR-10, the model is capacity-limited rather than operating in the excess-buffer regime. Higher-dimensional curves test whether data-level covariance predicts the timing of memorization onset once the representation is expressive enough. The visual point is the same increasing-then-decreasing regime seen in the aggregate curves, with relatively little irregular dimension-to-dimension variation once the low-dimensional bottleneck is past.

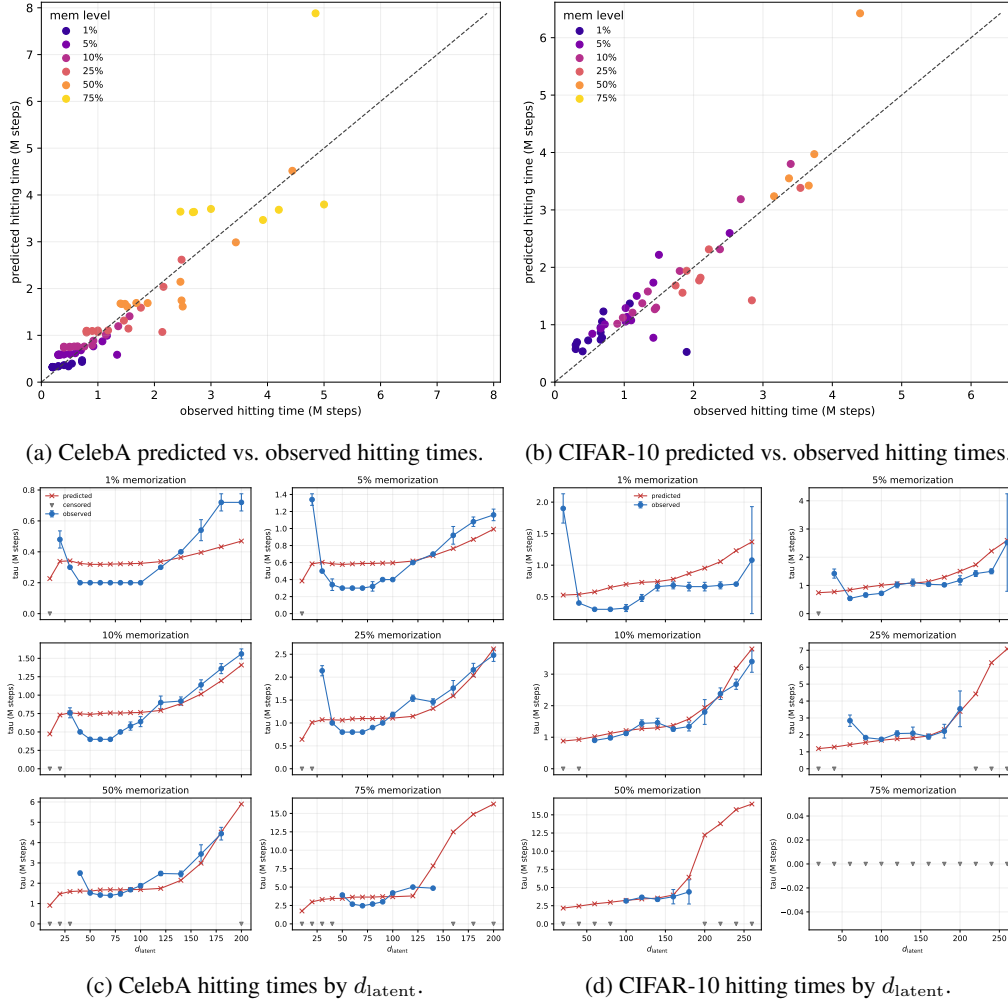


Figure 38: Hitting-time summaries from the spectral predictor. The predicted versus observed panels test whether a single globally calibrated spectral clock can explain multiple memorization thresholds across dimensions. The τ -versus- d_{latent} panels show how each fixed memorization level shifts as the latent dimension changes. These summaries are intentionally more compressed than the curve overlays: censoring, low-dimensional VAE bottlenecks, and nonmonotone empirical curves all become single plotted points or missing thresholds. Agreement in these plots therefore indicates a stronger form of predictive alignment than visual agreement in the small multiples alone. Disagreement at very small d_{latent} is expected when the VAE is below the dataset’s effective intrinsic dimension, because the diffusion model lacks a faithful latent representation on which the excess-buffer spectral argument could operate. The cleanest agreement should occur after the approximate intrinsic threshold, where increasing d_{latent} mainly adds buffer and shifts the memorization hitting times later.

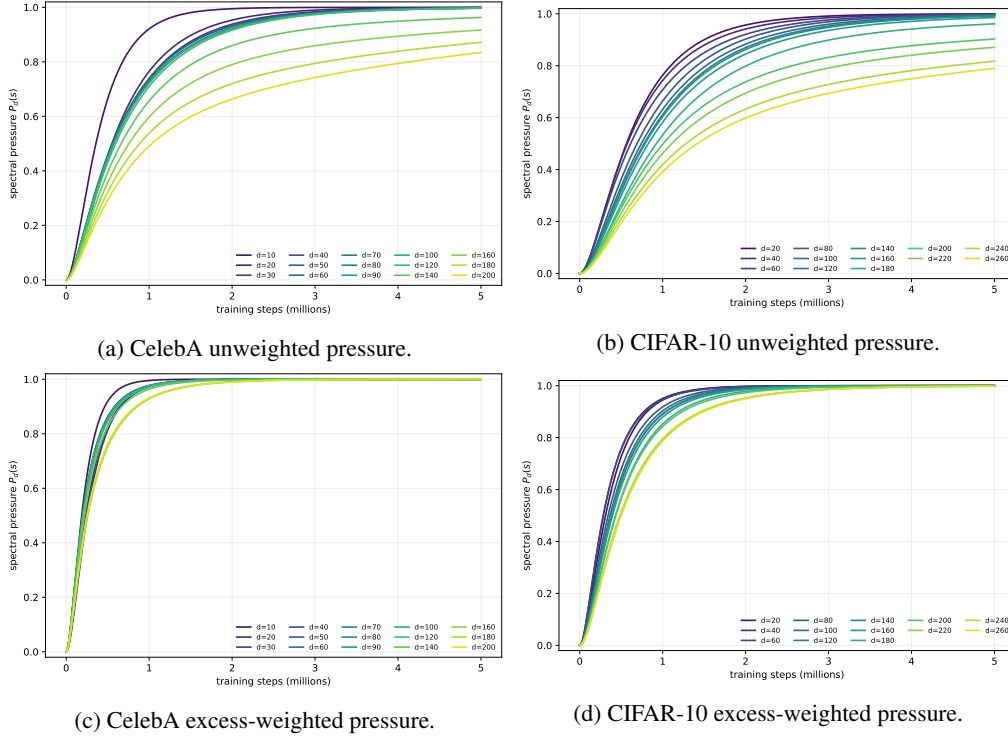


Figure 39: Spectral pressure curves used to generate predicted hitting times. For each d_{latent} , the curve accumulates mode absorption as a function of the calibrated training-time variable using the eigenvalues of the diffused latent covariance. The unweighted pressure treats all covariance eigenmodes equally, while the excess-weighted pressure gives more mass to modes whose eigenvalues rise above the isotropic diffusion floor. If the spectral picture is correct, dimensions with lower pre-activation variance q_d (equivalently, more added low-variance coordinates) should have smaller sample-specific eigenvalues $g(q_d)/n$ and later memorization; the unweighted pressure reproduces this only through the growing number of floor eigenvalues, i.e. through posterior collapse rather than through data-carrying modes.

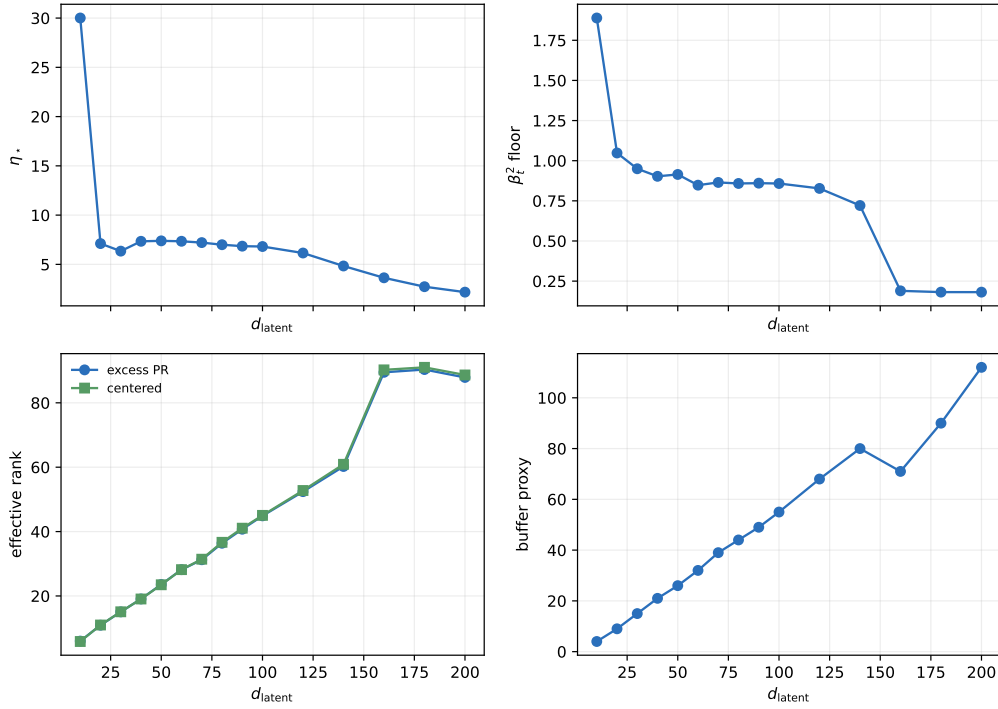


Figure 40: CelebA data-level spectral summaries across latent dimension. The panels summarize the encoded training-set covariance through eigenvalue statistics, effective rank, residual-mass proxies, and the pre-activation variance $q_d = \text{tr } M_t / d_{\text{latent}}$. These quantities are computed before training the diffusion model, so they isolate what the VAE representation and data subset contribute to the predicted memorization schedule. Their trends help explain why the spectral predictor becomes more accurate once d_{latent} is large enough for the VAE to preserve identity while q_d keeps falling even after the effective rank saturates (CelebA $d_{\text{latent}} \geq 160$). In other words, the data-level spectra mark the approximate transition from the increasing representation-capacity regime to the decreasing memorization regime.

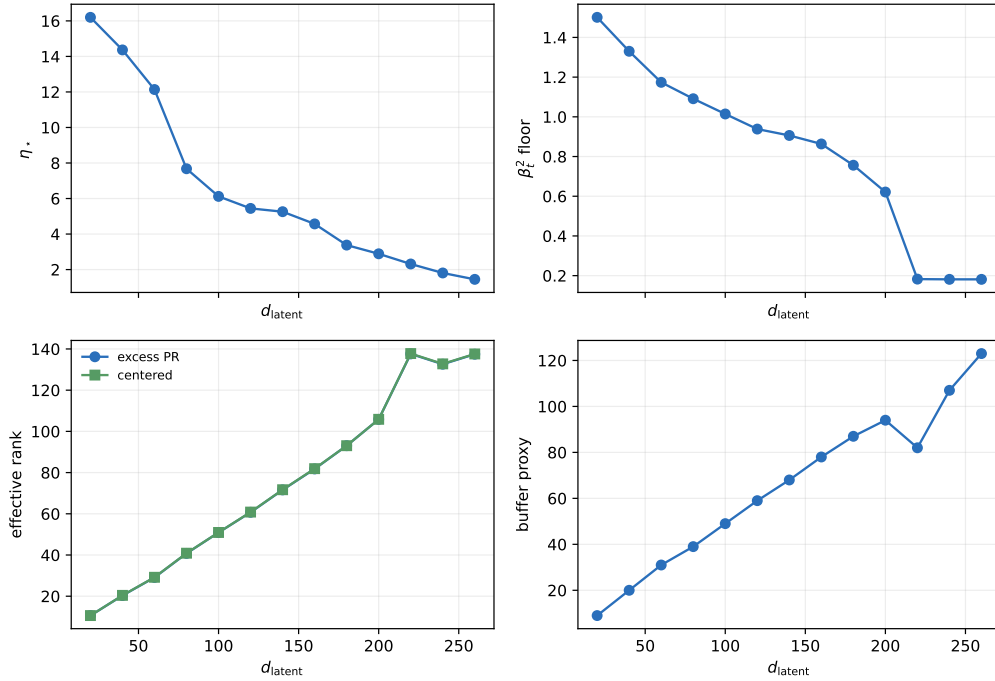


Figure 41: CIFAR-10 data-level spectral summaries across latent dimension. The same covariance statistics are computed before diffusion training, isolating the contribution of the VAE representation and selected training subset. The comparison with CelebA shows which trends are dataset-specific and which are shared consequences of increasing the latent buffer. The shared qualitative pattern is the same: an approximate intrinsic-dimension transition followed by a high-dimensional excess-buffer regime with delayed memorization.

$a_*(q)^2 = \mathbb{E}[\tanh(\sqrt{q}z)^2] - a_1(q)^2q$, $z \sim \mathcal{N}(0, 1)$ (Pennington & Worah, 2017; Benigni & P      , 2021; Goldt et al., 2022; Hu & Lu, 2023; Mei & Montanari, 2022).

Related work and scope. Bonnaire et al. (2025) derive a spectral description for noise-averaged random features with general Gaussian clean-data covariance; George et al. (2026) treat Gaussian subspace data. The gradient-flow identities below are finite-size algebra given U . The edge scales in Appendix I.2 are additional approximations, not those theorems evaluated at already-diffused covariance atoms. The operator distinctions are recorded next.

I.1.1 OPERATOR DEFINITIONS AND RELATION TO PRIOR THEORY

For fixed training data and W , distinguish

$$U_k^{(1)} = \frac{1}{n} \sum_{\mu} \phi(x_t^{\mu,k}) \phi(x_t^{\mu,k})^{\top}, \quad U^{(K)} = \frac{1}{K} \sum_{k=1}^K U_k^{(1)}, \quad U = \mathbb{E}_{\xi} U_k^{(1)}.$$

Exactly, $\text{rank } U_k^{(1)} \leq \min(p, n)$ and $\text{rank } U^{(K)} \leq \min(p, Kn)$. The first has at least $p - n$ zeros when $p > n$, with equality only at full row rank. Neither bound forces $p - n$ zeros in the expectation. The released `compute_U` uses $K = 50$ by default; the fixed- q control uses $K = 40$. Both estimate U , but their eigenvalues are finite-Monte-Carlo quantities. Fresh-noise SGD has the mean Euler recursion of Proposition I.2; its mean loss also contains parameter-variance terms and need not equal the loss of the mean iterate.

An exact decomposition follows from conditional covariance. With $m_{\mu} = \mathbb{E}_{\xi} \phi(x_t^{\mu})$,

$$\begin{aligned} U &= B + C, \\ B &= \frac{1}{n} \sum_{\mu} m_{\mu} m_{\mu}^{\top}, \\ C &= \frac{1}{n} \sum_{\mu} \text{Cov}_{\xi}(\phi(x_t^{\mu})), \\ B, C &\succeq 0, \quad \text{rank } B \leq n. \end{aligned} \tag{10}$$

This gives interlacing bounds relative to C , but no constant floor unless further structure of C is established. Thus the cuts at $d_{\text{intrinsic}}, d_{\text{latent}}, n$ are reporting conventions for $U^{(K)}$, not dimensions of invariant signal/sample subspaces proved by rank alone.

In Bonnaire et al. (2025, Theorem 3.2 and Appendix C.5), the Gaussian-equivalent noise-averaged operator has a floor weight $1 - (n + d_{\text{latent}})/p$ in the stated overparameterized regime with $p > n + d_{\text{latent}}$. The associated $n + d_{\text{latent}}$ constraints differ from the single-copy rank count. Their Theorem 3.1 uses the clean Gaussian covariance, and its derivation uses replica reasoning. A two-atom Gaussian surrogate here would use clean atoms $s^2/d_{\text{intrinsic}} + \sigma_{\text{sig}}^2$ and $\sigma_{\perp}^2$; the source already includes diffusion. Our mixture is not Gaussian, and fixed- $d_{\text{intrinsic}}$ spikes have vanishing spectral mass, so an additional outlier argument is needed. We therefore do not identify our four index groups with their spectral components.

George et al. (2026, Appendix B) construct a five-block pencil with dimensions $(p, d_{\text{latent}} - d_{\text{intrinsic}}, d_{\text{intrinsic}}, n, d_{\text{intrinsic}})$ for Gaussian subspace data; Appendix C uses a six-block construction. Pencil dimensions are not counts of spectral bands. The $\sigma_{\perp} \rightarrow 0$ comparison below is qualitative rather than a derived equality of the two models.

I.1.2 STANDING ASSUMPTIONS

Assumption I.1 (Standing hypotheses). (A1) *Fixed diffusion time.* $t > 0$ is fixed, $\Delta_t = 1 - e^{-2t}$; the readout is trained at this single t (repository: $t = 0.01$ for the RFNN, $t = 0.1$ for the real-data spectral clock).

(A2) *Two feature operators; over-parametrisation.* Write $U_k^{(1)} = \frac{1}{n} \sum_{\mu} \phi(x_t^{\mu,k}) \phi(x_t^{\mu,k})^{\top}$ for the *single-draw* operator built from one noise draw $\eta^{\mu,k}$ per training point, and $U = \mathbb{E}_{\eta} U_k^{(1)}$

for its expectation. Then $\text{rank } U_k^{(1)} \leq \min(p, n)$ exactly (this is the machine-precision cliff at index n in single-draw four-bulk spectra), whereas U is generically of full rank p : its $p - n$ lowest eigenvalues form a *tail block* $\mathcal{Z}$ that is *contiguous* with the bottom of the sample block (repository spectra, 50 draws, code units: $\lambda_n/\lambda_{n+1} = 1.00$ at $d_{\text{latent}} = 20$, 1.7 at $d_{\text{latent}} = 200$; a $(d_{\text{latent}}, p, n) = (20, 1280, 500)$ simulation gives tail max 1.79×10^{-6} against sample min 1.81×10^{-6} , stable from 50 to 400 draws). Since the training loop draws fresh noise at every step, the mean training trajectory follows the flow generated by U , not $U^{(1)}$ (Proposition I.2); all dynamical statements below refer to U . We require $p \geq n$ for the tail block to exist and for the four-block picture; the repository’s rule $p = 64d_{\text{latent}}$ violates this at $d_{\text{latent}} = 5$ ($p = 320 < n = 500$), where $\mathcal{M} = (d_{\text{latent}}, p]$ and $\mathcal{Z} = \emptyset$; Lemma I.3 is unchanged there.

(A3) *Data model and scaling regime.* $x^\mu = Q(m_{c(\mu)} + \xi^\mu)$ with k centres of norm s in a fixed $d_{\text{intrinsic}}$ -dimensional subspace, $\xi^\mu \sim \mathcal{N}(0, \text{diag}(\sigma_{\text{sig}}^2 I_{d_{\text{intrinsic}}}, \sigma_\perp^2 I_{d_{\text{latent}} - d_{\text{intrinsic}}}))$, Q Haar. The finite sweep holds $(k, s, d_{\text{intrinsic}}, \sigma_{\text{sig}}, \sigma_\perp, n)$ fixed and varies d_{latent} at fixed $\psi_p = p/d_{\text{latent}}$ (repository: 64). This is not a proportional asymptotic limit: $\psi_n = n/d_{\text{latent}}$ is not fixed; we require $\psi_n > 1$ so that the sample block is nonempty, and every statement that separates sample from population modes degrades as $\psi_n \downarrow 1$ (repository: $\psi_n = 2.5$ at $d_{\text{latent}} = 200$). Because $d_{\text{intrinsic}}$ is fixed, the signal block is a finite-rank (BBP-type) spike set, not a bulk.

(A4) *Operating point through q .* $q = q(d_{\text{latent}}) = \text{tr } M_t / d_{\text{latent}} \in [\beta_t^2, q(d_{\text{intrinsic}})]$ (Lemma I.6); $a_1(q) = \mathbb{E}[\tanh'(\sqrt{q}z)] \in (0, 1]$ and $a_*(q)^2 = \mathbb{E}[\tanh(\sqrt{q}z)^2] - a_1(q)^2 q > 0$ are always evaluated at $q(d_{\text{latent}})$ and never treated as d_{latent} -independent. No condition $s/\sqrt{d_{\text{latent}}} \ll 1$ is used.

(A5) *Block ordering.* The sorted spectrum of U is partitioned by index into $\mathcal{S} = \{1, \dots, d_{\text{intrinsic}}\}$, $\mathcal{N} = \{d_{\text{intrinsic}} + 1, \dots, d_{\text{latent}}\}$, $\mathcal{M} = \{d_{\text{latent}} + 1, \dots, n\}$, $\mathcal{Z} = \{n + 1, \dots, p\}$, with $\lambda_{\min}^{\mathcal{S}} > \lambda_{\max}^{\mathcal{N}}$ and $\lambda_{\min}^{\mathcal{N}} > \lambda_{\max}^{\mathcal{M}}$ (repository: $\lambda_{\min}^{\mathcal{N}}/\lambda_{\max}^{\mathcal{M}} = 7.7$ at $d_{\text{latent}} = 20$, 12 at $d_{\text{latent}} = 200$). No gap is assumed between $\mathcal{M}$ and $\mathcal{Z}$. In the isotropic case $\sigma_\perp = \sigma_{\text{sig}}$ the first inequality is dropped and $\mathcal{S} \cup \mathcal{N}$ is one population block of d_{latent} modes.

(A6') *Weighted spectral alignment (used only for the gap).* Let $\bar{U} = \mathbb{E}[\phi\phi^\top]$, $\bar{V} = \mathbb{E}[y y^\top]$, $\bar{u}_{ij} = v_i^\top \bar{U} v_j$, $\bar{u}_i = \bar{u}_{ii}$, and define the *alignment coefficient* $\gamma_i := \langle a_i^*, \bar{V} v_i \rangle / (\lambda_i \pi_i)$ for $\pi_i > 0$ ($\gamma_i = 1$ for a mode whose empirical optimum equals its population optimum, $\gamma_i = 0$ for a pure sample mode). Let $G_{\text{thr}} := \min\{G_\infty, 0.01d_{\text{latent}}/\Delta_t\}$ be the smallest gap level that matters (Remark I.12). We assume, uniformly in $T \geq 0$,

$$\begin{aligned} \left| \frac{1}{2} \sum_{i \neq j} (1 - e_i)(1 - e_j) \bar{u}_{ij} \langle a_i^*, a_j^* \rangle \right| &= o(G_{\text{thr}}), \\ \sum_{i \in \mathcal{S} \cup \mathcal{N}} \lambda_i \pi_i \left(\left| \frac{\bar{u}_i}{\lambda_i} - 1 \right| + 2|\gamma_i - 1| \right) &= o(G_{\text{thr}}). \end{aligned}$$

This is a hypothesis stated in the *mass-weighted* form that the exact gap formula requires: the target masses are large ($\Pi_{\mathcal{N}} \approx 1.1 \times 10^4$, $\Pi_{\mathcal{M}} \approx 2.4 \times 10^7$, $\Pi_{\mathcal{Z}} \approx 6 \times 10^8$ at $(d_{\text{latent}}, p, n) = (20, 1280, 500)$, since $a_i^* = V v_i / \lambda_i$), so an unweighted $o(1)$ in $\bar{u}_{ij}$ would not control the gap; at $d_{\text{latent}} = 20$, $\lambda^{\mathcal{N}} \Pi_{\mathcal{N}} \approx 60$ against $0.01d_{\text{latent}}/\Delta_t = 10$, so the population blocks must be aligned to relative accuracy $\ll 10\%$. Nothing is assumed about γ_i on $\mathcal{M} \cup \mathcal{Z}$: those coefficients are measured (Definition I.11).

Status. Not a theorem. (A2): the rank bound of the single-draw operator is exact linear algebra ($U_k^{(1)}$ is a sum of n rank-one terms); contiguity of the tail is read from the $\sigma_\perp = 0.5$ RFNN spectra (50 draws): $(\lambda_n, \lambda_{n+1})$ in code units = $(0.00239, 0.00238)$ at $d_{\text{latent}} = 20$, $(0.0122, 0.0072)$ at 60, $(0.0557, 0.0323)$ at 200. (A5) is the repository’s four-bulk observation. (A6') is a hypothesis to be tested by computing $\bar{U}, \bar{V}$ on held-out data (Prediction 2 below).

Numerical status of (A2). At $(d_{\text{latent}}, p, n, t) = (20, 1280, 500, 0.01)$ with the repository’s `compute_U`: one draw gives tail eigenvalues $\leq 10^{-17}$; 50 and 400 draws give an identical tail

(max 1.79×10^{-6} , mean 5.0×10^{-7} , 0.11% of $\text{tr } \mathbf{U}$), so this is the true $\mathbb{E}_\eta$ tail and not Monte-Carlo noise. The tail top equals the sample-block bottom (1.81×10^{-6} ; the Marchenko–Pastur lower edge $a_*^2(1 - \sqrt{n/p})^2/n = 4.6 \times 10^{-6}$ lies above the realised minimum). Under the clock $T = 0.02 \times \text{steps}$, $\lambda_{\mathcal{Z}} T \approx 0.003\text{--}0.011$ at 3×10^5 steps, and because $a_i^* = V v_i / \lambda_i$ with $\|V P_{\mathcal{Z}}\|_F^2 / \|V\|_F^2 = 4.3 \times 10^{-4}$, the tail target mass is $\Pi_{\mathcal{Z}} \approx 6 \times 10^8$ against $\Pi_{\mathcal{M}} \approx 2.4 \times 10^7$. The tail is therefore slow but not inert in Frobenius norm; it is inert in *loss* only to the extent of Proposition I.10’s bound (weights $\bar{u}_i, \lambda_i \lesssim 2 \times 10^{-6}$).

I.1.3 MEAN DYNAMICS, GRADIENT FLOW AND RESIDUAL SPECTRAL ERROR

Proposition I.2 (Mean Euler recursion of fresh-noise full-batch SGD; proved). *Let the training step be $A_{k+1} = A_k - h(A_k \mathbf{U}_k^{(1)} - V_k^{(1)})$ with $h = 2\Delta_t \eta_{\text{lr}} / d_{\text{latent}}$ for the coordinate-averaged logged loss and $(\mathbf{U}_k^{(1)}, V_k^{(1)})$ built from the k -th noise draw, $V_k^{(1)} = \frac{1}{n} \sum_{\mu} y^{\mu, k} \phi(x_t^{\mu, k})^\top$, draws i.i.d. across k , $A_0 = 0$, momentum 0. Then $\bar{A}_k := \mathbb{E}[A_k]$ satisfies exactly $\bar{A}_{k+1} = \bar{A}_k - h(\bar{A}_k \mathbf{U} - V)$ with $\mathbf{U} = \mathbb{E} \mathbf{U}_k^{(1)}$, $V = \mathbb{E} V_k^{(1)}$: the mean trajectory is the Euler discretisation of the flow of Lemma I.3 with $\delta T = h$.*

Proof. A_k is a function of draws $1, \dots, k-1$ only, hence independent of $(\mathbf{U}_k^{(1)}, V_k^{(1)})$. Taking expectations, $\mathbb{E}[A_k \mathbf{U}_k^{(1)}] = \mathbb{E}[A_k] \mathbb{E}[\mathbf{U}_k^{(1)}] = \bar{A}_k \mathbf{U}$ and $\mathbb{E}[V_k^{(1)}] = V$. The recursion is linear in A_k , so no further approximation is involved. (Numerically, the mean over 400 replicas of fresh-noise SGD at $(d_{\text{latent}}, p, n) = (4, 30, 12)$ matches the Euler scheme of the $\mathbb{E}_\eta$ flow to 1%, the Monte-Carlo error.) $\square$

Lemma I.3 (Closed-form gradient flow; proved). *Let $L_n(A) = \frac{1}{2n} \sum_{\mu} \mathbb{E}_\eta \|\phi(x_t^\mu) - y^\mu\|^2$ with $y^\mu = -\eta^\mu / \sqrt{\Delta_t}$, $\mathbf{U} = \frac{1}{n} \sum_{\mu} \mathbb{E}_\eta [\phi(x_t^\mu) \phi(x_t^\mu)^\top] = \sum_{i=1}^p \lambda_i v_i v_i^\top$ (eigenvalues sorted, $\lambda_i \geq 0$), $V = \frac{1}{n} \sum_{\mu} \mathbb{E}_\eta [y^\mu \phi(x_t^\mu)^\top]$. Then (i) every row of V lies in $\text{range}(\mathbf{U})$, so $A^* := V \mathbf{U}^+$ is the minimum-norm minimiser of L_n and $A^* \mathbf{U} = V$; (ii) the gradient flow $\dot{A} = -\nabla_A L_n(A) = -(A \mathbf{U} - V)$ from $A(0) = 0$ has the unique solution $A(T) = A^*(I - e^{-\mathbf{U}T})$; (iii) in the eigenbasis, $a_i(T) := A(T) v_i = a_i^*(1 - e^{-\lambda_i T})$ with $a_i^* = V v_i / \lambda_i$ for $\lambda_i > 0$, and $a_i(T) \equiv 0$ for $\lambda_i = 0$. Full rank does not imply active targets: a positive-eigenvalue mode is active only if $V v_i \neq 0$. For the reported tail at $d_{\text{latent}} = 20$, rates from 5×10^{-7} (mean) to 1.8×10^{-6} (maximum) give $\lambda_i T \approx 0.003\text{--}0.011$ at 3×10^5 default repository steps. (iv) The repository’s coordinate-averaged logged objective is $(2\Delta_t / d_{\text{latent}}) L_n$. With learning rate η_{lr} and zero momentum, its mean Euler increment is $h = 2\Delta_t \eta_{\text{lr}} / d_{\text{latent}}$, so $T = h \times \text{steps}$, faithful as a flow approximation when $h \lambda_1 \ll 1$. The default $\eta_{\text{lr}} = 0.01 d_{\text{latent}} / \Delta_t$ gives $T = 0.02 \times \text{steps}$; for $\tau_{\text{log}} = \eta_{\text{lr}} \times \text{steps}$, $T = (2\Delta_t / d_{\text{latent}}) \tau_{\text{log}}$.*

Proof. (i) If $u \perp \text{range}(\mathbf{U})$ then $0 = u^\top \mathbf{U} u = \frac{1}{n} \sum_{\mu} \mathbb{E}_\eta (u^\top \phi(x_t^\mu))^2$, hence $u^\top \phi(x_t^\mu) = 0$ a.s. for every μ , hence $V u = 0$. Thus $V = V \mathbf{U} \mathbf{U}^+$ and $A^* \mathbf{U} = V$. Since $L_n(A) = \frac{1}{2} \text{tr}(A \mathbf{U} A^\top) - \text{tr}(A V^\top) + c_n = \frac{1}{2} \text{tr}((A - A^*) \mathbf{U} (A - A^*)^\top) - \frac{1}{2} \text{tr}(A^* \mathbf{U} A^{*\top}) + c_n$, A^* minimises L_n and is the unique minimiser with rows in $\text{range}(\mathbf{U})$. (ii) $\frac{d}{dT} (A - A^*) = -(A - A^*) \mathbf{U}$ is linear with constant coefficients, so $A(T) - A^* = -A^* e^{-\mathbf{U}T}$. (iii) Multiply on the right by v_i ; the numerical tail exponents use $T = 6000$. (iv) Differentiating the coordinate-averaged objective gives $(2\Delta_t / d_{\text{latent}})(A \mathbf{U} - V)$, hence the mean Euler multiplier is $1 - h \lambda_i$. Fresh draws give the mean recursion of Proposition I.2, not equality of each stochastic trajectory to the flow. $\square$

Definition I.4 (Residual spectral error and target masses). Let $\pi_i := \|A^* v_i\|^2 = \|A^* P_i\|_F^2$, $P_i = v_i v_i^\top$, be the target mass in mode i , $i = 1, \dots, p$ ($\lambda_i = 0$ implies $\pi_i = 0$, while $V v_i = 0$ can also give zero target mass at positive λ_i ; for the full-rank $\mathbf{U}$ of (A2) the tail masses are the largest of all, $\pi_i = \|V v_i\|^2 / \lambda_i^2$). For a block $\mathcal{B} \subseteq \{1, \dots, p\}$ define

$$E_{\mathcal{B}}(T) := \sum_{i \in \mathcal{B}} \pi_i e^{-2\lambda_i T}, \quad \Pi_{\mathcal{B}} := E_{\mathcal{B}}(0) = \sum_{i \in \mathcal{B}} \pi_i, \quad \mathcal{L}_{\mathcal{B}}(T) := \frac{1}{2} \sum_{i \in \mathcal{B}} \lambda_i \pi_i e^{-2\lambda_i T},$$

and $E(T) = E_{\{1..p\}}(T)$. Because $\Pi_{\mathcal{Z}} \gg \Pi_{\mathcal{M}} \gg \Pi_{\mathcal{N}} \gg \Pi_{\mathcal{S}}$ (ratios $\sim 10^1$ – 10^3 at $d_{\text{latent}} = 20$: $\Pi_{\mathcal{S}} = 44$, $\Pi_{\mathcal{N}} = 1.1 \times 10^4$, $\Pi_{\mathcal{M}} = 2.4 \times 10^7$, $\Pi_{\mathcal{Z}} = 6 \times 10^8$ in a (20, 1280, 500) simulation), the *global* $E(T)$ is dominated by the tail and is not a useful observable; every timescale below is defined through a block-restricted $E_{\mathcal{B}}$ or through loss-weighted quantities, in which mode i enters with weight $\lambda_i \pi_i = \|V v_i\|^2 / \lambda_i$ or $\lambda_i^2 \pi_i = \|V v_i\|^2$.

Proposition I.5 (Exactness of the residual spectral error; proved). *Under Lemma I.3, for all $T \geq 0$: $\|A(T) - A^*\|_F^2 = E(T)$ and $L_n(A(T)) - L_n(A^*) = \mathcal{L}_{\{1..p\}}(T) = \frac{1}{2} \sum_{i \leq p} \lambda_i \pi_i e^{-2\lambda_i T}$. For any block $\mathcal{B}$, $\|A(T)P_{\mathcal{B}}\|_F^2 = \sum_{i \in \mathcal{B}} \pi_i (1 - e^{-\lambda_i T})^2$, so the observable absorption deficit $D_{\mathcal{B}}(T) := \Pi_{\mathcal{B}} - \|A(T)P_{\mathcal{B}}\|_F^2$ satisfies $E_{\mathcal{B}}(T) \leq D_{\mathcal{B}}(T) \leq 2\sqrt{\Pi_{\mathcal{B}} E_{\mathcal{B}}(T)}$. In particular the tail block satisfies $\|A(T)P_{\mathcal{Z}}\|_F^2 \leq T^2 \|VP_{\mathcal{Z}}\|_F^2$ and $\mathcal{L}_{\mathcal{Z}}(0) - \mathcal{L}_{\mathcal{Z}}(T) \leq T \|VP_{\mathcal{Z}}\|_F^2$ for all T .*

Proof. By Lemma I.3(ii), $A(T) - A^* = -\sum_i e^{-\lambda_i T} (A^* v_i) v_i^\top$; the v_i are orthonormal so $\|A(T) - A^*\|_F^2 = \sum_i e^{-2\lambda_i T} \pi_i$. The loss identity follows from $L_n(A) - L_n(A^*) = \frac{1}{2} \text{tr}((A - A^*)U(A - A^*)^\top)$. For the absorption, $A(T)P_{\mathcal{B}} = \sum_{i \in \mathcal{B}} (1 - e^{-\lambda_i T}) (A^* v_i) v_i^\top$, and $D_{\mathcal{B}}(T) = \sum_{i \in \mathcal{B}} \pi_i (2e_i - e_i^2)$ with $e_i = e^{-\lambda_i T}$; since $e_i^2 \leq 2e_i - e_i^2 \leq 2e_i$, $E_{\mathcal{B}} \leq D_{\mathcal{B}} \leq 2 \sum_{\mathcal{B}} \pi_i e_i \leq 2\sqrt{\Pi_{\mathcal{B}} E_{\mathcal{B}}}$ by Cauchy-Schwarz. Tail bounds: $(1 - e^{-x})^2 \leq x^2$ and $1 - e^{-2x} \leq 2x$ with $\pi_i \lambda_i^2 = \|V v_i\|^2$. Brute-force check at (4, 30, 12): all three identities hold to six decimals at $T = 0.3, 2, 10$. $\square$

I.1.4 OPERATING POINT

Lemma I.6 (Operating point along the sweep; proved). *Under (A3), with $M_t = \frac{1}{n} \sum_{\mu} \mathbb{E}[x_t^\mu x_t^{\mu\top}]$, $\alpha_t^2 = e^{-2t}(s^2/d_{\text{intrinsic}} + \sigma_{\text{sig}}^2) + \Delta_t$ and $\beta_t^2 = e^{-2t}\sigma_{\perp}^2 + \Delta_t$,*

$$\mathbb{E} q(d_{\text{latent}}) = \frac{\mathbb{E} \text{tr } M_t}{d_{\text{latent}}} = \beta_t^2 + \frac{e^{-2t}(s^2 + d_{\text{intrinsic}}(\sigma_{\text{sig}}^2 - \sigma_{\perp}^2))}{d_{\text{latent}}} = \beta_t^2 + \frac{d_{\text{intrinsic}}(\alpha_t^2 - \beta_t^2)}{d_{\text{latent}}}, \quad (11)$$

so $q(d_{\text{latent}})$ decreases monotonically to $\beta_t^2 \geq \Delta_t > 0$ as $d_{\text{latent}} \rightarrow \infty$ (never to 0). Along it $a_1(q(d_{\text{latent}}))$ is increasing in d_{latent} (proved) and $a_*(q(d_{\text{latent}}))^2$ is decreasing in d_{latent} (verified numerically for $q \in [0.01, 20]$; see proof). At $t = 0.01$, $\sigma_{\perp} = 0.5$ (Gauss-Hermite, 200 nodes): $\beta_t^2 = 0.2649$, $\alpha_t^2 = 2.7644$, and

d_{latent}	5	20	60	200	∞
q	2.764	0.890	0.473	0.327	0.265
$a_1(q)^2$	0.180	0.393	0.541	0.625	0.674
$a_*(q)^2$	0.0785	0.0235	0.0090	0.0047	0.0030

At $\sigma_{\perp} = 0.01$ the sweep reaches only $q(200) = 0.0885$ ($a_1^2 = 0.854$, $a_*^2 = 2.6 \times 10^{-4}$, $18\times$ below its $\sigma_{\perp} = 0.5$ value); the limit is $q(\infty) = \beta_t^2 = 0.0199$. As $q \rightarrow 0$, $a_1(q) = 1 - q + 2q^2 + O(q^3)$ and $a_*(q)^2 = \frac{2}{3}q^3 + O(q^4)$; the $O(q^4)$ term is large ($\frac{2}{3}q^3$ overshoots a_*^2 by 16% at $q = 0.02$ and by 77% at $q = 0.0885$), so the expansion is not quantitatively usable at any point of either sweep and the tabulated values should be used instead.

Proof. $\mathbb{E} \text{tr } M_t = e^{-2t} \mathbb{E} \|x^\mu\|^2 + d_{\text{latent}} \Delta_t$ and $\mathbb{E} \|x^\mu\|^2 = s^2 + d_{\text{intrinsic}} \sigma_{\text{sig}}^2 + (d_{\text{latent}} - d_{\text{intrinsic}}) \sigma_{\perp}^2$ (Q orthogonal). Divide by d_{latent} and regroup; the second form uses $d_{\text{intrinsic}} \alpha_t^2 = e^{-2t}(s^2 + d_{\text{intrinsic}} \sigma_{\text{sig}}^2) + d_{\text{intrinsic}} \Delta_t$. $a_1(q) = \mathbb{E}[1 - \tanh^2(\sqrt{q}z)]$ is decreasing in q because $\tanh^2(\sqrt{q}z)$ is increasing in q pointwise; hence $a_1(q(d_{\text{latent}}))$ increases with d_{latent} . For a_*^2 : it equals $\sum_{k \geq 3} c_k(q)^2$ with $c_k(q) = \mathbb{E}[\tanh(\sqrt{q}z) \text{He}_k(z)] / \sqrt{k!}$ (Hermite coefficients; $c_1^2 = a_1^2 q$ by Stein's lemma), and each $c_k(q)^2$ is not obviously monotone, so we do not claim a proof; on a 200-node Gauss-Hermite grid a_*^2 is strictly increasing in q on $[0.01, 20]$ and bounded by 1. Expansion: $\tanh u = u - u^3/3 + 2u^5/15 + O(u^7)$ gives $\mathbb{E} \tanh^2(\sqrt{q}z) = q - 2q^2 + \frac{17}{3}q^3 + O(q^4)$, $a_1 = 1 - q + 2q^2 + O(q^3)$, $a_1^2 q = q - 2q^2 + 5q^3 + O(q^4)$, $a_*^2 = \frac{2}{3}q^3 + O(q^4)$. Numerical check: a_*^2 vs $\frac{2}{3}q^3 = 6.2 \times 10^{-7}$ vs 6.7×10^{-7} ($q = 0.01$), 4.6×10^{-6} vs 5.3×10^{-6} (0.02), 3.5×10^{-4} vs 6.7×10^{-4} (0.1). The identity $a_1^2 q + a_*^2 = \mathbb{E} \tanh^2(\sqrt{q}z)$ holds exactly by definition and is the trace check of Appendix I.2. $\square$

Trace check against the released spectra. $\text{tr } U/p = 0.545 \rightarrow 0.209$ over $d_{\text{latent}} = 5 \rightarrow 200$ versus $\mathbb{E} \tanh^2(\sqrt{q}z) = 0.576 \rightarrow 0.209$; the sample-block mean is 0.72 – $0.89 \times a_*^2/n$ (Jensen dispersion of

the per-sample q_μ); the noise-dimension medians $5.22 \rightarrow 9.15$ (code units) versus the leading-order $a_1^2 \beta_t^2 \psi_p = 4.76 \rightarrow 10.6$ overshoot by 10–15% at large d_{latent} and saturate from $d_{\text{latent}} = 100$. This is why no edge and no timescale below is d_{latent} -independent.

I.1.5 CLOCKS

Remark I.7 (Three clocks and their conversion). The normalized feature operator has $\lambda = \lambda_{\text{code}}/p$, whereas the coordinate-averaged logged objective is $(2\Delta_t/d_{\text{latent}})L_n$. For zero-momentum SGD at the default learning rate $0.01d_{\text{latent}}/\Delta_t$,

$$T = \frac{2\Delta_t}{d_{\text{latent}}} \tau_{\log} = 0.02s, \quad s_i^{\text{e-fold}} = \frac{p}{0.02\lambda_{i,\text{code}}} = \frac{3200d_{\text{latent}}}{\lambda_{i,\text{code}}} \quad (p = 64d_{\text{latent}}).$$

Here s is the step count and $\tau_{\log} = \eta_{\text{lr}} s$. The exact mean-Euler e-fold count is $-1/\log(1 - 0.02\lambda_{i,\text{code}}/p)$ when its multiplier lies in $(0, 1)$. Using the rounded released eigenvalues, signal e-fold counts are $788 \rightarrow 8663$ over $d_{\text{latent}} = 5 \rightarrow 200$; noise-dimension counts are $12403 \rightarrow 301887$ over $20 \rightarrow 200$; index-based sample-top counts are $16719 \rightarrow 3699422$ over $5 \rightarrow 200$. Flow and default-step clocks therefore have the same dimension trend: signal and sample-top clocks grow about $11\times$ and $221\times$, respectively. These are spectral parameter e-folds, not observed stopping times. Ratios within a run are unchanged by conversion. Learning-rate overrides or momentum require their own conversion. The coordinate-summed v3 loss has a different clock (Remark I.40); it must not be applied to the v2 saved sweeps.

I.1.6 GENERALISATION TIME

Definition I.8 (Generalisation time). For a tolerance $\varepsilon \in (0, 1)$,

$$\tau_{\text{gen}}(\varepsilon) := \inf\{T \geq 0 : E_{\mathcal{S}}(T) \leq \varepsilon \Pi_{\mathcal{S}}\}, \quad \tau_{\text{pop}}(\varepsilon) := \inf\{T \geq 0 : E_{\mathcal{S} \cup \mathcal{N}}(T) \leq \varepsilon \Pi_{\mathcal{S} \cup \mathcal{N}}\}.$$

τ_{gen} is the time at which the signal-subspace score (the part that determines sample quality) is absorbed to relative tolerance ε ; τ_{pop} additionally requires the noise-dimension block (target $s_{\mathcal{N}}^*(x) = -x_{\mathcal{N}}/\beta_t^2$, exactly linear) to be absorbed. In the isotropic case only τ_{pop} is defined. Both are block functionals of $\mathcal{S}$ (resp. $\mathcal{S} \cup \mathcal{N}$) only and are unaffected by the tail block $\mathcal{Z}$ or by the sample block.

Proposition I.9 (τ_{gen} is set by the slowest signal mode; proved given the block scale of Appendix I.2). Let $\mathcal{B} \in \{\mathcal{S}, \mathcal{S} \cup \mathcal{N}\}$, $\lambda_{\min}^{\mathcal{B}} = \min\{\lambda_i : i \in \mathcal{B}, \pi_i > 0\}$ attained at i_o , and $\theta_{\mathcal{B}} := \log(\Pi_{\mathcal{B}}/\pi_{i_o}) \geq 0$. Then for every $\varepsilon \in (0, 1)$,

$$\frac{\log(1/\varepsilon) - \theta_{\mathcal{B}}}{2\lambda_{\min}^{\mathcal{B}}} \leq \tau_{\mathcal{B}}(\varepsilon) \leq \frac{\log(1/\varepsilon)}{2\lambda_{\min}^{\mathcal{B}}}, \quad (12)$$

and for the observable deficit $D_{\mathcal{B}}$, $\tau_{\mathcal{B}}^D(\varepsilon) := \inf\{T : D_{\mathcal{B}}(T) \leq \varepsilon \Pi_{\mathcal{B}}\}$ satisfies

$$\frac{\log(1/\varepsilon) - \theta_{\mathcal{B}}}{\lambda_{\min}^{\mathcal{B}}} \leq \tau_{\mathcal{B}}^D(\varepsilon) \leq \frac{\log(2/\varepsilon)}{\lambda_{\min}^{\mathcal{B}}}.$$

The slowest signal mode is not the block scale $a_1(q)^2 \alpha_t^2/d_{\text{latent}}$ but

$$\lambda_{\min}^{\mathcal{S}} = \frac{a_1(q(d_{\text{latent}}))^2 (e^{-2t} \ell_{\min} + \Delta_t)}{d_{\text{latent}}} (1 + o(1)), \quad (13)$$

where $\ell_{\min}$ is the smallest eigenvalue of the realized signal-subspace second moment $(\hat{C} + \hat{S})_{\mathcal{S}}$. Its finite-sample value need not exceed σ_{sig}^2 ; that lower bound applies to the population matrix. Holding the realized live coordinates fixed makes $\ell_{\min}$ dimension-independent. The conditional edge approximation yields $\tau_{\text{gen}} \propto d_{\text{latent}}/[a_1(q(d_{\text{latent}}))^2 (e^{-2t} \ell_{\min} + \Delta_t)]$. On the released spectra its e-fold scale grows about $11\times$ over $d_{\text{latent}} = 5 \rightarrow 200$, in both flow units and default repository steps ($788 \rightarrow 8663$). These are e-fold scales; tolerance hitting times also depend on the target weights in the displayed bounds.

Proof. Upper bound: $E_{\mathcal{B}}(T) \leq \Pi_{\mathcal{B}} e^{-2\lambda_{\min}^{\mathcal{B}} T}$, so $T = \log(1/\varepsilon)/(2\lambda_{\min}^{\mathcal{B}})$ is admissible. Lower bound: $E_{\mathcal{B}}(T) \geq \pi_{i_o} e^{-2\lambda_{\min}^{\mathcal{B}} T} > \varepsilon \Pi_{\mathcal{B}}$ for $T < (\log(1/\varepsilon) - \theta_{\mathcal{B}})/(2\lambda_{\min}^{\mathcal{B}})$. For $D_{\mathcal{B}}$ use

$\pi_{i_0} e^{-\lambda_{\min} T} \leq D_{\mathcal{B}}(T) \leq 2\Pi_{\mathcal{B}} e^{-\lambda_{\min} T}$ (proof of Proposition I.5); the factor 2 is genuinely attained (one mode, $\pi = \lambda = 1$: $D(\log(1/\varepsilon)) = 2\varepsilon - \varepsilon^2 > \varepsilon$), which is why $\log 2$ sits in the upper bound. The formula for $\lambda_{\min}^{\mathcal{S}}$ is the linear-response (Gaussian-equivalent) block scale $a_1(q)^2 \times (\text{data eigenvalue})/d_{\text{latent}}$ applied eigenvalue by eigenvalue inside the finite-rank signal block, where the data eigenvalues of M_t are $e^{-2t}\ell_j + \Delta_t$, ℓ_j the eigenvalues of $(\hat{C} + \hat{S})_{\mathcal{S}}$, whose population mean is $s^2/d_{\text{intrinsic}} + \sigma_{\text{sig}}^2$; the edge approximation and its limitations are discussed in Remarks I.16 and I.17, checked here numerically: (20, 1280, 500) simulation, signal mean 0.0526 (scale 0.0545), min 0.0314. The step-count numbers use Lemma I.3(iv) with code eigenvalues $\lambda_{\min, \text{code}}^{\mathcal{S}} = 20.3$ ($d_{\text{latent}} = 5$) and 73.9 ($d_{\text{latent}} = 200$). $\square$

Why the signal block. The noise-dimension target is exactly linear, $s_{\mathcal{N}}^*(x_t) = -x_{t,\mathcal{N}}/\beta_t^2$ (cluster means have no $\mathcal{N}$ -component; Appendix I.4), with per-coordinate second moment $1/\beta_t^2 = 3.8$ at $\sigma_{\perp} = 0.5$, whereas the signal-subspace target has per-coordinate second moment at most $1/(e^{-2t}\sigma_{\text{sig}}^2 + \Delta_t) = 1.0$ (convexity of Fisher information for the Gaussian mixture); hence for $d_{\text{latent}} \gg d_{\text{intrinsic}}$ the reducible population loss is dominated by $\mathcal{N}$ and a test-loss plateau measures τ_{pop} (rate $\lambda_{\min}^{\mathcal{N}} \simeq a_1(q)^2 \beta_t^2/d_{\text{latent}}$) rather than τ_{gen} (rate $\lambda_{\min}^{\mathcal{S}}$). Sample quality (score error in the signal subspace, FID after decoding) is a functional of the $\mathcal{S}$ -block score only, so τ_{gen} is defined through $\mathcal{S}$. The block $\mathcal{N}$ is the buffer: its absorption costs time and buys no sample quality.

I.1.7 GENERALISATION GAP AND MEMORISATION TIMES

Proposition I.10 (Exact spectral generalisation gap; exact formula proved, reduction under (A6')). *Let $L(A) = \frac{1}{2}\mathbb{E}\|A\phi(x_t) - y\|^2$ be the population objective (its held-out estimate is the logged test loss), $\bar{U} = \mathbb{E}[\phi\phi^\top]$, $\bar{V} = \mathbb{E}[y\phi^\top] = \mathbb{E}[s^*(x_t)\phi(x_t)^\top]$, and $G(T) := [L(A(T)) - L_n(A(T))] - [L(0) - L_n(0)]$. With $e_i = e^{-\lambda_i T}$, $\bar{u}_{ij} = v_i^\top \bar{U} v_j$, γ_i as in (A6'), exactly*

$$G(T) = \frac{1}{2} \sum_{i,j \leq p} (1 - e_i)(1 - e_j) \bar{u}_{ij} \langle a_i^*, a_j^* \rangle - \sum_{i \leq p} (1 - e_i) \gamma_i \lambda_i \pi_i + \frac{1}{2} \sum_{i \leq p} \lambda_i \pi_i (1 - e_i^2). \quad (14)$$

Define the diagonal per-mode contribution $g_i(T) := \frac{1}{2}\pi_i [\bar{u}_i(1 - e_i)^2 - 2\gamma_i \lambda_i(1 - e_i) + \lambda_i(1 - e_i^2)]$, which vanishes identically when $\bar{u}_i = \lambda_i$ and $\gamma_i = 1$. Under (A6'),

$$G(T) = \sum_{i \in \mathcal{M} \cup \mathcal{Z}} g_i(T) + o(G_{\text{thr}}) \quad \text{uniformly in } T.$$

For the sub-block $\mathcal{M}_{\text{blk}} := \{i \in \mathcal{M} : \gamma_i \leq \gamma_0\}$ (Definition I.11) write $G_{\text{blk}}(T) := \frac{1}{2} \sum_{i \in \mathcal{M}_{\text{blk}}} \pi_i [\bar{u}_i(1 - e_i)^2 + \lambda_i(1 - e_i^2)]$; then $|\sum_{i \in \mathcal{M}_{\text{blk}}} g_i - G_{\text{blk}}| \leq \sum_{i \in \mathcal{M}_{\text{blk}}} |\gamma_i| \lambda_i \pi_i (1 - e_i)$, and G_{blk} is nonnegative, increasing, $G_{\text{blk}}(0) = 0$, $G_{\infty}^{\text{blk}} = \frac{1}{2} \sum_{i \in \mathcal{M}_{\text{blk}}} \pi_i (\bar{u}_i + \lambda_i)$, $G'_{\text{blk}}(0) = \sum_{i \in \mathcal{M}_{\text{blk}}} \lambda_i^2 \pi_i$, and

$$G_{\infty}^{\text{blk}} (1 - e^{-\lambda_{\min}^{\text{blk}} T})^2 \leq G_{\text{blk}}(T) \leq G_{\infty}^{\text{blk}} (1 - e^{-2\lambda_{\max}^{\text{blk}} T}). \quad (15)$$

The tail contribution is bounded for all T by $|\sum_{i \in \mathcal{Z}} g_i(T)| \leq \frac{1}{2} \|VP_{\mathcal{Z}}\|_F^2 (T^2 \sup_{i \in \mathcal{Z}} \bar{u}_i + 2T(1 + \sup_{i \in \mathcal{Z}} |\gamma_i|))$.

Proof. Write $A(T) = \sum_{i \leq p} (1 - e_i) a_i^* v_i^\top$ (Lemma I.3). Then $\text{tr}(A\bar{U}A^\top) = \sum_{i,j} (1 - e_i)(1 - e_j) \bar{u}_{ij} \langle a_i^*, a_j^* \rangle$, $\text{tr}(A\bar{V}^\top) = \sum_i (1 - e_i) \langle a_i^*, \bar{V} v_i \rangle = \sum_i (1 - e_i) \gamma_i \lambda_i \pi_i$, $\text{tr}(AUA^\top) = \sum_i \lambda_i \pi_i (1 - e_i)^2$ and $\text{tr}(AV^\top) = \sum_i (1 - e_i) \langle a_i^*, V v_i \rangle = \sum_i \lambda_i \pi_i (1 - e_i)$ since $V v_i = \lambda_i a_i^*$. The additive constants cancel in G ; the empirical part is $\sum_i \lambda_i \pi_i [(1 - e_i) - \frac{1}{2}(1 - e_i)^2] = \frac{1}{2} \sum_i \lambda_i \pi_i (1 - e_i^2)$. (Brute-force check at (4, 30, 12): six decimals at $T = 0.3, 2, 10$.) Under (A6') the off-diagonal double sum is $o(G_{\text{thr}})$ and, for $i \in \mathcal{S} \cup \mathcal{N}$, $|g_i(T)| \leq \frac{1}{2} \pi_i \lambda_i (|\bar{u}_i/\lambda_i - 1| + 2|\gamma_i - 1|)$ (use $(1 - e_i)^2 \leq 1$, $1 - e_i \leq 1$ and $(1 - e_i)^2 - 2(1 - e_i) + (1 - e_i^2) = 0$), whose sum is $o(G_{\text{thr}})$. The error bound uses the triangle inequality. It reduces to $\gamma_0 \sum \lambda_i \pi_i$ only under the additional condition $|\gamma_i| \leq \gamma_0$ on this block; the one-sided cutoff alone does not ensure it. Properties of G_{blk} : each bracket is 0 at $T = 0$, increasing in T , and tends to $\bar{u}_i + \lambda_i$; $\frac{d}{dT}(1 - e_i)^2|_0 = 0$, $\frac{d}{dT}(1 - e_i^2)|_0 = 2\lambda_i$. Bounds: $(1 - e)^2 \leq 1 - e^2$ on $[0, 1]$, so the normalised bracket lies in $[(1 - e_i)^2, 1 - e_i^2]$; multiply by $\frac{1}{2} \pi_i (\bar{u}_i + \lambda_i)$ and sum. Tail: $(1 - e^{-x})^2 \leq x^2$, $1 - e^{-x} \leq x$, $1 - e^{-2x} \leq 2x$ with $\pi_i \lambda_i^2 = \|V v_i\|^2$. $\square$

Definition I.11 (Spike/bulk split of the sample block; memorisation times). Fix $\gamma_0 \in (0, \frac{1}{2}]$. Split $\mathcal{M} = \mathcal{M}_{\text{spk}} \cup \mathcal{M}_{\text{blk}}$ with $\mathcal{M}_{\text{spk}} := \{i \in \mathcal{M} : \gamma_i > \gamma_0\}$ (population-aligned sample modes: their empirical optimum a_i^* has a nonnegligible projection on the population target, but absorption lowers the diagonal population contribution only while $\bar{u}_i(1 - e_i) < \gamma_i \lambda_i$; cross-mode terms must also be controlled) and $\mathcal{M}_{\text{blk}}$ the remainder. For clustered data $\mathcal{M}_{\text{spk}}$ contains the cluster-residual spikes above the Marchenko–Pastur edge ($\lambda_{\max}^{\mathcal{M}}/\bar{\lambda}^{\mathcal{M}} = 21.5, 15.4, 5.2, 3.0, 1.8$ at $d_{\text{latent}} = 5, 20, 60, 100, 200$ on the repository spectra), and its size is expected to grow as $\psi_n \downarrow 1$.

(i) *Onset*: $\tau_{\text{mem}}^{\text{onset}} := 1/\lambda_{\max}^{\text{blk}}$, the e-folding time of the fastest *bulk* sample mode. The index-based statistic $1/\lambda_{\max}^{\mathcal{M}}$ (used in the tables) is a lower bound on it, with equality when $\mathcal{M}_{\text{spk}} = \emptyset$.

(ii) *Gap-based*: for $\delta \in (0, 1)$, $\tau_{\text{mem}}^{\text{gap}}(\delta) := \inf\{T : G_{\text{blk}}(T) \geq \delta G_{\infty}^{\text{blk}}\}$. By Proposition I.10,

$$\frac{\log \frac{1}{1-\delta}}{2\lambda_{\max}^{\text{blk}}} \leq \tau_{\text{mem}}^{\text{gap}}(\delta) \leq \frac{\log \frac{1}{1-\sqrt{\delta}}}{\lambda_{\min}^{\text{blk}}},$$

so the onset controls the earliest time the gap can reach a given fraction of its asymptote and the slowest bulk mode its completion; the position of $\tau_{\text{mem}}^{\text{gap}}(\delta)$ within the interval depends on the within-bulk distribution of $(\lambda_i, \pi_i, \bar{u}_i)_{i \in \mathcal{M}_{\text{blk}}}$. The tail block $\mathcal{Z}$ enters G through the bound of Proposition I.10. At 3×10^5 default steps one must evaluate its absorbed mass at $T = 6000$; values evaluated at $T = 120000$ do not describe that horizon. Small eigenvalues alone do not justify dropping target-weighted tail contributions.

(iii) *Fraction-level (placeholder, completed in Appendices I.5 and I.6)*: the paper’s primary statistic is the memorised fraction $F(T) = \frac{1}{n} \sum_{\mu} \mathbf{1}[\rho_{\mu}(T) < 1/3]$ with ρ_{μ} the generated-point nearest-neighbour ratio of Bonnaire et al. (2025). Its spectral surrogate is $\hat{F}(T) = \frac{1}{n} \sum_{\mu} \mathbf{1}[m_{\mu}(T) \geq \vartheta]$, $m_{\mu}(T) = \sum_{i \in \mathcal{M}_{\text{blk}}} \pi_{i,\mu} (1 - e^{-\lambda_i T})^2 / \sum_{i \in \mathcal{M}_{\text{blk}}} \pi_{i,\mu}$, where $\pi_{i,\mu}$ is the share of mode i ’s target mass attributable to training point μ ; a single sample edge makes $\hat{F}$ a step function, so the map $\mu \mapsto (\pi_{i,\mu})_i$ and the spread of λ_i over $\mathcal{M}_{\text{blk}}$ are required and are modelled conditionally in Appendix I.6.

The displayed bounds are Proposition I.10 inverted. The spike ratios are $\lambda_{\max}^{\mathcal{M}}/\text{mean}(\lambda_{\mathcal{M}})$ from the released $\sigma_{\perp} = 0.5$ spectra with the sample block cut at index n (index-based; γ_i has not yet been computed, see Prediction 5).

Consequences. (i) Under (A6’) the population blocks contribute nothing to the gap at the level G_{thr} : the model’s coefficient on a population direction converges to the population-optimal one, so train and test loss fall together on $\mathcal{S} \cup \mathcal{N}$ (the “both curves drop” phase of Section 4). (ii) A pure sample mode ($\gamma_i = 0$) raises the test loss by $\frac{1}{2} \bar{u}_i \pi_i (1 - e^{-\lambda_i T})^2$ and lowers the train loss by $\frac{1}{2} \lambda_i \pi_i (1 - e^{-2\lambda_i T})$; a population-aligned sample mode ($\gamma_i \approx 1$, $\bar{u}_i \approx \lambda_i$) lowers both. The repository test-loss minima occur at 35k, 55k and 90k steps for $d_{\text{latent}} = 40, 100, 200$. At $d_{\text{latent}} = 200$ the slowest noise-dimension e-fold is about 302k steps. Those minima therefore do not establish that population modes have finished learning, or identify aligned sample modes; held-out alignment measurements are needed. The squared form $(1 - e^{-\lambda_i T})^2$ is the origin of the shape used heuristically in the released predictor (Section 5), but there the eigenvalues are those of M_t , not of the sample block, a discrepancy addressed in Appendix I.6. (iii) $\tau_{\text{mem}}^{\text{onset}}$ and τ_{gen} are single-eigenvalue functionals, so $\tau_{\text{mem}}^{\text{onset}}/\tau_{\text{gen}}(\varepsilon) = 2\lambda_{\min}^{\mathcal{S}}/(\lambda_{\max}^{\text{blk}} \log(1/\varepsilon)) (1 + O(\theta_{\mathcal{S}}/\log(1/\varepsilon)))$ is clock-free. On the released spectra with the *index-based* sample top, $\lambda_{\min}^{\mathcal{S}}/\lambda_{\max}^{\mathcal{M}} = 34, 63, 143, 211, 279, 340, 416, 426$ for $d_{\text{latent}} = 10, 20, 40, 60, 80, 100, 150, 200$ (monotone; $d_{\text{latent}} = 5$ gives 21 but there $p < n$); since $\lambda_{\max}^{\text{blk}} \leq \lambda_{\max}^{\mathcal{M}}$ these are lower bounds on the bulk-onset ratio, whose monotonicity is to be re-established after the γ -cut (Prediction 5). No statement of the form “ $\tau_{\text{mem}} - \tau_{\text{gen}} \geq (d_{\text{latent}} - d_{\text{intrinsic}}) \times (\text{a timescale})$ ” follows from a diagonal flow, since modes decay in parallel; the buffer is a statement about the ratio of two eigenvalues (Appendix I.3).

I.1.8 MEASURED PROXIES

Remark I.12 (Correspondence with the measured proxies; sketched). (a) *Gap proxy*. The logged RFNN loss is per-coordinate, $\ell = \|\sqrt{\Delta_t} s_A + \eta\|^2/d_{\text{latent}}$, so $\ell_{\text{test}} - \ell_{\text{train}} = (2\Delta_t/d_{\text{latent}})G(T) +$

const and “gen-gap > 0.02 ” is the crossing $G(T) = 0.01d_{\text{latent}}/\Delta_t$: an *absolute* threshold, i.e. $\tau_{\text{mem}}^{\text{gap}}(\delta_{\text{eff}})$ with $\delta_{\text{eff}} = 0.01d_{\text{latent}}/(\Delta_t G_\infty)$ varying with d_{latent} through G_∞ (final gaps $0.019 \rightarrow 0.093$ over $d_{\text{latent}} = 5 \rightarrow 200$). Moreover the logged train loss uses one fresh noise draw per evaluation and the test loss one fixed draw, so the logged gap carries a noise floor of order $\sqrt{2/(nd_{\text{latent}})}$: 0.028 ($d_{\text{latent}} = 5$), 0.014 (20), 0.0045 (200), comparable to the 0.02 threshold for $d_{\text{latent}} \leq 20$; this, together with δ_{eff} , accounts for the erratic $\tau_{\text{mem}}^{\text{gap} > 0.02} = 20, 15, 25, 10, 10\text{k}$ steps at $d_{\text{latent}} = 5..60$. Any re-extraction should use a many-draw gap or the exact $G(T)$ of Proposition I.10. (b) *Test-loss proxy*. “Test loss within 5% of its minimum” fires at the first evaluation (5k-step grid) for every d_{latent} . This does not imply absorption of all population modes: at $d_{\text{latent}} = 200$ the signal and slowest noise-dimension e-folds are about 8.7k and 302k steps. The 1% and 2% crossings are 15/30/50k and 10/20/35k at $d_{\text{latent}} = 40/100/200$, and minima occur at 35/55/90k. Interpreting these aggregate losses requires target weights and alignment; they cannot be assigned to sample modes from the clock alone. (c) For the MLP and real-data sweeps the memorised fraction $F(T)$ crossing 0.01 (synthetic) or a fixed level (real) is the proxy for Definition I.11(iii). Ratios $\tau_{\text{mem}}/\tau_{\text{gen}}$ are clock-independent; absolute step counts carry the factor 0.02^{-1} of Lemma I.3(iv) (Remark I.7).

Concretely, the plotting code defines $\tau_{\text{gen}}^{\text{meas}} = \text{first step with test loss} \leq 1.05 \times \text{its run minimum}$; $\tau_{\text{mem}}^{\text{meas}} = \text{first step with memorised fraction} > 0.01$ when available (MLP, real data), otherwise first step with gen-gap > 0.02 (RFNN). In the RFNN logs (seed 42, evaluation every 5 000 steps) $\tau_{\text{gen}}^{\text{meas}} \leq 10^4$ steps for every d_{latent} ; $\tau_{\text{mem}}^{\text{meas}}$ moves from 2×10^4 ($d_{\text{latent}} = 5$) to 8×10^4 ($d_{\text{latent}} = 200$) and the final gap from 0.019 to 0.093; the theory attributes the modest growth to the d_{latent} -dependence of G_∞ entering δ_{eff} and to the single-draw noise floor $\sqrt{2/(nd_{\text{latent}})}$, and predicts a much steeper growth for a relative-threshold, many-draw extraction on $\mathcal{M}_{\text{blk}}$.

Fraction-level definition. The fraction $F(T)$ is a per-training-point statistic. Writing $\pi_i = \sum_\mu \pi_{i,\mu}$ for a decomposition of each bulk sample mode’s target mass over training points, point μ is memorised once the absorbed share $m_\mu(T)$ of Definition I.11(iii) exceeds a level ϑ set by the nearest-neighbour test; under an additional sampler-attachment assumption (the center condition of Lemma I.48 does not establish pathwise attachment), memorisation at fixed t is modelled as a question of how much bulk sample-mode mass has been absorbed. A single edge makes m_μ identical across μ and $\hat{F}$ a step; the required spread of $(\lambda_i, \pi_{i,\mu})$ over $\mathcal{M}_{\text{blk}}$ is modelled under additional assumptions in Appendices I.5 and I.6.

Predictions and how to test them.

1. τ_{gen} in flow time is set by $\lambda_{\text{min}}^S = a_1(q)^2(e^{-2t}\ell_{\text{min}} + \Delta_t)/d_{\text{latent}}$ and grows with d_{latent} ($10.9 \times$ over $d_{\text{latent}} = 5..200$), and in default repository steps it also grows ($788 \rightarrow 8663$). Test: log the signal-block absorption deficit D_S every ~ 50 steps, fit its late-time exponential rate, and compare with $0.02\lambda_{\text{min,code}}^S/p$ per step; compare λ_{min}^S across d_{latent} with (13) using ℓ_{min} from the realised $(\hat{C} + \hat{S})_S$. Zero free parameters.
2. *Zero-parameter reconstruction of the logged curves as expected dynamics:* with U, V the $\mathbb{E}_\eta$ operators, the train-loss excess of the mean flow is $\frac{1}{2} \sum_i \lambda_i \pi_i e^{-2\lambda_i T}$ and its gap is (14) with $T = 0.02 \times \text{steps}$, tail block included. Test: re-run one RFNN configuration saving A every 500 steps; compute $U, V, \bar{U}, \bar{V}$ (held-out data, many draws), $A^*, \pi_i, \gamma_i, \bar{u}_{ij}$; overlay the closed forms on the logged curves converted by $2\Delta_t/d_{\text{latent}}$. The mean-iterate loss must be distinguished from the mean loss; discretization, parameter variance, and Monte Carlo error must be controlled before testing the alignment approximation.
3. *The absolute-threshold proxy gen-gap > 0.02 conflates onset with asymptote and is contaminated by the single-draw noise floor; a relative threshold ($\delta = 0.2$ of the many-draw asymptotic gap) on $\mathcal{M}_{\text{blk}}$ will show a much steeper growth with d_{latent} than the logged $4 \times$.* Test: re-extract with ≥ 50 -draw train and test losses at each evaluation and compare growth factors with $3200d_{\text{latent}}/\lambda_{\text{max,code}}^{\text{blk}}$.
4. *Operating point:* $a_*(q)^2$ falls $0.0785 \rightarrow 0.0047$ over $d_{\text{latent}} = 5..200$ at $\sigma_\perp = 0.5$ and equals 2.6×10^{-4} at $d_{\text{latent}} = 200$ for $\sigma_\perp = 0.01$, so every memorisation timescale in flow units is pushed out by more than an order of magnitude at $\sigma_\perp = 0.01$ relative to 0.5 at equal d_{latent} . Test: compare $\text{tr } U/p$ minus the data-block mass across the two $\sigma_\perp$ sweeps with $a_*(q(d_{\text{latent}}))^2$ (expect $0.7\text{--}0.9 \times a_*^2/n$ for the sample-block mean).

5. Alignment coefficients γ_i on held-out data will be $1 + o(1)$ on $\mathcal{S} \cup \mathcal{N}$, exceed $\frac{1}{2}$ on a set $\mathcal{M}_{\text{spk}}$ of $O(k)$ top sample modes at small d_{latent} whose share grows as $\psi_n \rightarrow 1$, and be $O(1/\sqrt{n})$ on $\mathcal{M}_{\text{blk}}$; their contribution to the test-loss minima must be evaluated with the full spectral formula rather than inferred from a single e -fold. Test: compute $\bar{U}, \bar{V}$ on 10^4 held-out points, histogram γ_i by block at $d_{\text{latent}} \in \{5, 20, 60, 200\}$; sweep k at fixed n, d_{latent} to check $|\mathcal{M}_{\text{spk}}| \sim k$.

Open gaps. (1) The split $\mathcal{M}_{\text{spk}}/\mathcal{M}_{\text{blk}}$ is a definition plus Prediction 5: γ_i has not been computed on the repository’s data, so the quoted ratio numbers ($34 \rightarrow 426$, monotone) are index-based lower bounds on the bulk onset, and whether monotone growth of $\tau_{\text{mem}}^{\text{onset}}/\tau_{\text{gen}}$ survives the γ -cut is open. The tail bound in Proposition I.10 is stated through $\|VP_{\mathcal{Z}}\|_F^2$ and $\sup_{\mathcal{Z}} \bar{u}_i$, which have not been evaluated against the threshold $0.01d_{\text{latent}}/\Delta_t$. (2) The rank- n cliff is a single-draw fact; the released 50-draw spectra are full rank with a soft step ≤ 1.7 at index n , and the main text must say which operator each figure shows. (3) (A6’) is a hypothesis; the required relative accuracy on the population blocks is $\sim 10\%$ at $d_{\text{latent}} = 20$ and tighter at larger d_{latent} ($\Pi_{\mathcal{N}} \sim d_{\text{latent}}^2/(a_1^2\beta_t^4)$), and it may fail as $\psi_n \rightarrow 1$, consistent with the test loss still falling at 90k steps at $d_{\text{latent}} = 200$. (4) The $(1 + o(1))$ in (13) is not quantified here; the finite-rank BBP correction is Remark I.17’s. (5) Monotonicity of a_*^2 in q is numerical on $[0.01, 20]$ only. (6) The $d_{\text{latent}} = 5$ point violates $p \geq n$; all anchors there are flagged and the monotone ratio is quoted from $d_{\text{latent}} = 10$. (7) The regime $n \leq d_{\text{latent}}$ (memorisation returning at large d_{latent} , Appendix I.8) is outside (A3): there $\mathcal{M}$ is empty and $G_{\text{blk}} \equiv 0$; the growth of $\mathcal{M}_{\text{spk}}$ as $\psi_n \rightarrow 1$ is the natural bridge but is not derived. (8) Real VAE latents have continuous spectra with posterior-collapsed coordinates at exactly Δ_t ; the index-based partition of (A5) needs the data-driven cut of Appendix I.7. (9) The clock $T = 0.02 \times \text{steps}$ holds for full-batch SGD with the stated learning-rate rule and momentum 0; the MLP sweeps use Adam and no clock statement is made for them. (10) The $O(\theta_S/\lambda_{\min}^S)$ term in Proposition I.9 scales like τ_{gen} itself unless $\theta_S = \log(\Pi_S/\pi_{i_0})$ is small; for 10 random centres in $d_{\text{intrinsic}} = 5$ it is not, so $\tau_{\text{gen}}(\varepsilon)$ is pinned to $\lambda_{\min}^S$ only up to the factor $1 - \theta_S/\log(1/\varepsilon)$. (11) The sample-block mean is $0.72\text{--}0.89 \times a_*^2/n$ (Jensen dispersion of per-sample q_μ), so a_1, a_* should be defined with μ -averaged Gaussian expectations or the trace check is 10–30% off in the sample block; the noise-dimension median also overshoots by 10–15% at large d_{latent} . (12) The γ_0 -cut is not canonical; the bounds in Definition I.11 carry an explicit $\sum |\gamma_i| \lambda_i \pi_i (1 - e_i)$ error that has not been evaluated.

I.2 FOUR SPECTRAL BLOCKS OF THE FEATURE CORRELATION MATRIX

Throughout this subsection $\phi(x) = p^{-1/2} \tanh(Wx/\sqrt{d_{\text{latent}}})$ with $W_{ij} \sim \mathcal{N}(0, 1)$ i.i.d.; the released code uses $\tanh(\tilde{W}x)$ with $\tilde{W} = W/\sqrt{d_{\text{latent}}}$ and no $p^{-1/2}$, so $U_{\text{code}} = pU$ and all eigenvalues quoted from the repository are p times the theory values; timescales in theory units are p/λ_{code} . $U = \frac{1}{n} \sum_{\mu} \mathbb{E}_{\xi}[\phi(x_t^{\mu})\phi(x_t^{\mu})^{\top}]$ is the ξ -averaged matrix ($K = 50$ draws in `compute_U`); $U^{(1)} = \Phi^{\top}\Phi/n$ denotes the single-copy matrix with $x_t^{\mu} = e^{-t}x^{\mu} + \sqrt{\Delta_t}\xi^{\mu}$, $\Phi \in \mathbb{R}^{n \times p}$ the feature matrix of one diffused copy per point, so that $U^{(1)}$ shares its nonzero spectrum with the Gram matrix $G = \Phi\Phi^{\top}/n$ and $\text{rank } U^{(1)} \leq n$. Write $q_{\mu} = \|x_t^{\mu}\|^2/d_{\text{latent}}$ for the pre-activation variance of point μ and $q = \frac{1}{n} \sum_{\mu} q_{\mu} \simeq \text{tr } M_t/d_{\text{latent}}$, which for the synthetic family is explicit (Lemma I.6),

$$q(d_{\text{latent}}) = e^{-2t} \left[\frac{s^2 + d_{\text{intrinsic}}\sigma_{\text{sig}}^2}{d_{\text{latent}}} + \sigma_{\perp}^2 \left(1 - \frac{d_{\text{intrinsic}}}{d_{\text{latent}}}\right) \right] + \Delta_t \xrightarrow{d_{\text{latent}} \rightarrow \infty} e^{-2t}\sigma_{\perp}^2 + \Delta_t. \quad (16)$$

Let $\sigma_r = a_*(q)^2/n$ and $\kappa_{\text{MP}} = p^{-1}(\sqrt{n - d_{\text{latent}}} + \sqrt{p - d_{\text{latent}}})^2$. Split $M_t = e^{-2t}(\hat{C} + \hat{S}) + \Delta_t I$ into its $d_{\text{intrinsic}} \times d_{\text{intrinsic}}$ signal block $e^{-2t}(\hat{C} + \hat{S}_{\parallel}) + \Delta_t I$ and its null block $M_{\perp} = e^{-2t}\hat{S}_{\perp} + \Delta_t I$, where $\hat{S}_{\perp}$ is $\sigma_{\perp}^2$ times a white Wishart matrix of aspect ratio $\gamma_{\perp} = (d_{\text{latent}} - d_{\text{intrinsic}})/n$; set $\gamma_p = n/p$.

I.2.1 STEP 1: THE FEATURE KERNEL IS EXACTLY A HERMITE SERIES

Lemma I.13 (Kernel of the tanh features; proved). *Fix the diffused training points $x_t^1, \dots, x_t^n$ and let $q_{\mu} = \|x_t^{\mu}\|^2/d_{\text{latent}}$, $\rho_{\mu\nu} = \langle x_t^{\mu}, x_t^{\nu} \rangle / (d_{\text{latent}} \sqrt{q_{\mu} q_{\nu}})$. For $w \sim \mathcal{N}(0, I_{d_{\text{latent}}})$ the pre-activations*

$g_\mu = \langle w, x_t^\mu \rangle / \sqrt{d_{\text{latent}}}$ are jointly Gaussian with $\text{Var } g_\mu = q_\mu$ and $\text{Corr}(g_\mu, g_\nu) = \rho_{\mu\nu}$, and

$$K_{\mu\nu} := \mathbb{E}_w [\tanh(g_\mu) \tanh(g_\nu)] = \sum_{k \text{ odd}} c_k(q_\mu) c_k(q_\nu) \rho_{\mu\nu}^k, \quad (17)$$

$$c_k(q) = \mathbb{E} [\tanh(\sqrt{q}z) \text{He}_k(z)] / \sqrt{k!},$$

with only odd k because $\tanh$ is odd. By Stein's identity $c_1(q) = \sqrt{q} \mathbb{E} [\tanh'(\sqrt{q}z)] = \sqrt{q} a_1(q)$, so the $k = 1$ term equals $a_1(q_\mu) a_1(q_\nu) \langle x_t^\mu, x_t^\nu \rangle / d_{\text{latent}}$, and

$$K = A_1 \frac{X_t X_t^\top}{d_{\text{latent}}} A_1 + K_{\text{res}}, \quad A_1 = \text{diag}(a_1(q_\mu)), \quad (18)$$

where $(K_{\text{res}})_{\mu\mu} = a_*(q_\mu)^2$ and $(K_{\text{res}})_{\mu\nu} = \sum_{k \geq 3 \text{ odd}} c_k(q_\mu) c_k(q_\nu) \rho_{\mu\nu}^k = O(\rho_{\mu\nu}^3)$ for $\mu \neq \nu$.

Proof. Joint Gaussianity is immediate since g is linear in w . Expanding $\tanh(\sqrt{q} \cdot)$ in normalized Hermite polynomials and using Mehler's identity $\mathbb{E} [\text{He}_k(g_\mu / \sqrt{q_\mu}) \text{He}_l(g_\nu / \sqrt{q_\nu})] = \delta_{kl} k! \rho_{\mu\nu}^k$ gives (17); oddness kills even k (in particular $c_0 = 0$: the features are centred, so no rank-one $\mathbf{1}\mathbf{1}^\top$ term appears, unlike for ReLU). Stein's identity $\mathbb{E} [f(\sqrt{q}z)z] = \sqrt{q} \mathbb{E} [f'(\sqrt{q}z)]$ gives c_1 . The diagonal of K_{res} is $\mathbb{E} \tanh(\sqrt{q_\mu}z)^2 - a_1(q_\mu)^2 q_\mu = a_*(q_\mu)^2$ by definition. $\square$

For the synthetic family, $\rho_{\mu\nu} = O(d_{\text{latent}}^{-1/2}) + O(s^2 \langle m_{c(\mu)}, m_{c(\nu)} \rangle / (d_{\text{latent}} q))$ for points in different clusters (the generator uses $d_{\text{intrinsic}}$ orthogonal centres plus $k - d_{\text{intrinsic}}$ random unit centres, so cross-cluster centre overlaps are $O(s^2)$, not zero) and $\rho_{\mu\nu} \simeq s^2 / (d_{\text{latent}} \sqrt{q_\mu q_\nu})$ for the same cluster (0.65 at $d_{\text{latent}} = 5$, 0.5 at $d_{\text{latent}} = 20$). The off-diagonal part of K_{res} is $O(\rho^3)$: K_{res} is diagonal-dominant with a same-cluster component of rank $\lesssim k$ and top eigenvalue $\simeq (n/k) c_3(q)^2 \rho_{\text{same}}^3$ ($c_3^2 = 0.052, 0.033, 0.019, 0.011, 0.006, 0.004$ at $d_{\text{latent}} = 5, 10, 20, 40, 100, 200$; $\approx 10 a_*^2$ at $d_{\text{latent}} = 5$, $\approx 5 a_*^2$ at $d_{\text{latent}} = 20$). This is the precise form of the failure of the orthonormality ansatz $v_\mu^\top v_\nu = \delta_{\mu\nu} + o(1)$ for clustered data. The mixture is not Gaussian, but Lemma I.13 needs only Gaussianity over W , which is exact.

I.2.2 STEP 2: GAUSSIAN EQUIVALENCE AND THE SPIKED-COVARIANCE FORM

Assumption I.14 (Gaussian surrogate for the feature spectrum; conditional). Let $G = \Phi \Phi^\top / n$. (a) [Bulk] In the limit $p, n, d_{\text{latent}} \rightarrow \infty$ with $\gamma_p = n/p$, $\gamma_\perp = (d_{\text{latent}} - d_{\text{intrinsic}})/n$ fixed and $\max_\mu q_\mu$ bounded, assume that the empirical spectral distribution of G coincides with that of

$$G^{\text{GE}} = \frac{1}{n} K^{1/2} \frac{Z Z^\top}{p} K^{1/2}, \quad Z \in \mathbb{R}^{n \times p} \text{ i.i.d. } \mathcal{N}(0, 1), \quad (19)$$

a sample covariance with population K/n and p samples. (b) [Outliers] The $d_{\text{intrinsic}}$ largest eigenvalues (a finite-rank set) and the $d_{\text{latent}} - d_{\text{intrinsic}}$ noise-dimension eigenvalues of G are located by the same replacement to relative accuracy $o(1)$. (c) [ξ -average] The same holds for U with the Gram of the Kn noise draws in place of K .

The cited results motivate this surrogate but do not, without checking their hypotheses, prove (a)–(c) for our mixture and repeated-noise design. Louart et al. (2018) require control of the input operator norm; bounds on individual q_μ alone do not supply it. Liao & Couillet (2018) use normalized Gaussian mixtures with bounded pre-normalization means and vanishing pairwise overlaps, not $O(1)$ same-cluster overlaps. Fan & Wang (2020) impose approximate orthogonality. Learning-observable equivalence (Goldt et al., 2022; Hu & Lu, 2023; Mei & Montanari, 2022) and nonlinear spectral laws (Pennington & Worah, 2017; Benigni & Pécché, 2021) do not automatically imply outlier or blockwise operator convergence. Repeated draws share a training point; equivalence for their joint Gram matrix and the passage to $K \rightarrow \infty$ are additional assumptions. We retain this as a conditional approximation, not an imported theorem.

Identity in distribution and the correct perturbation model. In the homogeneous case $a_1(q_\mu) \equiv a_1$, let $L = [a_1 X_t / \sqrt{d_{\text{latent}}}, K_{\text{res}}^{1/2}] \in \mathbb{R}^{n \times (d_{\text{latent}} + n)}$, so $LL^\top = K$. For $\tilde{Z} \in \mathbb{R}^{(d_{\text{latent}} + n) \times p}$

Gaussian, each column of $L\tilde{Z}$ is $\mathcal{N}(0, K)$, hence $L\tilde{Z}/\sqrt{p} \stackrel{d}{=} K^{1/2}Z/\sqrt{p}$. Writing $\tilde{Z} = [W^\top; \Theta]$, the $p \times p$ picture

$$U^{(1)} \simeq \frac{a_1^2}{p d_{\text{latent}}} W \hat{M}_t W^\top + \frac{1}{p n} \Theta^\top K_{\text{res}} \Theta + \text{cross}, \quad \hat{M}_t = X_t^\top X_t / n, \quad (20)$$

is therefore the same object as (19), and the data directions are spikes of the *population* covariance $K/n = D + R$ of a Wishart-type matrix (multiplicative model), not additive rank- d_{latent} perturbations of a Wishart matrix; the cross terms in (20) are what the multiplicative model resolves. Two consequences that an additive model gets wrong: the data part $a_1^2 W \hat{M}_t W^\top / (p d_{\text{latent}})$ has the nonzero spectrum of $\hat{M}_t^{1/2} (W^\top W / p) \hat{M}_t^{1/2} a_1^2 / d_{\text{latent}}$, so every data block is multiplied by a white Wishart of ratio d_{latent}/p (Step 3); and the outlier shift and residual mass follow the spiked-covariance rules of Remark I.17 (Step 4).

I.2.3 STEP 3: SPECTRUM OF K/n , THE DATA BLOCKS AND THE FOUR COUNTS

Lemma I.15 (Single-copy rank and population-kernel interlacing; proved). *Let $D = A_1 X_t X_t^\top A_1 / (d_{\text{latent}} n) \succeq 0$, $r = \text{rank } D \leq d_{\text{latent}}$, and $R = K_{\text{res}}/n \succeq 0$. At least $n - r$ eigenvalues of $D + R$ lie in $[\lambda_{\min}(R), \lambda_{\max}(R)]$ and at most r exceed $\lambda_{\max}(R)$. For $p \geq n$, $U^{(1)}$ has at least $p - n$ zero eigenvalues, exactly that many when Φ has full row rank. The average obeys $\text{rank } U^{(K)} \leq \min(p, Kn)$. These facts do not establish a sample-block count or floor multiplicity for U .*

Proof. For decreasingly ordered eigenvalues, PSD rank- r interlacing gives $\lambda_i(R) \leq \lambda_i(D + R) \leq \lambda_{i-r}(R)$ for $i > r$. The feature-rank statements follow from $U^{(1)} = \Phi^\top \Phi / n$ and $U^{(K)} = \Phi_{\text{all}}^\top \Phi_{\text{all}} / (Kn)$. Passing from the population kernel to finite features, and then to noise averaging, needs the separate assumptions in Assumption I.14. $\square$

Data directions and the W factor. The nonzero eigenvalues of the data term are those of $\hat{M}_t^{1/2} (W^\top W / p) \hat{M}_t^{1/2} a_1^2 / d_{\text{latent}}$ up to the spread of $a_1(q_\mu)$ across points. After ξ -averaging $\hat{M}_t \rightarrow M_t = e^{-2t}(\hat{C} + \hat{S}) + \Delta_t I$, which is block diagonal in the (rotated) signal/null split up to $O(n^{-1/2})$ empirical cross-covariances: the signal block $e^{-2t}(\hat{C} + \hat{S}_{\parallel}) + \Delta_t I_{d_{\text{intrinsic}}}$ has $d_{\text{intrinsic}}$ eigenvalues with mean α_t^2 , and the null block $M_{\perp} = e^{-2t}\hat{S}_{\perp} + \Delta_t I$ has $d_{\text{latent}} - d_{\text{intrinsic}}$ eigenvalues, $\hat{S}_{\perp}$ being $\sigma_{\perp}^2$ times a white Wishart with n samples in $d_{\text{latent}} - d_{\text{intrinsic}}$ dimensions (Marchenko–Pastur, ratio $\gamma_{\perp}$, support $\sigma_{\perp}^2(1 \pm \sqrt{\gamma_{\perp}})^2$). The factor $W_{\perp}^\top W_{\perp} / p$ is an independent white Wishart of ratio $(d_{\text{latent}} - d_{\text{intrinsic}})/p$, asymptotically free of $M_{\perp}$, so block (ii) is $[e^{-2t}\sigma_{\perp}^2 \text{MP}(\gamma_{\perp}) + \Delta_t] \boxtimes \text{MP}(d_{\text{latent}}/p)$ scaled by a_1^2/d_{latent} ; the trace of a product of free elements is the product of traces, so the surrogate mean tends to $a_1^2\beta_t^2/d_{\text{latent}}$, and the support is $\frac{a_1^2}{d_{\text{latent}}}[e^{-2t}\sigma_{\perp}^2(1 \pm \sqrt{\gamma_{\perp}})^2 + \Delta_t]$ at leading order in d_{latent}/p with relative edge corrections $O(\sqrt{d_{\text{latent}}/p})$. Direct check (repository generator, $d_{\text{latent}} = 40$, single copy): noise-dim min/median/max 1.05/2.07/4.07, 4.63/8.11/13.47, 18.96/33.5/54.9 at $p = 16d_{\text{latent}}, 64d_{\text{latent}}, 256d_{\text{latent}}$ versus the leading-order formula 1.20/2.05/3.25, 4.81/8.19/13.00, 19.25/32.75/52.0: the median is p -independent to 1%, the edges move by $-13\%/+25\%$ at $d_{\text{latent}}/p = 1/16$ and by $\pm 4\%$ at $1/64$. For the finite-rank signal spikes the W factor contributes only $O(p^{-1/2})$ (signal-min/ p : 0.0161, 0.0156, 0.0158 across the three widths). The signal block at fixed $d_{\text{intrinsic}} = 5$, $k = 10$ is a finite set of eigenvalues of the realized centre geometry, not a self-averaging bulk: $a_1^2(p/d_{\text{latent}})\lambda_j(M_t)$ computed from the seed-42 training set reproduces all five signal eigenvalues to 2–5% at every d_{latent} (e.g. $d_{\text{latent}} = 200$: measured 167.8, 135.4, 99.8, 87.1, 73.9 vs 168.8, 141.4, 99.4, 88.0, 75.1; $d_{\text{latent}} = 5$: 46.6, 41.9, 29.7, 22.0, 20.3 vs 47.2, 39.5, 27.3, 24.3, 20.2), whereas the mean-only value $a_1^2\alpha_t^2\psi_p$ overshoots the median by 10–20%. For the single-copy matrix Δ_t enters the Wishart variance instead and the leading-order support is $\frac{a_1^2}{d_{\text{latent}}}\beta_t^2(1 \pm \sqrt{\gamma_{\perp}})^2$ (Remark I.21).

Residual directions. $R = K_{\text{res}}/n$ has $\text{tr } R = \frac{1}{n} \sum_{\mu} a_*(q_{\mu})^2$, which equals $a_*(q)^2$ to $< 1\%$ for $d_{\text{latent}} \geq 20$ (0.0733 vs 0.0785 at $d_{\text{latent}} = 5$). If the residual is homogeneous ($R = \sigma_r I$) the sample block is the plain Marchenko–Pastur law of Remark I.17(b), with top edge $\kappa_{\text{MP}}\sigma_r$. In general the block is the MP–Silverstein image of $\text{spec}(P_{\perp} R P_{\perp})$ with $p - d_{\text{latent}}$ effective samples, whose width

is set by the spread of $a_*(q_\mu)^2$ (range 0.0036–0.153 at $d_{\text{latent}} = 5$, 0.0017–0.035 at $d_{\text{latent}} = 40$, 0.0024–0.0089 at $d_{\text{latent}} = 200$, because q_μ inherits the cross term $2\langle m_{c(\mu)}, \xi^\mu \rangle / d_{\text{latent}}$ of the mixture) and by the same-cluster component of Step 1; both shrink with d_{latent} . We do not have a closed form for the resulting top edge $\kappa_{\text{top}}\sigma_r$; it is characterised by this transform and measured in Table 3, where it decreases from 19.0 to 1.45 and reaches $\kappa_{\text{MP}} = 1.31$ at $d_{\text{latent}} = 200$ once the $\approx 11\%$ inflation caused by the $K = 50$ average is removed. On the $p = d_{\text{latent}} + n + 300$ spectra ($K = 500$) the measured values 14.5, 12.1, 8.0, 3.10, 2.00 at $d_{\text{latent}} = 10, \dots, 200$ compare with $\kappa_{\text{MP}} = 3.14, 3.07, 2.94, 2.59, 2.08$; the additive-model value $(1 + \sqrt{\gamma_p})^2 = 2.91$ at $d_{\text{latent}} = 200$ would wrongly put the measured edge below the homogeneous one.

Remark I.16 (Four-group spectral approximation; conditional). Assume $p \geq n > d_{\text{latent}} > d_{\text{intrinsic}}$, $d_{\text{intrinsic}}$ and $t > 0$ fixed, and take $d_{\text{latent}}, n, p \rightarrow \infty$ with $\gamma_\perp, \gamma_p$ fixed, under Assumption I.14. Let $\varepsilon = \bar{k} \Delta_t / q$ with $\bar{k} = \sum_{k \geq 3} k c_k(q)^2 / \sum_{k \geq 3} c_k(q)^2 \in [3.2, 3.9]$. Partition the measured $U^{(K)}$ by index as in (A5). The following are proposed scales when these groups separate, not proven spectral components of U :

- (i) $d_{\text{intrinsic}}$ *signal outliers*, $\lambda_j^{\text{sig}} = \frac{a_1(q)^2}{d_{\text{latent}}} [e^{-2t} \lambda_j(\hat{C} + \hat{S}_\parallel) + \Delta_t] (1 + O(p^{-1/2})) + \sigma_r (1 + O(n/p))$, population mean $a_1(q)^2 \alpha_t^2 / d_{\text{latent}}$;
- (ii) $d_{\text{latent}} - d_{\text{intrinsic}}$ *noise-dimension eigenvalues*, those of $\frac{a_1(q)^2}{d_{\text{latent}}} M_\perp^{1/2} (W_\perp^\top W_\perp / p) M_\perp^{1/2}$, i.e. $[e^{-2t} \sigma_\perp^2 \text{MP}(\gamma_\perp) + \Delta_t] \boxtimes \text{MP}(d_{\text{latent}}/p)$ scaled by a_1^2/d_{latent} : approximate mean $a_1(q)^2 \beta_t^2 / d_{\text{latent}}$, support $\frac{a_1(q)^2}{d_{\text{latent}}} [e^{-2t} \sigma_\perp^2 (1 \pm \sqrt{\gamma_\perp})^2 + \Delta_t]$ at leading order in d_{latent}/p , edges corrected by $O(\sqrt{d_{\text{latent}}/p})$, median p -independent;
- (iii) $n - d_{\text{latent}}$ *sample eigenvalues*: the Marchenko–Pastur–Silverstein image, with $p - d_{\text{latent}}$ effective samples, of the residual kernel $P_\perp K_{\text{res}} P_\perp / n$ (diagonal $a_*(q_\mu)^2$, $P_\perp$ off the data directions); mean $\sigma_r(1 - d_{\text{latent}}/p)(1 - \varepsilon)$, top edge $\kappa_{\text{top}}\sigma_r$, where $\kappa_{\text{top}} = \kappa_{\text{MP}}$ for a homogeneous residual and a single copy, and in general is set by the spread of $a_*(q_\mu)^2$ and the same-cluster part of K_{res} ;
- (iv) $p - n$ *residual-floor eigenvalues* of total mass $\simeq \varepsilon a_*(q)^2$; for a single diffused copy these are exact zeros (Remark I.21).

The homogeneous single-copy surrogate obeys the following algebraic bookkeeping identity; its proposed allocation among blocks is only approximate:

$$\begin{aligned}
 & \underbrace{d_{\text{intrinsic}} \frac{a_1^2 \alpha_t^2}{d_{\text{latent}}} + (d_{\text{latent}} - d_{\text{intrinsic}}) \frac{a_1^2 \beta_t^2}{d_{\text{latent}}}}_{= a_1(q)^2 q} \\
 & + \underbrace{d_{\text{latent}} \sigma_r \left(1 + \frac{n - d_{\text{latent}}}{p}\right)}_{\text{outlier shifts}} + \underbrace{(n - d_{\text{latent}}) \sigma_r \left(1 - \frac{d_{\text{latent}}}{p}\right)}_{\text{sample bulk}} \\
 & = a_1(q)^2 q + a_*(q)^2 = \mathbb{E}[\tanh(\sqrt{q}z)^2],
 \end{aligned} \tag{21}$$

while $\mathbb{E}_W \text{tr } U = \frac{1}{n} \sum_\mu \mathbb{E}_{\xi, z} \tanh(\sqrt{q_\mu} z)^2$ exactly, which differs from $\mathbb{E} \tanh(\sqrt{q}z)^2$ by a Jensen correction of 2–5% over the sweep. All data-side eigenvalues scale as $a_1(q(d_{\text{latent}}))^2 / d_{\text{latent}}$ and are p -independent at leading order in d_{latent}/p (the code normalization $U_{\text{code}} = pU$ reintroduces $\psi_p = p/d_{\text{latent}}$); the sample block scales as $a_*(q(d_{\text{latent}}))^2 / n$.

Rationale and limits. The Hermite kernel split is exact over W (Lemma I.13). Transferring it to finite-feature eigenvalues uses Assumption I.14; splitting the data term further neglects empirical signal–null cross terms and finite-width corrections. The projected-Wishart model in Remark I.17 approximates the residual only in a strong-spike regime. The noise-averaging argument below is a covariance heuristic, not a proof preserving the single-copy counts. In particular, the factors $(1 - \varepsilon)$ and the tail mass are approximate and the heterogeneous top-edge coefficient is measured. Agreement of a trace identity cannot establish any of these individual block claims.

I.2.4 STEP 4: SPIKED-COVARIANCE OUTLIERS AND THE TRACE CHECK

Remark I.17 (Spiked-covariance reference model and strong-spike approximation). Let $G = \Sigma^{1/2}(ZZ^\top/p)\Sigma^{1/2}$ with Z iid Gaussian and $\Sigma = D + \sigma_r I$, where D has eigenvalues $\theta_j > 0$. (a) In the classical fixed-rank limit, $n, p \rightarrow \infty$ at fixed $\gamma_p = n/p < 1$, a supercritical spike has limiting location

$$\lambda(\theta_j) = (\theta_j + \sigma_r) \left(1 + \frac{\gamma_p \sigma_r}{\theta_j}\right), \quad \theta_j > \sigma_r \sqrt{\gamma_p}. \quad (22)$$

Subcritical spikes approach the MP upper edge. This is the spiked sample-covariance result (Baik et al., 2005; Baik & Silverstein, 2006); more general perturbation results also have specific hypotheses (Benaych-Georges & Nadakuditi, 2011). It is not a theorem for a growing rank- d_{latent} nonlinear data block. (b) In a separate strong-spike approximation, write the rows as Z_1 (spiked) and Z_2 (residual), and project off the row space of Z_1 . The projected matrix $(\sigma_r/p)Z_2 P^\perp Z_2^\top$ is exactly Wishart conditional on Z_1 , with $p-d_{\text{latent}}$ effective samples. Its expected trace is $(n-d_{\text{latent}})\sigma_r(1-d_{\text{latent}}/p)$ and its asymptotic upper edge is $\sigma_r \kappa_{\text{MP}}$. The finite eigenvalues of G approach this projected model when all spike strengths diverge at fixed matrix dimensions, as follows from the Schur complement. At finite spike strengths it is not the exact restriction of G , and interlacing alone does not identify the eigenvalues. The inferred outlier mass and finite-ratio edge formulas below are therefore approximations; no uniform joint-limit error is established here.

Numerical check of (b) (homogeneous, $n = 500$, $d_{\text{latent}} = 40$, $p = 2560$, 5 repetitions): bulk mass/ $(\sigma_r(n-d_{\text{latent}})) = 0.9840$ vs $1-d_{\text{latent}}/p = 0.9844$; top edge/ $\sigma_r = 1.987$ vs $\kappa_{\text{MP}} = 2.005$, excluding the additive-model value $(1 + \sqrt{\gamma_p})^2 = 2.079$. Individual outliers fluctuate by $\theta_j \sqrt{2/p}$, so the per-outlier shift (σ_r vs $1.18\sigma_r$) is not resolvable in a single run; this comparison tests the approximation in (b), rather than proving an exact block law. In code units the threshold is $\theta_c p \in [0.03, 0.1]$ across the sweep while the smallest noise-dimension eigenvalue is 2–5 at $\sigma_\perp = 0.5$: every data direction is an outlier by a margin $\geq 20\times$, and the shift σ_r is 0.15–0.25% of $\lambda_{\min}^{\text{sig}}$ ($a_*^2 p/n = 0.05$ – 0.12 vs 20–74), which is the operational content of “the data-side eigenvalues are p -independent” beyond the W factor of Step 3.

Trace arithmetic. For fixed W , $\text{tr} U = \frac{1}{n} \sum_\mu \mathbb{E}_\xi \|\phi(x_t^\mu)\|^2$ exactly. Averaging also over W gives $\mathbb{E}_W \text{tr} U = \frac{1}{n} \sum_\mu \mathbb{E}_{\xi, z} \tanh(\sqrt{q_\mu} z)^2 = \frac{1}{n} \sum_\mu \mathbb{E}_\xi [a_1(q_\mu)^2 q_\mu + a_*(q_\mu)^2]$. On the released spectra this heterogeneous prediction matches $\text{tr} U_{\text{code}}/p$ to 1.5% at every d_{latent} (0.548, 0.449, 0.361, 0.292, 0.233, 0.209 predicted vs 0.545, 0.456, 0.363, 0.292, 0.233, 0.209 measured, $d_{\text{latent}} = 5, \dots, 200$), while the homogeneous value $\mathbb{E} \tanh(\sqrt{q} z)^2 = a_1(q)^2 q + a_*(q)^2$ is 2–5% high at small d_{latent} because $q \mapsto \mathbb{E} \tanh(\sqrt{q} z)^2$ is concave (Jensen; $\text{sd}_\mu q_\mu = 1.37$ at $d_{\text{latent}} = 5$, 0.04 at $d_{\text{latent}} = 200$). On the block side, with $\text{tr} M_t = d_{\text{intrinsic}} \alpha_t^2 + (d_{\text{latent}} - d_{\text{intrinsic}}) \beta_t^2 = d_{\text{latent}} q$ and Remark I.17(b), the identity (21) holds because $d_{\text{latent}}(n-d_{\text{latent}})/p - (n-d_{\text{latent}})d_{\text{latent}}/p = 0$. An edge formula that divides every block by $\psi_p = p/d_{\text{latent}}$ would make the left side $O(d_{\text{latent}}/p)$ and fails this check at $d_{\text{latent}} = 40$ by a factor 64. Block-level closure is weaker than trace-level closure on the released spectra: the noise-dim block sum at $d_{\text{latent}} = 40$ is 0.118 vs $a_1^2(d_{\text{latent}} - d_{\text{intrinsic}}) \beta_t^2 / d_{\text{latent}} = 0.114$, but the sample block carries only 0.57, 0.65, 0.72, 0.77, 0.80, 0.81 of $a_*^2(1-d_{\text{latent}}/n)$ at $d_{\text{latent}} = 5, \dots, 200$ (theory units, 0.0088 vs 0.0115 at $d_{\text{latent}} = 40$). The factor $(1-d_{\text{latent}}/p)$ accounts for 1.6%, the ξ -average floor for 0–20% (growing with d_{latent}), and heterogeneity of the trace split for $< 1\%$; the remaining deficit, largest at small d_{latent} , has the size and d_{latent} -dependence of the same-cluster component of K_{res} , which lives in the span of the cluster-indicator vectors and is therefore absorbed by the signal outliers rather than by the sample block. We have not verified this decomposition and record it as open.

I.2.5 STEP 5: SEPARATION CONDITIONS AND LIMITS

Corollary I.18 (Separation conditions; sketched). *Under Remark I.16, at leading order in d_{latent}/p the blocks are disjoint iff*

$$(\text{signal} \succ \text{noise-dim}) \quad \lambda_{\min}(\hat{C} + \hat{S}_{\parallel}) > \sigma_\perp^2 (1 + \sqrt{\gamma_\perp})^2, \quad (23)$$

$$(\text{noise-dim} \succ \text{sample}) \quad \frac{n}{d_{\text{latent}}} \frac{a_1(q)^2}{a_*(q)^2} \left[e^{-2t} \sigma_\perp^2 (1 - \sqrt{\gamma_\perp})^2 + \Delta_t \right] > \kappa_{\text{top}}. \quad (24)$$

In (23) the factors a_1^2 , $1/d_{\text{latent}}$ and Δ_t cancel exactly; with population means as proxies it reads $(\alpha_t^2 - \Delta_t)/(\beta_t^2 - \Delta_t) = (s^2/d_{\text{intrinsic}} + \sigma_{\text{sig}}^2)/\sigma_{\perp}^2 > (1 + \sqrt{\gamma_{\perp}})^2$ (11.2 versus ≤ 2.7 for $d_{\text{latent}} \leq 240$ at $\sigma_{\perp} = 0.5$): signal/null separation is governed by the anisotropy against a Marchenko–Pastur spread of ratio d_{latent}/n , and p never enters. In (24) the left side carries the explicit factor n/d_{latent} , the de-saturation ratio a_1^2/a_*^2 (which grows from 2.3 to 134 over the sweep) and a bracket that closes as $d_{\text{latent}} \rightarrow n$ or $\sigma_{\perp} \rightarrow 0$, with an $O(\sqrt{d_{\text{latent}}/p})$ correction from the W factor (order one for $p = d_{\text{latent}} + n + 300$ at $d_{\text{latent}} = 200$, where $d_{\text{latent}}/p = 0.2$).

Proof. Compare $\lambda_{\min}^{\text{sig}}$ with the upper edge of block (ii) and the lower edge of block (ii) with $\kappa_{\text{top}}\sigma_r$, using the edge formulas of Remark I.16; the common factor a_1^2/d_{latent} and the additive Δ_t cancel in the first comparison, and $\alpha_t^2 - \Delta_t = e^{-2t}(s^2/d_{\text{intrinsic}} + \sigma_{\text{sig}}^2)$, $\beta_t^2 - \Delta_t = e^{-2t}\sigma_{\perp}^2$. The realized $\lambda_{\min}(\hat{C} + \hat{S}_{\parallel}) \approx 1.7\text{--}1.9$ exceeds $0.25(1 + \sqrt{\gamma_{\perp}})^2 \leq 0.7$. $\square$

The measured gap $\lambda_{\min}^{\text{nd}}/\lambda_{\max}^{\text{samp}}$ is 4.4, 7.7, 13, 18, 12 ($p = 64d_{\text{latent}}$) and 5.1, 5.9, 8.2, 11.4, 6.1 ($p = d_{\text{latent}} + n + 300$) at $d_{\text{latent}} = 10, 20, 40, 100, 200$: non-monotone; it opens because a_1^2/a_*^2 grows and κ_{top} falls faster than n/d_{latent} shrinks, then closes because $(1 - \sqrt{\gamma_{\perp}})^2 \rightarrow 0$ and $n - d_{\text{latent}} \rightarrow 0$ as $d_{\text{latent}} \rightarrow n$; at $d_{\text{latent}} = 500 = n$ the released spectra show $\lambda_{\min}^{\text{nd}} = 0.77$ (code units) and at $d_{\text{latent}} = 1000$ the noise-dim median has collapsed to 1.00.

Remark I.19 (Isotropic limit; sketched). For $\sigma_{\perp} = \sigma_{\text{sig}}$ condition (23) requires $(s^2/d_{\text{intrinsic}} + 1) = 2.8 > (1 + \sqrt{\gamma_{\perp}})^2$, which fails as soon as $\gamma_{\perp} \gtrsim 0.1$: blocks (i)–(ii) merge into a single population bulk with $d_{\text{intrinsic}}$ BBP-type outliers at its top, qualitatively analogous to the population/sample separation of Bonnaire et al. (2025), without identifying their component weights. Block (ii) does not disappear: at $d_{\text{latent}} = 100$, $\sigma_{\perp} = 1$ ($q = 1.088$, $a_1^2 = 0.348$) its leading-order lower edge $a_1^2 p [e^{-2t}(1 - \sqrt{\gamma_{\perp}})^2 + \Delta_t]/d_{\text{latent}} = 7.40$ (measured 7.4–7.5 in a direct simulation with the repository generator, $p = 6400$, $K = 50$; W -factor correction $\approx -4\%$) sits $12\times$ above the sample top 0.61, with signal median 61.8 (predicted 61.6) and noise-dim max 44.8 (predicted 45.5). The $d_{\text{latent}} - d_{\text{intrinsic}}$ noise-dimension modes therefore *exist* in the isotropic case and remain separated from the sample bulk by (24); anisotropy adds the internal split of ρ_2 into (i) and (ii), not the existence of (ii).

Remark I.20 ($\sigma_{\perp} \rightarrow 0$; sketched). As $\sigma_{\perp} \rightarrow 0$ the noise-dimension population eigenvalues collapse onto the diffusion floor $\theta_{\perp} = a_1^2 \Delta_t/d_{\text{latent}}$, which is comparable to the top edge $\kappa_{\text{top}}\sigma_r$ of the heterogeneous sample bulk ($\kappa_{\text{top}} \approx 10\text{--}16$ at $d_{\text{latent}} \leq 40$). By Remark I.17 such directions are not outliers far above the bulk but are repelled to just above its edge, so blocks (ii) and (iii) merge, which is the smooth data/sample transition observed at $\sigma_{\perp} = 0.01$: noise-dim min/median/max 0.71/0.87/1.08, 0.58/0.68/0.96, 0.64/0.81/1.00 (code units, $d_{\text{latent}} = 10, 20, 40$) versus the floor $a_1^2 \Delta_t p/d_{\text{latent}} = 0.38, 0.57, 0.76$ and the sample top 0.66, 0.58, 0.56; the gap ratio is 1.08, 1.01, 1.14. The $\sigma_{\perp} = 0$ subspace model of George et al. (2026) provides a qualitative comparison: the $d_{\text{latent}} - d_{\text{intrinsic}}$ null directions are no longer separated from the sample block.

Relation to prior spectral results. The prior spectral results motivate the surrogate; they do not prove the four-group decomposition for the present mixture. In particular, clean covariance atoms must be used in Bonnaire’s equations, and a vanishing-mass fixed-rank signal requires an additional outlier analysis. George’s subspace model is a related limiting setting, not an equality obtained by setting one variance to zero in our index partition. See Appendix I.1.1 for the operator, normalization and pencil distinctions.

I.2.6 THE ξ -AVERAGED MATRIX USED IN THE CODE

Remark I.21 (Single copy vs. ξ -average; rank exact, edges approximate). For $U^{(1)}$ the approximation of Remark I.16 uses with $\hat{M}_t$ in place of M_t (so block (ii) has leading-order support $\frac{a_1^2}{d_{\text{latent}}} \beta_t^2 (1 \pm \sqrt{\gamma_{\perp}})^2$: lower edge 1.50 vs 2.17 code units at $d_{\text{latent}} = 200$; measured 4.58 vs 4.52 at $d_{\text{latent}} = 40$, $K = 1$), $\varepsilon = 0$, and, at full row rank, exactly $p - n$ zeros ($\lambda_{n+1}/\lambda_n \sim 3 \times 10^{-6}$ in the float32 code path, 10^{-16} in float64): a machine-precision cliff at index n . On the released $K = 50$ spectra there is no visible drop at index n for $d_{\text{latent}} \leq 20$ ($\lambda_n/\lambda_{n+1} = 1.01$; $\gamma_p \geq 0.4$ pushes the lower MP edge of the sample block to ≈ 0) and a drop by 1.32, 1.68, 1.73 at $d_{\text{latent}} = 40, 100, 200$; there is no drop at index $d_{\text{latent}} + n$ at any d_{latent} .

The released `compute_U` averages $\Phi_k^\top \Phi_k / n$ over $K = 50$ independent noise draws, so $\text{rank } \mathbf{U}^{(K)} \leq \min(p, Kn)$ and the zeros of block (iv) are replaced by a floor. A heuristic for its scale uses Lemma I.13: two draws of the same point have $\rho \simeq 1 - \Delta_t/q$, so their residual kernel is $\sum_{k \geq 3} c_k^2 (1 - \Delta_t/q)^k \simeq a_*^2 (1 - \varepsilon)$ with $\varepsilon = \bar{k} \Delta_t/q$, $\bar{k} = \sum_{k \geq 3} k c_k^2 / \sum_{k \geq 3} c_k^2 = 3.9, 3.7, 3.5, 3.3, 3.2, 3.2$ at $d_{\text{latent}} = 5, \dots, 200$. The residual Gram block of one point is therefore $\frac{a_*^2}{Kn} [(1 - \varepsilon) \mathbf{1}\mathbf{1}^\top + \varepsilon I]$: one coherent mode of weight $a_*^2 (1 - \varepsilon)/n$ that belongs to the sample block, and $K - 1$ incoherent modes of weight $a_*^2 \varepsilon / (Kn)$ each, total $\simeq \varepsilon a_*^2$ over the training set, which populate the $p - n$ floor when $p - n \gg n$ and stay inside the sample block otherwise. Measured tail mass $\sum_{i > n} \lambda_i / (p a_*^2) = 0.001, 0.017, 0.065, 0.139, 0.196$ at $d_{\text{latent}} = 10, \dots, 200$ vs $\varepsilon = 0.05, 0.08, 0.11, 0.16, 0.19$ (agreement at $d_{\text{latent}} \geq 100$, where $p - n \geq 5900$). For $p = d_{\text{latent}} + n + 300$ the ratio $\lambda_n / \lambda_{n+1}$ is 1.00–1.01 throughout. The average also inflates the sample top by $\approx 11\%$ at $d_{\text{latent}} = 40$ ($0.371 \rightarrow 0.413$, code units), so the κ_{top} column of Table 3 is ξ -averaged data compared with a single-copy formula. The proposed data-term approximation changes through $\hat{M}_t \rightarrow M_t$, which is why Δ_t enters (ii) as an exact shift rather than through the Wishart variance. (Appendix I.3 writes the same coherent fraction as $\theta(q) = \kappa_{\text{res}}(q - \Delta_t) / \kappa_{\text{res}}(q)$, the exact Hermite-series form of $1 - \varepsilon$.)

I.2.7 NUMERICAL CHECKS AND TIMESCALES

Table 3 compares the released $\mathbf{U}_{\text{code}}$ spectra (seed 42, $\sigma_\perp = 0.5$, $p = 64d_{\text{latent}}$, $K = 50$) with the parameter-free predictions above; $q(d_{\text{latent}})$ is (16) and a_1, a_*, c_k are evaluated by 200-node Gauss–Hermite quadrature. The noise-dimension block is reproduced to 2–8% in median, lower and upper edge (max: 6.44/5.69, 9.52/8.98, 13.8/13.0, 19.75/19.68, 27.2/26.7 measured/predicted at $d_{\text{latent}} = 10, \dots, 200$) once its MP shape at ratio $\gamma_\perp$ is used; mean-only predictions overshoot the median by up to 15% at $d_{\text{latent}} = 200$. The signal eigenvalues are reproduced to 2–5% by the realized centre geometry. On the $p = d_{\text{latent}} + n + 300$ spectra ($K = 500$) the theory-unit noise-dim medians are 0.0083/0.0074, 0.0051/0.0052, 0.0030/0.0033, 0.0014/0.0016, 0.0007/0.0008 (measured/predicted), confirming p -independence of the median. In theory units (p/λ_{code}) the gradient-flow timescales on the Table spectra are $\tau_{\text{gen}} = 1/\lambda_{\text{min}}^{\text{sig}} = 15.8, 20.1, 30.5, 47.7, 94.8, 173$ ($10.96\times$, equal to $40 \times (0.180/0.625) \times (1.76/1.85)$): the d_{latent}/a_1^2 factor times the realized λ_5 ratio, with no residual), $\tau_{\text{mem}}(\text{onset}) = 1/\lambda_{\text{max}}^{\text{samp}} = 334, 688, 1918, 6843, 32249, 73857$ ($221\times = 16.7$ from $a_*^2 \times 13.1$ from κ_{top}) and $\tau_{\text{mem}}/\tau_{\text{gen}} = 21.2, 34.3, 62.9, 143, 340, 426$, monotone ($20.1\times$). A fixed- μ_1 theory that ignores the drift of $q(d_{\text{latent}})$ predicts flat block positions (64.9 and 6.22 in code units) and is rejected by the same data.

Table 3: Predicted vs. measured block statistics of $\mathbf{U}_{\text{code}} = p\mathbf{U}$ ($\sigma_\perp = 0.5$, $p = 64d_{\text{latent}}$, $n = 500$, $K = 50$, $t = 0.01$, seed 42). Predictions use $q(d_{\text{latent}})$ from (16) and no fitted parameter. Signal min: fifth signal eigenvalue (sets τ_{gen}) vs. $a_1^2(p/d_{\text{latent}})\lambda_5(M_t)$ of the realized training set; noise-dim median and min: MP median and lower edge at ratio $\gamma_\perp$ (leading order in d_{latent}/p); κ_{top} : sample-bulk top in units of $\sigma_r p$, vs. the homogeneous single-copy value κ_{MP} (the $K = 50$ average inflates the measured value by $\approx 10\%$); $\text{tr } \mathbf{U}$: measured vs. $\frac{1}{n} \sum_\mu \mathbb{E} \tanh^2(\sqrt{q_\mu} z)$ (homogeneous value in parentheses). [†]At $d_{\text{latent}} = 5$, $p = 320 < n$: block (iv) is empty, the sample block has $p - d_{\text{latent}}$ modes, and κ_{top} is reported in the same units for comparability only.

d_{latent}	q	a_1^2	a_*^2	signal min		noise-dim median		noise-dim min		κ_{top}		$\text{tr } \mathbf{U}$	
				meas.	pred.	meas.	pred.	meas.	pred.	meas.	κ_{MP}	meas.	pred.
5 [†]	2.76	0.180	0.0785	20.3	20.2	—	—	—	—	19.0	5.00	0.545	0.548 (0.576)
10	1.52	0.281	0.0447	31.9	31.5	5.22	4.75	4.10	3.92	16.3	3.49	0.456	0.449 (0.470)
20	0.89	0.393	0.0235	42.0	44.2	7.03	6.60	5.16	4.71	11.1	2.57	0.363	0.361 (0.373)
40	0.58	0.494	0.0125	53.6	55.9	8.36	8.19	4.90	4.81	5.85	2.01	0.292	0.292 (0.297)
100	0.39	0.586	0.0064	67.5	67.8	9.32	9.34	3.57	3.66	2.41	1.54	0.233	0.233 (0.235)
200	0.33	0.625	0.0047	73.9	75.1	9.22	9.31	2.12	2.17	1.45	1.31	0.209	0.209 (0.209)

Predictions and how to test them.

1. Block (ii) is $\text{MP}(\gamma_\perp) \boxtimes \text{MP}(d_{\text{latent}}/p)$ scaled by $a_1(q)^2/d_{\text{latent}}$: its median is p -independent (to $\sim 1\%$), its edges move by $O(\sqrt{d_{\text{latent}}/p})$, and its lower edge sinks toward $a_1^2 \Delta_t/d_{\text{latent}}$ as $d_{\text{latent}} \rightarrow n$. Test: compare noise-dim medians across $p \in \{16, 64, 256\}d_{\text{latent}}$ at

$d_{\text{latent}} = 40$ (single copy); run an n -sweep at fixed $d_{\text{latent}} = 40$ ($n = 250 \dots 2000$) and check the lower edge follows $(1 - \sqrt{(d_{\text{latent}} - 5)/n})^2$.

2. *The sample-bulk top edge homogenises to $\kappa_{\text{MP}}\sigma_r$ from above as d_{latent} grows; the small- d_{latent} excess is set by the spread of $a_*(q_\mu)^2$ and a same-cluster component of size $\sim (n/k)c_3(q)^2\rho_{\text{same}}^3$; the K -draw average inflates the top by $\sim 10\%$.* Test: compare the sample-block distribution with the MP–Silverstein transform of $\{a_*(q_\mu)^2\}$; regenerate with $k = 1$ at fixed norm; recompute with $K = 1$.
3. $\tau_{\text{mem}}(\text{onset}) = 1/(\kappa_{\text{top}}\sigma_r)$ *decomposes into the $a_*(q(d_{\text{latent}}))^2$ trajectory ($16.7\times$, closed form) and the homogenisation of κ_{top} ($13.1\times$, characterised); a different $\sigma_\perp$ changes the predicted a_*^2 curve before any run.* Test: regress $\log \tau_{\text{mem}}(\text{onset})$ on $\log a_*^2$ and $\log \kappa_{\text{top}}$; predict $a_*(q(d_{\text{latent}}))^2$ for $\sigma_\perp \in \{0.1, 0.3, 0.01\}$ and compare the sample-top eigenvalue on existing spectra.
4. $\tau_{\text{gen}} = d_{\text{latent}}/(a_1(q(d_{\text{latent}}))^2[e^{-2t}\lambda_5(\hat{C} + \hat{S}_\parallel) + \Delta_t])$ *in theory units; a_1^2 rising $0.18 \rightarrow 0.625$ makes the growth $\sim 11\times$ over $d_{\text{latent}} = 5..200$ rather than $40\times$.* Test: plot $\tau_{\text{gen}}(d_{\text{latent}})a_1(q)^2/d_{\text{latent}}$ against $1/(e^{-2t}\lambda_5 + \Delta_t)$ of the realized training set; repeat over centre-geometry seeds.
5. *Memorisation returns as $d_{\text{latent}} \rightarrow n$ because the noise-dim/sample gap closes and the sample block empties; the gap ratio is non-monotone with a peak near $d_{\text{latent}} = 100$.* Test: check the $d_{\text{latent}} = 500, 1000$ spectra for the noise-dim minimum and the drop at index n ; train RFNNs at $d_{\text{latent}} \in \{300, 400, 480\}$ and check the fraction rises.
6. *With one diffused copy $\mathbf{U}$ has exactly $p - n$ zeros; with the K -draw average the block becomes a floor of mass $\sim \bar{k}(\Delta_t/q)a_*^2$, visible as a $1.3\text{--}1.8\times$ drop at index n only for $\gamma_p \leq 0.2$.* Test: recompute $\mathbf{U}$ with $K \in \{1, 50\}$ and $t \in \{0.001, 0.01, 0.05\}$; the tail mass should scale as Δ_t/q .
7. *Trace-level bookkeeping closes to 1.5% ; block-level bookkeeping does not: the sample block carries $0.57\text{--}0.81$ of its predicted mass.* Test: decompose K_{res} of the realized training set into diagonal, same-cluster and cross-cluster parts and check whether the same-cluster mass accounts for the deficit.
8. *As $\sigma_\perp \rightarrow 0$ blocks (ii) and (iii) merge smoothly.* Test: sweep $\sigma_\perp \in \{0.01, 0.05, 0.1, 0.25, 0.5\}$ at $d_{\text{latent}} = 40$ and track $\lambda_{\text{min}}^{\text{nd}}/\lambda_{\text{max}}^{\text{samp}}$; it should cross 1 where $a_1^2 e^{-2t} \sigma_\perp^2 (1 - \sqrt{\gamma_\perp})^2 / d_{\text{latent}} \sim (\kappa_{\text{top}} - 1)\sigma_r$.

Open gaps. (1) The four groups are index conventions for the measured $\mathbf{U}^{(K)}$; only the single-copy rank bound is exact. Existing figure colors and archived block summaries must be interpreted with their stated cuts. (2) The finite-rank BBP limit and the strong-spike projected-Wishart approximation are distinct regimes; the latter has no uniform error bound here. (3) Bonnaire’s floor weight is $1 - (n + d_{\text{latent}})/p$ in its stated regime, not the single-copy rank deficit (Appendix I.1.1). (4) The sample group carries only $0.57\text{--}0.81$ of the proposed residual mass; the same-cluster explanation is unverified. (5) Gaussian-surrogate hypotheses, outlier transfer and noise averaging remain to be established for this mixture. (6) κ_{top} is measured, not derived; its apparent homogenization is an observation. (7) The proposed tail mass $\bar{k}\Delta_t/q$ agrees better at large d_{latent} than small d_{latent} . (8) The sweep fixes $n = 500$ rather than taking a proportional limit, and $d_{\text{latent}} = 5$ has $p < n$. (9) Some archived configuration files contain stale dimension metadata; numerical claims must use the audited run mapping.

I.3 THE BUFFER AS A RATIO OF SPECTRAL EDGES

The edge-based identities in this subsection are exact when expressed in measured eigenvalues. Substituting the scales from Remark I.16 is a conditional surrogate approximation; its $o(1)$ notation is not a proved error rate for the finite-mixture sweep. No spectral component count is inferred for the noise expectation from single-copy rank.

Setting. We work with the RFNN of (1) trained by the gradient flow (3) at fixed t , with $\mathbf{U}$ as in (2). Write $\mathbf{U} = \sum_{i=1}^p \lambda_i v_i v_i^\top$ with $\lambda_1 \geq \dots \geq \lambda_p \geq 0$, and use the a-priori index blocks $I_{\text{sig}} = \mathcal{S} = \{1, \dots, d_{\text{intrinsic}}\}$, $I_{\text{noise}} = \mathcal{N} = \{d_{\text{intrinsic}} + 1, \dots, d_{\text{latent}}\}$, $I_{\text{samp}} = \mathcal{M} = \{d_{\text{latent}} + 1, \dots, n\}$,

$I_{\text{null}} = \mathcal{Z} = \{n+1, \dots, p\}$ of (A5). Data and notation are as in Appendix I.1; in addition $\alpha_0^2 := s^2/d_{\text{intrinsic}} + \sigma_{\text{sig}}^2$ (so $\alpha_t^2 = e^{-2t}\alpha_0^2 + \Delta_t$), $\psi_{\perp} := (d_{\text{latent}} - d_{\text{intrinsic}})/n$ ($= \gamma_{\perp}$ of Appendix I.2) and $\gamma := n/p$. “ d_{latent} sweep” means d_{latent} varies with $(d_{\text{intrinsic}}, n, k, s, \sigma_{\text{sig}}, \sigma_{\perp}, t)$ fixed and p either $\propto d_{\text{latent}}$ or fixed $\geq n$. Eigenvalues in the released code are $p \times$ the ones here; all ratios are normalisation-free. The nonzero spectrum of the Monte-Carlo estimate of U with N_{mc} draws equals that of the $(nN_{\text{mc}}) \times (nN_{\text{mc}})$ Gram $\Phi\Phi^{\top}/(nN_{\text{mc}})$, which is how the simulations of this subsection (“sim”: $N_{\text{mc}} = 4$, seed 0) are computed; “repo” refers to the released $\sigma_{\perp} = 0.5$ RFNN spectra ($p = 64d_{\text{latent}}, N_{\text{mc}} = 50$, seed 42) unless another set is named.

Definition I.22 (ε -absorption times). Let $a_i(T) := A(T)v_i \in \mathbb{R}^{d_{\text{latent}}}$ and $a_i^* := A^*v_i$, so that $a_i(T) - a_i^* = (a_i(0) - a_i^*)e^{-\lambda_i T}$ (Lemma I.3). For $\varepsilon \in (0, 1)$ the ε -absorption time of mode i is $\tau_i(\varepsilon) := \lambda_i^{-1} \log(1/\varepsilon)$. Define

$$\begin{aligned}\tau_{\text{gen}}(\varepsilon) &:= \max_{i \in I_{\text{sig}}} \tau_i(\varepsilon) = \frac{\log(1/\varepsilon)}{\lambda_{\min}^{\text{sig}}}, \\ \tau_{\text{mem}}^{\text{on}}(\varepsilon) &:= \min_{i \in I_{\text{samp}}} \tau_i(\varepsilon) = \frac{\log(1/\varepsilon)}{\lambda_{\max}^{\text{samp}}}, \\ R &:= \frac{\tau_{\text{mem}}^{\text{on}}}{\tau_{\text{gen}}} = \frac{\lambda_{\min}^{\text{sig}}}{\lambda_{\max}^{\text{samp}}}, \\ R_{\text{full}} &:= \frac{\lambda_{\min}^{\text{noise}}}{\lambda_{\max}^{\text{samp}}},\end{aligned}\tag{25}$$

with $\lambda_{\min}^{\text{sig}} := \lambda_{d_{\text{intrinsic}}}$, $\lambda_{\min}^{\text{noise}} := \lambda_{d_{\text{latent}}}$, $\lambda_{\max}^{\text{samp}} := \lambda_{d_{\text{latent}}+1}$. τ_{gen} is the time to learn the score on the *signal subspace*; the noise-dimension modes, which carry the exactly linear null score (Appendix I.4), are slower, so $R_{\text{full}} \leq R$ measures the onset relative to the whole data block. Both ratios are independent of ε and of the learning rate. These are the per-mode versions of Definitions I.8 and I.11: they coincide with the block functionals there up to the factor 2 in (12) and the θ_B correction, and $\tau_{\text{mem}}^{\text{on}}$ is the index-based statistic that lower-bounds $\tau_{\text{mem}}^{\text{onset}}$ of Definition I.11(i).

Proof. Each mode is a scalar exponential with rate λ_i ; the max (resp. min) of $\lambda_i^{-1} \log(1/\varepsilon)$ over a block is attained at its smallest (resp. largest) eigenvalue, and $\log(1/\varepsilon)$ cancels in the ratios. $R_{\text{full}} \leq R$ because $\lambda_{d_{\text{latent}}} \leq \lambda_{d_{\text{intrinsic}}}$. $\square$

I.3.1 STANDING HYPOTHESES

Assumption I.23 (Inputs to the buffer corollary; conjectured). (H1) (*Gaussian equivalence at block scale.*) With $a_1(q)$, $a_*(q)^2$ as before, write $\phi(x) = a_1(q)(pd_{\text{latent}})^{-1/2}Wx + a_*(q)p^{-1/2}z(x)$, where $z(x) \in \mathbb{R}^p$ has zero mean, unit-variance coordinates, is uncorrelated with Wx , and has Gram $\mathbb{E}_W[z(x)^\top z(x')]/p = \kappa_{\text{res}}(x \cdot x'/d_{\text{latent}})/a_*(q)^2$,

$$\begin{aligned}\kappa_{\text{res}}(\rho) &= \sum_{\ell \text{ odd} \geq 3} c_\ell(q)^2 (\rho/q)^\ell, \\ c_\ell(q) &:= \frac{\mathbb{E}[\tanh(\sqrt{q}z) \text{He}_\ell(z)]}{\sqrt{\ell!}}, \\ \kappa_{\text{res}}(q) &= a_*(q)^2\end{aligned}\tag{26}$$

(the Hermite series of Lemma I.13). We assume $U = U_{\text{lin}} + U_{\text{res}} + E$ with $U_{\text{lin}} = a_1(q)^2(d_{\text{latent}}p)^{-1}WM_tW^\top$, $U_{\text{res}} = a_*(q)^2(np)^{-1}\sum_\mu \mathbb{E}_\xi[z(x_t^\mu)z(x_t^\mu)^\top]$, and E negligible at the scale of each block: $\|P_{\text{data}}EP_{\text{data}}\| = o(a_1^2/d_{\text{latent}})$, $\|P_{\text{samp}}EP_{\text{samp}}\| = o(\theta a_*^2/n)$. A plain $o(1)$ operator-norm statement would be vacuous (every block is $o(1)$). This is stronger than the ESD-level Gaussian-equivalence theorems (Goldt et al., 2022; Hu & Lu, 2023; Mei & Montanari, 2022), is of the type proved for the extreme eigenvalues of the conjugate kernel with i.i.d. inputs by Benigni & Pécché (2021), and is assumed here for the k -centre mixture; its trace consequence is verified below.

(H2) (ξ -averaging.) Two independent diffusion draws of the same x^μ have $\mathbb{E}[x_t^\mu \cdot x_t^{\mu'}]/d_{\text{latent}} = e^{-2t} \|x^\mu\|^2/d_{\text{latent}} = q - \Delta_t + o(1)$; hence the ξ -averaged residual feature $\tilde{z}_\mu := \mathbb{E}_\xi z(x_t^\mu)$

has coherent self-kernel $\kappa_{\text{res}}(q - \Delta_t) = \theta(q)a_*(q)^2$, $\theta(q) := \kappa_{\text{res}}(q - \Delta_t)/\kappa_{\text{res}}(q)$ (Lemma I.24(v)). We assume the incoherent residual mass $(1 - \theta)a_*^2$, spread over p modes (exact $\mathbb{E}_\xi$) or $n(N_{\text{mc}} - 1)$ modes (N_{mc} draws), lies below the sample block: $(1 - \theta)/\min(p, nN_{\text{mc}}) \ll \theta/n$. (On the repo's $N_{\text{mc}} = 50$ spectra $\lambda_{n+1}/\lambda_n = 0.48$ – 0.76 for $d_{\text{latent}} \geq 40$: the index- n step is real but not sharp at finite N_{mc} .)

(H3) (*Asymptotic regime.*) $d_{\text{latent}}, n, p \rightarrow \infty$ with $\psi_n = n/d_{\text{latent}}$, $\psi_p = p/d_{\text{latent}}$, q as parameters; along the sweep these move (ψ_n from 100 to ≈ 1), so no statement is uniform in d_{latent} , and finite-size corrections are $O(d_{\text{latent}}^{-1/2})$ relative ($\|x^\mu\|^2/d_{\text{latent}}$ fluctuates by $\approx 25\%$ at $d_{\text{latent}} = 5$: trace deficit 5% there, $< 1\%$ for $d_{\text{latent}} \geq 100$). $p \geq n > d_{\text{latent}}$ is needed for the block counts; for $p = 64d_{\text{latent}}$ it fails at $d_{\text{latent}} \leq 7$.

(H4) (*Data-side edges; Appendix I.2.*) Let $\tilde{\nu}_1 \geq \dots \geq \tilde{\nu}_{d_{\text{intrinsic}}}$ be the top $d_{\text{intrinsic}}$ eigenvalues of the full empirical covariance $\hat{C} + \hat{S}$ and $\nu_1 \geq \dots \geq \nu_{d_{\text{intrinsic}}}$ those of its $d_{\text{intrinsic}} \times d_{\text{intrinsic}}$ signal block (law independent of d_{latent} ; mean α_0^2 ; measured $\nu_{\min}/\alpha_0^2 = 0.39$ – 0.46). Then $\lambda_i = a_1(q)^2(e^{-2t}\tilde{\nu}_i + \Delta_t)/d_{\text{latent}}(1 + o(1))$ for $i \in I_{\text{sig}}$, with the spiked-covariance law (Baik et al., 2005)

$$\tilde{\nu}_i = \nu_i \left(1 + \frac{\psi_\perp \sigma_\perp^2}{\nu_i - \sigma_\perp^2} \right) (1 + o(1)) \quad \text{if } \nu_i > \sigma_\perp^2 (1 + \sqrt{\psi_\perp}), \quad (27)$$

and I_{noise} occupies $a_1(q)^2[e^{-2t}\sigma_\perp^2(1 \pm \sqrt{\psi_\perp})^2 + \Delta_t]/d_{\text{latent}}$ (Marchenko–Pastur law of the empirical null covariance centred at β_t^2 ; verified to 3–6% at both edges at $p = 64d_{\text{latent}}$; the edges move by $\approx 10\%$ between $p = 850$ and 3200 at $d_{\text{latent}} = 50$, the medians by $< 0.5\%$).

(H5) (*Block separation.*) $\lambda_{\min}^{\text{noise}} > \lambda_{\max}^{\text{sam}}$. On the repo's spectra this holds at $\sigma_\perp = 0.5$ ($R_{\text{full}} = 4.4$ – 18.8 over $d_{\text{latent}} = 10$ – 200) and fails at $\sigma_\perp = 0.01$ ($R_{\text{full}} = 1.0$ – 1.15 at every $d_{\text{latent}} = 8$ – 40), where the noise-dimension block sits on the diffusion floor $a_1^2\Delta_t/d_{\text{latent}}$ and merges with the sample block (Remark I.20). All statements about $\tau_{\text{mem}}^{\text{on}}$, R , R_{full} are restricted to $\sigma_\perp = 0.5$ (and the isotropic control); at $\sigma_\perp = 0.01$ only τ_{gen} is predicted.

Lemma I.24 (Gaussian-equivalent coefficients along the sweep; proved except as noted). *Assume the data construction above with $\alpha_0^2 > \sigma_\perp^2$, $n \rightarrow \infty$ (so $\text{tr } \hat{C} \rightarrow s^2$, $\text{tr } \hat{S} \rightarrow d_{\text{intrinsic}}\sigma_{\text{sig}}^2 + (d_{\text{latent}} - d_{\text{intrinsic}})\sigma_\perp^2$) and $\Delta_t < q$.*

(i) $q(d_{\text{latent}}) = q_\infty + c_q/d_{\text{latent}}$, $q_\infty := e^{-2t}\sigma_\perp^2 + \Delta_t$, $c_q := e^{-2t}d_{\text{intrinsic}}(\alpha_0^2 - \sigma_\perp^2) > 0$; q decreases strictly from $q_{\max} := q(d_{\text{intrinsic}}) = \alpha_t^2$ to q_∞ (this is (11) rewritten).

(ii) $a_1(q) = \mathbb{E}[\text{sech}^2(\sqrt{q}z)]$ is strictly decreasing in q , $a_1 \leq 1$, $a_1(q)^2 = 1 - 2q + O(q^2)$; $q a_1(q)^2 = c_1(q)^2$ is strictly increasing in q ; hence $a_1(q(d_{\text{latent}}))^2$ is strictly increasing in d_{latent} with limit $a_1(q_\infty)^2$.

(iii) $a_*(q)^2 = \sum_{\ell \text{ odd} \geq 3} c_\ell(q)^2$ and $a_*(q)^2 = \frac{2}{3}q^3(1 - 8q + \frac{266}{5}q^2 + O(q^3))$ as $q \rightarrow 0$ (a small- q expansion; numerical values below use quadrature).

(iv) (Trace identity) $a_1(q)^2q + a_*(q)^2 = \mathbb{E}[\tanh^2(\sqrt{q}z)] = \text{tr } \mathbf{U} (1 + O(d_{\text{latent}}^{-1/2}))$; equivalently $d_{\text{intrinsic}}\lambda^{\text{sig}} + (d_{\text{latent}} - d_{\text{intrinsic}})\lambda^{\text{noise}} + \sum_{I_{\text{sam}}} \lambda_i + (\text{floor}) = \mathbb{E} \tanh^2(\sqrt{q}z) = O(1)$, which any proposed edge must satisfy.

(v) (ξ -averaging factor) $\theta(q) := \kappa_{\text{res}}(q - \Delta_t)/\kappa_{\text{res}}(q)$ satisfies $0 < \theta(q) \leq (1 - \Delta_t/q)^3$, with equality to leading order as $q \rightarrow 0$, and $\theta \rightarrow 1$ as $\Delta_t/q \rightarrow 0$.

For the repo sweep ($t = 0.01$, $\Delta_t = 0.0198$, $d_{\text{intrinsic}} = 5$, $s = 3$, $\sigma_{\text{sig}} = 1$, $n = 500$; 120-node Gauss–Hermite quadrature) at $\sigma_\perp = 0.5$: $c_q = 12.50$, $q_\infty = 0.265$, $d^* := c_q/q_\infty = 47$, $a_1(q_\infty)^2/a_1(q_{\max})^2 = 3.73$, and

d_{latent}	5	10	20	50	100	200	300	400	480
q	2.764	1.515	0.890	0.515	0.390	0.327	0.307	0.296	0.291
$a_1(q)^2$	0.180	0.281	0.393	0.521	0.586	0.625	0.640	0.647	0.651
$a_*(q)^2 (\times 10^3)$	78.5	44.7	23.5	10.4	6.44	4.67	4.11	3.85	3.72
$\theta(q)$	0.969	0.951	0.925	0.880	0.847	0.821	0.811	0.805	0.802
$\mathbb{E} \tanh^2(\text{pred.})$	0.576	0.470	0.373	0.278	0.235	0.209	0.200	0.195	0.193
$\text{tr } U \text{ (sim.)}$	0.551	0.458	0.365	0.271	0.233	0.208	0.200	0.196	0.193

(repo: $\text{tr } U = 0.545, 0.456, 0.363, 0.233, 0.209$ at $d_{\text{latent}} = 5, 10, 20, 100, 200$). At $\sigma_{\perp} = 0.01$: $c_q = 13.72$, $q_{\infty} = 0.0199$, $d^* = 690$, slope ratio 5.35, $\theta = 0.80, 0.66, 0.47$ at $d_{\text{latent}} = 50, 100, 200$. At $\sigma_{\perp} = 1$: $c_q = 8.82$, $q_{\infty} = 1$, $d^* = 8.8$, slope ratio 2.04, $\theta \geq 0.93$.

Proof. (i) $\text{tr } M_t = e^{-2t} [d_{\text{intrinsic}} \alpha_0^2 + (d_{\text{latent}} - d_{\text{intrinsic}}) \sigma_{\perp}^2] + d_{\text{latent}} \Delta_t$; divide by d_{latent} . (ii) sech^2 is even and strictly decreasing in $|x|$, so for $q < q'$ and every $z \neq 0$, $\text{sech}^2(\sqrt{q}z) > \text{sech}^2(\sqrt{q'}z)$; integrate. The expansion uses $\text{sech}^2 x = 1 - x^2 + \frac{2}{3}x^4 - \dots$. By Stein's lemma $c_1(q) = \mathbb{E}[z \tanh(\sqrt{q}z)] = \sqrt{q} a_1(q)$, so $q a_1(q)^2 = c_1(q)^2 = (\mathbb{E}[|z| \tanh(\sqrt{q}|z|)])^2$, increasing in q since $\tanh$ is. Compose with (i). (iii) $\tanh$ is odd so even Hermite coefficients vanish; Parseval gives $\mathbb{E} \tanh^2(\sqrt{q}z) = \sum_{\ell} c_{\ell}^2$, hence $a_*^2 = \sum_{\ell \geq 3} c_{\ell}^2$. From $\tanh x = x - \frac{x^3}{3} + \frac{2x^5}{15} - \frac{17x^7}{315} + \dots$, $x^3 = \text{He}_3 + 3\text{He}_1$, $x^5 = \text{He}_5 + 10\text{He}_3 + 15\text{He}_1$, $x^7 = \text{He}_7 + 21\text{He}_5 + 105\text{He}_3 + 105\text{He}_1$; $c_3(q) = -\frac{2}{\sqrt{6}} q^{3/2} (1 - 4q + 17q^2 + O(q^3))$, so $c_3^2 = \frac{2}{3} q^3 (1 - 8q + 50q^2 + O(q^3))$, while $c_5^2 = \frac{32}{15} q^5 + O(q^6)$; adding it gives the coefficient $266/5$ in a_*^2 . For the third-Hermite term alone, quadrature: $c_3^2 / [\frac{2}{3} q^3 (1 - 8q + 50q^2)] = 0.998, 0.960, 0.73$ at $q = 0.02, 0.05, 0.1$. (iv) $\text{tr } U = n^{-1} \sum_{\mu} \mathbb{E} \|\phi(x_t^{\mu})\|^2 = \mathbb{E} \tanh^2(w \cdot x_t^{\mu} / \sqrt{d_{\text{latent}}})$ with $w \cdot x_t^{\mu} / \sqrt{d_{\text{latent}}} \sim \mathcal{N}(0, q_{\mu})$, $q_{\mu} = \|x_t^{\mu}\|^2 / d_{\text{latent}} = q(1 + O(d_{\text{latent}}^{-1/2}))$; Parseval again. (v) $\theta = \sum_{\ell \geq 3} c_{\ell}^2 (1 - \Delta_t/q)^{\ell} / \sum_{\ell \geq 3} c_{\ell}^2$ is a weighted average of $(1 - \Delta_t/q)^{\ell}$, $\ell \geq 3$ odd, with $0 < 1 - \Delta_t/q < 1$, hence $\leq (1 - \Delta_t/q)^3$; as $q \rightarrow 0$ the weight concentrates on $\ell = 3$ by (iii). $\square$

Remark I.25 (Numerical monotonicity of a_*^2 ; conjectured). On $q \in [0.01, 4]$, $a_*(q)^2$ and $\theta(q)a_*(q)^2$ are strictly increasing in q (quadrature on a 400-point grid; proved only for $q < 0.05$ via Lemma I.24(iii)). Hence θa_*^2 is strictly decreasing in d_{latent} along the sweep: $0.0761 \rightarrow 0.00383$ over $d_{\text{latent}} = 5 \rightarrow 200$ at $\sigma_{\perp} = 0.5$. We use this only as a numerical input.

I.3.2 SCALING OF τ_{gen}

Proposition I.26 (τ_{gen} scales with d_{latent} ; conditional on (H1), (H3), (H4)). With $\tilde{\nu}_{\min} := \tilde{\nu}_{d_{\text{intrinsic}}}$ and $\nu_{\min} := \nu_{d_{\text{intrinsic}}}$,

$$\begin{aligned} \tau_{\text{gen}}(\varepsilon) &= \frac{d_{\text{latent}} \log(1/\varepsilon)}{a_1(q(d_{\text{latent}}))^2 (e^{-2t} \tilde{\nu}_{\min}(d_{\text{latent}}) + \Delta_t)} (1 + o(1)), \\ q(d_{\text{latent}}) &= q_{\infty} + \frac{c_q}{d_{\text{latent}}}, \\ \tilde{\nu}_{\min} &= \nu_{\min} \left(1 + \frac{\psi_{\perp} \sigma_{\perp}^2}{\nu_{\min} - \sigma_{\perp}^2} \right). \end{aligned} \tag{28}$$

The prefactor $d_{\text{latent}}/a_1(q(d_{\text{latent}}))^2$ is strictly increasing in d_{latent} , and for $\psi_{\perp} \sigma_{\perp}^2 \ll \nu_{\min} - \sigma_{\perp}^2$ (i.e. $d_{\text{latent}} \ll n(\nu_{\min} - \sigma_{\perp}^2)/\sigma_{\perp}^2 \approx 3.6n$ at $\sigma_{\perp} = 0.5$)

$$\frac{d_{\text{latent}} \log(1/\varepsilon)}{a_1(q_{\infty})^2 (e^{-2t} \nu_{\min} + \Delta_t)} \lesssim \tau_{\text{gen}}(\varepsilon) \lesssim \frac{d_{\text{latent}} \log(1/\varepsilon)}{a_1(q_{\max})^2 (e^{-2t} \nu_{\min} + \Delta_t)}, \tag{29}$$

with slope ratio $a_1(q_{\infty})^2/a_1(q_{\max})^2$ (3.73 at $\sigma_{\perp} = 0.5$, 5.35 at $\sigma_{\perp} = 0.01$). For $d_{\text{latent}} \lesssim d^*$ the growth is sublinear (a_1^2 still rising); for $d^* \ll d_{\text{latent}} \ll 3.6n$ it is linear with slope $\log(1/\varepsilon)/[a_1(q_{\infty})^2 (e^{-2t} \nu_{\min} + \Delta_t)]$; beyond, the BBP factor (linear in d_{latent}) makes it sublinear again, with saturation at $n(\nu_{\min} - \sigma_{\perp}^2) \log(1/\varepsilon)/[a_1(q_{\infty})^2 e^{-2t} \nu_{\min} \sigma_{\perp}^2]$ as long as the spike stays above threshold ($d_{\text{latent}} \lesssim 13n$).

Proof. By Definition I.22, $\tau_{\text{gen}} = \log(1/\varepsilon)/\lambda_{d_{\text{intrinsic}}}$, and by (H4) $\lambda_{d_{\text{intrinsic}}} = a_1(q)^2 (e^{-2t} \tilde{\nu}_{\min} + \Delta_t)/d_{\text{latent}}$ ($1 + o(1)$). The law of $\nu_{\min}$ is d_{latent} -free (signal coordinates are drawn before the

embedding; $WQ^\top \stackrel{d}{=} W$ makes $\text{spec } U$ invariant under Q); the d_{latent} -dependence of $\tilde{\nu}_{\min}$ is the BBP factor (27). Substituting Lemma I.24(i) gives (28). *Monotonicity of $g(d) := d/a_1(q(d))^2$* : $d = c_q/(q - q_\infty)$, so $g = c_q/h(q)$ with $h(q) := (q - q_\infty)a_1(q)^2 = c_1(q)^2 - q_\infty a_1(q)^2$; c_1^2 is increasing and $-q_\infty a_1^2$ is increasing (Lemma I.24(ii)), so h is increasing in q , g decreasing in q , and since q decreases in d , g increases in d . The sandwich follows from $a_1(q_{\max})^2 \leq a_1(q(d))^2 \leq a_1(q_\infty)^2$ and $\tilde{\nu}_{\min} = \nu_{\min}(1 + o(1))$ in the stated regime; the large- d_{latent} statements substitute $\psi_\perp = (d_{\text{latent}} - d_{\text{intrinsic}})/n$ into (27). Numerically, $\tilde{\nu}_{\min}/\nu_{\min} = 1.01, 1.03, 1.06, 1.11, 1.19, 1.21, 1.23$ at $d_{\text{latent}} = 10, 50, 100, 200, 300, 400, 480$ ($\sigma_\perp = 0.5$), and (28) reproduces the simulated $1/\lambda_{\min}^{\text{sig}}$ to 1–4% (Table 4); on the repo’s $d_{\text{latent}} = 500 \rightarrow 1000$ spectra τ_{gen} grows $1.28\times$ versus $1.62\times$ predicted with $\nu_{\min} = 0.41\alpha_0^2$ and $2\times$ for linear growth. $\square$

I.3.3 SCALING OF THE ONSET

Proposition I.27 (Onset of the sample block; sketched, small- d_{latent} factor g open). *Assume (H1)–(H3), (H5), clusters of sizes n_c , and within-cluster overlaps $\rho_c = e^{-2t}x^\mu \cdot x^\nu/d_{\text{latent}}$ with $\mathbb{E}\rho_c = e^{-2t}s^2/d_{\text{latent}}$ and non-vanishing relative fluctuations $O(\sigma_{\text{sig}}/s)$. Let $\bar{Z} = [\bar{z}_1, \dots, \bar{z}_n] \in \mathbb{R}^{p \times n}$ and $\bar{K} := \mathbb{E}_W[\bar{Z}^\top \bar{Z}]/(p\theta a_*^2)$ (unit diagonal; $\bar{K}_{\mu\nu} = \kappa_{\text{res}}(\rho_{\mu\nu})/\kappa_{\text{res}}(q - \Delta_t)$). To leading order:*

(a) (Mass and mean.) *The sample block is the compression of the n -mode Wishart $U_{\text{res}}^{\text{coh}} = \theta a_*^2(np)^{-1} \bar{Z} \bar{Z}^\top$ onto the orthogonal complement of the d_{latent} data-block eigenvectors; it has $n - d_{\text{latent}}$ modes, mean eigenvalue $\theta(q)a_*(q)^2/n$ and total mass $(1 - d_{\text{latent}}/n)\theta a_*^2$.*

(b) (Edges by interlacing.) *Let $\mu_1 \geq \dots \geq \mu_n$ be the eigenvalues of $U_{\text{res}}^{\text{coh}}$ in units of $\theta a_*^2/n$. Then $\mu_{d_{\text{latent}}+1} \leq \lambda_{\max}^{\text{samp}} n/(\theta a_*^2) \leq \mu_1$. For $\bar{K} = I$, $\mu_1 = (1 + \sqrt{\gamma})^2$ and $\mu_{d_{\text{latent}}+1}$ is the Marchenko–Pastur quantile of order $1 - d_{\text{latent}}/n$; for $\bar{K} = I + \sum_c r_c \mathbf{1}_c \mathbf{1}_c^\top + E$, $r_c = \kappa_{\text{res}}(\rho_c)/\kappa_{\text{res}}(q - \Delta_t)$, a cluster spike detaches when $\ell_c := 1 + n_c r_c > 1 + \sqrt{\gamma}$, at $\mu_1 = \ell_c(1 + \gamma/(\ell_c - 1))$ (Baik et al., 2005).*

(c) (Onset.) *With $g(d_{\text{latent}}) := \lambda_{\max}^{\text{samp}} n/(\theta a_*^2)$,*

$$\lambda_{\max}^{\text{samp}} = \frac{\theta(q(d_{\text{latent}})) a_*(q(d_{\text{latent}}))^2}{n} g(d_{\text{latent}}) (1 + o(1)), \quad (30)$$

$$\mu_{d_{\text{latent}}+1} \leq g(d_{\text{latent}}) \leq \max \left\{ (1 + \sqrt{\gamma})^2, \max_c \ell_c (1 + \frac{\gamma}{\ell_c - 1}) \right\},$$

$$\tau_{\text{mem}}^{\text{on}}(\varepsilon) = \frac{n \log(1/\varepsilon)}{\theta(q(d_{\text{latent}})) a_*(q(d_{\text{latent}}))^2 g(d_{\text{latent}})} (1 + o(1)). \quad (31)$$

θa_*^2 is strictly decreasing in d_{latent} (Remark I.25), so $\tau_{\text{mem}}^{\text{on}}$ grows at least like $1/(\theta a_*^2)$ once g has settled. At $\sigma_\perp = 0.5$, g itself falls from ≈ 5 – 6.5 ($d_{\text{latent}} \leq 20$) to ≈ 1 ($d_{\text{latent}} \geq 200$) and below 1 for $d_{\text{latent}}/n \gtrsim 0.6$ (interlacing), adding ≈ 3 – $4\times$. We have no closed form for g at small d_{latent} : the spike bound in (b) with the full kernel gives 2.5, 2.9, 2.7, 1.7, 1.1, 1.0 at $d_{\text{latent}} = 5, 10, 20, 50, 100, 200$ against observed 5.2, 6.6, 5.1, 3.2, 1.7, 1.25, and the top sample eigenvector has squared overlap only 0.23–0.29 with the k cluster indicators at $d_{\text{latent}} \leq 20$ (null $\approx k/n = 0.02$), 0.17 at $d_{\text{latent}} = 50$, 0.03 at $d_{\text{latent}} = 200$. The remainder is attributed to the finite tensor rank of the cubic residual features (He_3 features span $\leq D_3 = \binom{d_{\text{latent}}+2}{3} = 35, 220, 1540$ dimensions at $d_{\text{latent}} = 5, 10, 20$, versus $n = 500$), which makes $\bar{K}$ rank-deficient for $d_{\text{latent}} \lesssim 13$ and far from I well beyond; an MP description needs at least $n \ll D_3$. In this subsection $g(d_{\text{latent}})$ is an empirically calibrated factor (it is $\kappa_{\text{top}}/\theta$ in the notation of Remark I.16). The cluster-spike ratio $n_c r_c$ is roughly constant for $d_{\text{latent}} \lesssim d^*$ ($\rho_c/q \rightarrow s^2/[d_{\text{intrinsic}}(\alpha_0^2 - \sigma_\perp^2)] = 0.71$ while $q \approx c_q/d_{\text{latent}}$; $n_c r_c = 10.1, 8.4, 5.6, 2.0$ at $d_{\text{latent}} = 5, 10, 20, 50$) and decays like $(d^*/d_{\text{latent}})^3$ only beyond, where it is below threshold.

Proof sketch. Under (H1)–(H2), $U = U_{\text{lin}} + U_{\text{res}}^{\text{coh}} + (\text{floor}) + E$; being functions of the same n points does not imply a rank- n bound for their noise average. The compression below is a surrogate assumption; neither its dimension nor its mass follows from that argument. Let P project onto the top- d_{latent} eigenvectors of U ($\approx$ range of $WM_t^{1/2}$ by (H4)), $P^\perp = I - P$. The eigenvalues of U below the top d_{latent} are exactly those of $P^\perp U P^\perp$ on $\text{ran } P^\perp$, and $P^\perp U_{\text{lin}} P^\perp \approx 0$, the

approximation identifies the sample group with the spectrum of the compression $P^\perp U_{\text{res}}^{\text{coh}} P^\perp$ of an n -mode Wishart by a projector of corank d_{latent} ; Cauchy interlacing gives $\mu_{i+d_{\text{latent}}} \leq \mu_i^{\text{samp}} \leq \mu_i$, whence (b) with $i = 1$; the trace, with the compressed-out directions carrying a d_{latent}/n fraction of the residual mass on average (they are uncorrelated with $\bar{z}$ to leading order), gives (a). Conditionally on the data the rows of $\bar{Z}$ across features are i.i.d. with covariance $\theta a_*^2 \bar{K}$, so $\bar{Z}^\top \bar{Z}/p$ is a sample covariance of p observations with population $\bar{K}$: the MP and BBP formulas follow from the Gaussian replacement under their respective hypotheses (Couillet & Debbah, 2011; Baik et al., 2005). Covariance $\bar{K} = I$ alone does not establish these laws for the nonlinear feature rows. $\bar{K}$ is a kernel matrix whose entries are a Hadamard-power series ($\ell \geq 3$) in the overlaps, related to kernel random matrices (Karoui, 2010), without importing that paper as a theorem for the present finite tensor-rank correction; the cross-cluster entries are $O(d_{\text{latent}}^{-3})$ individually but not negligible collectively at small d_{latent} because of the rank bound $D_3 + D_5 + \dots$. Neglected: the $U_{\text{lin}} - U_{\text{res}}$ interaction (uncorrelated, not independent), non-Gaussianity of the residuals for the mixture, and the fluctuations of ρ_c (cross terms $m_c \cdot \xi^\nu / d_{\text{latent}} \sim \mathcal{N}(0, s^2 \sigma_{\text{sig}}^2 / d_{\text{latent}}^2)$ are a fixed $\approx 34\%$ of $\mathbb{E}\rho_c$, so $\mathbb{E}r_c$ exceeds $r_c(\mathbb{E}\rho_c)$ by up to ≈ 1.7 for the cubic term). Checks (Table 4): per-mode mean $\times n / (\theta a_*^2) = 0.60, 0.67, 0.78, 0.88, 0.97, 1.01, 1.03, 1.05, 1.07$ (sim, $d_{\text{latent}} = 5, \dots, 480$) and $0.92, 0.68, 0.78, 0.86, 0.91, 0.93, 0.95, 0.97, 0.98$ (repo, $d_{\text{latent}} = 5, 10, 20, 40, 60, 80, 100, 150, 200$; 0.80 at $d_{\text{latent}} = 200$ if θ is omitted); in the $p = n + d_{\text{latent}} + 300$ sweep $g/(1 + \sqrt{\gamma})^2 = 6.1, 4.8, 4.1, 2.9, 1.2, 0.83$ at $d_{\text{latent}} = 5, 10, 20, 40, 100, 200$, the value below 1 at $d_{\text{latent}} = 200$ ($d_{\text{latent}}/n = 0.4$, $\gamma = 0.5$) being the interlacing shortfall. $\square$

I.3.4 THE BUFFER COROLLARY

Corollary I.28 (The buffer is the ratio $\lambda_{\text{min}}^{\text{sig}}/\lambda_{\text{max}}^{\text{samp}}$; conditional). *Under the hypotheses of Propositions I.26 and I.27,*

$$R(d_{\text{latent}}) = \frac{\tau_{\text{mem}}^{\text{on}}}{\tau_{\text{gen}}} = \frac{\lambda_{\text{min}}^{\text{sig}}}{\lambda_{\text{max}}^{\text{samp}}} \quad (32)$$

$$= \frac{n}{d_{\text{latent}}} \cdot \frac{a_1(q(d_{\text{latent}}))^2}{\theta(q(d_{\text{latent}})) a_*(q(d_{\text{latent}}))^2} \cdot \frac{e^{-2t} \tilde{\nu}_{\text{min}}(d_{\text{latent}}) + \Delta_t}{g(d_{\text{latent}})} (1 + o(1)),$$

independent of ε and of the learning rate. d_{latent} enters explicitly through the $1/d_{\text{latent}}$ normalisation of the data-side edges (the linear mass $a_1^2 q$ is shared among d_{latent} coordinates by the $d_{\text{latent}}^{-1/2}$ input scaling), through $q(d_{\text{latent}})$, and through g and $\tilde{\nu}_{\text{min}}$; it never multiplies an absorption time. R opens with d_{latent} exactly when $a_1^2(e^{-2t} \tilde{\nu}_{\text{min}} + \Delta_t)/(\theta a_*^2 g)$ grows faster than n/d_{latent} shrinks.

Proof. Divide (31) by (28). With $g \equiv 1$, (32) gives 56, 102, 212, 328, 391, 362, 287, 288, 284 at $d_{\text{latent}} = 5, \dots, 480$ against simulated 11, 15, 41, 101, 231, 290, 262, 298, 312: accurate to $\leq 15\%$ for $d_{\text{latent}} \geq 200$, over-predicting by the factor g below. $\square$

Proposition I.29 (Two phases of the buffer; conjectural). *Let $d^* = c_q/q_\infty$ ($= 47$ at $\sigma_\perp = 0.5$), suppose $(1 + \sqrt{\gamma})^2$ varies by at most a constant along the sweep, $d_{\text{latent}} < n$, and $\sigma_\perp = 0.5$. (i) Opening phase $d_{\text{latent}} \lesssim 4d^*$: $a_1^2/(\theta a_*^2)$ grows from 2.4 to 163 over $d_{\text{latent}} = 5 \rightarrow 200$, outpacing the $40\times$ fall of n/d_{latent} and the $3.5\times$ fall of $(1 + \sqrt{\gamma})^2$, and g collapses from ≈ 5 – 6.5 to ≈ 1 ; R increases. (ii) Saturated phase $d_{\text{latent}} \gg d^*$: $a_1, a_*, \theta \rightarrow$ their q_∞ values and*

$$R(d_{\text{latent}}) \simeq \frac{n}{d_{\text{latent}}} \cdot \frac{a_1(q_\infty)^2 (e^{-2t} \tilde{\nu}_{\text{min}}(d_{\text{latent}}) + \Delta_t)}{\theta(q_\infty) a_*(q_\infty)^2 g(d_{\text{latent}})} \quad (33)$$

is predicted to decrease, roughly like n/d_{latent} tempered by the BBP growth of $\tilde{\nu}_{\text{min}}$ and the interlacing decrease of g . Hence R is conjectured non-monotone with a broad maximum at $d_{\text{latent}} = \Theta(d^*) - \Theta(n)$. This is not demonstrated: simulated $R = 290, 262, 298, 312$ at $d_{\text{latent}} = 200, 300, 400, 480$ is flat, and the repo's spectra with a non-empty sample block stop at $d_{\text{latent}} = 200$ with R still rising (341, 419, 428 at $d_{\text{latent}} = 100, 150, 200$). What is observed to close is R_{full} : as $d_{\text{latent}} \rightarrow n$ the noise-dimension lower edge $a_1^2[e^{-2t} \sigma_\perp^2 (1 - \sqrt{\psi_\perp})^2 + \Delta_t]/d_{\text{latent}} \rightarrow a_1^2 \Delta_t/d_{\text{latent}}$ approaches $\lambda_{\text{max}}^{\text{samp}}$ ($R_{\text{full}} = 3.1, 6.3, 12.5, 17.2, 11.3, 6.1, 3.7, 2.4$ at $d_{\text{latent}} = 10, \dots, 480$, sim; $4.4 \rightarrow 18.8 \rightarrow 12.5$ over $d_{\text{latent}} = 10 \rightarrow 80 \rightarrow 200$, repo) while $|I_{\text{samp}}| = n - d_{\text{latent}} \rightarrow 0$. The empirical return of memorization at large d_{latent} (an MLP sweep, Appendix I.8) is at most a candidate consequence of this; it is not explained by R , which stays ≈ 300 .

Proof sketch. (ii) is substitution of $q = q_\infty + O(1/d_{\text{latent}})$ into (32); with $g \equiv 1$ it gives 391, 362, 287, 288, 284 at $d_{\text{latent}} = 100, \dots, 480$. (i): numbers from Lemma I.24 and Table 4. The closing statement uses the MP lower edge of (H4) (verified to 3–10% for $d_{\text{latent}} \geq 20$) and the mode count of Lemma I.15. $\square$

Corollary I.30 (Fixed-energy sweep; sketched). *For the headline control ($p = 1800$ fixed, $\sigma_\perp^2 = E_\perp/(d_{\text{latent}} - d_{\text{intrinsic}})$, $E_\perp = 3.75$), $q(d_{\text{latent}}) = \Delta_t + e^{-2t}(d_{\text{intrinsic}}\alpha_0^2 + E_\perp)/d_{\text{latent}}$: $q_\infty = \Delta_t = 0.0198$, $c_q = 17.4$, $d^* = 878 > n$. The whole feasible sweep is in the opening phase, θ falls to 0.54 at $d_{\text{latent}} = 200$, block separation holds (the noise-dimension block sits at $a_1^2[e^{-2t}E_\perp(1 \pm \sqrt{\psi_\perp})^2/(d_{\text{latent}} - d_{\text{intrinsic}}) + \Delta_t]/d_{\text{latent}} \gg \theta a_*^2 g/n$), and (32) predicts a monotonically increasing R over $d_{\text{latent}} \leq 200$, consistent with the monotone growth reported for that sweep ($R = 16.7 \rightarrow 889$ over $d_{\text{latent}} = 5 \rightarrow 200$, recomputed from the repository’s spectra in clean figures/p.fixed.energy.fixed, $p = 1800$, $n = 500$).*

Table 4: Own simulation of (2) via its Gram ($t = 0.01$, $\sigma_\perp = 0.5$, $n = 500$, $k = 10$, $s = 3$, $p = 64d_{\text{latent}}$, $N_{\text{mc}} = 4$, seed 0; theory normalisation, arbitrary time units). “pred.” is (28) with $\tilde{\nu}_{\min}$ from the same data; $1/\lambda_+^{\text{MP}}$ uses $\theta a_*^2(1 + \sqrt{\gamma})^2/n$; g here is $\lambda_{\text{max}}^{\text{samp}} n/(\theta a_*^2)/(1 + \sqrt{\gamma})^2$; “ov.” is the squared overlap of the top sample eigenvector with the cluster indicators; $R_{g=1}$ is (32) with $g = 1$. Repo values ($N_{\text{mc}} = 50$, $d_{\text{latent}} = 5, 10, 20, 40, 60, 80, 100, 150, 200$): $1/\lambda_{\min}^{\text{sig}} = 16, 20, 31, 48, 63, 81, 95, 137, 173$; $1/\lambda_{\text{max}}^{\text{samp}} = 333, 684, 1927, 6866, 13457, 22778, 32303, 57587, 73819$; $g = 3.9, 4.9, 4.5, 3.1, 2.6, 2.0, 1.7, 1.3, 1.23$; $R = 21, 34, 63, 144, 212, 280, 341, 419, 428$; $R_{\text{full}} = -, 4.4, 7.8, 13.0, 15.6, 18.8, 18.1, 18.0, 12.5$.

d_{latent}	q	$\theta a_*^2 (\times 10^3)$	$1/\lambda_{\min}^{\text{sig}}$	pred.	$1/\lambda_{\text{max}}^{\text{samp}}$	$1/\lambda_+^{\text{MP}}$	g	ov.	mean- $n/\theta a_*^2$	R	$R_{g=1}$	R_{full}
10	1.515	42.5	33.0	32.4	506	3315	6.55	0.29	0.67	15	102	3.1
20	0.890	21.7	41.1	41.1	1700	8708	5.12	0.29	0.78	41	212	6.3
50	0.515	9.14	86.7	85.6	8743	28096	3.21	0.17	0.88	101	328	12.5
100	0.390	5.45	145.6	143.2	33629	56016	1.67	0.16	0.97	231	391	17.2
200	0.327	3.83	250.7	251.5	72657	90985	1.25	0.03	1.01	290	362	11.3
300	0.307	3.33	384.4	387.3	100699	111143	1.10	0.01	1.03	262	287	6.1
400	0.296	3.10	431.1	431.7	128379	124287	0.97	0.02	1.05	298	288	3.7
480	0.291	2.98	464.2	465.3	145021	131962	0.91	0.01	1.07	312	284	2.4

I.3.5 REMARKS

Remark I.31 (Why the count-times-timescale bound is a non-sequitur). An earlier draft asserted $\tau_{\text{mem}} - \tau_{\text{gen}} \geq (d_{\text{latent}} - d_{\text{intrinsic}})/\lambda_{\min}^{\text{noise}}$, as if the readout had to traverse each noise-dimension mode in turn. Under $\dot{a}_i = -\lambda_i(a_i - a_i^*)$ all modes decay simultaneously; the time at which sample mode j reaches tolerance ε is $\lambda_j^{-1} \log(1/\varepsilon)$ whatever the number or rates of the other modes. Adding $d_{\text{latent}} - d_{\text{intrinsic}}$ intermediate modes changes $\tau_{\text{mem}}^{\text{on}}$ only insofar as it changes $\lambda_{\text{max}}^{\text{samp}}$ (through $q(d_{\text{latent}})$, θ , g) and τ_{gen} through the $1/d_{\text{latent}}$ normalisation and $\tilde{\nu}_{\min}$. A mode count can never multiply a timescale in a diagonal linear system, and the asserted bound is violated at every $d_{\text{latent}} \geq 20$ on the repo’s own spectra. The correct statement is the ratio (32).

Remark I.32 (Onset versus median). The sample block has $n - d_{\text{latent}}$ modes of mean $\theta a_*^2/n$. At small d_{latent} the residual Gram is rank-deficient ($D_3 < n$ for $d_{\text{latent}} \leq 13$) and spiked, so most of the mass sits in a few top modes and a heavy lower tail: median $\times n/(\theta a_*^2) = 0.037, 0.072, 0.28, 0.57, 0.72, 0.80, 0.84, 0.92, 0.95$ at $d_{\text{latent}} = 5, \dots, 200$ on the repo ($\lambda_{\text{max}}^{\text{samp}}/\text{median} \approx 540$ at $d_{\text{latent}} = 5$, ≈ 2 at $d_{\text{latent}} = 200$). As d_{latent} grows the block becomes a proper Marchenko–Pastur bulk and the median rises toward the mean, while $\lambda_{\min}^{\text{sig}} \propto a_1^2 \tilde{\nu}_{\min}/d_{\text{latent}}$ falls. Therefore $\lambda_{\min}^{\text{sig}}/\text{median}(\lambda_{\text{max}}^{\text{samp}})$ shrinks with d_{latent} (repo: $11350 \rightarrow 798$, $14\times$; sim: $5750 \rightarrow 540$ over $d_{\text{latent}} = 10 \rightarrow 200$) while R grows (repo: $21 \rightarrow 428$). The buffer is a claim about the *onset* of sample-specific learning; the time to absorb the bulk of the sample modes, which governs the memorization fraction, is a different quantity (Appendices I.5 and I.6).

Remark I.33 (The isotropic control). Set $\sigma_\perp = \sigma_{\text{sig}}$. *Structure:* the $d_{\text{latent}} - d_{\text{intrinsic}}$ noise-dimension modes still exist, at $a_1^2[e^{-2t}\sigma_{\text{sig}}^2(1 \pm \sqrt{\psi_\perp})^2 + \Delta_t]/d_{\text{latent}}$ (verified to 3%), between the $d_{\text{intrinsic}}$ signal spikes and the sample block; what isotropy removes is the factor $\sigma_{\text{sig}}^2/\sigma_\perp^2$ separating the signal and noise-dimension blocks, so the data side becomes a single Marchenko–Pastur bulk with $d_{\text{intrinsic}}$ BBP spikes, the ρ_2 of Bonnaire et al. (2025) with outliers: the *four-bulk structure* is

absent, not the noise-dimension modes (Remark I.19). *Mechanism:* de-saturation is essentially absent ($c_q = 8.8$, $q_\infty = 1$, $d^* = 8.8$; $a_1^2/(\theta a_*^2)$ rises only $2.4 \rightarrow 13$ over $d_{\text{latent}} = 5 \rightarrow 200$), and (32) with $g \equiv 1$ gives $R = 56, 80, 95, 89, 74, 58, 50$ at $d_{\text{latent}} = 5, 10, 20, 50, 100, 200, 300$, a decline beyond $d_{\text{latent}} \approx 20$ driven by n/d_{latent} and only partly offset by the strong BBP lift of the signal spikes ($\tilde{\nu}_{\min}/\nu_{\min} = 1.15, 1.12, 1.52, 1.77, 2.20, 2.93$ at $d_{\text{latent}} = 10, \dots, 300$). The simulated R is $11, 21, 49, 83, 71, 60, 55$: it rises $7\times$ to $d_{\text{latent}} \approx 50$ because g collapses ($5.2, 4.0, 1.9, 1.1, 1.0, 0.9$), then follows the predicted decline. Under isotropy R is therefore predicted non-monotone with a peak near $d_{\text{latent}} \approx 50$; the earlier claim that de-saturation still makes R grow is withdrawn. A direct simulation of (2) at $\sigma_\perp = \sigma_{\text{sig}} = 1$ ($n = 500$, $p = 4096$, $t = 0.01$, three seeds; `next_steps/scripts/e5_isotropic.py`) gives $\lambda_{\min}^S/\lambda_{\max}^M = 14, 30, 58, 87, 77, 60$ at $d_{\text{latent}} = 5, 10, 20, 40, 80, 160$: a rise to a peak near $d_{\text{latent}} \approx 40$ followed by the predicted decline. (An earlier “ $10.7\times$ growth of the gap” was a difference of code-unit eigenvalues, which carry the factor $p = 64d_{\text{latent}}$; in theory units the same four values are $1.50, 2.22, 2.45, 1.61$ at $d_{\text{latent}} = 10, 20, 40, 100$, rising then falling.) With no low-dimensional structure at all ($x \sim \mathcal{N}(0, I_{d_{\text{latent}}})$, same settings) the data-block/sample-block ratio falls monotonically from 77 to 6 over $d_{\text{latent}} = 5 \rightarrow 160$.

Remark I.34 (Norm-fixed control; simulated). Holding the per-coordinate input variance $q_{\text{in}} = \mathbb{E}\|x\|^2/d_{\text{latent}}$ fixed as d_{latent} grows separates the mode count from de-saturation. With $\sigma_\perp^2(d_{\text{latent}}) = (0.57d_{\text{latent}} - 14)/(d_{\text{latent}} - d_{\text{intrinsic}})$, which pins $\text{tr } U$ at 0.29 (baseline: $0.29 \rightarrow 0.20$) and coincides with the $\sigma_\perp = 0.5$ sweep at $d_{\text{latent}} = 40$, the same RFNN ($n = 500$, $p = 4096$, $t = 0.01$, three seeds; `next_steps/scripts/e4_norm_fixed.py`) gives $R = \lambda_{\min}^S/\lambda_{\max}^M = 92, 140, 158, 136, 119, 100$ at $d_{\text{latent}} = 40, 60, 80, 120, 160, 240$, against 92, 145, 201, 266, 287, 291 for the baseline: a $1.09\times$ change versus $3.17\times$, while the noise-dimension count grows $6.7\times$. The decomposition is the one (32) predicts: $\lambda_{\min}^S$ falls $4.3\text{--}4.4\times$ in both sweeps (the $1/d_{\text{latent}}$ of the signal edge at fixed a_1), whereas the sample top falls $13.6\times$ in the baseline and only $4.8\times$ at fixed q ; the missing $2.8\times$ is what (32) attributes to the shrinkage of $a_*(q)^2$ as q falls from 0.56 to 0.30. The mode count enters nowhere.

Remark I.35 (Relation to prior work). The spectral studies of Bonnaire et al. (2025); George et al. (2026) motivate (H4), but do not establish its four index groups for our mixture; the covariance and operator distinctions are in Appendix I.1.1. At fixed d_{latent} , (31) gives $\tau_{\text{mem}}^{\text{on}} \propto n/(\theta a_*^2 g)$, consistent with the linear-in- n memorization time of Bonnaire et al. (2025); the d_{latent} -dependence through $q(d_{\text{latent}})$, θ , $\tilde{\nu}_{\min}$ and g , the ratio form (32), and the training-time reading for $\sigma_\perp > 0$ are the new content.

Remark I.36 (What this subsection does not predict). R concerns two spectral edges under gradient flow at fixed t . It does not by itself predict (a) the memorization fraction at a fixed step budget T , which depends on how many sample modes have $\lambda_i^{-1} \log(1/\varepsilon) < T$, i.e. on the within-block spread of the sample spectrum, on the map from gradient-flow time to optimizer steps (Remark I.7), and on the drift of the NN-ratio statistic with ambient dimension; (b) whether generation quality is matched across d_{latent} at the comparison time, since τ_{gen} grows $\approx 11\times$ over $d_{\text{latent}} = 5 \rightarrow 200$ and concerns the signal subspace only; (c) the MLP and real-VAE settings, whose spectra are continuous rather than two-atom and where a_1, a_*, θ must be replaced by learned-feature analogues (Appendix I.7); (d) anything at $\sigma_\perp = 0.01$ beyond τ_{gen} , since block separation fails there (H5).

Predictions and how to test them.

1. $\tau_{\text{gen}} a_1(q(d_{\text{latent}}))^2(e^{-2t}\tilde{\nu}_{\min}(d_{\text{latent}}) + \Delta_t)$ is linear in d_{latent} through the origin with slope $\log(1/\varepsilon)$, independent of d_{latent} , $\sigma_\perp$ and p . Test: from the saved spectra extract $\lambda_{d_{\text{intrinsic}}}$, rebuild the data with the saved seed to get $\tilde{\nu}_{\min}$, and plot $d_{\text{latent}}/(\lambda_{d_{\text{intrinsic}}} a_1(q)^2(e^{-2t}\tilde{\nu}_{\min} + \Delta_t))$ against d_{latent} ; both width rules and both $\sigma_\perp$ should collapse onto one line. Without the BBP shift the collapse fails at the 10–25% level for $d_{\text{latent}} \geq 200$.
2. τ_{gen} grows linearly only for $d^* \ll d_{\text{latent}} \ll 3.6n$; beyond, the BBP lift makes it sublinear. Test: $\tau_{\text{gen}}(1000)/\tau_{\text{gen}}(500)$ on the existing spectra should be ≈ 1.6 , not 2 (observed 1.28 at $p = 64d_{\text{latent}}$, 1.59 at $p = n + d_{\text{latent}} + 300$; the residual 20% is unexplained).
3. The sample-block per-mode mean equals $\theta(q)a_*(q)^2/n$ with $\theta \approx (1 - \Delta_t/q)^3$. Test: $\text{mean}(\lambda_{d_{\text{latent}}+1..n}n/a_*(q)^2)$ should equal $\theta(q) = 0.97, 0.95, 0.93, 0.88, 0.85, 0.82$ at $d_{\text{latent}} = 5, 10, 20, 50, 100, 200$; at $\sigma_\perp = 0.01$ the factor is 0.47 at $d_{\text{latent}} = 200$.

4. $\lambda_{\max}^{\text{samp}}$ lies between $\mu_{d_{\text{latent}}+1}$ and μ_1 of an n -mode Wishart of scale $\theta a_*^2/n$; for $d_{\text{latent}} \gtrsim 150$ it is within $\sim 25\%$ of the MP edge and falls below it when d_{latent}/n and n/p are both $O(1)$; at small d_{latent} it exceeds the edge by $4\text{--}6.5\times$, of which only $\sim 25\%$ is cluster-coherent. Test: rebuild $\mathbf{U}$ with eigenvectors, compute the squared overlap of the $(d_{\text{latent}}+1)$ -th eigenvector with the k cluster indicators.
5. $\tau_{\text{mem}}^{\text{on}}$ grows by more than two orders of magnitude over $d_{\text{latent}} = 5..200$: $\sim 20\times$ from θa_*^2 , $\sim 3.5\times$ from $(1 + \sqrt{\gamma})^2$ at $p = 64d_{\text{latent}}$, $\sim 3\text{--}4\times$ from the collapse of g . Test: tabulate $1/\lambda_{d_{\text{latent}}+1}$ against $n/(\theta a_*^2(1 + \sqrt{n/p})^2)$ and read off g .
6. R is conjectured non-monotone at $\sigma_{\perp} = 0.5$ with a broad maximum between $d^* \sim 50$ and $\sim n$. Test: extend the sweep to $d_{\text{latent}} \in \{300, 400, 480\}$ with $N_{\text{mc}} = 50$ and several seeds, and to $n \in \{1000, 2000\}$; the peak should scale with n .
7. R_{full} is non-monotone with a peak near $d_{\text{latent}} \sim 60\text{--}100$ and closes to $O(1)$ as $d_{\text{latent}} \rightarrow n$; this, with the vanishing count $n - d_{\text{latent}}$, is the candidate mechanism for the return of memorization. Test: $\lambda_{d_{\text{latent}}}/\lambda_{d_{\text{latent}}+1}$ on new $d_{\text{latent}} \in \{300, 400, 480\}$ runs; then test whether the RFNN memorization fraction returns where $R_{\text{full}} \lesssim 3$.
8. R is independent of ε and of the learning rate. Test: re-extract at two plateau thresholds and two learning rates; the ratio is unchanged while the individual times scale.
9. The median-based ratio decreases with d_{latent} (opposite to the onset ratio). Verified on the repo: $\text{median} \cdot n/(\theta a_*^2) = 0.037 \rightarrow 0.95$ and R_{median} shrinks $14\times$.
10. Isotropic control: R rises $\sim 6\times$ to a peak near $d_{\text{latent}} \approx 40$ and then declines like n/d_{latent} ; the noise-dimension modes exist but are not separated from the signal. Status: the rise-then-decline is confirmed by the three-seed simulation in Remark I.33 ($14 \rightarrow 87 \rightarrow 60$). Remaining test: trained runs at $\sigma_{\perp} = 1$, $d_{\text{latent}} \in \{5, \dots, 300\}$; check $\tilde{\nu}_{\min}/\nu_{\min} = 1.5, 1.8, 2.2$ at $d_{\text{latent}} = 50, 100, 200$.
11. Changing p at fixed d_{latent} leaves τ_{gen} unchanged (to $\sim 2\%$) and changes $\tau_{\text{mem}}^{\text{on}}$ by at most the MP-edge factor $[(1 + \sqrt{n/p_2})/(1 + \sqrt{n/p_1})]^2$. Repo: ratio $0.95, 1.01, 1.09, 1.26, 1.39, 1.31$ at $d_{\text{latent}} = 5, \dots, 200$ vs MP factor $0.63, \dots, 2.04$; τ_{gen} differs by $1\text{--}7\%$.
12. At $\sigma_{\perp} = 0.01$ block separation fails, so $\tau_{\text{mem}}^{\text{on}}$ and R are undefined while τ_{gen} is still predicted. Repo: $\lambda_{d_{\text{latent}}}/\lambda_{d_{\text{latent}}+1} = 1.0\text{--}1.15$; $1/\lambda_5$ grows $2.4\times$ over $d_{\text{latent}} = 5..40$ vs predicted $2.4\times$.

Open gaps. (1) $g(d_{\text{latent}})$ at small d_{latent} has no derivation: the cluster-spike model with the full residual kernel accounts for $\sim$ half ($2.5\text{--}2.9\times$) and the top eigenvector is only $\sim 25\%$ cluster-coherent; the remainder is attributed to the finite tensor rank D_3 of the cubic residual features without a controlled derivation, so Proposition I.27 is sketched with the small- d_{latent} formula explicitly open. (2) The turnover of R (Proposition I.29) is conjectural: the $N_{\text{mc}} = 4$ simulation is flat over $d_{\text{latent}} = 200..480$ and the repo has no spectra with a non-empty sample block beyond $d_{\text{latent}} = 200$; only R_{full} is observed to close, and the link to the return of memorization is untested in the RFNN. (4) The machine-precision cliff at index n does not hold for the finite- N_{mc} $\mathbf{U}$ ($\lambda_{n+1}/\lambda_n = 0.48\text{--}0.76$ for $d_{\text{latent}} \geq 40$); why the Monte-Carlo excess modes sit only $\sim 2\times$ below the sample-block bottom rather than $\sim 100\times$ is not understood. (5) The rise-then-decline of R under isotropy is confirmed by simulation (Remark I.33); no trained $\sigma_{\perp} = 1$ run exists in the repository. (6) At $\sigma_{\perp} = 0.01$ block separation fails outright; the fixed-energy control's R numbers are recomputed ($16.7 \rightarrow 889$) but its mechanism is not analysed here. (7) The p -dependence of the onset is a one-sided interlacing bound, partially falsified as an equality (1.39 vs 1.86 at $d_{\text{latent}} = 100$). (8) All $o(1)$ statements are proportional-asymptotic with $O(d_{\text{latent}}^{-1/2})$ corrections; nothing is uniform in the sweep. (9) Monotonicity of a_*^2 and θa_*^2 in q is numerical. (10) The BBP shift (H4) is verified to $1\text{--}4\%$ but is an input from spiked-covariance theory, not derived within the RFNN pencil; in the isotropic case the closed-form factor diverges near threshold and the empirical top- $d_{\text{intrinsic}}$ eigenvalues of $\hat{C} + \hat{S}$ must be used directly. (11) Main-text statements asserting d_{latent} -independent τ_{gen} , sequential traversal of the four bulks, “no buffer in the isotropic case”, or sample indices $[d_{\text{latent}}+1, d_{\text{latent}}+n]$ are inconsistent with this subsection and must be revised.

I.4 THE EXACTLY SOLVABLE LINEAR SCORE MODEL

This subsection isolates the part of the problem that is exactly solvable: a linear score model on the block-anisotropic mixture. Throughout, $t > 0$ is fixed and expectations are over the noised population distribution p_t unless a subscript n indicates the empirical one. Two values of t occur: the MLP is evaluated at $t = 0.1$, while the RFNN is trained *and* evaluated at $t_{\text{fixed}} = 0.01$. Statements are basis-free; proofs are written in the rotated frame $Q^\top x$ where the signal subspace is coordinate-aligned. *Convention.* The risks here carry no factor $\frac{1}{2}$, so the flow is $\dot{B} = -2(BM_t + I)$ and parameters decay as $e^{-2\lambda T}$; in the convention of Lemma I.3 (risk with $\frac{1}{2}$) the same trajectory is $e^{-\lambda T'}$ with $T' = 2T$. Ratios of clocks are convention-free.

Definition I.37 (Setting for the linear score model). Fix $t > 0$ and write $x_t = (x_{t,S}, x_{t,N})$ in the rotated frame $Q^\top x_t$, with $x_{t,S} \in \mathbb{R}^{d_{\text{intrinsic}}}$ (signal block) and $x_{t,N} \in \mathbb{R}^{d_{\text{latent}} - d_{\text{intrinsic}}}$ (null block). Under the data model $x = m_c + \xi$, $c \sim \text{Unif}[k]$, $m_c \in \text{span}(S)$, $\xi \sim \mathcal{N}(0, \text{diag}(\sigma_{\text{sig}}^2 I_{d_{\text{intrinsic}}}, \sigma_{\perp}^2 I_{d_{\text{latent}} - d_{\text{intrinsic}}}))$, and $x_t = e^{-t}x + \sqrt{\Delta_t}\eta$, the noised marginal is the Gaussian mixture

$$p_t = \frac{1}{k} \sum_c \mathcal{N}(e^{-t}m_c, \Sigma_t), \quad \Sigma_t = \text{diag}(\gamma_t^2 I_{d_{\text{intrinsic}}}, \beta_t^2 I_{d_{\text{latent}} - d_{\text{intrinsic}}}), \quad (34)$$

with $\gamma_t^2 = e^{-2t}\sigma_{\text{sig}}^2 + \Delta_t$ and $\beta_t^2 = e^{-2t}\sigma_{\perp}^2 + \Delta_t$. Its *uncentered* second moment is $M_t = \mathbb{E}[x_t x_t^\top] = \text{diag}(M_{t,S}, \beta_t^2 I_{d_{\text{latent}} - d_{\text{intrinsic}}})$, $M_{t,S} = e^{-2t}(C + \sigma_{\text{sig}}^2 I) + \Delta_t I$, $C = \frac{1}{k} \sum_c m_c m_c^\top$, with eigenvalues $\lambda_1^S \geq \dots \geq \lambda_{d_{\text{intrinsic}}}^S \geq \gamma_t^2$; they all equal $\alpha_t^2 = e^{-2t}(s^2/d_{\text{intrinsic}} + \sigma_{\text{sig}}^2) + \Delta_t$ only for isotropic centers $C = (s^2/d_{\text{intrinsic}})I$, and in general α_t^2 is their trace average (exact because $\text{tr } C = s^2$). M_t is the second moment, not the covariance, of x_t ; the two differ on the signal block by $e^{-2t}\bar{m}\bar{m}^\top$, $\bar{m} = \frac{1}{k} \sum_c m_c$ ($\|\bar{m}\| = 1.15$ for the repo's centers), and it is M_t that enters the dynamics. Given a training set $\{x^\mu\}_{\mu \leq n}$, the *empirical* second moment is $\widehat{M}_t = e^{-2t} \frac{1}{n} \sum_\mu x^\mu x^{\mu\top} + \Delta_t I$. The *linear score model* is $s_B(x) = Bx$, $B \in \mathbb{R}^{d_{\text{latent}} \times d_{\text{latent}}}$, trained by gradient flow $\dot{B} = -\nabla_B L$ from $B(0) = B_0$ on either the population score-matching risk $L_t(B) = \mathbb{E}_{p_t} \|Bx_t - s^*(x_t)\|^2$, $s^* = \nabla \log p_t$, or the empirical denoising risk $L_{t,n}(B) = \frac{1}{n} \sum_\mu \mathbb{E}_\eta \|Bx_t^\mu + \eta/\sqrt{\Delta_t}\|^2$. The population denoising objective equals L_t up to a B -independent constant (Vincent, 2011), so it generates the same flow; $L_{t,n}$ with $\mathbb{E}_\eta$ is the mean path of full-batch SGD with a fresh noise draw per step (Proposition I.2), which is what the code runs (momentum 0).

Table 5: Constants of the linear model at the two evaluation times used by the code ($\sigma_{\text{sig}} = 1$, $s = 3$, $d_{\text{intrinsic}} = 5$, $k = 10$, centers of seed 42: $\text{eig } C = \{0.907, 1.033, 1.203, 2.491, 3.367\}$). $\mathcal{J}_{t,S}$ and R_{irr} are 4×10^5 -sample Monte-Carlo values.

	MLP, $t = 0.1$	RFNN, $t = 0.01$
Δ_t, e^{-2t}	0.1813, 0.8187	0.0198, 0.9802
γ_t^2	1.000	1.000
λ_1^S	1.742, 1.846, 1.985, 3.039, 3.756	1.889, 2.013, 2.179, 3.441, 4.300
α_t^2 (trace average)	2.474	2.764
$\text{tr } M_{t,S}^{-1}, \mathcal{J}_{t,S}, R_{\text{irr}}$	2.215, 3.582, 1.367	2.008, 3.686, 1.678
$\beta_t^2, 1/\beta_t^2, 1/(4\beta_t^2)$ at $\sigma_{\perp} = 0.5$	0.386, 2.59, 0.648	0.2649, 3.78, 0.944
$\beta_t^2, 1/\beta_t^2, 1/(4\beta_t^2)$ at $\sigma_{\perp} = 0.01$	0.1814, 5.51, 1.379	0.0199, 50.3, 12.56

Lemma I.38 (The null block of the score is exactly linear; the best linear score is $-M_t^{-1}$; proved). *Under Definition I.37:*

(a) (exact null block) For every x_t and every $t > 0$,

$$s_N^*(x_t) = -x_{t,N}/\beta_t^2, \quad (35)$$

while $s_S^*(x_t) = -\gamma_t^{-2}(x_{t,S} - e^{-t} \sum_c w_c(x_{t,S}) m_c)$ with softmax weights $w_c \propto \exp(-\|x_{t,S} - e^{-t} m_c\|^2/2\gamma_t^2)$; this is affine if all centers coincide and linear only if all $m_c = 0$. The weights depend on x_t only through $x_{t,S}$.

(b) (Stein identity) $\mathbb{E}_{p_t}[s^*(x_t) x_t^\top] = -I_{d_{\text{latent}}}$.

(c) (best linear score) L_t is a strictly convex quadratic with unique minimizer $B_t^* = -M_t^{-1}$, which is block diagonal; on the null block $B_{t,N}^* = -\beta_t^{-2}I$ reproduces (35) exactly. The irreducible risk is supported on the signal block and equals

$$R_{\text{irr}}(t) := \min_B L_t(B) = \mathbb{E}\|s_\mathbb{S}^*\|^2 - \text{tr } M_{t,\mathbb{S}}^{-1} \in \left[0, \frac{d_{\text{intrinsic}}}{\gamma_t^2} - \sum_{i=1}^{d_{\text{intrinsic}}} \frac{1}{\lambda_i^5}\right], \quad (36)$$

with $R_{\text{irr}} = 0$ iff $p_t = \mathcal{N}(0, M_t)$ (a centered Gaussian). For the repo's centers (Table 5) a Monte-Carlo evaluation gives $R_{\text{irr}} = 1.37$ at $t = 0.1$ and 1.68 at $t = 0.01$, independent of d_{latent} .

Proof. (a) Because $m_{c,N} = 0$ and Σ_t is block diagonal, each mixture component factorizes, $\mathcal{N}(x; e^{-t}m_c, \Sigma_t) = \mathcal{N}(x_\mathbb{S}; e^{-t}m_c, \gamma_t^2 I) \mathcal{N}(x_\mathbb{N}; 0, \beta_t^2 I)$, and the second factor is common to all c . Hence $p_t(x) = p_{t,\mathbb{S}}(x_\mathbb{S}) \mathcal{N}(x_\mathbb{N}; 0, \beta_t^2 I)$ and $\nabla_\mathbb{N} \log p_t = -x_\mathbb{N}/\beta_t^2$. The signal-block formula is the gradient of log of the $d_{\text{intrinsic}}$ -dimensional mixture $p_{t,\mathbb{S}}$; with a single center m it is $-\gamma_t^{-2}(x_\mathbb{S} - e^{-t}m)$, affine with nonzero intercept unless $m = 0$. (b) $\mathbb{E}[s^* x^\top] = \int \nabla p_t(x) x^\top dx = -\int p_t(x) \nabla(x^\top) dx = -I$, integrating by parts coordinate-wise; the boundary term $p_t(x)x^\top$ vanishes at infinity because p_t has Gaussian tails. (c) Expanding and using (b),

$$L_t(B) = \text{tr}(BM_t B^\top) - 2 \text{tr}(B \mathbb{E}[x_t s^{*\top}]) + \mathbb{E}\|s^*\|^2 = \text{tr}(BM_t B^\top) + 2 \text{tr } B + \mathcal{J}_t,$$

where $\mathcal{J}_t := \mathbb{E}\|s^*\|^2 = \text{tr } J(p_t)$ is the Fisher information of p_t . Since $M_t \succeq \Delta_t I \succ 0$, L_t is strictly convex with $\nabla_B L_t = 2BM_t + 2I$, so $B_t^* = -M_t^{-1}$ and $L_t(B_t^*) = \mathcal{J}_t - \text{tr } M_t^{-1}$. Independence of $x_{t,\mathbb{S}}$ and $x_{t,\mathbb{N}}$ together with $\mathbb{E}x_{t,\mathbb{N}} = 0$ makes M_t block diagonal, hence so is B_t^* ; on the null block $B_{t,N}^* x_{t,N} = -x_{t,N}/\beta_t^2 = s_N^*$ by (a), so the null block contributes 0 and $\mathcal{J}_t - \text{tr } M_t^{-1} = \mathbb{E}\|s_\mathbb{S}^*\|^2 - \text{tr } M_{t,\mathbb{S}}^{-1}$. Nonnegativity is $L_t \geq 0$, with equality iff $s^*(x) = -M_t^{-1}x$ a.e. iff $p_t = \mathcal{N}(0, M_t)$. For the upper bound, write $p_{t,\mathbb{S}} = \sum_c w_c q_c$ with $q_c = \mathcal{N}(e^{-t}m_c, \gamma_t^2 I)$; by joint convexity of $(a, b) \mapsto \|a\|^2/b$ on $\mathbb{R}^{d_{\text{intrinsic}}} \times \mathbb{R}_{>0}$, $\|\nabla p_{t,\mathbb{S}}\|^2/p_{t,\mathbb{S}} \leq \sum_c w_c \|\nabla q_c\|^2/q_c$; integrating, $\mathbb{E}\|s_\mathbb{S}^*\|^2 \leq \sum_c w_c \text{tr } J(q_c) = d_{\text{intrinsic}}/\gamma_t^2$. The Monte-Carlo values use 4×10^5 samples of $p_{t,\mathbb{S}}$ with the seed-42 centers and the exact softmax score. $\square$

Proposition I.39 (Closed-form gradient flow and per-mode risk; proved). *Let $M_t = \sum_{i=1}^{d_{\text{latent}}} \lambda_i v_i v_i^\top$ be the spectral decomposition. Under Definition I.37, for every B_0 and $T \geq 0$,*

$$B(T) = -M_t^{-1} + (B_0 + M_t^{-1}) e^{-2M_t T}, \quad (37)$$

and for every B , $L_t(B) = L_t(B_t^*) + \text{tr}((B - B_t^*)M_t(B - B_t^*)^\top)$, so the excess risk $\mathcal{E}(T) := L_t(B(T)) - L_t(B_t^*)$ decomposes over the eigenmodes of M_t as

$$\mathcal{E}(T) = \sum_{i=1}^{d_{\text{latent}}} \lambda_i e^{-4\lambda_i T} \|(B_0 + M_t^{-1})v_i\|^2 \xrightarrow{B_0=0} \mathcal{E}(T) = \sum_{i=1}^{d_{\text{latent}}} \frac{e^{-4\lambda_i T}}{\lambda_i}. \quad (38)$$

Consequently, for $B_0 = 0$: (i) the risk carried by mode i decays with the mode clock $\tau_i = 1/(4\lambda_i)$ (the parameter error on that mode decays with $1/(2\lambda_i)$; all ratios of clocks are the same); (ii) the risk initially carried by mode i is its target energy $\mathbb{E}\langle v_i, B_t^* x_t \rangle^2 = 1/\lambda_i$, so the slowest modes are also the largest; (iii) if B_0 is block diagonal (in particular $B_0 = 0$) then so is $B(T)$ for all T : the signal and null blocks never interact during population training.

Proof. From the proof of Lemma I.38(c), $\dot{B} = -2(BM_t + I)$. Set $E(T) = B(T) + M_t^{-1}$; then $\dot{E} = -2EM_t$, with unique solution $E(T) = E(0)e^{-2M_t T}$, which is (37). Substituting $B = B_t^* + E$ into $L_t(B) = \text{tr}(BM_t B^\top) + 2 \text{tr } B + \mathcal{J}_t$ gives $L_t(B) = L_t(B_t^*) + 2 \text{tr}(EM_t B_t^{*\top}) + 2 \text{tr } E + \text{tr}(EM_t E^\top) = L_t(B_t^*) + \text{tr}(EM_t E^\top)$, since $B_t^* = -M_t^{-1}$ makes $2 \text{tr}(EM_t B_t^{*\top}) = -2 \text{tr } E$. Hence $\mathcal{E}(T) = \text{tr}(E(0)e^{-2M_t T} M_t e^{-2M_t T} E(0)^\top) = \sum_i \lambda_i e^{-4\lambda_i T} \|E(0)v_i\|^2$. With $B_0 = 0$, $E(0)v_i = v_i/\lambda_i$, giving (38). (i) is read off from $e^{-4\lambda_i T}$ versus $E(T)v_i = e^{-2\lambda_i T} E(0)v_i$. (ii): $\mathbb{E}\langle v_i, B_t^* x_t \rangle^2 = \lambda_i^{-2} v_i^\top M_t v_i = 1/\lambda_i$. (iii): M_t and M_t^{-1} are block diagonal, so (37) preserves block diagonality. (Checked against an Euler integration at $d_{\text{latent}} = 6$: $\max |B_{\text{Euler}} - B_{\text{closed}}| = 3 \times 10^{-4}$ at $dt = 10^{-3}$.) $\square$

Remark I.40 (Discrete steps, loss normalization, and the code's clocks). Gradient descent with step η on $c_{\text{loss}} \cdot L$ (the code multiplies the risk by a normalization c_{loss}) gives $B_k + M_t^{-1} = (B_0 + M_t^{-1})(I - 2\eta_{\text{eff}} M_t)^k$ with $\eta_{\text{eff}} = \eta c_{\text{loss}}$, i.e. (38) with $e^{-4\lambda_i T} \rightarrow (1 - 2\eta_{\text{eff}} \lambda_i)^{2k}$; it is stable iff $\eta_{\text{eff}} < 1/\lambda_{\text{max}}$ and reduces to the flow with $T = \eta_{\text{eff}} k$ when $2\eta_{\text{eff}} \lambda_{\text{max}} \ll 1$, so that $\tau_i = 1/(4\eta_{\text{eff}} \lambda_i)$ steps. The repo's two RFNN scripts share $\eta = 0.01 d_{\text{latent}}/\Delta_t$ but differ in c_{loss} : the `code/v3` path sums the residual $\|\sqrt{\Delta_t} s + \eta\|^2$ over dimensions and averages over the batch, so $c_{\text{loss}} = \Delta_t$, whereas the script that produced the saved sweeps divides additionally by d_{latent} :

$$T = 0.01 d_{\text{latent}} k \text{ (code/v3)}, \quad T = 0.01 k \text{ (saved sweeps)}. \quad (39)$$

Stability holds in both: $2\eta_{\text{eff}} \lambda_{\text{max}}(\mathbf{U}) \approx 0.02 a_1(q)^2 \lambda_{\text{max}}^S \approx 0.05$ (v3) and ≤ 0.01 (v2). Both use momentum 0 and full batch with fresh η each step, so the realized trajectory is this ODE plus zero-mean fluctuations (Proposition I.2). The draft's (3) writes $\dot{A} = -(A - A_*)\mathbf{U}$ with per-mode factor $e^{-\lambda_i T}$: its T is twice ours and its factor is the *parameter* clock, so the risk clock $1/(4\lambda_i)$ here is $1/(2\lambda_i)$ in the draft's units (Remark I.7 uses the v2 clock $T = 0.02 \times \text{steps}$ in those units). Adaptive optimizers (the MLP's Adam) are not covered.

Corollary I.41 (Population risk versus (d_{latent}, T) in the block-anisotropic model; proved). *With $B_0 = 0$ and population training, the excess risk of Proposition I.39 is*

$$\mathcal{E}(T; d_{\text{latent}}) = \underbrace{\sum_{i=1}^{d_{\text{intrinsic}}} \frac{e^{-4\lambda_i^S T}}{\lambda_i^S}}_{\text{signal block}} + \underbrace{(d_{\text{latent}} - d_{\text{intrinsic}}) \frac{e^{-4\beta_t^2 T}}{\beta_t^2}}_{\text{null block}}, \quad (40)$$

and the per-dimension score error is $\text{err}_{\infty}(T; d_{\text{latent}}) = [\mathcal{E}(T; d_{\text{latent}}) + R_{\text{irr}}]/d_{\text{latent}}$. Writing $a(T) = \sum_i e^{-4\lambda_i^S T}/\lambda_i^S + R_{\text{irr}}$ and $b(T) = e^{-4\beta_t^2 T}/\beta_t^2$, both independent of d_{latent} :

- (a) (affine law) $d_{\text{latent}} \cdot \text{err}_{\infty}(T; d_{\text{latent}}) = a(T) + (d_{\text{latent}} - d_{\text{intrinsic}}) b(T)$ is affine in d_{latent} at every T , with slope $b(T)$ decaying at the single rate $4\beta_t^2$.
- (b) (monotone in d_{latent}) $\text{err}_{\infty}(T; d_{\text{latent}}) = b(T) + (a(T) - d_{\text{intrinsic}} b(T))/d_{\text{latent}}$ is monotone in d_{latent} at every T : increasing toward the plateau $b(T)$ while $a(T) < d_{\text{intrinsic}} b(T)$, which holds at $T = 0$ whenever $\sigma_{\perp} < \sigma_{\text{sig}}$ since $a(0) \leq d_{\text{intrinsic}}/\gamma_t^2 < d_{\text{intrinsic}} b(0)$, and decreasing like $R_{\text{irr}}/d_{\text{latent}}$ for T beyond the d_{latent} -independent crossover $T_{\times}$ solving $a(T_{\times}) = d_{\text{intrinsic}} b(T_{\times})$. The late regime and $T_{\times}$ are $n = \infty$ statements; at the d_{latent}/n of the sweeps they are replaced by Proposition I.43(d).
- (c) (null dominance) For $\lambda_i^S \equiv \alpha_i^2$, $\mathcal{E}_N/\mathcal{E}_S = \frac{d_{\text{latent}} - d_{\text{intrinsic}}}{d_{\text{intrinsic}}} \cdot \frac{\alpha_t^2}{\beta_t^2} \cdot e^{4(\alpha_t^2 - \beta_t^2)T}$, increasing in both d_{latent} and T ; with the actual centers replace α_t^2 by the slowest signal eigenvalue λ_{min}^S for the late-time rate.
- (d) (ε -time) The null block's per-dimension excess risk falls below ε at $T_{\varepsilon} = (4\beta_t^2)^{-1} \log(1/(\beta_t^2 \varepsilon))$, independent of d_{latent} at $n = \infty$.

Proof. Insert the block spectrum of M_t into (38) and add the irreducible term of (36); (a)–(d) are algebraic rearrangements. In (b), $\partial_{d_{\text{latent}}} \text{err}_{\infty} = -(a - d_{\text{intrinsic}} b)/d_{\text{latent}}^2$ has the sign of $d_{\text{intrinsic}} b - a$, independent of d_{latent} ; $a(0) = \mathcal{J}_{t,S} \leq d_{\text{intrinsic}}/\gamma_t^2$ by Lemma I.38(c). In (c) divide the two terms of (40). $\square$

Remark I.42 (Isotropic control and the diffusion floor). If $\sigma_{\perp} = \sigma_{\text{sig}}$ then $\beta_t^2 = \gamma_t^2 (= 1$ for all t when $\sigma_{\text{sig}} = 1$): the null block still exists, still carries energy $(d_{\text{latent}} - d_{\text{intrinsic}})/\gamma_t^2$ and still has its own clock $1/(4\gamma_t^2)$; what disappears is only the separation λ_i^S/β_t^2 of its clock from the signal clocks. If instead $\sigma_{\perp} \rightarrow 0$ then $\beta_t^2 \rightarrow \Delta_t$: the null block sits on the diffusion floor, its target energy per dimension $1/\Delta_t$ is set by the noising process alone and its clock $1/(4\Delta_t)$ is the slowest attainable at that t . In this limit the empirical $\widehat{M}_t$ has null block $\Delta_t I$ up to $e^{-2t} \sigma_{\perp}^2 (1 + \sqrt{c})^2 \leq 2.5 \times 10^{-4}$ (relative 1% at $\sigma_{\perp} = 0.01$), so the finite- n effects of Proposition I.43 are small but nonzero: the $\sigma_{\perp} = 0.01$ sweep approximates this corollary.

Zero-initialization score error. With $B_0 = 0$ the predicted per-dimension score error at the first evaluation is $[\mathcal{J}_{t,S} + (d_{\text{latent}} - d_{\text{intrinsic}})/\beta_t^2]/d_{\text{latent}}$ with the Monte-Carlo $\mathcal{J}_{t,S}$ of Table 5 (a point prediction, no fitted parameters). At $t = 0.1$, $\sigma_{\perp} = 0.5$: 0.716 ($d_{\text{latent}} = 5$), 1.654 (10), 2.122 (20), 2.357 (40), 2.497 (100), 2.544 (200), limit 2.59; at $\sigma_{\perp} = 0.01$: 0.716, 3.115, 4.315, 4.914, 5.274, 5.394, limit 5.51. At the RFNN's $t = 0.01$, $\sigma_{\perp} = 0.5$: 0.737, 2.256, 3.016, 3.396, 3.624, 3.700, limit 3.78; at $\sigma_{\perp} = 0.01$: 0.737, 25.5, 37.9, 44.1, 47.8, 49.0, limit 50.3. The code's first logged evaluation is step 1, which moves the top signal mode by a factor $(1 - 2\eta_{\text{eff}}\lambda_{\text{max}})^2 \approx 0.9$ and the null modes by less than 1%.

Proposition I.43 (Empirical-risk flow: finite- n null band, overshoot, and late-time floor; (a)–(c) proved, (d) leading order). *Let $\hat{B}(T)$ be gradient flow on $L_{t,n}$ from $B_0 = 0$ and let $M_t = \text{diag}(M_{t,S}, \beta_t^2 I)$ be the population second moment.*

(a) (exact) $\nabla_B L_{t,n} = 2(B\hat{M}_t + I)$, hence $\hat{B}(T) = -\hat{M}_t^{-1}(I - e^{-2\hat{M}_t T}) \rightarrow -\hat{M}_t^{-1}$, and the population excess risk of the trained model is exactly

$$\mathcal{E}_n(T) = \text{tr}(E(T)M_tE(T)^{\top}), \quad E(T) = M_t^{-1} - \hat{M}_t^{-1}(I - e^{-2\hat{M}_t T}). \quad (41)$$

(b) (null block) Let $\hat{M}_{t,N}$ be the null-null principal block of $\hat{M}_t$, with eigenvalues $\hat{\lambda}_i^N$. Dropping the signal-null cross block of $\hat{M}_t$, the null contribution to (41) is

$$\mathcal{E}_{n,N}(T) = \beta_t^2 \sum_{i=1}^{d_{\text{latent}} - d_{\text{intrinsic}}} \left(\frac{1}{\beta_t^2} - \frac{1 - e^{-2\hat{\lambda}_i^N T}}{\hat{\lambda}_i^N} \right)^2. \quad (42)$$

A mode with $\hat{\lambda}_i^N \geq \beta_t^2$ decays monotonically to its floor $\beta_t^2(1/\hat{\lambda}_i^N - 1/\beta_t^2)^2$; a mode with $\hat{\lambda}_i^N < \beta_t^2$ first reaches zero population error at $T_i^0 = (2\hat{\lambda}_i^N)^{-1} \log[\beta_t^2/(\beta_t^2 - \hat{\lambda}_i^N)]$ and then rises to the same floor: the empirical fit overshoots the population score on under-sampled null directions.

(c) (Marchenko–Pastur band) $\hat{M}_{t,N} = e^{-2t}\sigma_{\perp}^2 \mathcal{W} + \Delta_t I$ with $\mathcal{W} = \frac{1}{n}X_N^{\top}X_N$ a white Wishart matrix of aspect ratio $c = (d_{\text{latent}} - d_{\text{intrinsic}})/n$. As $n \rightarrow \infty$ with c fixed its spectrum follows the shifted, scaled Marchenko–Pastur law (Marchenko & Pastur, 1967; Couillet & Debbah, 2011), with edges

$$\hat{\lambda}_{\pm}^N = e^{-2t}\sigma_{\perp}^2(1 \pm \sqrt{c})^2 + \Delta_t, \quad c = \frac{d_{\text{latent}} - d_{\text{intrinsic}}}{n}. \quad (43)$$

For $0 < c \leq 1$ the null risk clocks have asymptotic band $[1/(4\hat{\lambda}_{+}^N), 1/(4\hat{\lambda}_{-}^N)]$. At fixed $t > 0$ the slow edge tends to $1/(4\Delta_t)$ as $c \uparrow 1$, a finite limit (12.625 at $t = 0.01$). For $c > 1$ the shifted MP law additionally has an atom at Δ_t , which sets the global slow edge. Interlacing gives indexed bounds, not the same bracket for all bottom full-matrix eigenvalues; globally $\hat{M}_t \succeq \Delta_t I$.

(d) (late-time floor) As $T \rightarrow \infty$,

$$\begin{aligned} \text{err}_n(\infty; d_{\text{latent}}) &= \frac{R_{\text{irr}} + \mathcal{E}_S^{\text{est}}}{d_{\text{latent}}} + \frac{d_{\text{latent}} - d_{\text{intrinsic}}}{d_{\text{latent}}} F_t(c), \\ F_t(c) &= \beta_t^2 \int \left(\frac{1}{\hat{\lambda}} - \frac{1}{\beta_t^2} \right)^2 d\mu_{t,c}^{\text{MP}}(\hat{\lambda}) = c \frac{(e^{-2t}\sigma_{\perp}^2)^2}{\beta_t^6} + O(c^2), \end{aligned} \quad (44)$$

with $\mathcal{E}_S^{\text{est}} = O(d_{\text{intrinsic}}/n)$ the signal-block estimation error. F_t vanishes at $c = 0$ and at $\sigma_{\perp} \rightarrow 0$. Numerical evaluation in the studied range gives growth with c and a late-time U-shaped curve in d_{latent} (the $R_{\text{irr}}/d_{\text{latent}}$ decrease at small d_{latent} , the $F_t(c)$ increase at large d_{latent}), and at large d_{latent}/n the trained model is worse on the null block than the zero model (per null dimension $F_t > 1/\beta_t^2$ at $d_{\text{latent}} \geq 200$, $t = 0.01$).

Proof. (a) $\nabla_B L_{t,n} = \frac{2}{n} \sum_{\mu} \mathbb{E}_{\eta}[(Bx_t^{\mu} + \eta/\sqrt{\Delta_t})x_t^{\mu\top}] = 2B\hat{M}_t + \frac{2}{\sqrt{\Delta_t}} \frac{1}{n} \sum_{\mu} \mathbb{E}_{\eta}[\eta x_t^{\mu\top}] = 2(B\hat{M}_t + I)$, using $\mathbb{E}_{\eta}[\eta x_t^{\mu\top}] = \sqrt{\Delta_t}I$ and $\frac{1}{n} \sum_{\mu} \mathbb{E}_{\eta}[x_t^{\mu} x_t^{\mu\top}] = \hat{M}_t$. The ODE $\dot{B} = -2(B\hat{M}_t + I)$ is solved

as in Proposition I.39 with $M_t \rightarrow \widehat{M}_t$. The population excess of any B is $\text{tr}((B + M_t^{-1})M_t(B + M_t^{-1})^\top)$ by the exact expansion in that proposition, and $\widehat{B}(T) + M_t^{-1} = E(T)$. (b) With the cross block dropped, $\widehat{M}_t$ and $E(T)$ are block diagonal; on the null block $M_t = \beta_t^2 I$ and $E(T)$ is diagonal in the eigenbasis of $\widehat{M}_{t,N}$ with entries $1/\beta_t^2 - g_i(T)$, $g_i(T) = (1 - e^{-2\widehat{\lambda}_i T})/\widehat{\lambda}_i$, giving (42). g_i increases from 0 to $1/\widehat{\lambda}_i$; if $\widehat{\lambda}_i < \beta_t^2$ then $1/\widehat{\lambda}_i > 1/\beta_t^2$ and $g_i(T_i^0) = 1/\beta_t^2$ at the stated T_i^0 , after which $(1/\beta_t^2 - g_i)^2$ increases to $(1/\beta_t^2 - 1/\widehat{\lambda}_i)^2$. Numerically (seed 42, $n = 500$, $\sigma_\perp = 0.5$, $t = 0.01$, $d_{\text{latent}} = 200$) the full- $\widehat{M}_t$ evaluation of (41) gives a late floor of 4.81 per dimension versus 4.50 with the cross block dropped, and the null eigenvalues move from $[0.0554, 0.661]$ to $[0.0544, 0.658]$. (c) In the rotated frame the null coordinates of the training points are i.i.d. $\mathcal{N}(0, \sigma_\perp^2 I_{d_{\text{latent}} - d_{\text{intrinsic}}})$, independent of the signal coordinates and of the labels, so $\frac{1}{n} X_N^\top X_N = \sigma_\perp^2 \mathcal{W}$ with $\mathcal{W}$ white Wishart; the MP law with variance parameter 1 has support $[(1 - \sqrt{c})^2, (1 + \sqrt{c})^2]$, mean 1 and variance c . Interlacing: for a principal submatrix, $\lambda_{k+d_{\text{intrinsic}}}(\widehat{M}_t) \leq \lambda_k(\widehat{M}_{t,N}) \leq \lambda_k(\widehat{M}_t)$. At $n = 500$ the measured null band and the MP edges (43) agree to 1–3% (Table 6). (d) Let $T \rightarrow \infty$ in (b) and average over the MP law. Expanding $1/\widehat{\lambda} - 1/\beta_t^2 = -(\widehat{\lambda} - \beta_t^2)/\beta_t^4 + O((\widehat{\lambda} - \beta_t^2)^2)$ and using $\mathbb{E}(\widehat{\lambda} - \beta_t^2)^2 = (e^{-2t} \sigma_\perp^2)^2 \text{Var}_{\text{MP}} = (e^{-2t} \sigma_\perp^2)^2 c$ gives the leading term (at $t = 0.1$, $d_{\text{latent}} = 40$: 0.051 versus the exact 0.060). On the signal block $\widehat{B}_S(\infty) = -\widehat{M}_{t,S}^{-1}$ with $\widehat{M}_{t,S} - M_{t,S} = O_P(n^{-1/2})$ of fixed rank $d_{\text{intrinsic}}$, so its excess is $O(d_{\text{intrinsic}}/n)$. Monotonicity of F_t in c and the U-shape are verified numerically across the sweep (Table 6): the finite- n term overtakes $R_{\text{irr}}/d_{\text{latent}}$ at $d_{\text{latent}} \approx 40$ ($t = 0.1$) and $d_{\text{latent}} \approx 20$ ($t = 0.01$). $\square$

Table 6: Finite- n null block of $\widehat{M}_t$ ($n = 500$, $\sigma_\perp = 0.5$, seed 42) and the resulting late-time population excess per dimension of the trained linear model (full $\widehat{M}_t$, including the signal-block estimation error). MP edges from (43).

d_{latent}	20	40	100	200	240
$c = (d_{\text{latent}} - 5)/500$	0.03	0.07	0.19	0.39	0.47
$t = 0.01$: null band [min, med, max]	[0.195, 0.263, 0.361]	[0.156, 0.261, 0.410]	[0.098, 0.248, 0.516]	[0.054, 0.233, 0.658]	[0.045, 0.225, 0.707]
MP edges	[0.187, 0.357]	[0.152, 0.412]	[0.098, 0.525]	[0.054, 0.666]	[0.044, 0.716]
slow-edge clock / $1/(4\beta_t^2)$	1.4	1.7	2.6	4.8	5.9
late excess per dim	0.11	0.32	1.21	4.81	7.4
$t = 0.1$: null band [min, med, max]	[0.327, 0.385, 0.466]	[0.295, 0.380, 0.507]	[0.247, 0.369, 0.596]	[0.210, 0.358, 0.714]	[0.203, 0.353, 0.755]
late excess per dim	0.02	0.066	0.167	0.365	0.45
$R_{\text{irr}}/d_{\text{latent}}$ ($t = 0.1 / 0.01$)	0.068/0.084	0.034/0.042	0.014/0.017	0.007/0.008	0.006/0.007

Corollary I.44 (Random-feature readout: substitute $M_t \rightarrow U$; (a) proved, (b) sketched). Let $L_n(A) = \frac{1}{n} \sum_\mu \mathbb{E}_\eta \|A\phi(x_t^\mu) + \eta/\sqrt{\Delta_t}\|^2$ and $V := -\frac{1}{n} \sum_\mu \mathbb{E}_\eta [\eta \phi(x_t^\mu)^\top] / \sqrt{\Delta_t} \in \mathbb{R}^{d_{\text{latent}} \times p}$. Let $U = \sum_i \lambda_i(U) u_i u_i^\top$ and $A_\star := VU^+$.

(a) (exact) Gradient flow on L_n from $A_0 = 0$ gives $A(T) = A_\star(I - e^{-2UT})$ and

$$L_n(A(T)) - \min_A L_n = \sum_{i: \lambda_i(U) > 0} \lambda_i(U) e^{-4\lambda_i(U)T} \|A_\star u_i\|^2, \quad (45)$$

so every mode of U has clock $\tau_i = 1/(4\lambda_i(U))$ and modes in $\ker U$ never move. For a single fixed noise draw rank $U \leq n$, so at least $p - n$ such modes exist (the cliff at index n); for the $\mathbb{E}_\eta$ -averaged U the eigenvalues beyond index n are not exactly zero but $O(10^{-6})$, with clocks far beyond any training horizon. This is Proposition I.39 with $M_t \rightarrow U$ and $M_t^{-1} \rightarrow A_\star$ (and Lemma I.3 in the present convention).

(b) (data-side identification, sketched) Under the Gaussian-equivalence decomposition $\phi(x) = Gx + a_\star(q)z(x)$, $G = a_1(q)W/\sqrt{pd_{\text{latent}}}$, restricted to the linear part $\phi_{\text{lin}} = Gx$ and with $W^\top W = pI$, the d_{latent} nonzero eigenvalues of $U_{\text{lin}} = G\widehat{M}_t G^\top$ are

$$\lambda_i(U) = \frac{a_1(q)^2}{d_{\text{latent}}} \widehat{\lambda}_i(\widehat{M}_t), \quad q = \frac{\text{tr } \widehat{M}_t}{d_{\text{latent}}} \simeq \beta_t^2 + \frac{d_{\text{intrinsic}}(\alpha_t^2 - \beta_t^2)}{d_{\text{latent}}}, \quad (46)$$

and the population excess risk carried by these modes is that of the empirical-risk linear model, Proposition I.43, with every clock multiplied by $d_{\text{latent}}/a_1(q)^2$:

$$\mathcal{E}^{\text{RF}}(T; d_{\text{latent}}) = \mathcal{E}_n(a_1(q)^2 T / d_{\text{latent}}), \quad (47)$$

so that the null clock is the band

$$\begin{aligned} \tau_{\mathbf{N}}^{\text{RF}}(d_{\text{latent}}) &\in \frac{d_{\text{latent}}}{4 a_1(q)^2} \left[\frac{1}{\widehat{\lambda}_{+}^{\mathbf{N}}}, \frac{1}{\widehat{\lambda}_{-}^{\mathbf{N}}} \right], \\ \text{median} &\approx \frac{d_{\text{latent}}}{4 a_1(q)^2 \beta_t^2} \text{ (MP median}/\beta_t^2 = 0.88 \text{ at } d_{\text{latent}} = 200), \end{aligned} \quad (48)$$

with $\lambda_{\pm}^N$ from (43). In gradient-flow units of L_n the median clock grows essentially linearly in d_{latent} , partly offset by the de-saturation of $\tanh(q \downarrow, a_1(q)^2 \uparrow)$: at $t = 0.01$, $\sigma_{\perp} = 0.5$ it is $26.2 \rightarrow 302$ over $d_{\text{latent}} = 5 \rightarrow 200$ ($11.5\times$), and its slow edge is a further $4.8\times$ longer at $d_{\text{latent}} = 200$. In training steps the code’s learning rate $0.01 d_{\text{latent}}/\Delta_t$ removes the factor d_{latent} ((39)): $\tau_{\text{N}}^{\text{RF}} = 25/(a_1(q)^2 \beta_t^2)$ steps in code/v3, decreasing $524 \rightarrow 151$ across the sweep, and $25d_{\text{latent}}/(a_1(q)^2 \beta_t^2) = 2620 \rightarrow 30190$ steps for the saved sweeps. The trace of (46) over the data-side modes is $a_1(q)^2 q$, the linear part of $\mathbb{E} \tanh^2(\sqrt{q} z) = a_1(q)^2 q + a_*(q)^2$; the remaining mass $a_*(q)^2$ lives on the $n - d_{\text{latent}}$ sample modes, which are invisible to this corollary.

Proof. (a) L_n is a quadratic in A with $\nabla_A L_n = 2(AU - V)$, so $\dot{A} = -2(AU - V)$. The rows of V lie in the row space of U : if $u \in \ker U$ then $\frac{1}{n} \sum_{\mu} \mathbb{E}_{\eta} (\phi(x_{\mu}^T)^\top u)^2 = 0$, so $\phi(x_{\mu}^T)^\top u = 0$ a.s. and $Vu = 0$. Hence $V = A_* U$ with $A_* = VU^\top$; $E := A - A_*$ obeys $\dot{E} = -2EU$ and $E(T) = -A_* e^{-2UT}$. Expanding $L_n(A_* + E) = L_n(A_*) + \text{tr}(EUE^\top)$ (the cross term vanishes because $A_* U = V$) gives (45); modes with $\lambda_i(U) = 0$ contribute nothing and $A(T)u_i = 0$ for them. The rank statement for a single noise draw is $U = \frac{1}{n} \Phi^\top \Phi$ with $\Phi \in \mathbb{R}^{n \times p}$; for the $\mathbb{E}_{\eta}$ average, compute U at $d_{\text{latent}} = 20$, $p = 1280$, $t = 0.01$ gives eigenvalues 2.0×10^{-6} and 1.8×10^{-6} at indices $n - 1$ and n : no exact cliff, but clocks of order 10^5 flow units. (b) Empirical version, linear part. The empirical fixed point solves $A_* G \widehat{M}_t G^\top = -G^\top$ (the empirical Stein identity $\frac{1}{n} \sum_{\mu} \mathbb{E}_{\eta} [(\eta/\sqrt{\Delta_t}) x_{\mu}^{\top}] = I$ replaces Lemma I.38(b)). With $G^\top G = cI_{d_{\text{latent}}}$, $c = a_1(q)^2 W^\top W / (p d_{\text{latent}}) = a_1(q)^2 / d_{\text{latent}}$, right-multiplying by G gives $A_* G = -\widehat{M}_t^{-1}$. Also $U_{\text{lin}} G = G \widehat{M}_t G^\top G = c G \widehat{M}_t$, hence $e^{-2U_{\text{lin}} T} G = G e^{-2c \widehat{M}_t T}$ and the nonzero spectrum of U_{lin} is $c \widehat{\lambda}_i(\widehat{M}_t)$ with eigenvectors $G \widehat{v}_i / \|G \widehat{v}_i\|$, which is (46). The population excess risk of $s_{A(T)}$ on these modes is $\text{tr}((A(T)G + M_t^{-1})M_t(A(T)G + M_t^{-1})^\top)$ with $A(T)G + M_t^{-1} = M_t^{-1} - \widehat{M}_t^{-1}(I - e^{-2c \widehat{M}_t T}) = E(cT)$ in the notation of (41), which is (47); the band (48) is Proposition I.43(c) rescaled. The step conversions follow from Remark I.40. Consistency with the measured spectra: at $t = 0.01$, $\sigma_{\perp} = 0.5$, the idealized $q = \text{tr } M_t / d_{\text{latent}}$ runs $2.764 \rightarrow 0.327$ and $a_1(q)^2 = 0.180 \rightarrow 0.625$ over $d_{\text{latent}} = 5 \rightarrow 200$, exactly the values of Lemma I.6; the code-normalized noise-dim medians predicted by (46) with the MP median of $\widehat{M}_{t,N}$ are $64 a_1(q)^2 \text{med}(\widehat{\lambda}^N) = 4.77$ ($d_{\text{latent}} = 10$) and 9.33 ($d_{\text{latent}} = 200$) against measured 5.22 and 9.22 ; using the population β_t^2 instead gives 4.76 and 10.60 , so the 13% shortfall at $d_{\text{latent}} = 200$ is the MP median ($0.233/0.265 = 0.88$), not a finite- p effect. What is only sketched: (i) $W^\top W = pI$ holds in expectation with relative fluctuations $O(\sqrt{d_{\text{latent}}/p})$, which perturb the data-side eigenvalues by the same relative amount (the W factor of Appendix I.2); (ii) the residual features $a_*(q)z$ are uncorrelated with x under Gaussian equivalence (Pennington & Worah, 2017; Benigni & Pécché 2021; Goldt et al., 2022; Hu & Lu, 2023; Mei & Montanari, 2022) but not orthogonal to Gx sample by sample, so the data-side eigenvectors of the full U are those of U_{lin} only to leading order; the data-side eigenvalues of the full U measured with compute_U ($d_{\text{latent}} = 20$, $p = 1280$, $t = 0.01$) exceed (46) by 4–7% (null block $[3.96, 5.43, 7.63] \times 10^{-3}$ versus $[3.82, 5.13, 7.10] \times 10^{-3}$) and $\text{tr } U = 0.357$ versus $a_1^2 q + a_*^2 = 0.372$; this few-percent correction is consistent with row-wise fluctuations of the pre-activation variance $q_j = w_j^\top \widehat{M}_t w_j / d_{\text{latent}}$ (Jensen) and with the non-Gaussian mixture pre-activations $w_j^\top m_c / \sqrt{d_{\text{latent}}}$, both decreasing with d_{latent} , and is much larger than the sample-bulk mean $a_*(q)^2/n \approx 4 \times 10^{-5}$; (iii) Gaussian equivalence itself is proved for Gaussian or gaussian data with isotropic-plus-spike covariance; a rigorous version of (b) would follow from a Gaussian-equivalence theorem for the conjugate kernel with anisotropic input covariance (Hu & Lu, 2023; Bonnaire et al., 2025) evaluated at the two-atom spectral measure, and for data exactly on a subspace ($\sigma_{\perp} = 0$) the corresponding block structure appears in George et al. (2026). $\square$

Remark I.45 (What the linear model cannot and can see). Trained on the empirical denoising risk, the linear model converges to $-\widehat{M}_t^{-1}$, the score of the centered Gaussian with the empirical second

moment; it never reproduces training points, because it has no sample bulk. Memorization in the NN-ratio sense requires the residual features $a_*(q)z$, whose second moment is the $n - d_{\text{latent}}$ -mode sample bulk of U ; Corollary I.44(a) gives those modes the same clock law $1/(4\lambda_i(U))$ but with weights $\lambda_i \|A_* u_i\|^2$ that are not $1/\lambda_i$, and they are treated in Appendices I.3 and I.6. What the linear model *does* see is the finite-sample overfitting of the null block (Proposition I.43): the null directions are estimated from n points, the empirical eigenvalues spread over the MP band, and the fitted inverse overshoots on the under-sampled directions with a late-time population penalty $F_t(d_{\text{latent}}/n)$ that grows with d_{latent} . This is the linear-model form of the empirical observation that memorization returns as $d_{\text{latent}} \rightarrow n$ (Appendix I.8), and it lives entirely in the d_{latent} data-side modes.

Remark I.46 (Which block sets the generalization clock). Since $\beta_t^2 < \lambda_{\min}^S$ whenever $\sigma_{\perp} < \sigma_{\text{sig}}$ (0.265 versus 1.89 at $t = 0.01$), the slowest *data-side* modes are the null band, not the signal block: the population score error is learned on the clock band $[1/(4\hat{\lambda}_+^N), 1/(4\hat{\lambda}_-^N)]$ (linear model) or $d_{\text{latent}}/(4a_1(q)^2)$ times that (RFNN), ordered between the signal clocks and the sample-bulk onset. If τ_{gen} is defined from the signal edge (Definitions I.8 and I.22), the ordering $\tau_{\text{gen}} < \tau_N < \tau_{\text{mem}}$ holds; if it is defined from the score error, it is the slow edge of τ_N itself. At $t = 0.01$ the factor $d_{\text{latent}}/a_1(q)^2$ common to all data-side clocks grows $11.5\times$ over $d_{\text{latent}} = 5 \rightarrow 200$ in flow units, while the signal-edge τ_{gen} on the released spectra grows $11\times$ with the actual (non-isotropic, empirical) signal spectrum (Appendix I.2); the two are the same quantity only up to the spread of $\hat{\lambda}^S$ and the step conversion (39), which must be applied before any absolute comparison of step counts.

Predictions and how to test them (all for the corrected score-error metric; no saved value constrains them).

1. *Zero-initialization score error per latent dimension* is $[\mathcal{J}_S + (d_{\text{latent}} - d_{\text{intrinsic}})/\beta_t^2]/d_{\text{latent}}$ (values listed above). Test: log the score error at the first evaluation for the RFNN ($A_0 = 0$) and for an MLP whose last layer is zeroed, at $t = 0.01$ and $t = 0.1$ separately.
2. *Affine law*: $d_{\text{latent}} \cdot \text{err}(T)$ is affine in d_{latent} with a single slope $b(T)$ only where the null band is narrow ($d_{\text{latent}} \leq 40$ at $\sigma_{\perp} = 0.5$; the whole sweep at $\sigma_{\perp} = 0.01$); beyond, the slope is the MP average of (42), larger than $b(T)$ and d_{latent}/n -dependent. Test: fit lines at several fixed steps; check the fitted slope decays at the rate of (39); check linearity breaks upward at $d_{\text{latent}} \geq 100$, $\sigma_{\perp} = 0.5$, late steps.
3. *Shape*: monotone increasing in d_{latent} at early times, U-shaped at late times with minimum near $d_{\text{latent}} \sim 20$ –40 and the large- d_{latent} branch rising with $F_t(c)$; never n-shaped; non-monotone in steps at $d_{\text{latent}} \geq 100$ (early-stopping minimum ≈ 0.36 per dim at $d_{\text{latent}} = 200$, $t = 0.01$, floor 4.8). Test: plot err vs d_{latent} at a grid of steps and vs steps at $d_{\text{latent}} \geq 100$; compare late floors with Table 6.
4. $\sigma_{\perp} = 0.01$ *control*: the null block is $2.13\times$ ($t = 0.1$) or $13.3\times$ ($t = 0.01$) larger and slower than at $\sigma_{\perp} = 0.5$; slope ratio $b_{0.01}/b_{0.5} = (\beta_{0.5}^2/\beta_{0.01}^2)e^{+4(\beta_{0.5}^2 - \beta_{0.01}^2)T}$; no finite- n floor, so the late d_{latent} -dependence has opposite sign in the two sweeps.
5. *RFNN null band*: $d_{\text{latent}}\lambda^{\text{nd}}(U)/a_1(q)^2$ fills $e^{-2t}\sigma_{\perp}^2(1 \pm \sqrt{(d_{\text{latent}} - 5)/500})^2 + \Delta_t$ with medians 0.263, 0.261, 0.248, 0.233, 0.225 at $d_{\text{latent}} = 20, \dots, 240$, and per-mode risk decays as $e^{-4\lambda_i T}$. Test: rescale the saved noise-dim block by $1/(\psi_p a_1(q)^2)$ and overlay the MP band; fit per-mode risk decay in the eigenbasis of U to check the factor 4.
6. *Isotropic control*: the null block persists with energy $d_{\text{latent}} - 5$ and clock $1/4$ exactly; only its separation from the signal clocks survives; the finite- n band is four times wider in absolute terms and the late floor larger. Test: early slope decays at rate 4; late floor exceeds the $\sigma_{\perp} = 0.5$ floor at matched d_{latent} .

Open gaps. (1) All RFNN-facing constants are at $t_{\text{fixed}} = 0.01$ and the null block is carried as the empirical $\hat{M}_t$ with its MP band; the late-time shape is U-shaped in d_{latent} (not monotone at every T , and not “decreasing like $R_{\text{irr}}/d_{\text{latent}}$, never n-shaped” as an earlier draft said). (2) Proposition I.43(b) drops the signal–null cross block of $\hat{M}_t$ (1% on eigenvalues, 7–8% on the late floor at $d_{\text{latent}} = 200$); the exact (41) needs no such approximation, but a clean analytic bound on the cross-block contribution is not written out; monotonicity of $F_t(c)$ beyond leading order and the location of the U minimum are numerical. (3) Corollary I.44(b) needs (i) concentration of $W^T W/p$, (ii) control of the $a_*(q)z$ cross terms on the data-side eigenvectors of the full U (the +4–7% shift is attributed, without proof,

to row-wise q fluctuations and mixture pre-activations), (iii) a Gaussian-equivalence theorem with anisotropic covariance for the two-atom ρ_Σ . (4) The linear model has no sample bulk and never reproduces training points; it explains the generalizing data-side modes, the score-error curve and the finite- n null overfitting, not the NN-ratio fraction curve. (5) The MLP is trained with Adam, minibatches, $t \sim U[0.01, 3]$ and non-zero default init; every MLP-facing prediction is conjectural for it. (6) The clock is $1/(4\lambda_i)$ for the risk and $1/(2\lambda_i)$ for the parameters in this convention; absolute step counts must be converted with (39) before comparison.

I.5 BRIDGE FROM RESIDUAL SAMPLE-BULK ERROR TO NEAR-DUPLICATE GENERATION

This subsection connects the spectral quantities above to the memorisation *fraction* that the paper reports. It proves two elementary lemmas, gives the sketch of Proposition I.50, derives its corollaries, and records the empirical anchors on the paper’s own synthetic data. Two referee reports on this subsection could not be incorporated by revision; their substantive corrections are applied inline where they are unambiguous (marked “[R]”) and listed under Open gaps otherwise. Throughout, the sampler is the reverse VP-SDE $dx = (x + 2s_T) d\tau + \sqrt{2} dW$ ($\tau = -t$) run from $t_{\max} = 3$ to $t_{\min} = 0.01$ by Euler–Maruyama with 500 steps and no terminal denoising step, as in the released code; $\Delta_{t_{\min}} = 1 - e^{-0.02} = 0.0198$.

Definition I.47 (Empirical score, residual error, NN-ratio). Let $\{x^\mu\}_{\mu=1}^n \subset \mathbb{R}^{d_{\text{latent}}}$ be the training set and $\Delta_t = 1 - e^{-2t}$. The *exact empirical score* is the score of the noised empirical measure $\rho_t^{\text{emp}} = \frac{1}{n} \sum_{\mu} \mathcal{N}(e^{-t}x^\mu, \Delta_t I)$,

$$s_{\text{emp}}(x, t) = \sum_{\mu=1}^n w_{\mu}(x, t) \frac{e^{-t}x^\mu - x}{\Delta_t}, \quad w_{\mu}(x, t) = \frac{\exp(-\|x - e^{-t}x^\mu\|^2/2\Delta_t)}{\sum_{\nu} \exp(-\|x - e^{-t}x^\nu\|^2/2\Delta_t)} \quad (49)$$

(Scarvelis et al., 2025; Gu et al., 2025). For a score model s_T at training time T write $\zeta_T(x, t) := s_T(x, t) - s_{\text{emp}}(x, t)$ and define the *residual error at diffusion time t*

$$\mathcal{E}_t(T) := \frac{1}{n} \sum_{\mu} \mathbb{E}_{\eta} \|\zeta_T(e^{-t}x^\mu + \sqrt{\Delta_t}\eta, t)\|^2, \quad \eta \sim \mathcal{N}(0, I_{d_{\text{latent}}}), \quad (50)$$

and its per-sample version $\mathcal{E}_{t,\mu}(T)$. For the RFNN at its training time t , $\mathcal{E}_t(T) = \sum_i \lambda_i \|a_i(0) - a_i^*\|^2 e^{-2\lambda_i T} + \mathcal{E}_t(\infty)$ (Proposition I.5), where $\mathcal{E}_t(\infty)$ is the $L^2(\rho_t^{\text{emp}})$ projection error of s_{emp} onto the span of the p features (the gradient-flow fixed point is that projection; [R] this is unrelated to the $\lambda = 0$ modes); we split $\mathcal{E}_t = \mathcal{E}_t^{\text{sig}} + \mathcal{E}_t^{\text{null}} + \mathcal{E}_t^{\text{samp}} + \mathcal{E}_t(\infty)$ by block. Let $d_{\text{NN}}(\mu) = \min_{\nu \neq \mu} \|x^\mu - x^\nu\|$, G_{NN} its empirical CDF over μ , and $d_{\text{NN}}^{(r)}$ its r -quantile. For a generated point $\hat{x}$ with nearest and second-nearest training points NN_1, NN_2 , two NN-ratio statistics occur in the repository. The released sweeps (the 5M-step synthetic sweep and the real-data runs) use $\varrho_B(\hat{x}) = \|\hat{x} - \text{NN}_1\|/\|\hat{x} - \text{NN}_2\|$, both distances taken from the generated sample, the criterion of Bonnaire et al. (2025); an earlier code path used $\varrho(\hat{x}) = \|\hat{x} - \text{NN}_1\|/\|\text{NN}_1 - \text{NN}_2\|$, an alternative repository statistic, not a formula established by Somepalli et al. (2023). $\hat{x}$ is memorized iff the ratio is below $1/3$, and $F_{\text{mem}}(T) = \Pr[\varrho(\hat{x}) < 1/3]$. Statements below are written for the second form, whose criterion is $\|\delta\| < d_{\text{NN}}/3$ for $\hat{x} = x^\mu + \delta$. For the released ϱ_B , with δ nearly orthogonal to $x^\mu - x^\nu$ (large d_{latent}), $\|\hat{x} - \text{NN}_2\|^2 = d_{\text{NN}}^2 + \|\delta\|^2$ and $\varrho_B < 1/3 \iff \|\delta\|^2 < d_{\text{NN}}^2/8$: every constant 9 below becomes 8 at leading order, and at small d_{latent} the criterion is only sandwiched ($\|\delta\| < d_{\text{NN}}/4 \Rightarrow \text{memorized} \Rightarrow \|\delta\| < d_{\text{NN}}/2$).

Lemma I.48 (Finite-sample softmax concentration; proved here). Fix t , a training index μ , and $\delta \in (0, \frac{1}{2})$. Let $x = e^{-t}x^\mu + \sqrt{\Delta_t}\eta$ with $\eta \sim \mathcal{N}(0, I_{d_{\text{latent}}})$. If

$$e^{2t}\Delta_t \leq \frac{d_{\text{NN}}(\mu)^2}{32 \log(n/\delta)}, \quad (51)$$

then with probability at least $1 - 2\delta$ over η [R], $1 - w_{\mu}(x, t) \leq \delta^8 n^{-7}$, and on that event $s_{\text{emp}}(x, t) = (e^{-t}x^\mu - x)/\Delta_t + r$ with $\|r\| \leq \delta^8 n^{-7} \max_{\nu} \|e^{-t}(x^\mu - x^\nu)\|/\Delta_t$. Evaluating the log-weight ratio at $\eta = 0$ gives the sufficient condition $e^{2t}\Delta_t < d_{\text{NN}}(\mu)^2/(2 \log n)$ for $w_{\mu}(e^{-t}x^\mu, t) > \frac{1}{2}$, which for $t \ll 1$ reduces to $\Delta_t < d_{\text{NN}}^2/(2 \log n)$. This finite-sample sufficient bound is derived here; Biroli et al. (2024) study the related sampling-time collapse phenomenon. Since $e^{2t}\Delta_t = e^{2t} - 1$ is increasing in t , condensation at t_c implies condensation at every $t \leq t_c$.

Proof. For $\nu \neq \mu$ write $D_\nu = \|x^\mu - x^\nu\| \geq d_{\text{NN}}(\mu)$. From (49),

$$\log \frac{w_\nu}{w_\mu} = -\frac{\|x - e^{-t}x^\nu\|^2 - \|x - e^{-t}x^\mu\|^2}{2\Delta_t} = -\frac{e^{-2t}D_\nu^2}{2\Delta_t} - \frac{e^{-t}\langle \eta, x^\mu - x^\nu \rangle}{\sqrt{\Delta_t}},$$

using $x - e^{-t}x^\nu = e^{-t}(x^\mu - x^\nu) + \sqrt{\Delta_t}\eta$ and $x - e^{-t}x^\mu = \sqrt{\Delta_t}\eta$. The second term is Gaussian with standard deviation $e^{-t}D_\nu/\sqrt{\Delta_t}$; by a two-sided union bound, with probability $\geq 1 - 2\delta$ simultaneously $|\langle \eta, x^\mu - x^\nu \rangle| \leq D_\nu\sqrt{2\log(n/\delta)}$ for all ν . On this event

$$\log \frac{w_\nu}{w_\mu} \leq -\frac{e^{-t}D_\nu}{2\sqrt{\Delta_t}} \left(\frac{e^{-t}D_\nu}{\sqrt{\Delta_t}} - 2\sqrt{2\log(n/\delta)} \right).$$

Condition (51) says $e^{-t}d_{\text{NN}}/\sqrt{\Delta_t} \geq 4\sqrt{2\log(n/\delta)}$, so the bracket is at least $\frac{1}{2}e^{-t}D_\nu/\sqrt{\Delta_t}$ and $\log(w_\nu/w_\mu) \leq -e^{-2t}D_\nu^2/(4\Delta_t) \leq -8\log(n/\delta)$. Summing, $\sum_{\nu \neq \mu} w_\nu/w_\mu \leq n(\delta/n)^8$, hence $1 - w_\mu \leq \delta^8 n^{-7}$. The bound on $r = \sum_{\nu \neq \mu} w_\nu e^{-t}(x^\nu - x^\mu)/\Delta_t$ follows from $\sum_{\nu \neq \mu} w_\nu = 1 - w_\mu$. At $\eta = 0$ the cross term vanishes and $\sum_{\nu \neq \mu} w_\nu/w_\mu \leq (n-1)e^{-e^{-2t}d_{\text{NN}}^2/2\Delta_t} < 1$ if $e^{2t}\Delta_t < d_{\text{NN}}^2/(2\log n)$. $\square$

Lemma I.49 (Exact marginal law and a single-component endpoint approximation). *Let the reverse VP-SDE with the exact score s_{emp} be run from $x_{t_c} \sim \rho_{t_c}^{\text{emp}}$ to $t_{\min}$. Then $x_{t_{\min}} \sim \rho_{t_{\min}}^{\text{emp}} = \frac{1}{n} \sum_{\mu} \mathcal{N}(e^{-t_{\min}}x^\mu, \Delta_{t_{\min}}I)$ [R: no condensation hypothesis is needed for this; only the $\mathcal{N}(0, I)$ initialisation at $t_{\max} = 3$ is approximate]. For a separate idealized single-component probability-flow ODE, assume attachment to μ throughout, $y_{t_c} \sim \mathcal{N}(0, \Delta_{t_c}I)$, and set $r = 0$. A pointwise concentration bound does not prove this pathwise attachment. In that idealization the offset $y_t = x_t - e^{-t}x^\mu$ satisfies $y_{t_{\min}} = \sqrt{\Delta_{t_{\min}}/\Delta_{t_c}}y_{t_c}$. In either case the endpoint is x^μ up to a displacement of per-coordinate variance $\Delta_{t_{\min}}$ (plus the mean shrinkage $e^{-t_{\min}}x^\mu - x^\mu$). Consequently $\varrho(x_{t_{\min}}) \approx \sqrt{d_{\text{latent}}\Delta_{t_{\min}}}/d_{\text{NN}}(\mu)$ under concentration and unchanged nearest-neighbor identities this suggests the endpoint baseline, not an upper bound for arbitrary learned scores,*

$$F_{\text{mem}}^{\text{base}} = 1 - G_{\text{NN}}(3\sqrt{d_{\text{latent}}\Delta_{\text{term}}}) \quad (\text{SDE sampler stopped at } t_{\min}, d_{\text{latent}} \gg 1), \quad (52)$$

with $\Delta_{\text{term}} = \Delta_{t_{\min}}$ for the continuous-time sampler. [R] For the released Euler–Maruyama loop (500 steps on $[0.01, 3]$, $dt = 0.00598$, last-step contraction 0.63) the terminal per-coordinate variance of the condensed linear score is $\Delta_{\text{term}} = 0.0269 = 1.36\Delta_{t_{\min}}$ (exact variance recursion and a 4×10^5 -sample simulation; $1.17\times$ at 1000 steps, $1.03\times$ at 5000): discretisation raises the terminal variance and lowers the endpoint baseline. At $d_{\text{latent}} \lesssim 40$ the $\chi_{d_{\text{latent}}}^2$ spread of $\|\delta\|^2$ (relative s.d. $\sqrt{2/d_{\text{latent}}}$) is not negligible and (52) must be evaluated by Monte Carlo over the endpoint law rather than in the concentrated form.

Proof. The reverse SDE with the exact score transports the time- t_c marginal of the forward VP process started at ρ_0^{emp} to its time- $t_{\min}$ marginal (Song et al., 2021), which is the stated mixture; in the additional single-component idealization $s_{\text{emp}}(x, t) = -(x - e^{-t}x^\mu)/\Delta_t$ is the score of the single component $\mathcal{N}(e^{-t}x^\mu, \Delta_t I)$. For the ODE, $\frac{dx}{dt} = x + s_{\text{emp}}$ and $\frac{d}{dt}(e^{-t}x^\mu) = e^{-t}x^\mu$ give $\frac{dy}{dt} = -\frac{e^{-2t}}{\Delta_t}y$, and $\int_{t_{\min}}^{t_c} \frac{e^{-2t}}{1-e^{-2t}} dt = \frac{1}{2} \log(\Delta_{t_c}/\Delta_{t_{\min}})$. For (52): $\|x_{t_{\min}} - x^\mu\|^2 = \Delta_{\text{term}}\|\eta\|^2 + O(t_{\min}^2\|x^\mu\|^2) +$ a cross term of s.d. $2t_{\min}\sqrt{\Delta_{\text{term}}}\|x^\mu\|$, concentrating at $d_{\text{latent}}\Delta_{\text{term}}(1 + O(d_{\text{latent}}^{-1/2}))$; when this is below $d_{\text{NN}}(\mu)^2/9$, the attachment point is the closest training point, but identifying the second neighbor requires an additional ordering approximation. Replacing the random radius by its typical value and holding these identities fixed yields (52) as a concentrated baseline. Finite-dimensional evaluation should instead sample the full endpoint mixture and compute the actual nearest-neighbor statistic. None of these steps bounds the fraction for a different learned score. The E–M variance is obtained by iterating the time-dependent offset recursion $v_{k+1} = a_k^2 v_k + 2dt$, $a_k = 1 - (1 + e^{-2t_k})dt/\Delta_{t_k}$, evaluated along the code’s grid. $\square$

Proposition I.50 (Conditional second-moment approximation for the endpoint; sketched). *Fix a diffusion time $t \in [t_{\min}, t_c]$ at which the sampler makes its final attachment to a training point, and a training time T . Assume*

(B1) (controlled condensation remainder) *the stronger condition (51) holds at the attachment time. The contribution of r , including the complement of the concentration event, is negligible at the metric threshold as quantified in the sketch below;*

(B2) (concentrated residual) *[R] conditionally on μ , $\|\zeta_T(x_t, t)\|^2$ with $x_t = e^{-t}x^\mu + \sqrt{\Delta_t}\eta$ concentrates at $\mathcal{E}_{t,\mu}(T)$ over η , and its cross term with the terminal noise of Lemma I.49 is negligible (neither Gaussianity nor isotropy of ζ_T is used for (54); the isotropic form is used only for the distribution of $\|\delta\|^2$);*

(B3) (one-attachment reduction) *the endpoint of the learned-score sampler equals the Tweedie denoiser of the learned score applied at the attachment time t , plus the sampler's terminal noise of Lemma I.49.*

Then the generated point is $\hat{x} = x^\mu + \delta$ with $\mathbb{E}\|\delta\|^2 \simeq d_{\text{latent}}\Delta_{\text{eff},\mu}(T)$,

$$\Delta_{\text{eff},\mu}(T) = \Delta_{\text{term}} + c_t \mathcal{E}_{t,\mu}(T), \quad c_t = \frac{e^{2t}\Delta_t^2}{d_{\text{latent}}}, \quad (53)$$

This matches a second moment only, not the law of the exact empirical sampler. Under the concentration and neighbor-stability approximations, the alternative-metric threshold is estimated by

$$d_{\text{latent}} \Delta_{\text{eff},\mu}(T) < \frac{1}{9} d_{\text{NN}}(\mu)^2 \iff \mathcal{E}_{t,\mu}(T) < \frac{d_{\text{NN}}(\mu)^2 - 9d_{\text{latent}}\Delta_{\text{term}}}{9e^{2t}\Delta_t^2}, \quad (54)$$

and

$$F_{\text{mem}}(T) \simeq \frac{1}{n} \sum_{\mu=1}^n \mathbf{1} \left[d_{\text{NN}}(\mu) > 3\sqrt{d_{\text{latent}}\Delta_{\text{term}} + e^{2t}\Delta_t^2 \mathcal{E}_{t,\mu}(T)} \right] \xrightarrow{\mathcal{E}_{t,\mu} \equiv \mathcal{E}_t} 1 - G_{\text{NN}} \left(3\sqrt{d_{\text{latent}}\Delta_{\text{term}} + e^{2t}\Delta_t^2 \mathcal{E}_t(T)} \right). \quad (55)$$

The alternative candidate definition “condensation at the effective level”, $e^{2t}\Delta_{\text{eff},\mu}(T) < d_{\text{NN}}(\mu)^2/(2\log n)$, is implied by (54) whenever $d_{\text{latent}} > \frac{2}{9}e^{2t}\log n$ [R: $d_{\text{latent}} > 10.6$ at $t = t_c(200) = 1.02$, $n = 500$]: the metric threshold, not condensation, binds. [R] The constant c_t depends on the unstated attachment time t : $e^{2t}\Delta_t$ spans 0.020 ($t = t_{\min}$) to 6.7 ($t = 1.02$), so the proposition fixes the functional form $\Delta_{\text{eff}} = \Delta_{\text{term}} + c\mathcal{E}$ but not c (Open gaps, G2).

Sketch. Under (B1), write $s_{\text{emp}}(x_t, t) = -(x_t - e^{-t}x^\mu)/\Delta_t + r$, retaining the concentration remainder. Tweedie’s identity (Efron, 2011) gives the posterior mean for the true marginal score. For the learned score define the plug-in estimator $\hat{x}_T = e^t(x_t + \Delta_t s_T(x_t, t))$. Substitution gives the exact algebra $\hat{x}_T = x^\mu + e^t\Delta_t(\zeta_T + r)$. Consequently the error from dropping r obeys

$$\|\hat{x}_T - x^\mu\|^2 - e^{2t}\Delta_t^2\|\zeta_T\|^2 \leq e^{2t}\Delta_t^2(2\|\zeta_T\|\|r\| + \|r\|^2).$$

At a given point, Lemma I.48 controls r on its stated event. To use (53) requires this correction to be negligible at the metric threshold and negligible complementary-event contribution, in addition to (B2)–(B3). This is not implied merely by $w_\mu > 1/2$, nor does a one-step denoiser identity prove the multistep sampler approximation. $\chi_{d_{\text{latent}}}^2$ concentration of $\|\delta\|^2$ and the identification $\text{NN}_1 = x^\mu$, $\|\text{NN}_1 - \text{NN}_2\| = d_{\text{NN}}(\mu)$ (the first follows when $\|\delta\| < d_{\text{NN}}(\mu)/2$; the second additionally assumes that the nearest competitor to the training point remains the second neighbor of the generated point) give (54); averaging over the attachment index gives (55). Comparison of thresholds: (54) reads $\Delta_{\text{eff}} < d_{\text{NN}}^2/(9d_{\text{latent}})$, condensation reads $\Delta_{\text{eff}} < e^{-2t}d_{\text{NN}}^2/(2\log n)$.

Heuristic motivation for (B2). At $T \gg \tau_{\text{gen}}$ the signal and noise-dim modes are absorbed and $\zeta_T(x) = -(A^* - A(T))\phi(x) + (s_\infty - s_{\text{emp}})(x)$; restricting it to sample modes also neglects the projection error. The sample-bulk eigendirections are the feature-space images of the per-sample deviations ξ^μ with the linear (population) part removed; after the random rotation Q their output directions in $\mathbb{R}^{d_{\text{latent}}}$ are exchangeable across coordinates, and the residual at x_t is a sum over the $N_{\text{res}}(T) = \#\{i \in \mathcal{M} : \lambda_i T \lesssim 1\}$ unabsorbed modes. When $N_{\text{res}}(T) \gg 1$ and $d_{\text{latent}} \gg 1$ an additional central-limit approximation would make $\|\zeta_T\|^2$ concentrate. The approximation fails (i) early, when the residual is dominated by the $d_{\text{intrinsic}}$ signal modes (then $\mathcal{E}_t$ is far above the threshold and $F_{\text{mem}} = 0$ regardless), and (ii) at the very end, when only a few sample modes

remain: the residual is then low-rank and localized on a few training points, and (55) must be read with the per-sample $\mathcal{E}_{t,\mu}$. [R] In the “blend” model $\hat{x}_T = (1 - m)\hat{x}_{\text{pop}} + m\hat{x}_{\text{emp}}$ the Tweedie identity holds equally (on the null block $c_t\mathcal{E}_{t,\mu} = (1 - m)^2\Delta_t\sigma_{\perp}^2/\beta_t^2$ per coordinate, the same as $\|\delta\|^2/d_{\text{latent}}$ computed directly), so the two readings differ only through the choice of t and the multi-step accumulation (G2, G3), not through the form of the residual. $\square$

Corollary I.51 (Fraction curve and quantile-indexed memorization time; conditional on Proposition I.50). *Under Proposition I.50 with a μ -independent residual $\mathcal{E}_t(T)$ decreasing in T , consider the surrogate fraction of (55). Let $d_{[m]}$ be the m -th largest training nearest-neighbor distance, $m = \lceil qn \rceil$, and assume its threshold below is positive. Its crossing infimum at level q is*

$$\tau_{\text{mem}}(q) = \inf \left\{ T : \mathcal{E}_t(T) < \frac{d_{[qn]}^2 - 9d_{\text{latent}}\Delta_{\text{term}}}{9e^{2t}\Delta_t^2} \right\} : \quad (56)$$

within this common-residual surrogate, the most isolated points cross first. Strict inequalities are retained at ties; the infimum need not itself attain the level. In the following continuous-distance approximation, $d_{\text{NN}}^{(1-q)}$ denotes the corresponding population quantile. If $\mathcal{E}_t(T) = \mathcal{E}_t^{\text{samp}}(0)e^{-2\lambda_{\text{samp}}T}$ then

$$\tau_{\text{mem}}(q_2) - \tau_{\text{mem}}(q_1) = \frac{1}{\lambda_{\text{samp}}} \log \frac{d_{\text{NN}}^{(1-q_1)}}{d_{\text{NN}}^{(1-q_2)}} + O\left(\frac{d_{\text{latent}}\Delta_{\text{term}}}{\lambda_{\text{samp}}(d_{\text{NN}}^{(1-q)})^2}\right), \quad (57)$$

a difference in linear T . [R] Under this single-exponential, data-spread-only model, $T(5\%)/T(1\%) = 1 + \log(d_{\text{NN}}^{(0.99)}/d_{\text{NN}}^{(0.95)})/(\lambda_{\text{samp}}T(1\%)) \approx 1 + 0.15/(\lambda_{\text{samp}}T(1\%))$, the measured ratios 2.2–3.5 cannot refute this explanation without separately determining $\lambda_{\text{samp}}T(1\%)$. Within-bulk spectral spread and endpoint-radius spread are additional possible contributions. A single edge with a degenerate G_{NN} would give a step function.

Proof. For the empirical fraction $n^{-1}\#\{\mu : d_{\text{NN}}(\mu) > u\}$, reaching q is equivalent to $u < d_{[qn]}$, including ties. Substitute $u = 3\sqrt{d_{\text{latent}}\Delta_{\text{term}} + e^{2t}\Delta_t^2\mathcal{E}_t(T)}$. For (57) take logarithms of $3e^t\Delta_t\sqrt{\mathcal{E}_t^{\text{samp}}(0)}e^{-\lambda_{\text{samp}}T} = \sqrt{(d_{\text{NN}}^{(1-q)})^2 - 9d_{\text{latent}}\Delta_{\text{term}}}$ and expand in $9d_{\text{latent}}\Delta_{\text{term}}/(d_{\text{NN}}^{(1-q)})^2$. $\square$

Corollary I.52 (What τ_{mem} depends on through d_{NN} ; sketched). *For the synthetic data of Section 2, with labels uniform at random and the n/k -per-cluster Poisson heuristic for the signal part,*

$$\begin{aligned} d_{\text{NN}}(\mu)^2 &\approx \underbrace{C_{d_{\text{intrinsic}}} \sigma_{\text{sig}}^2 (n/k)^{-2/d_{\text{intrinsic}}} \kappa_{\mu}}_{\text{signal part}} \\ &+ \underbrace{2\sigma_{\perp}^2 (d_{\text{latent}} - d_{\text{intrinsic}}) \left(1 - O(\sqrt{\log n / (d_{\text{latent}} - d_{\text{intrinsic}})})\right)}_{\text{null part}}, \end{aligned} \quad (58)$$

with $C_{d_{\text{intrinsic}}}$ depending only on $d_{\text{intrinsic}}$ and κ_{μ} an $O(1)$ random factor with a fixed law. The displayed scale estimate is not a bound: exactly $\min_{\nu}(S_{\nu} + N_{\nu}) \geq \min_{\nu} S_{\nu} + \min_{\nu} N_{\nu}$, while replacing the separate minima by typical scales introduces further approximation. On the repository generator (seed 42, $\sigma_{\perp} = 0.5$) the median d_{NN}^2 is 6.60/14.72/40.0/85.0 at $d_{\text{latent}} = 20/40/100/200$ against the uncorrected sum 8.61/18.61/48.61/98.61, i.e. 15–30% lower, and the null part alone already exceeds the measured median. Hence: (i) if $(d_{\text{latent}} - d_{\text{intrinsic}})\sigma_{\perp}^2 \ll \sigma_{\text{sig}}^2 (n/k)^{-2/d_{\text{intrinsic}}}$ (e.g. $\sigma_{\perp} = 0.01$) the threshold in (56) scales as $n^{-2/d_{\text{intrinsic}}}$ and is independent of d_{latent} : all d_{latent} -dependence of τ_{mem} enters through $\mathcal{E}_t(T)$, and since a larger n lowers the threshold, $\partial \log \tau_{\text{mem}} / \partial \log n$ acquires the additive term $+\frac{1}{d_{\text{intrinsic}}\lambda_{\text{samp}}\tau_{\text{mem}}}$ [R: sign corrected; denser data memorizes later] beyond the spectral n -dependence, a $d_{\text{intrinsic}}$, not d_{latent} , exponent; (ii) if $(d_{\text{latent}} - d_{\text{intrinsic}})\sigma_{\perp}^2 \gg \sigma_{\text{sig}}^2 (n/k)^{-2/d_{\text{intrinsic}}}$ (e.g. $\sigma_{\perp} = 0.5$, $d_{\text{latent}} \gtrsim 20$), $d_{\text{NN}}^2 \simeq 2\sigma_{\perp}^2 d_{\text{latent}} \times (0.7\text{--}0.85)$, (54) becomes approximately a per-coordinate condition on Δ_{eff} , the spread $d_{\text{NN}}^{(0.9)}/d_{\text{NN}}^{(0.1)} \rightarrow 1$, and the same metric drives the NN-ratio of non-memorized samples toward 1; (iii) the endpoint baseline (52) collapses with d_{latent} in regime (i) and tends to 1 in regime (ii).

Proof. Within a cluster the signal coordinates of the $\approx n/k$ points are i.i.d. $\mathcal{N}(0, \sigma_{\text{sig}}^2 I_{d_{\text{intrinsic}}})$; the NN distance of m i.i.d. points from a smooth density in $d_{\text{intrinsic}}$ dimensions scales as $m^{-1/d_{\text{intrinsic}}}$ with a limiting law for the rescaled distance (Levina & Bickel, 2004; Facco et al., 2017). The null coordinates of two distinct points are independent $\mathcal{N}(0, \sigma_{\perp}^2 I_{d_{\text{latent}} - d_{\text{intrinsic}}})$, so their squared distance is $2\sigma_{\perp}^2 \chi_{d_{\text{latent}} - d_{\text{intrinsic}}}^2$; the minimum over $n - 1$ such variables is $2\sigma_{\perp}^2 (d_{\text{latent}} - d_{\text{intrinsic}})(1 - O(\sqrt{\log n / (d_{\text{latent}} - d_{\text{intrinsic}})}))$ by Gaussian concentration and a union bound. The minimum of a sum is at least the sum of the minima; different minimizers increase this discrepancy and cannot explain a downward shortfall against an uncorrected typical-scale estimate. The observed shortfall therefore concerns the approximations to the minima, not that inequality. $\square$

Remark I.53 (Real data: latent versus pixel space). For VAE latents the sampler runs in $\mathbb{R}^{d_{\text{latent}}}$ but ϱ is computed in pixel space after decoding with $D: \mathbb{R}^{d_{\text{latent}}} \rightarrow \mathbb{R}^P$. If $\ell \|z - z'\| \leq \|D(z) - D(z')\| \leq L \|z - z'\|$ on the relevant region, then $(\ell/L)\varrho_{\text{lat}} \leq \varrho_{\text{pix}} \leq (L/\ell)\varrho_{\text{lat}}$ (assuming the pixel-space NN_1, NN_2 are the decoded latent ones; otherwise the denominator needs a global minimum over training pairs). [R] This two-sided sandwich imposes no ordering between the two fractions at the common threshold $\frac{1}{3}$: it gives only $F_{\text{lat}}(\ell/(3L)) \leq F_{\text{pix}}(\frac{1}{3}) \leq F_{\text{lat}}(L/(3\ell))$. Two consequences survive: (a) L/ℓ is not controlled across the sweep (one VAE per d_{latent}), so pixel-space fractions at different d_{latent} are compared at effectively different latent thresholds; (b) posterior-collapsed coordinates ($\lambda_{\min} = \Delta_t$ exactly for $d_{\text{latent}} \geq 140$ on CelebA, Section 5) contribute to $d_{\text{latent}}\Delta_{\text{eff}}$ in latent space but are annihilated by D , so in regime (ii) the relevant dimension is the decoder-visible one ($\approx 0.5d_{\text{latent}}$, the measured effective rank), not d_{latent} .

Empirical anchors (synthetic data, $n = 500, k = 10, d_{\text{intrinsic}} = 5, s = 3, \sigma_{\text{sig}} = 1$; five replicate training sets). (1) Endpoint baseline (52) in the concentrated form with $\Delta_{\text{term}} = \Delta_{t_{\min}}$ at $\sigma_{\perp} = 0.5$: 0.676, 0.790, 0.870, 0.924, 0.950, 0.979, 0.998 for $d_{\text{latent}} = 5, 8, 10, 12, 15, 20, 25$ and ≈ 1 beyond. [R] With the code’s E–M terminal variance and metric, the exact empirical-score sampler run in the code’s own loop gives fractions 0.508/0.597/0.667/0.715/0.757/0.808/0.873/0.960 at $d_{\text{latent}} = 5/8/10/12/15/20/25/40$; these are empirical baseline values, not universal upper bounds on learned-score samplers. Measured final fractions of the 5M-step MLP sweep (seed 42): 0.290, 0.224, 0.168, 0.127, 0.079, 0.044, 0.017, 0.006, 0.004, 0.002 for $d_{\text{latent}} = 5, \dots, 40$ —all below both. At $\sigma_{\perp} = 0.01$ the concentrated baseline is 0.668, 0.364, 0.250, 0.165, 0.100, 0.034, 0.015, 0.008, 0.002 for the same d_{latent} , but the exact endpoint-law Monte Carlo gives 0.478, 0.349, $\dots$, 0.095 ($d_{\text{latent}} = 8, 10, \dots, 20$), 0.050 (25), 0.008 (40), and the E–M loop 0.531/0.235/0.041/0.001 at $d_{\text{latent}} = 5/10/20/40$; the available $\sigma_{\perp} = 0.01$ runs (older statistic, 300k steps) report fractions ≤ 0.006 and exactly 0 for $d_{\text{latent}} \geq 20$, which is consistent with but not a like-for-like test of the endpoint baseline. (2) d_{NN} deciles ($q_{10}/q_{50}/q_{90}$) at $\sigma_{\perp} = 0.5$: 0.72/1.10/1.63 ($d_{\text{latent}} = 5$), 2.13/2.56/3.00 (20), 8.75/9.16/9.56 (200); at $\sigma_{\perp} = 0.01$: 0.70/1.07/1.60, flat in d_{latent} . (3) median(d_{NN}) $\cdot n^{1/5} = 3.66, 3.72, 3.77, 3.76$ for $n = 100, 500, 1000, 5000$ at $\sigma_{\perp} = 0.01$. (4) Condensation times of the *exact* empirical score in the $\eta = 0$ form (median d_{NN}): $t_c = 0.046, 0.105, 0.21, 0.39, 0.72, 1.02$ for $d_{\text{latent}} = 5, 10, 20, 40, 100, 200$ at $\sigma_{\perp} = 0.5$, and 0.045 flat at $\sigma_{\perp} = 0.01$. [R] Typical- η condensation (median off-attachment mass $1 - w_{\mu} < 0.05$) occurs later: 0.053/0.19/0.315/0.757 at $d_{\text{latent}} = 5/20/40/200$; at the $\eta = 0$ value $t_c = 1.02$ ($d_{\text{latent}} = 200$) the median off-attachment mass is 0.83, i.e. not condensed. Either way the exact empirical measure condenses *later in t* (earlier in the reverse process) as d_{latent} grows; the observed delay of memorisation is a property of the learned residual $\mathcal{E}_t(T)$, not of the data. (5) Mean NN-ratio of generated samples 0.524 ($d_{\text{latent}} = 5$) $\rightarrow$ 0.868 (40) follows the training-set self-ratio trend 0.83 $\rightarrow$ 0.96 (distance concentration, regime (ii)). (6) Crossing times $T(1\%)/T(5\%)/T(10\%)$ in steps: 150k/350k/550k ($d_{\text{latent}} = 5$), 250k/550k/1.0M (8), 300k/700k/1.7M (10), 400k/1.0M/2.65M (12), 600k/2.1M/— (15), 1.05M (20), 2.55M (25).

Predictions and how to test them.

1. *One-constant fraction prediction (synthetic):* $F_{\text{mem}}(T) = 1 - G_{\text{NN}}(3\sqrt{d_{\text{latent}}}\Delta_{\text{term}} + c\mathcal{E}_t(T))$ with $\mathcal{E}_t(T)$ measured against the closed form (49) and G_{NN} from the training set; c is fixed by the attachment time (G2) and is the one constant. Test: requires re-running training with checkpoints (none are saved); curves for different d_{latent} should collapse; systematic over-prediction indicates (B2)/(B3) fail.

2. *Endpoint baseline at $\sigma_{\perp} = 0.01$* : compare longer learned-score runs with the exact-score sampler baseline, not an upper bound. Test: train to 5M steps and separately lower $t_{\min}$ or add a terminal Tweedie step; measure whether the baseline and learned fractions change together.
3. *$d_{\text{intrinsic-not-}d_{\text{latent}}}$ scaling of the data-side threshold*: $d_{\text{NN}} n^{1/d_{\text{intrinsic}}}$ is constant in the signal-dominated regime, so at $\sigma_{\perp} \rightarrow 0$ the threshold scales as $n^{-2/d_{\text{intrinsic}}}$ and is d_{latent} -independent; denser data memorises later by the extra term $+1/(d_{\text{intrinsic}} \lambda_{\text{samp}} \tau_{\text{mem}})$ in $\partial \log \tau_{\text{mem}} / \partial \log n$. Test: sweep $n \in \{100, 500, 1000, 5000\}$ and $d_{\text{intrinsic}} \in \{3, 5, 8\}$ at $\sigma_{\perp} = 0.01$.
4. *Fraction-curve width*: the data-level spread $d_{\text{NN}}^{(0.9)} / d_{\text{NN}}^{(0.1)}$ falls $2.28 \rightarrow 1.09$ over $d_{\text{latent}} = 5 \rightarrow 200$ at $\sigma_{\perp} = 0.5$ and stays at 2.3 at $\sigma_{\perp} = 0.01$, but the measured $T(5\%) / T(1\%) = 2.2\text{--}3.5$ alone cannot distinguish distance spread from spectral or endpoint-radius spread without estimating $\lambda_{\text{samp}} T(1\%)$. Test: extract $\tau_{\text{mem}}(q)$ for $q \in \{1, 2, 5\%\}$, normalise by the RFNN sample-bulk $1/\lambda$ and compare the residual spread with $\log(d_{\text{NN}}^{(0.99)} / d_{\text{NN}}^{(0.95)})$.
5. *The exact empirical score memorises at the endpoint baseline for every d_{latent}* , the opposite of the learned-score trend. Test: run the closed-form s_{emp} sampler for each d_{latent} ([R] done by a referee: $0.508 \rightarrow 0.960$ over $d_{\text{latent}} = 5 \rightarrow 40$ at $\sigma_{\perp} = 0.5$).

Open gaps. (G1) *Residual model*. (B2) is stated as a concentration hypothesis with its justification and failure windows, not proved. (G2) *Attachment time and multi-step accumulation*. c_t is not determined by the proposition: numerical integration of the offset equation with a persistent relative error $r^2 = \mathcal{E}_s \Delta_s / d_{\text{latent}}$ gives per-coordinate displacement variance (units $\Delta_{t_{\min}} r^2$) 1.02 for Tweedie at $t_{\min}$, 4.9/26/338 for Tweedie at $t_c = 0.046/0.21/1.02$, 0.61/2.64/8.42 ODE-integrated, and 1.18/2.58/3.57 SDE-integrated (the sampler actually used); the constant is d_{latent} -dependent through t_c and the earlier “accurate to a logarithmic factor” understates the spread. (G3) The RFNN spectrum is at $t = 0.01$ while attachment occurs near t_c ; relating $\mathcal{E}_{t_c}(T)$ to the fixed- t spectrum is not done. (G4) NN_1 equals the attachment point only for $\|\delta\| < d_{\text{NN}}/2$; the non-memorized tail of ϱ is not modelled. (G5) The projection floor $\mathcal{E}_t(\infty) > 0$ is not computed; the return of memorisation as $d_{\text{latent}} \rightarrow n$ is outside the bridge (Appendix I.8). (G6) The per-sample residual $\mathcal{E}_{t,\mu}$ is not derived from the sample-bulk modes; isolated points may carry the least-supported, smallest- λ_i modes, partially cancelling “most isolated first”. (G7) L/ℓ of the decoders is unknown. (G8) Lemma I.48 is rigorous with $32 \log(n/\delta)$; that condition is never met on the sampler interval where memorisation is largest ($e^{-t} d_{\text{NN}} / \sqrt{\Delta_t} = 7.4$ vs the required 16.5 at $d_{\text{latent}} = 5$, $t = t_{\min}$), so the sketch’s step “(B1) $\Rightarrow r$ negligible” is not a consequence of the lemma; the $\eta = 0$ form is not a typical- η criterion ($\mathbb{E}_{\eta}[w_{\nu}/w_{\mu}] = 1$ identically), and a typical- η criterion needs $e^{-2t} d_{\text{NN}}^2 / (2\Delta_t) \gtrsim 5.8 \log n$. (G9) The released sweeps use ϱ_{B} (Definition I.47) while the statements here carry the constant 9 of the other form; every baseline estimate quoted here shifts when $9 \rightarrow 8$. The older $\sigma_{\perp} = 0.01$ sweep also runs on a different step clock ((39)), so its anchors are not like-for-like. (G10) [R] The E–M terminal variance is $1.36 \Delta_{t_{\min}}$, not $\Delta_{t_{\min}}$, and the $\chi_{d_{\text{latent}}}^2$ spread is $O(1)$ at $d_{\text{latent}} \leq 40$; concentrated baseline estimates quoted to three digits are not meaningful there. (G11) [R] The zero-parameter test (Prediction 1) is not executable from the repository: no score-model checkpoints are saved. (G12) The label `prop:bridge` must be issued once; the main text should cite this statement rather than restate it.

I.6 FROM THE FROZEN-LATENT SPECTRUM TO A MEMORISATION-FRACTION PREDICTOR

Throughout, $\phi(x) = p^{-1/2} \tanh(Wx / \sqrt{d_{\text{latent}}})$ with $W_{ij} \sim \mathcal{N}(0, 1)$, $x_t = e^{-t}x + \sqrt{\Delta_t} \eta$, $M_t = \frac{1}{n} \sum_{\mu} \mathbb{E}[x_t^{\mu} x_t^{\mu \top}]$, $q_d = \text{tr}(M_t) / d_{\text{latent}}$, and $\mathbf{U} = \frac{1}{n} \sum_{\mu} \mathbb{E}[\phi(x_t^{\mu}) \phi(x_t^{\mu})^{\top}]$; “code units” means $p \times$ theory units. This subsection derives what the released spectral predictor (Section 5) actually measures, supplies the sample-bulk term it lacks, and resolves the sample bulk per training point, which is what a fraction-level prediction needs (Definition I.11(iii)).

I.6.1 GAUSSIAN-EQUIVALENCE COEFFICIENTS

Lemma I.54 (Identities proved; monotonicity checked numerically). *Let $\sigma = \tanh$, $z \sim \mathcal{N}(0, 1)$, $q > 0$, $a_1(q) = \mathbb{E}[\sigma'(\sqrt{q}z)]$, $a_*(q)^2 = \mathbb{E}[\sigma(\sqrt{q}z)^2] - a_1(q)^2 q$, and $\sigma(\sqrt{q}z) = \sum_{k \text{ odd}} c_k(q) \text{He}_k(z) / k!$ (unnormalised Hermite coefficients; c_k here equals $\sqrt{k!}$ times the c_k of*

Lemma I.13. Then (i) $\mathbb{E}[\sigma(\sqrt{q}z)] = \sqrt{q} a_1(q)$; (ii) $a_*(q)^2 = \mathbb{E}[(\sigma(\sqrt{q}z) - a_1(q)\sqrt{q}z)^2] \geq 0$; (iii) $a_1^2(q) = c_1^2$ and $a_*^2 = \sum_{k \text{ odd} \geq 3} c_k^2/k!$; (iv) $a_*^2 = \frac{2}{3}q^3 + O(q^4)$, numerically increasing on the evaluated grid in $[0.05, 3.5]$, with $a_*^2 = 0.0047, 0.0274, 0.0359, 0.0544, 0.0784$ and $a_1^2 = 0.623, 0.367, 0.320, 0.246, 0.180$ at $q = 0.33, 1, 1.247, 1.834, 2.76$; (v) for jointly Gaussian (z, z') with correlation $\varrho \in [0, 1]$,

$$g(q, \varrho) := \mathbb{E}[\sigma(\sqrt{q}z)\sigma(\sqrt{q}z')] - a_1(q)^2 q \varrho = \sum_{k \text{ odd} \geq 3} \frac{c_k(q)^2}{k!} \varrho^k \in [0, \varrho^3 a_*(q)^2],$$

increasing in ϱ with $g(q, 1) = a_*^2$. With $r_t(q) := g(q, 1 - \Delta_t/q)/a_*(q)^2$ (the $\theta(q)$ of Lemma I.24(v)): at $t = 0.1$, $r_t = 0.086, 0.505, 0.574, 0.675, 0.755$ at $q = 0.33, 1, 1.247, 1.834, 2.76$; at $t = 0.01$, $r_t = 0.822, 0.943, 0.969$ at $q = 0.33, 1.247, 2.76$. $g(q, 1 - \Delta_t/q)$ is numerically increasing on the evaluated grid for both t ; no continuous-interval proof is supplied.

Proof. (i) Stein's lemma. (ii) Expand and use (i). (iii) $\tanh$ is odd; orthogonality with $\mathbb{E}\text{He}_k^2 = k!$. (iv) $\mathbb{E}\sigma^2 = q - 2q^2 + \frac{17}{3}q^3 + O(q^4)$ and $a_1^2 q = q - 2q^2 + 5q^3 + O(q^4)$; the table is 120-node Gauss-Hermite quadrature and reproduces the $0.180 \rightarrow 0.625$ range of a_1^2 of Lemma I.6. (v) Mehler's formula $\mathbb{E}[\text{He}_k(z)\text{He}_l(z')] = \delta_{kl} k! \varrho^k$; the subtraction removes $k = 1$, oddness the even terms; each term is nonnegative and increasing in ϱ , and $\varrho^k \leq \varrho^3$. The r_t values agree between two-dimensional quadrature and the Hermite series truncated at $k = 39$ to four digits. $\square$

I.6.2 MODE CLOCKS UNDER GRADIENT FLOW

Proposition I.55 (Exact mode flow; conditional pointwise interpretation). Let $L(A) = \frac{1}{2n} \sum_{\mu} \mathbb{E}_{\xi} \|A\phi(x_t^{\mu}) - y_{\mu}(x_t^{\mu})\|^2$ for square-integrable targets y_{μ} , $U = \sum_i \lambda_i u_i u_i^{\top}$, $V = \frac{1}{n} \sum_{\mu} \mathbb{E}_{\xi} [y_{\mu} \phi^{\top}]$, and let $\dot{A} = -\nabla L(A)$, $A(0) = 0$, $T = \eta s$ ($s = \text{steps}$). With $a_i(T) = A(T)u_i$ and $a_i^* = Vu_i/\lambda_i$ ($\lambda_i > 0$),

$$a_i(T) = a_i^* (1 - e^{-\lambda_i T}), \quad L(T) - L(\infty) = \frac{1}{2} \sum_i \lambda_i \|a_i^*\|^2 e^{-2\lambda_i T}.$$

Mode i thus has absorbed amplitude $1 - e^{-\eta \lambda_i s}$, absorbed loss $1 - e^{-2\eta \lambda_i s}$ and learned-energy fraction $(1 - e^{-\eta \lambda_i s})^2$, ordered $(1 - e^{-x})^2 \leq 1 - e^{-x} \leq 1 - e^{-2x}$. Changing coordinates preserves the exponent only if the optimizer scale is transformed with them. For the repository's optimized A and normalized ϕ , $T = (2\Delta_t \eta_{\text{lr}}/d_{\text{latent}})s$ and $\lambda_i T = (2\Delta_t \eta_{\text{lr}}/d_{\text{latent}} p) \lambda_{i, \text{code}} s$. If u is a unit eigenvector localised on training point μ , $u \propto \bar{\rho}_{\mu} = \mathbb{E}_{\xi} \rho(x_t^{\mu})$ with eigenvalue $\Lambda_{\mu} = \|\bar{\rho}_{\mu}\|^2/n$ (Proposition I.59), and $\bar{y}_{\mu}$ is the sample-specific target, then at leading order $a^* = \bar{y}_{\mu}/\|\bar{\rho}_{\mu}\|$ and the model output at x_t^{μ} along u is $\bar{y}_{\mu}(1 - e^{-\Lambda_{\mu} T})$; the amplitude fraction is the fraction of the sample-specific score output produced at the training point.

Proof. $\nabla_A L = AU - V$; project on u_i and integrate (this is Lemma I.3). The loss identity is the quadratic expansion about $A^* = VU^+$. Units follow by differentiating the logged loss as in Remark I.7; reparametrizing A without changing the learning rate would define a different optimization. Meaning: only the $\nu = \mu$ term of V correlates with u at leading order, so $Vu = \bar{y}_{\mu} \bar{\rho}_{\mu}^{\top} \bar{\rho}_{\mu}/(n \|\bar{\rho}_{\mu}\|) = \bar{y}_{\mu} \|\bar{\rho}_{\mu}\|/n$ and $a(T) u^{\top} \bar{\phi}_{\mu} = a^* (1 - e^{-\Lambda_{\mu} T}) \|\bar{\rho}_{\mu}\|$. For the RFNN, each fresh-noise loss is an unbiased estimate of the averaged objective. The mean iterate follows its Euler recursion; the loss of that mean is not generally the expected loss of the stochastic iterates. $\square$

Remark I.56 (The square is not derived). The released clock $P_d(s) = \text{mean}_i (1 - e^{-\kappa \lambda_i (M_t)s})^2$ is the learned-energy fraction averaged over the eigenvalues of M_t rather than of U . For a single mode the three fractions are monotone reparametrisations of $\eta \lambda_i s$, so the exponent is absorbed into the threshold; it matters only through the average over modes, where the square suppresses slow modes at early times quadratically. A derivation would have to show that the NN-ratio memorisation statistic is a quadratic functional of the learned sample-specific displacement at small absorbed amplitude; we do not have one, and use the amplitude clock, which has the output-space meaning above.

I.6.3 DATA-SIDE RATES: WHAT $\text{spec}(M_t)$ LABELS

Proposition I.57 (Conditional data-term approximation). *In this subsection W denotes the already-scaled code weights and $\phi(x) = p^{-1/2} \tanh(Wx)$. Let $W_{ij} \sim \mathcal{N}(0, \sigma_w^2)$, $\gamma := d_{\text{latent}} \sigma_w^2$ ($\gamma = 1$ for the RFNN), $p/d_{\text{latent}} \rightarrow \infty$, and assume Gaussian equivalence of the $\tanh$ random-feature kernel for the input distribution (Pennington & Worah, 2017; Benigni & P    , 2021; Hu & Lu, 2023; Goldt et al., 2022). With concentrated input norms and a negligible nonlinear perturbation on the data-side eigenspaces, the candidate data-side eigenvalues are $\lambda_i^{\text{data}} \simeq \gamma a_1(\gamma q_d)^2 \lambda_i(M_t)/d_{\text{latent}} (1 + o(1))$, $i \leq d_{\text{latent}}$, with trace $a_1(\gamma q_d)^2 \gamma q_d$. The data-side amplitude clock is $1 - \exp[-\eta \gamma a_1(\gamma q_d)^2 \lambda_i(M_t)s/d_{\text{latent}}]$, so a corresponding effective κ would be $\eta \gamma a_1(\gamma q_d)^2/d_{\text{latent}}$ in theory units; for the default coordinate-averaged RFNN optimizer the per-step rate is $0.02 \gamma a_1(\gamma q_d)^2 \lambda_i(M_t)/d_{\text{latent}}$. The explicit dimension divisor remains (Remark I.7).*

Proof sketch. Decompose $\tanh(w^\top x) = a_1(q_x)w^\top x + r_w(x)$, $q_x = \gamma \|x\|^2/d_{\text{latent}}$; by Lemma I.54(i) the residual is uncorrelated with the linear part over w for fixed inputs; neglecting the realized operator cross term additionally requires an operator-norm approximation, assumed here. The linear block is $\sigma_w^2 W' \tilde{M}_t W'^\top/p$, where $W' = W/\sigma_w$ with $\tilde{M}_t = \frac{1}{n} \sum_\mu a_1(q_{x_t^\mu})^2 \mathbb{E} x_t^\mu x_t^{\mu\top}$; replacing $a_1(q_{x_t^\mu})$ by $a_1(\gamma q_d)$ is a heterogeneity correction of relative size $\text{cv}(q_\mu)|a_1'|q/a_1$, and the nonzero spectrum of $W' \tilde{M}_t W'^\top/p$ is that of $\tilde{M}_t(W'^\top W'/p) \rightarrow \tilde{M}_t$. This is a data-term approximation motivated by prior random-feature analyses (Bonnaire et al., 2025; George et al., 2026); it does not establish the full nonlinear spectrum (Remark I.16). Finite- p corrections are the $O(\sqrt{d_{\text{latent}}/p})$ Marchenko–Pastur broadening of $W'^\top W'/p$ (Marchenko & Pastur, 1967), consistent with the $< 0.5\%$ median motion under a $5.8\times$ change of p . Trace: $\gamma a_1^2 \text{tr} M_t/d_{\text{latent}} = a_1^2 \gamma q_d$. $\square$

Remark I.58 (What a clock built from $\text{spec}(M_t)$ measures). The d_{latent} eigenvalues of M_t label the signal and noise-dimension bulks of U , whose targets are population-score components. The $n - d_{\text{latent}}$ sample modes, which carry the sample-specific components and hence memorisation, are absent from $\text{spec}(M_t)$. The released $P_d(s)$ therefore measures absorption of the generalising modes and tracks memorisation only insofar as their clock co-varies with the sample-mode clock, which happens through q_d (Corollary I.62(iii)). The two clocks couple to posterior collapse through different channels: the sample-mode clock through q -dilution—the channel of the synthetic $\sigma_\perp \rightarrow 0$ buffer—and the released clock through the collapsed eigenvalues $\lambda_i = \Delta_t = 0.181$ themselves ($d_{\text{latent}} \geq 140$ CelebA, ≥ 160 CIFAR-10; excess $< 2 \times 10^{-3}$), its slowest modes, which dominate the late-time growth of $P_d(s)$ although their score target is $-x_i/\Delta_t$. CelebA’s effective rank saturates (89.5, 90.3, 87.9 at $d_{\text{latent}} = 160, 180, 200$) while memorisation keeps falling; a clock reading $\text{spec}(M_t)$ reproduces this only through the growing number of collapsed coordinates, a channel carrying no data-side information, whereas the sample-mode clock reproduces it through $q_d = 1.83 \rightarrow 1.25$.

I.6.4 PER-SAMPLE RESOLUTION OF THE SAMPLE BULK

Proposition I.59 (Residual covariance identity and conditional spectral approximation). *In the setting of Proposition I.57 with $\gamma = 1$ and $p > n$, let $\rho(x) = \phi(x) - p^{-1/2} a_1(q_x)Wx$, $q_\mu = e^{-2t} \|x^\mu\|^2/d_{\text{latent}} + \Delta_t$, $\varrho_\mu = 1 - \Delta_t/q_\mu$, $\bar{\rho}_\mu = \mathbb{E}_\xi \rho(x_t^\mu)$. Distinguish the single-draw $U^{(1)} = \frac{1}{n} \sum_\mu \phi(x_t^\mu) \phi(x_t^\mu)^\top$ from the noise-averaged U (the training loss; the $n_\xi \rightarrow \infty$ limit of $\text{compute-}U$, which averages $n_\xi = 50$ draws). Then:*

1. $\mathbb{E}_W \|\rho(x)\|^2 = a_*(q_x)^2$ exactly, with relative concentration $p^{-1/2}$, and $\mathbb{E}_{W, \xi} \|\rho(x_t^\mu)\|^2 = a_*(q_\mu)^2 (1 + O(d_{\text{latent}}^{-1}))$.
2. For $\varrho_{xy} = x^\top y / (d_{\text{latent}} \sqrt{q_x q_y})$: $\mathbb{E}_W [\rho(x)^\top \rho(y)] = \sum_{k \text{ odd} \geq 3} c_k(q_x) c_k(q_y) \varrho_{xy}^k / k! = O(\varrho_{xy}^3)$; two noise draws of the same point have overlap ϱ_μ , so $\mathbb{E}_W \|\bar{\rho}_\mu\|^2 = g(q_\mu, \varrho_\mu) (1 + O(d_{\text{latent}}^{-1/2})) = r_t(q_\mu) a_*(q_\mu)^2 (1 + O(d_{\text{latent}}^{-1/2}))$.
3. $U_* = \frac{1}{n} \sum_\mu \mathbb{E}_\xi [\rho \rho^\top] = U_{\text{loc}} + U_{\text{cloud}}$, $U_{\text{loc}} = \frac{1}{n} \sum_\mu \bar{\rho}_\mu \bar{\rho}_\mu^\top$, $U_{\text{cloud}} = \frac{1}{n} \sum_\mu \text{Cov}_\xi(\rho_\mu) \succeq 0$. For individual localization, additionally assume the off-diagonal Gram perturbation and cloud operator norm are small relative to the relevant diagonal eigenvalue gaps. These

conditions are not established here. Under them, $\mathbf{U}_{\text{loc}}$ has one eigenvalue per training point, eigenvector $\propto \bar{\rho}_\mu$,

$$\Lambda_\mu = \frac{r_t(q_\mu) a_*(q_\mu)^2}{n}, \quad (59)$$

and $\mathbf{U}_{\text{cloud}}$ has trace $\approx (1 - r_t(q_d))a_*(q_d)^2$ over up to p directions (at finite n_ξ , at most $\min\{p, (n_\xi - 1)n\}$), but this trace and rank bound alone gives no per-direction operator bound. A diffuse cloud of operator norm $O((1 - r_t)a_*^2/p)$ is an additional assumption. For $\mathbf{U}^{(1)}$ the same argument with $\varrho_\mu \rightarrow 1$ gives $\Lambda_\mu^{(1)} = a_*(\hat{q}_\mu)^2/n$, no cloud and rank $\leq n$. Trace check for both objects: $a_1^2 q_d + r_t a_*^2 + (1 - r_t)a_*^2 = \mathbb{E} \tanh^2(\sqrt{q_d} z)$.

4. A relative spread $\sqrt{n/p}$ and MP edges $(1 \pm \sqrt{n/p})^2$ are possible surrogate diagnostics, not consequences of random signs or entrywise bounds alone. Their use requires control of dependence and operator norms. Small off-diagonal entries do not imply individual eigenvectors when diagonal values are nearly degenerate. Without individual gaps, a quantile comparison requires a separate empirical or spectral-distribution check. The numerical comparisons below evaluate this approximation; they do not prove localization. The regime $p < n$ (including $d_{\text{latent}} = 5$ for $p = 64d_{\text{latent}}$) is outside the proposed pointwise attachment.

Proof sketch. (1) Condition on x ; row j contributes $p^{-1}[\tanh(\sqrt{q_x} z_j) - a_1 \sqrt{q_x} z_j]^2$, z_j i.i.d. standard Gaussian; Lemma I.54(ii) and the LLN over rows. Over ξ , $\|x_t^\mu\|^2/d_{\text{latent}}$ concentrates at q_μ with $O(d_{\text{latent}}^{-1/2})$ fluctuations. (2) Mehler, Lemma I.54(v); for two draws of one point $x_t(\xi)^\top x_t(\xi')/d_{\text{latent}} = e^{-2t}\|x^\mu\|^2/d_{\text{latent}} + O(d_{\text{latent}}^{-1/2})$, so $\varrho = \varrho_\mu$. (3) Variance identity $\mathbb{E}[\rho\rho^\top] = \bar{\rho}\bar{\rho}^\top + \text{Cov}(\rho)$; the nonzero eigenvalues of $\mathbf{U}_{\text{loc}}$ are those of the Gram $\bar{G}/n$, $\bar{G}_{\mu\nu} = \bar{\rho}_\mu^\top \bar{\rho}_\nu = \text{diag}(r_t a_*^2) + E$ with $|E_{\mu\nu}| = O(|\varrho_{\mu\nu}|^3)$, $\varrho_{\mu\nu} = e^{-2t} x^\mu^\top x^\nu / (d_{\text{latent}} \sqrt{q_\mu q_\nu})$. (4) The displayed covariance and Gram identities are exact; eigenvalue attachment additionally needs an operator perturbation bound, and eigenvector attachment needs eigenvalue gaps. Entrywise concentration alone establishes neither an MP law nor a semicircle bound here.

Numerical check (repository spectra, $\sigma_\perp = 0.5$, $p = 64d_{\text{latent}}$, $n = 500$, $t = 0.01$, $n_\xi = 50$, seed 42, code units). At $d_{\text{latent}} = 200$: sample-bulk median 0.093 vs. $pr_t a_*^2/n = 0.098$ (single-draw formula: 0.119); sample-bulk mass 28.9 vs. $(n - d_{\text{latent}})pr_t a_*^2/n = 29.4$; mass beyond index n 11.7 vs. cloud $(1 - r_t)pa_*^2 = 10.8$; $\lambda_{n-1}/\lambda_n = 1.7$. At $d_{\text{latent}} = 150, 100$: medians 0.077, 0.059 vs. 0.084, 0.070. Below $d_{\text{latent}} \approx 60$ the per-sample prediction fails progressively (0.032 vs. 0.057 at $d_{\text{latent}} = 40$; 0.0018 vs. 0.076 at $d_{\text{latent}} = 5$): the mass migrates into a few collective modes (Remark I.64). $\square$

Remark I.60 (Single-draw versus noise-averaged objects). The machine-precision rank cliff at index n is a property of $\mathbf{U}^{(1)}$. The saved spectra use $n_\xi = 50$ draws, so rank $\leq \min\{p, n_\xi n\} = p$: all p eigenvalues exceed 10^{-8} at every d_{latent} and λ_{n-1}/λ_n is 1.01 at $d_{\text{latent}} = 10, 20$ and 1.7 at $d_{\text{latent}} = 100, 200$ —a shoulder whose position and mass are those of the cloud. The training loss is the noise-averaged object, so under the additional localization assumptions the candidate rate of a training point is $r_t(q_\mu)a_*(q_\mu)^2/n$; r_t is 0.82–0.97 on the synthetic sweep ($t = 0.01$) and 0.09–0.76 at $t = 0.1$, the diffusion time of the real-data spectra. Cloud modes carry the part of the residual feature not reproducible across noise draws; their absorption must be evaluated from their eigenvalues and target weights at the correctly normalized training horizon.

I.6.5 THE TWO-SOURCE FRACTION PREDICTOR

Definition I.61. Given latents $\{z^\mu\}_{\mu \leq n}$, diffusion time t , an odd first-layer activation σ and gain $\gamma = d_{\text{latent}}\sigma_w^2$ read from the initialisation, set $q_\mu = \gamma(e^{-2t}\|z^\mu\|^2/d_{\text{latent}} + \Delta_t)$, $\varrho_\mu = 1 - \gamma\Delta_t/q_\mu$, $\Lambda_\mu = g(q_\mu, \varrho_\mu)/n$ (with g for σ), $d_{\text{NN}}(\mu) = \min_{\nu \neq \mu} \|z^\mu - z^\nu\|$ and $\mathcal{D}_{t_c} = \{\mu : d_{\text{NN}}(\mu)^2 > 2e^{2t_c}\Delta_{t_c} \log n\}$ (Biroli et al., 2024). The predicted memorised fraction at step s is

$$\hat{F}(s) = \frac{1}{n} \sum_{\mu} \mathbf{1}[\mu \in \mathcal{D}_{t_c}] \mathbf{1}[\Lambda_\mu s \geq C], \quad C = \log \frac{1}{1-\theta}/\eta,$$

optionally with the second indicator replaced by $\Phi(\log(\Lambda_\mu s/C)/\varsigma)$, $\varsigma = \sqrt{n/p}$ (not fitted). θ is the fraction of the sample-specific score output that must be produced for the sampler to return z^μ

(Proposition I.55). If the first layer also receives a fixed-norm time embedding $e(t) \in \mathbb{R}^{m_e}$ with fan-in init $\sigma_w^2 = c/(d_{\text{latent}} + m_e)$, then $q_\mu = c(e^{-2t}\|z^\mu\|^2 + d_{\text{latent}}\Delta_t + \|e(t)\|^2)/(d_{\text{latent}} + m_e)$ and $\varrho_\mu = 1 - c\Delta_t d_{\text{latent}}/((d_{\text{latent}} + m_e)q_\mu)$. The center-separation filter is a heuristic based on the zero-offset condition, not a pathwise guarantee. The intersection models a point as returned only if its sample-specific component is learned and wins the softmax competition of the empirical score (Lemma I.48); under approximate independence $\hat{F}(s) \approx (|\mathcal{D}_{t_c}|/n) \cdot \frac{1}{n} \#\{\mu : \Lambda_\mu s \geq C\}$. Hypotheses: $p > n$; lazy first layer; the decoder approximately preserves nearest-neighbour ordering ($\mathcal{D}_{t_c}$ is latent-space, the statistic is pixel-space; Remark I.53).

Corollary I.62 (Consequences of the heuristic predictor). *For the unsmoothed predictor of Definition I.61, hold $\mathcal{D}_{t_c}$, γ and t fixed. Order the eligible rates as $\Lambda_{[1]} \geq \dots \geq \Lambda_{[m]}$, $m = |\mathcal{D}_{t_c}|$. For $0 < f \leq m/n$, the first crossing is $s_f = C/\Lambda_{[fn]}$ (assuming positive rates). Under the numerically observed monotonicity of $g(q, 1 - \gamma\Delta_t/q)$, this orders eligible points by decreasing q_μ , hence latent norm. This is a consequence of the predictor, not a theorem about the trained sampler.*

At fixed gain and diffusion time, collapsed coordinates affect these candidate rates through q_μ and the corresponding $\varrho_\mu = 1 - \gamma\Delta_t/q_\mu$. In the CelebA mean-norm approximation at $d_{\text{latent}} = 200$, removing collapsed coordinates at unchanged gain raises q from 1.247 to 2.60 and increases g by about $2.7\times$. Rescaling gain to preserve q generally changes ϱ and therefore g ; invariance requires preserving both. The main synthetic fixed- q control keeps gain and diffusion time unchanged.

Proof. The unsmoothed threshold counts eligible rates satisfying $\Lambda_\mu s \geq C$; at least $\lceil fn \rceil$ must cross. The ordering by q additionally uses the stated monotonicity assumption. Dependence on gain follows directly from the definitions of q_μ and ϱ_μ . $\square$

Remark I.63 (Parameter count and fit sets). Released: one global κ (121-point $\log_{10}$ grid on $[-12, -4]$, step $1.17\times$) plus six PAVA threshold levels, 7 fitted numbers per dataset, fitted on *all uncensored* rows including pre-threshold dims ($d_{\text{latent}} = 20, 30$ CelebA; 40 CIFAR-10) with no hold-out; the main text's (κ, a, b) sigmoid is not what the code fits. Derived: γ (0 fitted), ς (0), $\mathcal{D}_{t_c}$ (0 or 1), one scalar C (the single unknown θ). All one-constant fits below use $C = \exp\{\text{median}_{d_{\text{latent}}} \log(\Lambda_{(f)} \hat{s}_f)\}$ on declared dim sets.

Practitioner recipe. Inputs: frozen encoder, $n \approx 10^3$ training latents $Z \in \mathbb{R}^{n \times d_{\text{latent}}}$, diffusion time t , the diffusion network's first-layer activation, fan-in and initialisation variance (giving γ, c, m_e), learning rate η , sampler condensation time t_c .

1. q_μ and ϱ_μ from Definition I.61 (surrogate: $q_\mu = e^{-2t}\|z^\mu\|^2/d_{\text{latent}} + \Delta_t$).
2. $g(q, \varrho)$ by two-dimensional Gauss–Hermite quadrature (40–80 nodes per axis): $\sum_{jk} w_j w_k \sigma(\sqrt{q}x_j) \sigma(\sqrt{q}(\varrho x_j + \sqrt{1 - \varrho^2}x_k)) - a_1(q)^2 q \varrho$ (for a non-odd σ subtract also $a_0^2, a_0 = \mathbb{E}\sigma(\sqrt{q}z)$); $\Lambda_\mu = g(q_\mu, \varrho_\mu)/n$.
3. $d_{\text{NN}}(\mu)$ in latent space; $\mathcal{D}_{t_c} = \{d_{\text{NN}}^2 > 2e^{2t_c}\Delta_t \log n\}$.
4. Calibrate C once as the exponential of the median over training dims of $\log(\Lambda_{(5\%)} \hat{s}_{5\%})$; report $\hat{s}_f$ for all levels and all d_{latent} from the same C .

Validation protocol and current status. *Leave-one- d_{latent} -out.* Fit C on all dims of a declared set but one; predict the held-out hitting times; score by $|\log \hat{s}_f/s_f|$. *Cross-dataset.* Fit C on CelebA, predict CIFAR-10. *Collapsed-coordinate test.* Retrain on latents with collapsed coordinates removed, with and without gain rescaling (Corollary I.62(iii)); requires new runs. *Ordering test.* Memorised training points should have larger $\|z^\mu\|$ early in training; requires logging the NN1 index.

Status on the hitting-time tables (Table 7) (mean $|\log \text{residual}|$, in-sample / leave-one-out; released 7-parameter in-sample on the same set). CelebA 5%, $d_{\text{latent}} \geq 50$ (11 dims): g -clock 0.309/0.407; a^2 -clock 0.347/0.449; $\tau \propto d_{\text{latent}}$ 0.125/0.125; released 0.339. CelebA 25%, $d_{\text{latent}} \geq 50$: 0.207/0.262; 0.246/0.308; 0.083/0.095; released 0.169. CelebA 5%, $d_{\text{latent}} \geq 70$ (9 dims): 0.277/0.384 vs. released 0.311. CelebA 5%, all uncensored $d_{\text{latent}} \geq 20$ (14): 0.346/0.417 vs. released 0.365, with $\tau \propto d_{\text{latent}}$ the worst model (0.407/0.477). CIFAR-10 5%, $d_{\text{latent}} \geq 60$ (11): 0.130/0.151 vs. released 0.146 ($\tau \propto d_{\text{latent}}$ 0.155/0.198); $d_{\text{latent}} \geq 100$: 0.120/0.145 vs. 0.116; $d_{\text{latent}} \geq 140$ (7): 0.112/0.136 vs. 0.106. CIFAR-10 25%, $d_{\text{latent}} \geq 60$ (8): 0.168/0.171 vs. 0.143. The g -clock is thus comparable to the released fit on CIFAR-10 and somewhat worse on CelebA, with

Table 7: Empirical hitting times on the real-data sweeps: first evaluation step (in millions, mean $\pm$ sd over seeds) at which the pixel-space memorized fraction exceeds the threshold; c = seeds censored at the 5M-step horizon. Source: `{celeba, cifar10}_empirical_tau_summary.csv` in the analysis repository. These are the “hitting-time tables” used throughout Appendices I.6 and I.7.

CelebA ($n = 1000$)				CIFAR-10 ($n = 1000$)			
d_{latent}	thr. 0.01	0.05	0.10	d_{latent}	thr. 0.01	0.05	0.10
10	– ($c=5$)	– ($c=5$)	– ($c=5$)	20	1.90 $\pm$ 0.19	– ($c=5$)	– ($c=5$)
20	0.48 $\pm$ 0.04	1.34 $\pm$ 0.05	– ($c=5$)	40	0.40 $\pm$ 0.00	1.42 $\pm$ 0.13	– ($c=5$)
30	0.30 $\pm$ 0.00	0.50 $\pm$ 0.00	0.76 $\pm$ 0.05	60	0.30 $\pm$ 0.00	0.54 $\pm$ 0.05	0.90 $\pm$ 0.00
40	0.20 $\pm$ 0.00	0.34 $\pm$ 0.05	0.50 $\pm$ 0.00	80	0.30 $\pm$ 0.00	0.66 $\pm$ 0.05	0.98 $\pm$ 0.04
50	0.20 $\pm$ 0.00	0.30 $\pm$ 0.00	0.40 $\pm$ 0.00	100	0.32 $\pm$ 0.04	0.72 $\pm$ 0.04	1.12 $\pm$ 0.04
60	0.20 $\pm$ 0.00	0.30 $\pm$ 0.00	0.40 $\pm$ 0.00	120	0.48 $\pm$ 0.04	1.02 $\pm$ 0.08	1.44 $\pm$ 0.09
70	0.20 $\pm$ 0.00	0.30 $\pm$ 0.00	0.40 $\pm$ 0.00	140	0.66 $\pm$ 0.05	1.10 $\pm$ 0.10	1.46 $\pm$ 0.11
80	0.20 $\pm$ 0.00	0.32 $\pm$ 0.04	0.50 $\pm$ 0.00	160	0.68 $\pm$ 0.04	1.04 $\pm$ 0.05	1.26 $\pm$ 0.05
90	0.20 $\pm$ 0.00	0.40 $\pm$ 0.00	0.58 $\pm$ 0.04	180	0.66 $\pm$ 0.05	1.02 $\pm$ 0.04	1.34 $\pm$ 0.11
100	0.20 $\pm$ 0.00	0.40 $\pm$ 0.00	0.64 $\pm$ 0.05	200	0.66 $\pm$ 0.05	1.18 $\pm$ 0.13	1.80 $\pm$ 0.32
120	0.30 $\pm$ 0.00	0.60 $\pm$ 0.00	0.90 $\pm$ 0.07	220	0.68 $\pm$ 0.04	1.42 $\pm$ 0.08	2.38 $\pm$ 0.15
140	0.40 $\pm$ 0.00	0.70 $\pm$ 0.00	0.92 $\pm$ 0.04	240	0.70 $\pm$ 0.00	1.50 $\pm$ 0.07	2.68 $\pm$ 0.13
160	0.54 $\pm$ 0.05	0.92 $\pm$ 0.08	1.14 $\pm$ 0.05	260	1.08 $\pm$ 0.68	2.52 $\pm$ 1.39	3.40 $\pm$ 0.24 ($c=1$)
180	0.72 $\pm$ 0.04	1.08 $\pm$ 0.04	1.36 $\pm$ 0.05	–	–	–	–
200	0.72 $\pm$ 0.04	1.16 $\pm$ 0.05	1.56 $\pm$ 0.05	–	–	–	–

no fitted response function and one constant. It accounts for $1.78\times$ of the observed $3.9\times$ (CelebA $d_{\text{latent}} = 50 \rightarrow 200$), $1.74\times$ of $2.9\times$ ($100 \rightarrow 200$) and $1.85\times$ of $2.1\times$ (CIFAR-10 $100 \rightarrow 240$; $d_{\text{latent}} = 260$ has $\tau_5 = 2520\text{k}$ with s.d. 1392k and is not used). The fitted C differs between datasets by $2.0\times$ (5%: 14.5 vs. 29.1) and $1.6\times$ (25%) with identical optimiser settings, so one-constant transfer fails and a dataset-dependent factor ($|\mathcal{D}_{t_c}|/n$, decoder distortion of d_{NN}) is missing. The width from the learning side alone is too narrow: at CelebA $d_{\text{latent}} = 200$ the Gaussian-moment estimate (mean 1.302, s.d. 0.2905 of $\|z\|^2/d_{\text{latent}}$; per-sample latents are not in the repository) gives a $1.46\times$ span between the levels matching 5% and 25% absolute memorisation (11% and 56% of the 0.448 asymptote) against the observed $\tau_{25}/\tau_5 = 2.14$.

Surrogate caveat. All real-data numbers use $\sigma = \tanh$, $\gamma = 1$, $m_e = 0$. The deployed network’s first layer is Linear($d_{\text{latent}} + 64, 1024$) with GELU on $[x_t; 32 \sin, 32 \cos]$ and PyTorch default init ($\sigma_w^2 = 1/(3 \text{fan_in})$), for which $q_\mu \approx (e^{-2t}\|z\|^2 + d_{\text{latent}}\Delta_t + 32)/(3(d_{\text{latent}} + 64)) = 0.36, 0.43, 0.36$ at CelebA $d_{\text{latent}} = 50, 100, 200$ and $0.43, 0.33$ at CIFAR-10 $d_{\text{latent}} = 100, 260$ —nearly d_{latent} -independent—and GELU is not odd (its $k = 0, 2$ Hermite terms add a rank-one spike and $O(q^2)$ off-diagonals; at $q = 1.834$ the proper GELU a_*^2 is 0.191 against 0.380 from the odd-activation formula). The derived predictor has therefore not been evaluated with the deployed network’s own first-layer statistics; one conjecture is that the surrogate reflects learned features, with only a loose analogy to one-step two-layer regression results (Ba et al., 2022); alternatively, the real-data d_{latent} -slope arises mainly on the dynamics side and from the NN-ratio detection drift.

Remark I.64 (Conjecture, numerically supported: cluster-collective modes at small d_{latent}). Code units, $\sigma_\perp = 0.5$, $p = 64d_{\text{latent}}$, $n = 500$, $t = 0.01$, seed 42. The per-sample picture predicts a sample-bulk median $pr_t a_*^2 / \min\{n, p\}$ rising $0.054 \rightarrow 0.098$ ($1.8\times$) and a top edge $pr_t a_*^2 (1 + \sqrt{n/p})^2 / n$ falling $0.193 \rightarrow 0.141$ ($1.4\times$) over $d_{\text{latent}} = 10 \rightarrow 200$ ($d_{\text{latent}} = 5$ has $p < n$). Observed: median $0.0039 \rightarrow 0.093$ ($24\times$), largest sample-bulk eigenvalue $0.930 \rightarrow 0.173$ ($5.4\times$; 0.957 at $d_{\text{latent}} = 5$); the top ten sample-bulk modes carry 47%, 35%, 22%, 6.3%, 5.4% of the sample-bulk mass at $d_{\text{latent}} = 5, 10, 20, 100, 200$ (flat reference 3.3%), with $\lambda_{\text{top}}/\lambda_{\text{median}} = 532, 236, 42, 3.4, 1.9$. At $d_{\text{latent}} \geq 100$ the per-sample+MP picture is quantitatively right; at small d_{latent} about $k = 10$ modes absorb half the mass, and the per-sample clock explains essentially none of the small- d_{latent} onset behaviour ($1.46\times$ predicted vs $221\times$ observed in code units over $d_{\text{latent}} = 5 \rightarrow 200$). Conjectured mechanism: same-cluster points have noise-averaged overlap $\varrho_c = e^{-2t} s^2 / (d_{\text{latent}} q_d) = 0.64, 0.58, 0.50, 0.23, 0.135$, so $\bar{G}$ acquires per cluster a block of n/k entries $g(q_d, \varrho_c)$ and k collective eigenvalues $\approx p[r_t a_*^2 + (n/k - 1)g(q_d, \varrho_c)]/n = 0.53, 0.50, 0.36, 0.11, 0.11$, reproducing the observed trend from $d_{\text{latent}} = 10$ within a uniform factor 1.6–1.9 (cross-cluster overlaps, also $O(1)$ at $d_{\text{latent}} = 5$, and the MP edge are omitted; this block can contribute at most $\sim 10\times$ at $d_{\text{latent}} = 5$, so the remainder is the g factor of Proposition I.27).

Such modes learn the nonlinear shape of a cluster and are generalising; if confirmed, the small- d_{latent} onset statistic is not a memorisation clock and the buffer claim must be re-based on the localised part of the sample bulk (the γ -cut of Definition I.11). Test: training-point-basis IPR of the top 20 sample-bulk eigenvectors (≈ 1 localised, $\approx k/n$ collective); requires re-running the sweep with eigenvectors saved.

Predictions and how to test them.

1. *Training points are memorised in decreasing order of latent norm.* Test: log the NN1 index of memorised samples; compare $\|z^\mu\|^2/d_{\text{latent}}$ of those training latents with the full distribution (norm-ranked ROC AUC > 0.5 early, $\rightarrow 0.5$ at saturation).
2. *Collapsed coordinates enter the memorisation clock only through q_μ :* dropping them at fixed gain makes CelebA $d_{\text{latent}} = 200$ memorise $\sim 2.7\times$ faster; gain rescaling at fixed q_μ also changes ϱ_μ , so the resulting curve must be recomputed; the released $P_d(s)$ moves in both cases. Test: re-encode and retrain with and without collapsed coordinates and with/without gain rescaling.
3. *The one-constant sample-mode clock $\tau_f(d_{\text{latent}}) = Cn/g(q_d, 1 - \Delta_t/q_d)$ reproduces hitting times with leave-one-out log-error comparable to the paper's 7-parameter in-sample fit but under-predicts the d_{latent} -slope ($1.78\times$ of $3.9\times$, CelebA $50 \rightarrow 200$).* Test: executable now from the hitting-time tables (Table 7); then add the detection-drift correction and the deployed-network q_μ .
4. *The learning-side width is too narrow ($1.46\times$ vs observed $2.14\times$ at CelebA $d_{\text{latent}} = 200$);* a second source (the d_{NN} quantile, Appendix I.5) is required. Test: requires per-sample latents.
5. *Data-side rate check:* on the synthetic RFNN the data-side eigenvalues scale as $64a_1(q_d)^2\lambda_i(M_t)$ in code units, so a per- d_{latent} refit of κ scales as $a_1(q_d)^2$. Test: regress log block medians on $\log a_1(q_d)^2$.
6. *Top sample-bulk eigenvectors at small d_{latent} are cluster-collective, $\sim k$ of them with eigenvalue $\sim p[r_t a_*^2 + (n/k - 1)g(q_d, \varrho_c)]/n$ within a factor 2.* Test: IPR of the top 20 sample-bulk eigenvectors.
7. *Cross-dataset transfer at the one-constant level currently fails (C differs $2.0\times$ at 5%, $1.6\times$ at 25%).* Test: fit C on CelebA, predict CIFAR-10; check whether $|\mathcal{D}_{t_c}|/n$ accounts for the offset.
8. *Recomputing $\mathbf{U}$ with one noise draw restores the rank cliff at n and raises the sample-bulk median by $1/r_t(q_d)$ ($1.22\times$ at $d_{\text{latent}} = 200$, $1.08\times$ at $d_{\text{latent}} = 20$).* Test: compute $\mathbf{U}$ with $n_\xi \in \{1, 50, 500\}$.

Open gaps. (1) The noise-averaged eigenvalue $r_t(q_\mu)a_*(q_\mu)^2/n$ replaces $a_*(q_\mu)^2/n$ throughout; at $t = 0.01$ (the synthetic spectra) r_t ranges only $0.97 \rightarrow 0.82$, at $t = 0.1$ (real data) $0.09 \rightarrow 0.76$. In code units the per-sample clock explains essentially none of the small- d_{latent} synthetic behaviour (Remark I.64). (2) Only unit-free eigenvalue ratios are used here; absolute step counts depend on the clock convention of (39). (3) The cliff at index n is a single-draw statement; the saved 50-draw spectra have full rank p . (4) Individual attachment requires Gram and cloud operator bounds together with separated eigenvalues; these are not verified for either the synthetic data or the real latents. At $n/p = 1000/1024$, a small-aspect-ratio localization approximation is especially unsupported. The quantile predictor is an empirical surrogate, not a proved spectral law. (5) The learning-side clock explains $1.7\text{--}1.9\times$ of the $2.1\text{--}3.9\times$ observed real-data slowdown; candidate sources of the rest: NN-ratio detection drift, the d_{latent} -dependence of $\mathcal{D}_{t_c}$, and a d_{latent} -proportional factor. (6) The deployed real-data first layer (GELU, fan-in $d_{\text{latent}} + 64$ with a fixed-norm time embedding, PyTorch default init, momentum 0.8 so $\eta_{\text{eff}} = \eta/(1 - 0.8)$) has $q_\mu \approx 0.36$ nearly independent of d_{latent} ; applied literally, the recipe predicts almost no d_{latent} -dependence from the first layer. This is the largest open issue for the real-data reading. (7) The threshold θ has an output-space meaning but is not related to the softmax condensation basin of the empirical score. (8) The square in the released clock is not derived (Remark I.56); the released fit uses 7 parameters with no hold-out (Remark I.63).

I.7 BEYOND TWO ATOMS: GENERAL LATENT SPECTRA, EFFECTIVE DIMENSION, AND POSTERIOR COLLAPSE

This subsection considers general latent spectra. Bonnaire’s general-covariance Gaussian analysis and George’s Gaussian subspace model motivate the comparison (Bonnaire et al., 2025; George et al., 2026); they do not establish it for non-Gaussian VAE latents. Here the one-copy matrix $U^{(1)}$ is denoted locally by U , and every nonlinear spectral conclusion is conditional on Assumption I.67. It must not be read as a conclusion about the noise expectation defined in Equation (2).

Setting. Let $Z \in \mathbb{R}^{n \times d_{\text{latent}}}$ hold the encoder means z^μ of the n training images, $\Sigma_0 := Z^\top Z/n = \sum_{i=1}^{d_{\text{latent}}} s_i v_i v_i^\top$ with $s_1 \geq \dots \geq s_{d_{\text{latent}}} \geq 0$, $x_t^\mu = e^{-t} z^\mu + \sqrt{\Delta_t} \eta^\mu$ with one diffused copy per training point, and

$$\begin{aligned} M_t &:= e^{-2t} \Sigma_0 + \Delta_t I_{d_{\text{latent}}}, \\ S_t &:= \frac{1}{n} \sum_{\mu} x_t^\mu x_t^{\mu\top}, \\ \mathbb{E}_\eta S_t &= M_t, \\ \lambda_i(M_t) &= e^{-2t} s_i + \Delta_t. \end{aligned} \tag{60}$$

M_t is the matrix used by the predictor of Section 5; its eigenvectors are those of Σ_0 , and Δ_t is the smallest value any of its eigenvalues can take. The feature second-moment matrix $U = n^{-1} \sum_{\mu} \phi(x_t^\mu) \phi(x_t^\mu)^\top$ is formed here from the same single copies (so $\text{rank } U \leq n$; this is $U^{(1)}$ of Appendix I.2), and its data-side spectrum is controlled by S_t , not by M_t ; the difference is a sampling broadening quantified in Lemma I.66. At $t = 0.1$, $\Delta_t = 0.1813$, $a_1(\Delta_t)^2 = 0.746$, $a_*(\Delta_t)^2 = 1.39 \times 10^{-3}$.

Definition I.65 (Latent spectral measure and effective dimensions). Let $\rho_{d_{\text{latent}}} := d_{\text{latent}}^{-1} \sum_i \delta_{s_i}$ be the empirical spectral measure of Σ_0 and

$$q[\rho_{d_{\text{latent}}}] := \frac{\text{tr } M_t}{d_{\text{latent}}} = e^{-2t} \int s \, d\rho_{d_{\text{latent}}}(s) + \Delta_t, \quad \hat{q} := \frac{\text{tr } S_t}{d_{\text{latent}}} = \frac{1}{n} \sum_{\mu} \frac{\|x_t^\mu\|^2}{d_{\text{latent}}}, \quad \mathbb{E} \hat{q} = q. \tag{61}$$

For $\delta > 0$ define the *floor count*, the *effective dimension*, and the *excess*

$$\begin{aligned} d_{\text{floor}}(\delta) &:= \#\{i : \lambda_i(M_t) \leq (1 + \delta)\Delta_t\}, \\ d_{\text{eff}} &:= \frac{(\sum_i (\lambda_i(M_t) - \Delta_t))^2}{\sum_i (\lambda_i(M_t) - \Delta_t)^2} = \frac{(\sum_i s_i)^2}{\sum_i s_i^2}, \\ d_{\text{exc}} &:= d_{\text{latent}} - d_{\text{eff}}. \end{aligned} \tag{62}$$

d_{eff} is the participation ratio of Σ_0 ; because Δ_t is known exactly, it involves no estimated floor. In the two-atom model, $d_{\text{eff}} = (d_{\text{intrinsic}} S + (d_{\text{latent}} - d_{\text{intrinsic}}) \sigma_\perp^2) / (d_{\text{intrinsic}} S^2 + (d_{\text{latent}} - d_{\text{intrinsic}}) \sigma_\perp^4)$ with $S = s^2/d_{\text{intrinsic}} + \sigma_{\text{sig}}^2$; this tends to $d_{\text{intrinsic}}$ as $\sigma_\perp \rightarrow 0$ but equals 76.6 at $(d_{\text{latent}}, \sigma_\perp) = (200, 0.5)$: a participation ratio counts a heavy excess block partially, which is why it is the wrong quantity to substitute for $d_{\text{intrinsic}}$ when $\sigma_\perp$ is not small. The released estimator of Section 5 (`effective_rank_excess`) takes the participation ratio of $(\lambda_i - \hat{\beta})_+$ with $\hat{\beta}$ the median of the lowest 30% of eigenvalues; on the *population* two-atom spectrum $\hat{\beta} = \beta_t^2$ and it returns $d_{\text{intrinsic}}$ exactly (on a one-copy finite- n spectrum the noise block is MP-spread, Lemma I.66, and it does not). On the real spectra $\hat{\beta}$ switches from 0.72 (CelebA, $d_{\text{latent}} = 140$) to 0.189, 0.182, 0.181 ($d_{\text{latent}} = 160, 180, 200$), so the estimator coincides with d_{eff} of (62) only where $\hat{\beta} = \Delta_t$, and its values on the two sides of that boundary measure different things. We recommend reporting d_{eff} and d_{floor} separately (Remark I.75).

Lemma I.66 (Sampling broadening of a collapsed block; (a)–(b) proved, (c) sketched). Let F be a set of d_F coordinates with $z_i^\mu = 0$ for all $i \in F$ and all μ , L its complement ($d_L = d_{\text{latent}} - d_F$), $H_F \in \mathbb{R}^{n \times d_F}$ the matrix of noises η_F^μ and $X_L \in \mathbb{R}^{n \times d_L}$ the matrix of diffused live coordinates.

- (a) $(S_t)_{FF} = \Delta_t H_F^\top H_F / n$ exactly, with H_F iid $\mathcal{N}(0, 1)$ independent of X_L . Writing $a_+ = \max(a, 0)$, for $u > 0$, with probability $\geq 1 - 2e^{-u^2/2}$, $\text{spec}(S_t)_{FF} \subset \Delta_t [(1 - \sqrt{d_F/n} - u/\sqrt{n})_+^2, (1 + \sqrt{d_F/n} + u/\sqrt{n})^2]$.

(b) (Interlacing) $\lambda_{d_L+k}(S_t) \leq \lambda_k((S_t)_{FF})$ for $k \leq d_F$; the d_F smallest eigenvalues of S_t lie below the upper edge in (a).

(c) (Schur complement) The floor eigenvalues λ of S_t solve $\det(\Delta_t n^{-1} H_F^\top (I - P_L(\lambda)) H_F - \lambda) = 0$ with $P_L(\lambda) := X_L(X_L^\top X_L - n\lambda)^{-1} X_L^\top$. Replacing $P_L(\lambda)$ by the orthogonal projector $P_L := P_L(0)$ onto $\text{col}(X_L)$ (relative error $\leq \lambda/(\lambda_{\min}((S_t)_{LL}) - \lambda)$), the matrix $\Delta_t n^{-1} H_F^\top (I - P_L) H_F$ is $\Delta_t \frac{n-d_L}{n}$ times a normalized Wishart($d_F, n - d_L$); hence, to leading order, the floor block of S_t is a Marchenko–Pastur bulk on

$$\Delta_t \left(1 - \frac{d_L}{n}\right) \left(1 \pm \sqrt{\frac{d_F}{n - d_L}}\right)^2. \quad (63)$$

The same statements hold with $\Delta_t \rightarrow \beta_t^2$ for the null block of the two-atom model, whose null coordinates are iid $\mathcal{N}(0, \beta_t^2)$ independent of the signal block.

Proof. (a) $x_{t,i}^\mu = \sqrt{\Delta_t} \eta_i^\mu$ for $i \in F$, and η_F is independent of (z^μ, η_L) ; the singular-value bound is Vershynin (2010, Cor. 5.35) applied to H_F . (b) Cauchy interlacing for the principal submatrix. (c) $\det(S_t - \lambda) = \det((S_t)_{LL} - \lambda) \det((S_t)_{FF} - \lambda - (S_t)_{FL}((S_t)_{LL} - \lambda)^{-1}(S_t)_{LF})$ and $(S_t)_{LF} = \sqrt{\Delta_t} X_L^\top H_F / n$, so the correction equals $\Delta_t n^{-1} H_F^\top X_L (X_L^\top X_L - n\lambda)^{-1} X_L^\top H_F = \Delta_t n^{-1} H_F^\top P_L(\lambda) H_F$. $I - P_L$ is a rank- $(n - d_L)$ orthogonal projector independent of H_F , so $(I - P_L) H_F$ is an $(n - d_L) \times d_F$ Gaussian matrix in a rotated basis, whose normalized Gram matrix has MP edges $(1 \pm \sqrt{d_F/(n - d_L)})^2$. $\square$

Numbers. CelebA $d_{\text{latent}} = 200$, $n = 1000$, $d_F \geq 30$: the block alone gives $[0.68, 1.38]\Delta_t$; with the projection factor $1 - 170/1000 = 0.83$ and aspect $30/830$, (63) gives $[0.54, 1.18]\Delta_t$. Synthetic null block ($d_F = 195$, $d_L = 5$, $n = 500$): $[0.14, 2.62]\beta_t^2$. Both dwarf the width- ε term of Proposition I.68 ($\varepsilon = 0.27$ at $p = 64d_{\text{latent}}$). The signal block ($d_{\text{intrinsic}} = 5$, aspect 0.01) has a sampling width of order $\sqrt{d_{\text{intrinsic}}/n} = 0.1$, of the same order as ε , which is why the signal edge statistic is sharp while the noise-dimension block is broad.

Assumption I.67 (Gaussian equivalence on latents; conjectured). As $d_{\text{latent}}, n, p \rightarrow \infty$ with $p > n > d_{\text{latent}}$ and $\varepsilon_0 := \sqrt{d_{\text{latent}}/p} + u/\sqrt{p} < 1$, the feature second-moment matrix admits the decomposition

$$\begin{aligned} U &= U^{\text{lin}} + U^{\text{res}}, \\ U^{\text{lin}} &= \frac{a_1(q)^2}{p d_{\text{latent}}} W S_t W^\top, \\ U^{\text{res}} &\succeq 0, \\ \text{tr } U^{\text{res}} &= a_*(q)^2 (1 + o(1)), \\ \|U^{\text{res}}\| &=: c_+ \frac{a_*(q)^2}{n}, \end{aligned} \quad (64)$$

with $q = q[\rho_{d_{\text{latent}}}]$ and $c_+ = O(1)$.

This is the Gaussian-equivalence decomposition of Appendices I.2 and I.3 (Bonnaire et al., 2025; Goldt et al., 2022; Hu & Lu, 2023; Mei & Montanari, 2022), now asserted for diffused latent inputs and written, as the one-copy U requires, with the sample second moment S_t . Three things make it an assumption rather than a theorem. Real latents are not Gaussian. The pre-diffusion $\|z^\mu\|^2/d_{\text{latent}}$ has coefficient of variation 0.22–0.23 across μ on CelebA for $d_{\text{latent}} \geq 160$ (hence $\|x_t^\mu\|^2/d_{\text{latent}}$ about 0.20), so q is the average of per-sample values q_μ and every edge below is correspondingly broadened ($\hat{q} = q$ to relative $0.2/\sqrt{n}$). And with one copy per point the residual $r_\mu = \phi(x_t^\mu) - a_1 W x_t^\mu / \sqrt{p d_{\text{latent}}}$ is correlated with $W x_t^\mu$: since $\text{rank } U \leq n$, $\text{rank } U^{\text{lin}} = d_{\text{latent}}$ and $U^{\text{res}} \succeq 0$, one has $\text{range } U^{\text{lin}} \subseteq \text{range } U$, so U^{res} is not an isotropic Wishart independent of W , and the Marchenko–Pastur value $(1 + \sqrt{n/p})^2$ is not the right reference for c_+ (Proposition I.27). We carry c_+ as a fitted $O(1)$ constant; on the synthetic sweep it is d_{latent} -dependent (it is $g(d_{\text{latent}})\theta$ of Appendix I.3).

Proposition I.68 (Conditional data-side eigenvalue bounds). *Under Assumption I.67, with probability at least $1 - 2e^{-u^2/2}$ over W ,*

$$\lambda_i(\mathbf{U}^{\text{lin}}) = \frac{a_1(q)^2}{d_{\text{latent}}} \lambda_i(S_t) (1 + \epsilon_i), \quad |\epsilon_i| \leq \varepsilon := 2\varepsilon_0 + \varepsilon_0^2, \quad i \leq d_{\text{latent}}, \quad (65)$$

$$\lambda_i(\mathbf{U}^{\text{lin}}) \leq \lambda_i(\mathbf{U}) \leq \lambda_i(\mathbf{U}^{\text{lin}}) + c_+ \frac{a_*(q)^2}{n}, \quad i \leq p. \quad (66)$$

Consequently the d_{latent} largest eigenvalues of $\mathbf{U}$ are, to relative accuracy $\varepsilon + c_+(a_*^2/a_1^2)(d_{\text{latent}}/n)/\lambda_i(S_t)$, the pushforward $(\lambda \mapsto a_1(q)^2 \lambda/d_{\text{latent}})_\#$ of the spectrum of S_t , which is the spectrum of M_t broadened by one-copy sampling (Lemma I.66). Under the single-copy rank bound, the intermediate $n - d_{\text{latent}}$ eigenvalues are at most $c_+ a_*^2/n$, and their total mass lies in

$$[\max\{0, \text{tr } \mathbf{U}^{\text{res}} - d_{\text{latent}} \|\mathbf{U}^{\text{res}}\|\}, \text{tr } \mathbf{U}^{\text{res}}].$$

At least $p - n$ eigenvalues vanish. These bounds do not identify an exact residual spectral law or equate its edge to the upper bound. The finite- W trace is $a_1^2 \text{tr}(S_t G)/d_{\text{latent}} + \text{tr } \mathbf{U}^{\text{res}}$ with $G = W^\top W/p$; the homogeneous Hermite trace is a surrogate expectation, not a finite-size equality.

Special cases. (a) Two-atom $\rho_{d_{\text{latent}}}$ with atoms α_t^2 ($d_{\text{intrinsic}}$ of them) and β_t^2 ($d_{\text{latent}} - d_{\text{intrinsic}}$): (65) with Lemma I.66 gives the signal block at $a_1(q)^2 \alpha_t^2/d_{\text{latent}}$ with sampling width of order $\sqrt{d_{\text{intrinsic}}/n}$ and the noise-dimension bulk on $a_1(q)^2 \beta_t^2(1 - d_{\text{intrinsic}}/n)(1 \pm \sqrt{(d_{\text{latent}} - d_{\text{intrinsic}})/(n - d_{\text{intrinsic}})})^2/d_{\text{latent}}$, median $\approx a_1(q)^2 \beta_t^2/d_{\text{latent}}$; this is the single-copy form of Remark I.16, whose noise-dimension medians $(5.22 \rightarrow 9.22$ in code units) the formula reproduces. (b) A general $\rho_{d_{\text{latent}}}$ gives a data-side continuum on $[a_1^2(e^{-2t} s_1 + \Delta_t)/d_{\text{latent}}, a_1^2(e^{-2t} s_1 + \Delta_t)/d_{\text{latent}}]$ together with a floor bulk of $d_{\text{floor}}(0)$ eigenvalues with median $\approx a_1(q)^2 \Delta_t/d_{\text{latent}}$ and relative half-width $\approx 2\sqrt{d_{\text{floor}}/n}$ by (63); the noise-dimension bulk is replaced by the part of the spectrum near the floor. (c) For CelebA at $d_{\text{latent}} = 200$, $n = 1000$, the additive term in (66) is $0.12 c_+$ of the floor median; for the synthetic noise-dimension edge at the t where $q = 0.33$ ($\beta_t^2 = 0.268$, $d_{\text{latent}} = 200$, $n = 500$) it is $0.011 c_+$.

Proof. (i) The nonzero eigenvalues of $W S_t W^\top / (p d_{\text{latent}})$ equal those of $d_{\text{latent}}^{-1} S_t^{1/2} G S_t^{1/2}$ with $G := W^\top W/p$, since AB and BA share their nonzero spectrum. By Vershynin (2010, Cor. 5.35), $\sqrt{p} - \sqrt{d_{\text{latent}}} - u \leq s_{\min}(W) \leq s_{\max}(W) \leq \sqrt{p} + \sqrt{d_{\text{latent}}} + u$ with probability $\geq 1 - 2e^{-u^2/2}$, hence $\text{spec } G \subset [(1 - \varepsilon_0)^2, (1 + \varepsilon_0)^2]$ and $\|G - I_{d_{\text{latent}}}\| \leq \varepsilon$; $\varepsilon_0 < 1$ makes $G \succ 0$. Ostrowski's theorem for the congruence $S_t^{1/2} G S_t^{1/2}$ gives $\lambda_i(S_t^{1/2} G S_t^{1/2}) = \theta_i \lambda_i(S_t)$ with $\theta_i \in [\lambda_{\min}(G), \lambda_{\max}(G)] \subset [1 - \varepsilon, 1 + \varepsilon]$. (ii) is Weyl's inequality for the positive-semidefinite perturbation $\mathbf{U}^{\text{res}}$. The counts follow from $\text{rank } \mathbf{U}^{\text{lin}} = d_{\text{latent}}$ and $\text{rank } \mathbf{U} \leq n$. Mass: by (66) the residual trace absorbed by the top d_{latent} eigenvalues is at most $d_{\text{latent}} c_+ a_*^2/n$. Trace: $\text{tr } \mathbf{U}^{\text{lin}} = a_1^2 \text{tr}(S_t G)/d_{\text{latent}} = a_1^2 \hat{q}(1 + O(\varepsilon))$ and (64). The lower tail-mass bound follows by bounding the top- d_{latent} sum by $\text{tr } \mathbf{U}^{\text{lin}} + d_{\text{latent}} \|\mathbf{U}^{\text{res}}\|$; the upper bound follows by PSD monotonicity. No exact residual law follows. $\square$

Definition I.69 (Timescales and buffer ratio for a general spectrum). Per-mode gradient-flow timescales are $\tau_i = 1/\lambda_i(\mathbf{U})$ (Lemma I.3). Fix $\theta \in (0, 1]$ and let $\lambda_{(\theta)}$ be the largest level such that the eigendirections of M_t at or above it carry a fraction θ of the latent variance, $\sum_{i: \lambda_i(M_t) \geq \lambda_{(\theta)}} s_i \geq \theta \sum_i s_i$. Define

$$\begin{aligned} \tau_{\text{gen}}^{(\theta)} &:= \frac{d_{\text{latent}}}{a_1(q)^2 \lambda_{(\theta)}}, \\ \tau_{\text{mem}}^{\text{onset}} &:= \frac{1}{\lambda_+^{\text{samp}}} = \frac{n}{c_+ a_*(q)^2}, \\ R^{(\theta)}[\rho_{d_{\text{latent}}}] &:= \frac{\tau_{\text{mem}}^{\text{onset}}}{\tau_{\text{gen}}^{(\theta)}} = \frac{n}{c_+ d_{\text{latent}}} \frac{a_1(q)^2}{a_*(q)^2} \lambda_{(\theta)}. \end{aligned} \quad (67)$$

For the two-atom model, $\lambda_{(\theta)} = \alpha_t^2$ and $\tau_{\text{gen}}^{(\theta)} = 1/\lambda_{\min}^{\text{signal}}$ (the statistic of Definition I.22) only for θ below the signal variance share $d_{\text{intrinsic}} S / (d_{\text{intrinsic}} S + (d_{\text{latent}} - d_{\text{intrinsic}}) \sigma_\perp^2)$, which is

0.22 at $(d_{\text{latent}}, \sigma_{\perp}) = (200, 0.5)$ and 0.05 at $\sigma_{\perp} = 1$; for larger θ , $\lambda_{(\theta)} = \beta_t^2$ and $\tau_{\text{gen}}^{(\theta)}$ is the noise-dimension timescale. On a continuous spectrum there is no canonical θ ; the results below are stated so that θ drops out. No count of coordinates appears in (67): d_{latent} enters through the fan-in $1/d_{\text{latent}}$, through $q[\rho_{d_{\text{latent}}}]$, and through c_+ (Remark I.31). *What the synthetic sweep says about c_+ .* With the Marchenko–Pastur guess $c_+ = (1 + \sqrt{n/p})^2$ and $p = 64d_{\text{latent}}$, the sweep $d_{\text{latent}} = 5 \rightarrow 200$ gives an a_*^2 ratio of 16.5 and a c_+ ratio of 3.5, i.e. a predicted onset growth of $58\times$ against the measured $221\times$: c_+ on clustered data carries a further $\approx 4\times$ d_{latent} -dependence (the g factor of Proposition I.27). Every real-data ratio below is therefore taken under the explicit hypothesis

$$(H_c) \quad c_+ \text{ is constant across the compared widths.} \quad (68)$$

Lemma I.70 (A collapsed coordinate sits exactly on the floor; proved). *Suppose coordinate i of the encoder means vanishes on the training set, $z_i^\mu = 0$ for all μ (vanishing encoder means). This condition alone does not establish posterior collapse of the full approximate posterior (Lucas et al., 2019). Then:*

- (a) $\sum_0 e_i = 0$ and $M_t e_i = \Delta_t e_i$: $\lambda = \Delta_t$ exactly, the minimum possible eigenvalue of M_t , with eigenvector e_i .
- (b) $x_{t,i}^\mu = \sqrt{\Delta_t} \eta_i$, and the exact score of the diffused empirical measure (Scarvelis et al., 2025; Gu et al., 2025; Biroli et al., 2024) has $s_i^*(x_t) = -x_{t,i}/\Delta_t$, a linear function of $x_{t,i}$ alone.
- (c) The coordinate contributes exactly $\Delta_t/d_{\text{latent}}$ to q .
- (d) Under Assumption I.67, the d_F collapsed coordinates contribute to U^{lin} the floor bulk $a_1(q)^2 \lambda(1 + \epsilon)/d_{\text{latent}}$ with λ in (63), the slowest data-side modes; the representative floor scale is $\tau_{\text{floor}} \simeq d_{\text{latent}}/(a_1(q)^2 \Delta_t)$ and the slowest floor mode is slower by the further factor $[(1 - d_L/n)(1 - \sqrt{d_F/(n - d_L)})^2]^{-1}$.
- (e) For a linear score model $s(x) = Bx$ trained by gradient flow on $\mathbb{E}\|Bx_t - s^*(x_t)\|^2$, column i of B evolves independently of all other entries: $B_{ii}(T) = -\Delta_t^{-1} + (B_{ii}(0) + \Delta_t^{-1})e^{-2\Delta_t T}$ and $B_{ji}(T) = B_{ji}(0)e^{-2\Delta_t T}$ for $j \neq i$ (the $\sigma_{\perp} = 0$ case of Proposition I.39).

Proof. (a) $(\Sigma_0)_{ij} = n^{-1} \sum_{\mu} z_i^\mu z_j^\mu = 0$ for every j . (b) The diffused empirical measure is $n^{-1} \sum_{\mu} \mathcal{N}(e^{-t} z^\mu, \Delta_t I)$ and every centre has zero i -th component, so the softmax weights $w_\mu(x) \propto \exp(-\|x - e^{-t} z^\mu\|^2/2\Delta_t)$ do not depend on $x_{t,i}$, the i -th marginal is $\mathcal{N}(0, \Delta_t)$ independent of the other coordinates, and $s_i^* = \sum_{\mu} w_\mu(x)(0 - x_{t,i})/\Delta_t = -x_{t,i}/\Delta_t$. (c) is (61). (d) is (65) with Lemma I.66(c). (e) The gradient with respect to column i is $2\mathbb{E}[(Bx_t - s^*)x_{t,i}] = 2(BM_t e_i - \mathbb{E}[s^* x_{t,i}])$. Here $M_t e_i = \Delta_t e_i$; for $j \neq i$, s_j^* depends only on $x_{t,-i}$, so $\mathbb{E}[s_j^* x_{t,i}] = \mathbb{E}[s_j^*] \mathbb{E}[x_{t,i}] = 0$; and $\mathbb{E}[s_i^* x_{t,i}] = -\mathbb{E}[x_{t,i}^2]/\Delta_t = -1$. Hence $\frac{d}{dT}(Be_i) = -2\Delta_t(Be_i + e_i/\Delta_t)$. $\square$

The contrast between (d) and (e) is a normalization: the RFNN rate carries the fan-in factor $1/d_{\text{latent}}$ of ϕ , the linear model does not. Ratios of timescales are normalization-free. (a) and (c) are exact statements about M_t ; the one-copy eigenvalue of a collapsed coordinate is not exactly Δ_t but MP-spread around it, (d).

Proposition I.71 (Effect of a block of n_{dead} collapsed coordinates; conditional on Assumption I.67, (68), and coefficient monotonicity). *Let a latent representation have d_0 live coordinates with spectrum ρ^{live} and $\Delta_t < q_0 = q[\rho^{\text{live}}] \leq 50$, and append n_{dead} collapsed coordinates, $d_{\text{latent}} = d_0 + n_{\text{dead}} < n < p$. Assume $a_*(q)^2$ is increasing over the relevant interval, as supported by the grid check below. All spectral scales below are conditional approximations within this finite-dimensional regime. Then, for any θ :*

- (i) $q(d_{\text{latent}}) = \Delta_t + \frac{d_0}{d_{\text{latent}}}(q_0 - \Delta_t)$, decreasing when $q_0 > \Delta_t$ and bounded below by Δ_t within $d_{\text{latent}} < n$; d_{eff} and $\lambda_{(\theta)}$ are unchanged; $d_{\text{floor}}(0)$ increases by n_{dead} .
- (ii) $a_1(q)$ is strictly decreasing in q , so $a_1(q(d_{\text{latent}}))^2$ increases with n_{dead} toward $a_1(\Delta_t)^2$; $a_*(q)^2$ is increasing in q on $[\Delta_t, 50]$ (assumed here, supported by a numerical grid check), so $a_*(q(d_{\text{latent}}))^2$ decreases toward $a_*(\Delta_t)^2$.

- (iii) Every live data-side eigenvalue is rescaled by $\frac{a_1(q(d_{\text{latent}}))^2}{a_1(q_0)^2} \frac{d_0}{d_{\text{latent}}}$, and $\tau_{\text{gen}}^{(\theta)}(d_{\text{latent}}) = \frac{d_{\text{latent}}}{a_1(q(d_{\text{latent}}))^2 \lambda_{(\theta)}} \geq \frac{d_{\text{latent}}}{a_1(\Delta_t)^2 \lambda_{(\theta)}}$ has the displayed linear lower bound; its relative growth need not be at least d_{latent}/d_0 ; the representative floor timescale is approximately $d_{\text{latent}}/(a_1(q)^2 \Delta_t)$, slower than $\tau_{\text{gen}}^{(\theta)}$ by $\lambda_{(\theta)}/\Delta_t$, and its lower edge falls by the further factor $(1 - d_0/n)(1 - \sqrt{n_{\text{dead}}/(n - d_0)})^2$.
- (iv) $\tau_{\text{mem}}^{\text{onset}}(d_{\text{latent}}) = n/(c_+ a_*(q(d_{\text{latent}}))^2)$ increases with n_{dead} but is bounded: $\tau_{\text{mem}}^{\text{onset}}(d_{\text{latent}})/\tau_{\text{mem}}^{\text{onset}}(d_0) \leq a_*(q_0)^2/a_*(\Delta_t)^2$.
- (v) $\frac{R^{(\theta)}(d_{\text{latent}})}{R^{(\theta)}(d_0)} = \frac{d_0}{d_{\text{latent}}} \frac{a_1(q)^2}{a_1(q_0)^2} \frac{a_*(q_0)^2}{a_*(q)^2}$, independent of θ and n (and of c_+ by (68)); and $R^{(\theta)}(d_{\text{latent}}) \leq \frac{n \lambda_{(\theta)}}{c_+ d_{\text{latent}}} \frac{a_1(\Delta_t)^2}{a_*(\Delta_t)^2}$ within the assumed regime. Sending $n_{\text{dead}} \rightarrow \infty$ at fixed n exits that regime; this bound does not establish an asymptotic limit there.

Collapsed coordinates affect the ratio through both q and the explicit fan-in factor $1/d_{\text{latent}}$, as well as finite-width spectral corrections. The competition between these factors must be evaluated within $d_{\text{latent}} < n$; eventual erosion outside that regime is not proved.

Proof. (i) Lemma I.70(c) applied n_{dead} times; $\lambda_{(\theta)}$ depends only on the live s_i . (ii) $a_1(q) = \mathbb{E}[\text{sech}^2(\sqrt{q}z)]$ is strictly decreasing (Lemma I.24(ii)). Monotonicity of a_*^2 on $[\Delta_t, 50]$ is checked by 200-node Gauss–Hermite quadrature on a 5000-point grid (minimum increment 3.3×10^{-5} ; sample values 0.0014, 0.031, 0.036, 0.043, 0.048, 0.054, 0.066, 0.074 at $q = 0.181, 1.11, 1.25, 1.47, 1.62, 1.83, 2.27, 2.58$; $a_*^2 \rightarrow 1 - 2/\pi$ as $q \rightarrow \infty$); all sweeps in this paper have $q \leq 2.6$. (iii)–(v) are substitutions into (65), (63) and (67); the bounds use $a_1(q)^2 \leq a_1(\Delta_t)^2$ and $a_*(q)^2 \geq a_*(\Delta_t)^2$ for $q \in [\Delta_t, 50]$. $\square$

Corollary I.72 (Flat effective dimension: what the mechanism predicts; conjectural). *Treat the RFNN surrogate at $t = 0.1$ with unit-variance first-layer weights as a proxy for the depth-5, width-1024 MLP actually trained (Remark I.73), and assume (68). On CelebA ($t = 0.1$, $n = 1000$) the measured spectra give $\text{tr } \Sigma_0 = 252.5, 255.2, 260.4$ at $d_{\text{latent}} = 160, 180, 200$ (increments 0.35, 0.14, 0.26 per added coordinate against a live mean $\text{tr } \Sigma_0/d_{\text{eff}} = 2.8\text{--}3.0$), estimator $d_{\text{eff}} = 89.5, 90.3, 87.9$, $\lambda_{\min} - \Delta_t = 6 \times 10^{-5}, 2 \times 10^{-5}, 1 \times 10^{-5}$, and the median of the lowest 30% of eigenvalues within 2.4×10^{-4} of Δ_t for $d_{\text{latent}} \geq 180$, so at least 15% of coordinates are on the floor. The live spectrum is flat: the hypotheses of Proposition I.71 hold to this accuracy. With $q = 1.473, 1.342, 1.247$, $a_1(q)^2 = 0.286, 0.305, 0.320$, $a_*(q)^2 = 0.0434, 0.0391, 0.0359$, the proposition predicts the factors in Table 8. If the memorization-fraction curve depends on training only through the sample bulk, uniformly rescaled by $a_*(q)^2$, then every iso-memorization hitting time scales by $a_*(q_{160})^2/a_*(q_d)^2$ and the final fraction at budget $T_{\max}$ is $F_d(T_{\max}) = F_{160}(T_{\max} a_*(q_d)^2/a_*(q_{160})^2)$, i.e. the $d_{\text{latent}} = 160$ curve read at 4.51M and 4.14M steps. On CIFAR-10 ($\text{tr } \Sigma_0 = 281\text{--}299$ for $d_{\text{latent}} = 160\text{--}260$; floor reached to 2×10^{-3} at $d_{\text{latent}} = 160, 3 \times 10^{-4}$ at 180, and to within 10^{-4} for $d_{\text{latent}} \geq 200$; $q = 1.621 \rightarrow 1.108$, $a_*^2 = 0.0480 \rightarrow 0.0312$) the same construction gives the CIFAR-10 rows.*

The absolute q depends on the first-layer weight scale of the surrogate (unit variance here; PyTorch fan-in initialization gives $q/3$). With $q \rightarrow q/3$ the CelebA 160 $\rightarrow$ 200 factors become $\tau_{\text{mem}} \times 1.33$, $R \times 1.14$ and the CIFAR-10 160 $\rightarrow$ 260 factor $\tau_{\text{mem}} \times 1.93$; the sign of every prediction is scale-free, the magnitude is not. The proposition’s hypotheses hold only marginally for the real model ($p = 1024$, $n = 1000$, $d_{\text{latent}} \leq 260$, $\varepsilon_0 \approx 0.4\text{--}0.5$), and its features are learned; hence the corollary is a conjectural prediction, not a consequence.

Remark I.73 (What the mechanism can and cannot account for). When d_{eff} is flat, added coordinates change q and the floor bulk, not any buffer count. The released `buffer_proxy` = $d_{\text{latent}} - \text{round}(d_{\text{eff}}) = 71, 90, 112$ enters no timescale. What the q -mechanism does deliver is the sign: hitting times lengthen and the final fraction falls as coordinates collapse. Quantitatively it accounts for $\log 1.21 / \log \{1.41, 1.34, 1.26\} = 0.55\text{--}0.82$ of the log-slowdown of CelebA hitting times, for $\log(0.559/0.608) / \log(0.448/0.608) = 0.28$ of the log-decline of the CelebA final fraction, and on CIFAR-10 for $\log 1.54 / \log 2.55 = 0.46$ of the hitting-time log-slowdown and

Table 8: Predicted (pure q -rescaling, Proposition I.71 with (H_c)) versus observed changes across the posterior-collapsed part of the real-data sweeps ($t = 0.1$, $n = 1000$, five seeds). Hitting-time ratios are relative to $d_{\text{latent}} = 160$; thresholds in parentheses. Final fractions at the 5M-step budget.

	quantity	predicted	observed
CelebA 160	$\rightarrow \tau_{\text{mem}}^{\text{onset}} / \text{hitting-time ratio}$	$\times 1.11$	$\times 1.17\text{--}1.33$ (thr. 0.01–0.5)
180	τ_{gen}, R	$\times 1.06, \times 1.05$	–
	final fraction	0.595	0.509
CelebA 160	$\rightarrow \tau_{\text{mem}}^{\text{onset}} / \text{hitting-time ratio}$	$\times 1.21$ ($\times 1.33$ with fan-in init)	$\times 1.33, 1.26, 1.37, 1.41$ (thr. 0.01, 0.05, 0.10, 0.25)
200	τ_{gen}, R	$\times 1.12, \times 1.08$	–
	final fraction	0.559	0.448
CIFAR-10 160	$\rightarrow \tau_{\text{mem}}^{\text{onset}} / \text{hitting-time ratio}$	$\times 1.54$ ($\times 1.93$ with fan-in init)	$\times 1.6$ (thr. 0.01), $\times 2.4$ (0.05, CI $\pm 1.7\text{M}$), $\times 2.7$ (0.10, one seed censored)
260	τ_{gen}, R	$\times 1.26, \times 1.22$	–
	final fractions ($d_{\text{latent}} = 180, \dots, 260$)	0.557, 0.552, 0.517, 0.505, 0.472	0.456, 0.346, 0.213, 0.183, 0.125

$\log(0.472/0.600)/\log(0.125/0.600) = 0.15$ of the final-fraction log-decline. The remainder is outside the surrogate. The real-data score model is a depth-5, width-1024 MLP with sinusoidal time features, trained by SGD (learning rate 10^{-3} , momentum 0.8) across $t \in [0.01, 3]$ on $n = 1000$ encoder means; $p = 1024 > n = 1000$, so an $n - d_{\text{latent}}$ residual bulk is only marginally accommodated, and its first-layer features are learned, so the RFNN quantities are a surrogate evaluated at one t . The residual bulk of clustered real features is not Marchenko–Pastur (and c_+ is d_{latent} -dependent even on the synthetic sweep); and the decoder’s response to imperfectly learned floor coordinates (the slowest data-side modes) is not modelled, although the FID worsening from 38.9 ($d_{\text{latent}} = 140$) to 47–50 ($d_{\text{latent}} = 160\text{--}200$) suggests it is not inert. Over $d_{\text{latent}} = 20\text{--}120$ on CelebA, q is flat (1.71–1.84, $a_*^2 = 0.051\text{--}0.055$) because $\text{tr } \Sigma_0$ grows by ≈ 2 per coordinate; there the surrogate predicts no change in $\tau_{\text{mem}}^{\text{onset}}$ and $\tau_{\text{gen}} \propto d_{\text{latent}}$, and the observed rise of memorization belongs to the decoder identity-threshold regime of Section 5. Finally, memorization is not a function of q alone across datasets: at $q \approx 1.25\text{--}1.29$ the final fractions are 0.45 (CelebA) and 0.21 (CIFAR-10). The pixel-space NN-ratio has no ambient-dimension drift across the d_{latent} sweep (decoded images have fixed dimension), but a decoder-dependent drift of its baseline has not been checked.

Remark I.74 (Encoder means versus posterior samples). The latents here are encoder means, for which a collapsed coordinate is identically zero. Had the diffusion model been trained on posterior samples $z \sim \mathcal{N}(\mu(x), \text{diag } \sigma^2(x))$, a collapsed coordinate ($\mu_i \equiv 0, \sigma_i \equiv 1$) would have $s_i = 1$ and $\lambda_i(M_t) = e^{-2t} + \Delta_t = 1$: it would be an isotropic noise-dimension coordinate with $\sigma_{\perp} = 1$, the isotropic control, rather than a floor coordinate, and the floor bulk would move up by $1/\Delta_t \approx 5.5$ at $t = 0.1$. Then $q = \Delta_t + [d_0(q_0 - \Delta_t) + n_{\text{dead}}e^{-2t}]/d_{\text{latent}}$ exceeds the mean-latent value, so $a_*(q)^2$ is larger and $\tau_{\text{mem}}^{\text{onset}}$ shorter, while $a_1(q)^2$ is smaller and τ_{gen} longer; R shrinks by both factors. The theory therefore predicts earlier onset memorization for sampled than for mean latents at large d_{latent} , with the caveat that the training latents are then a stochastic set, so the empirical-score target of Lemma I.70(b) and the memorization criterion itself change, not only q . Untested.

Remark I.75 (Reporting). For every real-data run we recommend reporting, alongside d_{latent} : d_{eff} (participation ratio of Σ_0 , floor at the known Δ_t), $d_{\text{floor}}(\delta)$ with $\delta = 2 \times 10^{-3}$, $\text{tr } \Sigma_0$, and $q[\rho_{d_{\text{latent}}}]$; and plotting memorization against d_{eff} and q as well as against d_{latent} . On the present sweeps the points with $d_{\text{latent}} \geq 160$ (CelebA) and $d_{\text{latent}} \geq 220$ (CIFAR-10) differ from one another in q and d_{floor} only, so any effect of d_{latent} there is a q effect, a c_+ effect, or is outside the theory.

Predictions and how to test them.

1. *Pure q -rescaling*: for CelebA $d_{\text{latent}} \geq 160$ every iso-memorization hitting time scales by $a_*(q_{160})^2/a_*(q_d)^2 = 1.11$ (180), 1.21 (200), threshold-independent, and the curves collapse under the step rescaling. Test: executable now from the released mean curves and hitting-time tables (observed 1.26–1.41, roughly threshold-independent, 5–20% above the prediction).

2. *The coordinates added beyond $d_{\text{latent}} \sim 140$ (CelebA) / ~ 160 (CIFAR-10) are posterior-collapsed: $d_{\text{floor}}(2 \times 10^{-3}) \geq 0.15d_{\text{latent}}$ and $\text{tr } \Sigma_0$ is flat. Test: regenerate the spectra from the VAE checkpoints (not in the local repository), count floor eigenvalues, and check the MP width (63).*
3. *Training on posterior samples instead of means at large d_{latent} increases onset memorization. Test: new training at CelebA $d_{\text{latent}} = 200$ on sampled latents.*
4. *Projecting out the floor coordinates restores memorization toward the $d_{\text{latent}} \sim 120$ –140 level, with τ_{mem} shortened by $a_*(q_{\text{live}})^2/a_*(q_{200})^2 \approx 0.7$ –0.8. Test: new training on the projected latents, decode by zero-padding.*
5. *At fixed nominal d_{latent} , changing the VAE KL capacity reorders memorization by q , not by d_{latent} . Test: two VAEs at $d_{\text{latent}} = 200$ with different capacity targets.*
6. *Memorization is not a function of q alone across datasets (already contradicted: 0.448 vs 0.213 at matched $q \approx 1.25$ –1.29); recorded as a falsification of the q -only reading.*
7. *c_+ is d_{latent} -dependent on clustered data ($\sim 13\times$ over $d_{\text{latent}} = 5..200$, not the MP $3.5\times$). Test: executable now from the saved synthetic spectra.*

Open gaps. (1) Assumption I.67 (Gaussian equivalence on real, non-Gaussian latents) is not proved; with one copy per point U^{res} is correlated with W , so c_+ is a fitted $O(1)$ constant, and on the synthetic spectra it carries a $\sim 4\times d_{\text{latent}}$ -dependence beyond MP; all real-data ratios are conditional on (H_c) and are sign-and-order-of-magnitude statements. (2) The real-data score model is a depth-5, width-1024 MLP (not depth-3/width-256 as an earlier draft said; the main text must match): $p = 1024 > n = 1000$, so the hypotheses $p > n > d_{\text{latent}}$ and $\varepsilon_0 < 1$ hold only marginally, the features are learned, and the RFNN quantities at $t = 0.1$ are a surrogate whose absolute q depends on the first-layer scale. (3) The pure time-rescaling hypothesis behind the final-fraction predictions is a modelling choice; it accounts for 0.28 (CelebA) and 0.15 (CIFAR-10) of the final-fraction log-decline and 0.46–0.82 of the hitting-time log-slowdown. (4) The quoted d_{eff} values are the released estimator’s and equal Definition I.65 only where $\hat{\beta} = \Delta_t$; the spectra, VAE checkpoints and diagnostics are absent from the local repository, so $d_{\text{floor}}(\delta)$ can only be bounded, and Predictions 2–5 require those assets or new training. (5) The θ -quantile τ_{gen} is a convention; only θ -free ratios are used. (6) The decoder’s response to floor coordinates and the pixel-space NN-ratio baseline drift are unmodelled.

I.8 REGIME BOUNDARIES AND SCOPE

This subsection delimits the four-bulk theory: the block counts and edges are leading-order statements under hypotheses (S1)–(S5) below, and several of the paper’s configurations sit at or beyond their boundaries. We first fix which feature second moment is meant, then treat three boundaries ($d_{\text{latent}} \rightarrow n$, $\sigma_{\perp} \rightarrow 0$, p close to $d_{\text{latent}} + n$), the fixed-null-energy control and feature learning; the closing Appendix I.9 lists the out-of-scope configurations explicitly.

Definition I.76 (Aspect ratios, null-data fraction, the two feature second moments, onset statistic). Let $D := d_{\text{latent}} - d_{\text{intrinsic}}$, $\psi_n := n/d_{\text{latent}}$, $\psi_p := p/d_{\text{latent}}$, $\psi := D/n$ (the $\gamma_{\perp}$ of Appendix I.2), and α_t^2, β_t^2 as before.

- (i) The *null-data fraction* $\rho_{\perp}(t) := e^{-2t}\sigma_{\perp}^2/\beta_t^2 \in [0, 1)$ is the share of null-block variance that comes from data rather than from diffusion noise.
- (ii) For the population second moment M_t the pre-activation variance obeys the exact identity

$$q = \frac{\text{tr } M_t}{d_{\text{latent}}} = \beta_t^2 + \frac{d_{\text{intrinsic}}(\alpha_t^2 - \beta_t^2)}{d_{\text{latent}}}, \quad q_{\infty} := \lim_{d_{\text{latent}} \rightarrow \infty} q = \beta_t^2 \quad (69)$$

(Lemma I.6). The empirical trace differs by $O(n^{-1/2})$, and the per-neuron variance $w_i^{\top} M_t w_i / d_{\text{latent}}$ fluctuates around q with relative width $O(d_{\text{latent}}^{-1/2})$; by Jensen on the concave $q \mapsto \mathbb{E} \tanh^2(\sqrt{q}z)$ this makes measured traces slightly low at small d_{latent} (3% at $d_{\text{latent}} = 10$).

- (iii) *Two feature second moments.* $U := \frac{1}{n} \sum_{\mu} \mathbb{E}_{\eta}[\phi(x_t^{\mu})\phi(x_t^{\mu})^{\top}]$ is the η -averaged object; it governs the gradient-flow ODE and is what the saved spectra estimate (Monte Carlo over 50

draws). $U^{(1)} := \Phi\Phi^\top/n$ is the single-draw Gram. All spectral statements below concern U unless $U^{(1)}$ is named.

(iv) λ_{onset} is the largest eigenvalue of U outside the span of the d_{latent} Gaussian-equivalent linear modes, i.e. the top of the residual (“sample”) bulk; $\tau_{\text{mem}}^{\text{onset}} := 1/\lambda_{\text{onset}}$ and $\tau_{\text{gen}} := 1/\lambda_{\text{min}}^{\text{sig}}$. For the RFNN these are spectral proxies: the RFNN runs log no sample-based memorization.

(v) For the null block, with $\hat{C}_\perp = \frac{1}{n} \sum_\mu \xi_\perp^\mu \xi_\perp^{\mu^\top}$ (rank $\min(D, n)$), $\hat{\Sigma}_{t,\perp} = e^{-2t} \hat{C}_\perp + \Delta_t I_D$ and $\Sigma_{t,\perp} = \beta_t^2 I_D$, the *sample-specificity* of the converged linear null score is

$$\varepsilon_\perp^2 := \frac{\mathbb{E} \|\hat{\Sigma}_{t,\perp}^{-1} - \Sigma_{t,\perp}^{-1}\|_F^2}{\|\Sigma_{t,\perp}^{-1}\|_F^2}. \quad (70)$$

Assumption I.77 (Four-bulk regime). (S1) *Frozen features.* W is fixed at initialization; only A is trained, from $A(0) = 0$, by gradient flow on score matching at fixed t , so each eigenmode of U decays independently at rate λ_i .

(S2) *Ordering.* $d_{\text{intrinsic}} < d_{\text{latent}} < n < p$.

(S3) *Spiked-plus-atom spectrum with data-dominated null block.* ρ_Σ consists of $d_{\text{intrinsic}}$ finite-rank spikes of mean α_0^2 (the k centers give distinct spike locations; measured spread $\approx 2.3\times$) and one atom $\sigma_\perp^2$ of weight D/d_{latent} ; and $\rho_\perp(t) \geq \frac{1}{2}$, i.e. $e^{-2t}\sigma_\perp^2 \geq \Delta_t$.

(S4) *Gaussian equivalence at leading order* (Goldt et al., 2022; Hu & Lu, 2023): $\phi(x) = a_1(q)Wx/\sqrt{d_{\text{latent}}p} + a_*(q)g(x)$ with g uncorrelated with x ; the signal–null cross covariance of the empirical $\hat{M}_t$ is neglected (block-diagonal approximation).

(S5) *Separation.* The noise-dim block lies at least one decade above λ_{onset} (Proposition I.82).

Under (S1)–(S5) the leading-order block positions (block *means*) are

$$\lambda^{\text{sig}} \asymp \frac{a_1(q)^2 \alpha_t^2}{d_{\text{latent}}}, \quad \lambda^{\text{nd}} \asymp \frac{a_1(q)^2 \beta_t^2}{d_{\text{latent}}}, \quad \bar{\lambda}^{\text{samp}} := \frac{a_*(q)^2}{n} \text{ (residual mass per training point)}, \quad (71)$$

with the trace check, valid for block means to leading order in d_{latent}/n and $1/n$,

$$d_{\text{intrinsic}} \lambda^{\text{sig}} + D \lambda^{\text{nd}} + a_*(q)^2 \approx \mathbb{E}[\tanh(\sqrt{q}z)^2] \quad (72)$$

(the exact form is (21)). The residual mass $a_*(q)^2$ is shared between the residual bulk (measured share 0.5–0.7) and $O(a_*^2/n)$ per linear mode; the residual bulk is heavy-tailed, so its median is not a location statistic and the check must use means. Measured ($\sigma_\perp = 0.5$, code units/ p): 0.456 vs $\mathbb{E} \tanh^2 = 0.470$ ($d_{\text{latent}} = 10$), 0.2915 vs 0.2974 (40), 0.2092 vs 0.2093 (200).

I.8.1 WHICH U : BLOCK COUNTS FOR THE TWO FEATURE SECOND MOMENTS

Remark I.78 (Index counts and data-term approximation under (S4)). Assume (S1), (S4) and $p > n$.

(a) *Single-draw Gram.* rank $U^{(1)} = n$ a.s., and the reporting convention partitions its non-zero spectrum as

$$\begin{aligned} N_{\text{sig}} &= d_{\text{intrinsic}}, \\ N_{\text{nd}} &= \min(d_{\text{latent}}, n) - d_{\text{intrinsic}}, \\ N_{\text{samp}} &= (n - d_{\text{latent}})_+, \\ N_{\text{null}} &= p - n, \end{aligned} \quad (73)$$

with a hard cliff at index n ; the residual bulk is empty for $\psi_n \leq 1$.

(b) *η -averaged U .* For the data-term approximation only, the linear part is $a_1(q)^2 W \hat{M}_t W^\top / (d_{\text{latent}} p)$ with $\hat{M}_t = e^{-2t} \hat{C} + \Delta_t I_{d_{\text{latent}}}$ of full rank d_{latent} for every d_{latent} . This motivates $d_{\text{intrinsic}}$ signal and D null directions in the linear term, without proving their separation in the full nonlinear matrix: for $D \leq n$ all D noise-dim modes lie in the data-broadened band of (79), and for $D > n$ exactly n of them do while $D - n$ sit on the

diffusion floor $a_1(q)^2 \Delta_t (1 \pm \psi_p^{-1/2})^2 / d_{\text{latent}}$. The residual part $\frac{a_*^2}{n} \sum_{\mu} \mathbb{E}_{\eta} [g(x_t^{\mu}) g(x_t^{\mu})^{\top}]$ has rank $> n$: there is no cliff at index n and no rank-null block, only a soft residual tail (measured $e_{n-1}/e_n = 1.32$ at $d_{\text{latent}} = 40$, 1.73 at $d_{\text{latent}} = 200$; 0.3% of the trace beyond index n).

- (c) *Eigenvectors (additional gap assumption)*. For isolated data-term eigenvalues with a spectral gap large compared with the perturbation, the noise-dim eigenvectors are $W_{\perp} v / \|W_{\perp} v\| + O(\psi_p^{-1/2})$ with v ranging over eigenvectors of the empirical noised null second moment; for U that matrix is $\hat{\Sigma}_{t,\perp}$, whose non-floor eigenvectors are those of $\hat{C}_{\perp}$, so they are determined by the training noise vectors $\{\xi_{\perp}^{\mu}\}$.

Data-term calculation and heuristic interpretation. (a) Under (S4), $\Phi = a_1(q) W X_t / \sqrt{d_{\text{latent}} p} + a_*(q) G$ with $X_t \in \mathbb{R}^{d_{\text{latent}} \times n}$ a single draw; $\text{rank}(W X_t) = \text{rank } X_t = \min(d_{\text{latent}}, n)$ a.s. (the $\sqrt{\Delta_t} \eta$ term makes X_t generic; W has full column rank since $p > d_{\text{latent}}$), of which $d_{\text{intrinsic}}$ directions carry the signal spikes. Φ has n columns, so $\text{rank } U^{(1)} \leq n$, with equality a.s. since G is generic; the residual therefore adds $n - \min(d_{\text{latent}}, n)$ directions and $p - n$ are annihilated (Lemma I.15). (b) $\mathbb{E}_{\eta} [x_t x_t^{\top}] = e^{-2t} x x^{\top} + \Delta_t I$, so the η -average of the linear Gram is $a_1^2 W \hat{M}_t W^{\top} / (d_{\text{latent}} p)$ with $\hat{M}_t \succeq \Delta_t I$ of full rank; its null block is $\hat{\Sigma}_{t,\perp} = e^{-2t} \sigma_{\perp}^2 \Xi \Xi^{\top} / n + \Delta_t I_D$ with $\Xi \in \mathbb{R}^{D \times n}$ standard Gaussian, whose $\min(D, n)$ non-trivial eigenvalues are $e^{-2t} \sigma_{\perp}^2 m_i + \Delta_t$ (m_i Wishart eigenvalues) and whose remaining $(D - n)_+$ eigenvalues equal Δ_t exactly. The residual η -average is a sum of n PSD matrices each of rank > 1 , which need not preserve the rank- n cliff; higher rank alone does not exclude an eigenvalue gap. (c) With $W_{\perp}^{\top} W_{\perp} / p = I_D + O(\psi_p^{-1/2})$ the eigenpairs of $W_{\perp} \hat{\Sigma}_{t,\perp} W_{\perp}^{\top}$ are $(\lambda_i p, W_{\perp} v_i)$ to that order; the isotropic Δ_t shift does not change the eigenvectors of $\hat{\Sigma}_{t,\perp}$ relative to $\hat{C}_{\perp}$. $\square$

I.8.2 THE BOUNDARY $d_{\text{latent}} \rightarrow n$

Proposition I.79 (Sample-specificity of the learned null score; proved in the block-diagonal approximation). *Model the null block of the score linearly, $s_{\perp}(x) = B_{\perp} x_{\perp}$, trained by gradient flow on denoising score matching over the n training points at fixed t , in the block-diagonal approximation (Proposition I.43). Then $B_{\perp}(T) \rightarrow -\hat{\Sigma}_{t,\perp}^{-1}$ (population target $-\Sigma_{t,\perp}^{-1} = -I_D / \beta_t^2$), and in the proportional limit $D, n \rightarrow \infty, D/n \rightarrow \psi$,*

$$\varepsilon_{\perp}^2 \rightarrow \int \frac{\rho_{\perp}^2 (m-1)^2}{(1 + \rho_{\perp} (m-1))^2} d\mu_{\psi}(m), \quad (74)$$

μ_{ψ} the Marchenko–Pastur law of ratio ψ with unit mean (Marchenko & Pastur, 1967). Consequently:

- (i) $\varepsilon_{\perp}^2 = \rho_{\perp}^2 \psi (1 - 2\rho_{\perp} \psi + O(\rho_{\perp}^2 \psi))$ for $\rho_{\perp} \psi \rightarrow 0$; $\varepsilon_{\perp}^2$ depends on (d_{latent}, n) only through $\psi = D/n$ and on $\sigma_{\perp}$ only through $\rho_{\perp}$. (The expansion is poor for $\rho_{\perp} \approx 1$, $\psi \gtrsim 0.1$; use (74) numerically there.)
- (ii) For $\psi > 1$ the atom of μ_{ψ} at 0 gives $\varepsilon_{\perp}^2 \geq (1 - \psi^{-1}) \rho_{\perp}^2 / (1 - \rho_{\perp})^2$: on the $D - n$ directions orthogonal to $\text{span}\{\xi_{\perp}^{\mu}\}$ the learned score is $-x_{\perp} / \Delta_t$ instead of $-x_{\perp} / \beta_t^2$, a contraction stronger by $(1 - \rho_{\perp})^{-1}$; the reverse dynamics confine generated null components to the span of the training noise vectors.
- (iii) $\varepsilon_{\perp}^2 \rightarrow 0$ as $\rho_{\perp} \rightarrow 0$ at fixed ψ : at $\sigma_{\perp} = 0.01$, $t = 0.01$, $\rho_{\perp} = 4.9 \times 10^{-3}$ and $\varepsilon_{\perp}^2 < 10^{-4}$ for all $d_{\text{latent}} \leq 1000$.

The neglected signal–null cross covariance $\frac{1}{n} \sum_{\mu} (m_{c(\mu)} + \xi_{\text{sig}}^{\mu}) \xi_{\perp}^{\mu \top}$ has entries of order $\alpha_t \beta_t / \sqrt{n}$ and contributes a sample-specific error of the same order as $\varepsilon_{\perp}^2$ when ψ is small.

Proof. With DSM target $y = -\eta / \sqrt{\Delta_t}$ and $x_t = e^{-t} x + \sqrt{\Delta_t} \eta$, the null-block loss over the empirical measure is $\mathbb{E} \|B_{\perp} x_{t,\perp} - y_{\perp}\|^2$; its gradient flow is $\dot{B}_{\perp} = -2(B_{\perp} \hat{\Sigma}_{t,\perp} + I_D)$ because $\mathbb{E}[x_{t,\perp} x_{t,\perp}^{\top}] = \hat{\Sigma}_{t,\perp}$ over the training points (cluster means have zero null component) and $\mathbb{E}[y_{\perp} x_{t,\perp}^{\top}] = -I_D$.

Hence $B_\perp(T) = -\hat{\Sigma}_{t,\perp}^{-1} + (B_\perp(0) + \hat{\Sigma}_{t,\perp}^{-1})e^{-2\hat{\Sigma}_{t,\perp}T}$. $\hat{C}_\perp = \sigma_\perp^2 \Xi \Xi^\top / n$, so its eigenvalues are $\sigma_\perp^2 m_i$ with $\{m_i\}$ Wishart eigenvalues of ratio ψ whose empirical law converges weakly to μ_ψ . Since $e^{-2t}\sigma_\perp^2 m + \Delta_t = \beta_t^2(1 + \rho_\perp(m-1))$,

$$\|\hat{\Sigma}_{t,\perp}^{-1} - \Sigma_{t,\perp}^{-1}\|_F^2 = \frac{1}{\beta_t^4} \sum_i \frac{\rho_\perp^2(m_i-1)^2}{(1 + \rho_\perp(m_i-1))^2}.$$

Dividing by $\|\Sigma_{t,\perp}^{-1}\|_F^2 = D/\beta_t^4$ gives the normalized trace of a bounded continuous function of $\hat{C}_\perp$ (denominator $\geq 1 - \rho_\perp > 0$ on $m \geq 0$), so weak convergence yields (74). (i): expand $(1 + \rho_\perp(m-1))^{-2}$ and use the MP central moments $\int(m-1)^2 d\mu_\psi = \psi$, $\int(m-1)^3 d\mu_\psi = \psi^2$, $\int(m-1)^4 d\mu_\psi = 2\psi^2 + \psi^3$. (ii): $\mu_\psi = (1 - \psi^{-1})\delta_0 + \psi^{-1}\tilde{\mu}$ for $\psi > 1$, and the integrand at $m = 0$ equals $\rho_\perp^2/(1 - \rho_\perp)^2$; the $m_i = 0$ eigenvectors are the complement of $\text{span}\{\xi_\perp^\mu\}$, on which $\hat{\Sigma}_{t,\perp} = \Delta_t I$. (iii): the integrand is $O(\rho_\perp^2)$ uniformly on the support once $\rho_\perp \leq \frac{1}{2}$. $\square$

Numerically, at $\sigma_\perp = 0.5$, $t = 0.01$ ($\rho_\perp = 0.925$, $n = 500$), (74) gives $\varepsilon_\perp^2 = 0.028, 0.103, 0.302, 0.638, 1.19$ at $d_{\text{latent}} = 20, 50, 100, 150, 200$; the first-order form $\rho_\perp^2 \psi$ gives $0.026, 0.077, 0.163, 0.248, 0.334$ and under-estimates from $d_{\text{latent}} \approx 100$ on because small eigenvalues of $\hat{C}_\perp$ are amplified by $(1 + \rho_\perp(m-1))^{-2}$. At $\sigma_\perp = 1$ ($\rho_\perp = 0.98$): $0.032, 0.122, 0.379, 0.867, 1.80$.

Proposition I.80 (Onset statistics across the $d_{\text{latent}} \rightarrow n$ boundary; sketched). Assume (S1), (S3), (S4) and $p > n$; U is the η -averaged second moment.

- (a) Residual-bulk onset. In the isotropic-residual approximation (columns $g(x_t^\mu)$ with uncorrelated entries of variance $1/p$),

$$\lambda_{\text{onset}}^{\text{iso}} = \frac{a_*(q)^2}{n} \left(1 + \sqrt{n/p}\right)^2, \quad \tau_{\text{mem}}^{\text{onset,iso}} = \frac{n}{a_*(q)^2(1 + \sqrt{n/p})^2}, \quad (75)$$

which depends on d_{latent} only through $q \downarrow \beta_t^2$ in $a_*(q)^2$. On the repo spectra ($\sigma_\perp = 0.5$, $p = 64d_{\text{latent}}$) the measured onset exceeds (75) by 4.6, 4.2, 2.8, 2.2, 1.7, 1.5, 1.1, 1.0 at $d_{\text{latent}} = 10, 20, 40, 60, 80, 100, 150, 200$: the excess is a small- d_{latent} effect of cluster-level residual spikes (the $g(d_{\text{latent}})$ of Proposition I.27, the collective modes of Remark I.64) and disappears at $d_{\text{latent}} \geq 150$, where (75) is quantitatively right (0.173 vs 0.171 code units at $d_{\text{latent}} = 200$). Because the excess itself decays, (75) under-predicts the d_{latent} -trend: the measured onset falls $107\times$ from $d_{\text{latent}} = 10$ to 200 ($221\times$ from $d_{\text{latent}} = 5$), the formula predicts $24\times$ ($\approx 60\times$).

- (b) Noise-dim top for $D > n$. The residual bulk does not empty at $\psi_n = 1$ (Remark I.78(b)); its top stays at (75). The noise-dim block keeps n data-broadened modes with top

$$\lambda_{\text{max}}^{\text{nd}} \asymp \frac{a_1(q)^2}{d_{\text{latent}}} \left[e^{-2t}\sigma_\perp^2(1 + \sqrt{\psi})^2 + \Delta_t \right] (1 + \psi_p^{-1/2})^2$$

$$\xrightarrow{\psi_n \rightarrow 0} \frac{a_1(\beta_t^2)^2 e^{-2t}\sigma_\perp^2(1 + \psi_p^{-1/2})^2}{n}, \quad (76)$$

so $1/\lambda_{\text{max}}^{\text{nd}} \rightarrow n/(a_1(\beta_t^2)^2 e^{-2t}\sigma_\perp^2(1 + \psi_p^{-1/2})^2)$: d_{latent} -independent and linear in n . Its ratio to the large- d_{latent} limit of $\tau_{\text{mem}}^{\text{onset,iso}}$ tends to $a_*(\beta_t^2)^2/(a_1(\beta_t^2)^2 e^{-2t}\sigma_\perp^2) = 0.019$ at $\sigma_\perp = 0.5$, $t = 0.01$ as $p/d_{\text{latent}}, p/n \rightarrow \infty$ (0.015 at $\psi_p = 64$). These modes are sample-specific in the sense of (70): their eigenvectors are fixed by $\{\xi_\perp^\mu\}$ and their learned content deviates from the population score by $\varepsilon_\perp^2 \geq \rho_\perp^2(1 - \psi^{-1})/(1 - \rho_\perp)^2$ (Proposition I.79(ii)). In this regime the earliest sample-specific modes are the noise-dim modes, learned $\approx 50\times$ faster than the residual bulk: the population/sample distinction of the noise-dim “buffer” dissolves.

Proof sketch. (a) Under (S4) the residual part of U is $\frac{a_*^2}{n} \sum_\mu \mathbb{E}_\eta[gg^\top]$; with uncorrelated columns of variance $1/p$ it is a_*^2/n times a Wishart matrix of ratio n/p with top edge

($1 + \sqrt{n/p}$)²; the residual bulk is this spectrum projected onto the complement of the linear span and by interlacing its top is at most the unprojected edge. The measured/iso ratios use the sample-bulk tops 0.93, 0.374, 0.198, 0.173 (code units, $d_{\text{latent}} = 10, 40, 100, 200$) against $p\lambda_{\text{onset}}^{\text{iso}} = 0.203, 0.133, 0.135, 0.171$ with $a_*(q)^2 = 0.0447, 0.0125, 0.00644, 0.00467$. (b) The null-block linear part $a_1^2 W_{\perp} \hat{\Sigma}_{t,\perp} W_{\perp}^{\top} / (pd_{\text{latent}})$ has non-zero eigenvalues equal to those of $a_1^2 \hat{\Sigma}_{t,\perp}^{1/2} (W_{\perp}^{\top} W_{\perp} / p) \hat{\Sigma}_{t,\perp}^{1/2} / d_{\text{latent}}$; the top edge of $\hat{\Sigma}_{t,\perp}$ is $e^{-2t} \sigma_{\perp}^2 (1 + \sqrt{\psi})^2 + \Delta_t$ (for $\psi > 1$ the non-zero Wishart eigenvalues lie in $[(\sqrt{\psi} - 1)^2, (\sqrt{\psi} + 1)^2]$) and that of $W_{\perp}^{\top} W_{\perp} / p$ is $(1 + \sqrt{D/p})^2$; the product of top edges is an upper-bound-based approximation, not generally the exact edge of the free multiplicative convolution. As $\psi_n \rightarrow 0$, $(1 + \sqrt{\psi})^2 / d_{\text{latent}} \rightarrow 1/n$ and $q \rightarrow \beta_t^2$. Numerics: $a_*(\beta_t^2)^2 = 3.09 \times 10^{-3}$, $a_1(\beta_t^2)^2 e^{-2t} \sigma_{\perp}^2 = 0.671 \times 0.245 = 0.164$. $\square$

Remark I.81 (Conjecture: return of memorization through the null block at large d_{latent}). Proposition I.80(b) shows that for $\psi_n \lesssim 1$ the fast noise-dim modes are already sample-specific. We conjecture that this is the channel through which memorization returns at large d_{latent} , and state what the conjecture does and does not contain.

- (i) *Scope of the linear null model.* The converged null score $-\hat{\Sigma}_{t,\perp}^{-1} x_{\perp}$ is linear; under the reverse dynamics it generates null components with the empirical covariance $\hat{C}_{\perp}$ (confined to $\text{span}\{\xi_{\perp}^{\mu}\}$ when $D > n$), i.e. $x_{\perp} \approx \Xi c / \sqrt{n}$ with $c \sim N(0, I_n)$, whose squared distance to the nearest training point is $\approx 2D\sigma_{\perp}^2 (1 - O(\sqrt{\log n/D}))$ and whose NN-ratio is ≈ 1 . This high-dimensional approximation predicts weak replication for the linear null score, not zero probability of a ratio below $1/3$. Empirical-score softmax concentration supplies a different route to replication (Biroli et al., 2024; Scarvelis et al., 2025; Gu et al., 2025); it is not necessary for an isolated metric-positive draw. Thus $\varepsilon_{\perp}$ is a *candidate diagnostic*, not an established necessary condition: it measures how much sample-specific null-block information the fast modes carry.
- (ii) *Conditional functional form.* If a metric-dependent threshold ε_c exists such that memorization appears once $\varepsilon_{\perp} \geq \varepsilon_c$, then, since $\varepsilon_{\perp}^2$ depends on (d_{latent}, n) only through ψ and on $\sigma_{\perp}$ only through $\rho_{\perp}$,

$$d_{\text{latent}}^{\text{reb}} = d_{\text{intrinsic}} + n \psi^{\text{reb}}(\rho_{\perp}), \quad \varepsilon_{\perp}^2(\rho_{\perp}, \psi^{\text{reb}}) = \varepsilon_c^2, \quad (77)$$

linear in n at fixed $(\sigma_{\perp}, t)$, absent as $\rho_{\perp} \rightarrow 0$, and earlier for larger $\rho_{\perp}$.

- (iii) *Evidence and confounds.* In the scaled-width MLP sweep (hidden $8d_{\text{latent}} = 800, 1200, 1600$; $\sigma_{\perp} = 0.5$, $n = 500, 3 \times 10^5$ steps) the memorization fraction is 0.0000, 0.0022, 0.0144 at $d_{\text{latent}} = 100, 150, 200$ and still rising ($0.004 \rightarrow 0.0144$ over the final 5×10^4 steps at $d_{\text{latent}} = 200$), with the mean NN-ratio falling $0.755 \rightarrow 0.713$; the matching $\sigma_{\perp} = 0.01$ scaled sweep gives 0 at all three with the NN-ratio rising $1.98 \rightarrow 2.59$ (distance concentration). In the paper’s *fixed-width* sweep (hidden 256) at $\sigma_{\perp} = 0.5$ the fraction is 0.0000 at every evaluation for $d_{\text{latent}} = 100, 150, 200$. Hence the rebound requires *both* $\rho_{\perp} \approx 1$ and capacity growing with d_{latent} : the $\sigma_{\perp}$ contrast at equal capacity rules out capacity as the sole cause, the fixed-width null rules out $\rho_{\perp}$ as the sole cause. Calibrating on the scaled sweep gives $\varepsilon_c^2 \in (0.30, 0.64)$ (the values at $d_{\text{latent}} = 100$ and 150) and $\psi^{\text{reb}}(0.925) \in (0.19, 0.29)$; this is a fit to those two points, not a test. Under (77) the discriminating predictions are $d_{\text{latent}}^{\text{reb}} \approx 55\text{--}80$ at $n = 250$ and $200\text{--}300$ at $n = 1000$ (same capacity schedule), and $d_{\text{latent}}^{\text{reb}} \in [88, 130]$ at $\sigma_{\perp} = 1$, $n = 500$.

I.8.3 THE BOUNDARY $\sigma_{\perp} \rightarrow 0$ AND THE MERGE CONDITION

Proposition I.82 ($\sigma_{\perp} \rightarrow 0$; separation threshold; sketched). Assume (S1), (S2), (S4); U is the η -averaged second moment.

- (i) As $\sigma_{\perp} \rightarrow 0$: $\beta_t^2 \rightarrow \Delta_t$, $\rho_{\perp} \rightarrow 0$, $q \rightarrow \Delta_t + d_{\text{intrinsic}}(\alpha_t^2 - \Delta_t)/d_{\text{latent}}$, all eigenvalues of $\hat{\Sigma}_{t,\perp}$ tend to Δ_t (no data-side broadening), and the noise-dim block sits on the diffusion floor

$$\lambda^{\text{nd}} \rightarrow \frac{a_1(q)^2 \Delta_t}{d_{\text{latent}}} (1 \pm \psi_p^{-1/2})^2, \quad (78)$$

broadened only by W (factor 1.65 at $\psi_p = 64$; measured 1.5–1.6 at $\sigma_\perp = 0.01$), with $N_{\text{nd}} = D$. These modes carry no data variance, the learned null score equals the population one ($\varepsilon_\perp \rightarrow 0$), and the data lie in the $d_{\text{intrinsic}}$ -subspace, the setting of the linear pencil of George et al. (2026) (Remark I.20).

- (ii) Including the data-side (ratio ψ) and width-side (ratio $D/p \approx \psi_p^{-1}$) Marchenko–Pastur broadenings, the lower edge of the noise-dim block is

$$\lambda_{\min}^{\text{nd}} \asymp \frac{a_1(q)^2}{d_{\text{latent}}} \left[e^{-2t} \sigma_\perp^2 (1 - \sqrt{\psi})^2 + \Delta_t \right] (1 - \psi_p^{-1/2})^2 \quad (\psi < 1), \quad (79)$$

and with $G := \log_{10}(\lambda_{\min}^{\text{nd}}/\lambda_{\text{onset}})$ the product-bound approximation predicts the one-decade criterion $G \geq 1$ of (S5) when

$$\sigma_\perp^2 \geq \frac{e^{2t}}{(1 - \sqrt{\psi})^2} \left[\frac{10 d_{\text{latent}} \lambda_{\text{onset}}}{a_1(q)^2 (1 - \psi_p^{-1/2})^2} - \Delta_t \right] =: (\sigma_\perp^*)^2. \quad (80)$$

- (iii) With the measured onsets of the repo spectra ($t = 0.01$, $p = 64d_{\text{latent}}$, $n = 500$, $\sigma_\perp = 0.5$): $\sigma_\perp^* = 0.91, 0.70, 0.50, 0.45, 0.41, 0.40, 0.42, 0.52$ at $d_{\text{latent}} = 10, 20, 40, 60, 80, 100, 150, 200$. Hence $\sigma_\perp = 0.5$ is below threshold for $d_{\text{latent}} \leq 20$ and at or above it for $d_{\text{latent}} \geq 40$, matching the measured $G < 1$ for $d_{\text{latent}} \leq 20$ and $G \geq 1$ for $d_{\text{latent}} \geq 30$ (measured $G = 0.64, 0.89, 1.12, 1.19, 1.27, 1.26, 1.26, 1.09$; predicted from (79): $0.51, 0.73, 0.99, 1.08, 1.14, 1.15, 1.11, 0.98$, the product-of-edges lower edge under-estimating $\lambda_{\min}^{\text{nd}}$ by $\approx 25\%$). $\sigma_\perp = 0.01$ lies below the floor ($\sigma_\perp^2 = 5 \times 10^{-3} \Delta_t$) and the measured separation there is $G = 0.004\text{--}0.056$ decades: a three-bulk spectrum. The four-bulk picture applies only for $\sigma_\perp \gtrsim \sigma_\perp^*$.

Proof sketch. (i) Substitute in Definition I.76 and (71); $\varepsilon_\perp^2 \rightarrow 0$ by Proposition I.79(iii); the null block of $\hat{M}_t$ is $e^{-2t} \hat{C}_\perp + \Delta_t I_D \rightarrow \Delta_t I_D$, so the only remaining broadening is that of $W_\perp^\top W_\perp/p$, with edges $(1 \pm \sqrt{D/p})^2$. (ii) The non-zero eigenvalues of $a_1^2 W_\perp \hat{\Sigma}_{t,\perp} W_\perp^\top / (p d_{\text{latent}})$ equal those of $a_1^2 \hat{\Sigma}_{t,\perp}^{1/2} (W_\perp^\top W_\perp/p) \hat{\Sigma}_{t,\perp}^{1/2} / d_{\text{latent}}$; the lower edges of the two factors are $e^{-2t} \sigma_\perp^2 (1 - \sqrt{\psi})^2 + \Delta_t$ and $(1 - \sqrt{D/p})^2$, and multiplying their lower edges gives a conservative product-bound approximation for this data term, not generally the exact free-convolution edge. The displayed separation threshold is therefore a diagnostic based on that approximation; it is not an if-and-only-if theorem for the full feature matrix. Solve $\lambda_{\min}^{\text{nd}} \geq 10 \lambda_{\text{onset}}$ for $\sigma_\perp^2$. (iii) Code units are $p \times$ theory units. $d_{\text{latent}} = 10$: $\lambda_{\text{onset}} = 0.93/640 = 1.45 \times 10^{-3}$, $a_1(q)^2 = 0.281$, $\psi = 0.01$, giving $(\sigma_\perp^*)^2 = e^{0.02} (0.9)^{-2} [100 \cdot 1.45 \times 10^{-3} / (0.281 \cdot 0.766) - 0.0198] = 0.83$. $d_{\text{latent}} = 40$: $\lambda_{\text{onset}} = 0.374/2560 = 1.46 \times 10^{-4}$, $a_1(q)^2 = 0.494$, $\psi = 0.07$, giving $(\sigma_\perp^*)^2 = 1.0202 \cdot 1.849 \cdot [400 \cdot 1.46 \times 10^{-4} / (0.494 \cdot 0.766) - 0.0198] = 0.256$, $\sigma_\perp^* = 0.50$. The remaining entries are the same evaluation at the other d_{latent} . Measured G uses only the noise-dim minimum and the sample-bulk top (index $d_{\text{latent}} + 1$) of the released block summaries (whose block partition otherwise follows the superseded cumulative count): $\sigma_\perp = 0.5$: 0.669, 0.644, 0.784, 0.888, 0.996, 1.12, 1.19, 1.27, 1.25, 1.26, 1.09 at $d_{\text{latent}} = 8, 10, 15, 20, 30, 40, 60, 80, 100, 150, 200$; $\sigma_\perp = 0.01$: 0.045, 0.032, 0.019, 0.006, 0.004, 0.056 at $d_{\text{latent}} = 8, \dots, 40$. $\square$

I.8.4 $p = d_{\text{latent}} + n + 300$: ψ_p DECREASING TOWARD 1

Remark I.83 ($p = d_{\text{latent}} + n + 300$). This configuration was run only at $d_{\text{latent}} = 20$ (the $d_{\text{intrinsic}}$ sweep); the statements for $d_{\text{latent}} \geq 40$ are predictions. One has $p > n$ for all d_{latent} and $\psi_p = 1 + 800/d_{\text{latent}} \geq 4.3$ for $d_{\text{latent}} \leq 240$; the margin $\psi_p - 1 - \psi_n = 300/d_{\text{latent}}$ of the proportional asymptotics of Bonnaire et al. (2025) stays positive but is not uniform in d_{latent} . The block means (71) are p -independent (a $5.8 \times$ change of p moves them by $< 0.5\%$). What changes is the block width: the factor $(1 \pm \sqrt{D/p})^2$ in (79) has $\sqrt{D/p} = 0.22$ at $d_{\text{latent}} = 40$ (noise-dim block spanning a factor 2.4) and 0.44 at $d_{\text{latent}} = 200$ (factor 6.8, i.e. 0.83 decades from W alone), against 0.125 (factor 1.65) at $\psi_p = 64$. (S5) then requires $G \geq 1$ after this broadening, which the product-of-edges estimate predicts to fail at $\sigma_\perp = 0.5$ for $d_{\text{latent}} \gtrsim 100$. On the residual side $n/p \rightarrow 0.5$ at $d_{\text{latent}} = 200$ and the MP lower edge $(1 - \sqrt{n/p})^2 = 0.086$ pushes the slowest residual modes toward zero (the random-features interpolation peak of Mei & Montanari, 2022); this stretches the time to complete memorization but leaves λ_{onset} within $(1 + \sqrt{n/p})^2 \leq 2.9$ of the residual mean.

I.8.5 THE FIXED-NULL-ENERGY CONTROL

Proposition I.84 ($\sigma_{\perp}^2 = c/D$; sketched). Assume (S1), (S2), (S4) and $\sigma_{\perp}^2 = c/D$ with $c > 0$ fixed. Then

$$\beta_t^2 = \Delta_t + \frac{e^{-2t}c}{D}, \quad \rho_{\perp} = \frac{e^{-2t}c}{e^{-2t}c + \Delta_t D}, \quad q = \Delta_t + \frac{d_{\text{intrinsic}}(\alpha_t^2 - \Delta_t) + e^{-2t}c}{d_{\text{latent}}}, \quad (81)$$

$$\lambda^{\text{sig}} \asymp \frac{a_1(q)^2 \alpha_t^2}{d_{\text{latent}}}, \quad \lambda^{\text{nd}} \asymp \frac{a_1(q)^2}{d_{\text{latent}}} \left(\Delta_t + \frac{e^{-2t}c}{D} \right), \quad \bar{\lambda}^{\text{samp}} = \frac{a_*(q)^2}{n}, \quad (82)$$

with crossover $D_c := e^{-2t}c/\Delta_t \approx 49.5c$ at $t = 0.01$. For $D \gg D_c$ the control is spectrally the $\sigma_{\perp} \rightarrow 0$ limit of Proposition I.82: (i) the noise-dim block collapses onto the diffusion floor and the four-bulk separation is lost; (ii) $\tau_{\text{gen}} = d_{\text{latent}}/(a_1(q)^2 \alpha_t^2)$ keeps its linear growth, with $a_1(q)^2 \rightarrow a_1(\Delta_t)^2 = 0.96$ (vs. 0.63 at $\sigma_{\perp} = 0.5$, $d_{\text{latent}} = 200$); (iii) $a_*(q)^2 = \frac{2}{3}q^3 + O(q^4)$ collapses faster than in the $\sigma_{\perp} = 0.5$ sweep, so the frozen-feature delay of the residual bulk persists (cf. Corollary I.30); (iv) $\varepsilon_{\perp}^2 \approx \rho_{\perp}^2 \psi \approx c^2 e^{-4t}/(\Delta_t^2 D n) \rightarrow 0$, so no null-block rebound (Remark I.81) is predicted. The control thus separates the count D acting through q -dilution (retained) from the null data variance acting through $\rho_{\perp}$ (removed): the frozen-feature theory predicts that the memorization delay survives the control while the four-bulk structure and the rebound do not.

Proof sketch. Substitute $\sigma_{\perp}^2 = c/D$ into Definition I.76 and (71); the trace check (72) holds by (69). (i) is Proposition I.82(i)–(ii). (ii) α_t^2 is $\sigma_{\perp}$ -independent and a_1 is decreasing in q ; $a_1(0.0198)^2 = 0.962$. (iii) is Lemma I.6. (iv) is Proposition I.79(i). The prediction that the delay persists is a frozen-feature statement, not a statement about the trainable MLP. $\square$

I.8.6 FEATURE LEARNING

Remark I.85 (Conjecture: one-step feature learning). Everything above is a frozen-feature (S1) result. For the trainable MLP we conjecture the following, modelled on the one-gradient-step analysis of Ba et al. (2022). Let A first be trained for a time T_0 (so that $A \neq 0$; at $A(0) = 0$ the W -gradient vanishes identically), then take one step $W_1 = W_0 - \eta \nabla_W L$ with

$$\nabla_W L = \frac{1}{n\sqrt{d_{\text{latent}}p}} \sum_{\mu} \left[(A^{\top} r_{\mu}) \odot \tanh'(W_0 x_t^{\mu} / \sqrt{d_{\text{latent}}}) \right] x_t^{\mu \top}, \quad r_{\mu} = s_A(x_t^{\mu}) - y_{\mu}.$$

The update is a sum of rank-one terms with row space in $\text{span}\{x_t^{\mu}\}$ whose signal- and null-block energies scale as α_t^2 and β_t^2 per dimension, so the spike it adds to the conjugate kernel $U_1 = \frac{1}{n} \sum_{\mu} \mathbb{E}_{\eta} \phi_1 \phi_1^{\top}$ loads on the signal subspace with per-dimension null leakage $O(\beta_t^2/\alpha_t^2)$ (0.096 at $\sigma_{\perp} = 0.5$, 0.007 at 0.01). Because $\mathbb{E}[r x_t^{\top}] = A \mathbb{E}[\phi x_t^{\top}] + I$ contains an isotropic term from the DSM target, the *total* null energy of the spike, $D\beta_t^2$ against $d_{\text{intrinsic}}\alpha_t^2$, can be comparable at large d_{latent} ; whether this shifts the noise-dim block is what (P-i) asks to be measured. **(P-i)** After the step the signal-block eigenvalues increase by $O(\eta)$ (the cross term of $\Phi_1 \Phi_1^{\top}$ is first order in the step; a rank- $\leq d_{\text{intrinsic}}$ spike, BBP-type, Baik et al., 2005) while the noise-dim edges move by at most $O(\eta\beta_t^2/\alpha_t^2)$: τ_{gen} decreases, the noise-dim timescale does not. **(P-ii)** Under continued training the first layer acquires components aligned with individual training points (the nonlinear part of the score target is the softmax assignment of the empirical score, Biroli et al., 2024; Scarvelis et al., 2025), raising the top of the residual bulk and lowering $\tau_{\text{mem}}^{\text{onset}}$; the buffer in absolute steps, $\tau_{\text{mem}}^{\text{onset}} - \tau_{\text{gen}}$, is then *shorter* than the frozen-feature value even though (P-i) alone lengthens it in ratio. Status: conjecture. The saved RFNN pre/post spectra cannot test it: W is frozen there and the pre/post block medians agree to $< 0.5\%$, which only confirms (S1). The test requires logging the first-layer conjugate-kernel spectrum of the MLP at initialization and during training (compute_U applies verbatim to W_1); the MLP runs saved only configurations and metrics.

Predictions and how to test them.

1. The large- d_{latent} return of memorization is a null-block effect that additionally requires capacity: it appears at $\rho_{\perp} \approx 1$ ($\sigma_{\perp} = 0.5$) with hidden = $8d_{\text{latent}}$, is absent at $\rho_{\perp} \approx 0$ ($\sigma_{\perp} = 0.01$) at the same scaled capacity, and is absent at $\rho_{\perp} \approx 1$ with fixed hidden = 256. Test: rerun $d_{\text{latent}} \in \{100, 150, 200\}$ at hidden = 1600 fixed for $\sigma_{\perp} \in \{0.01, 0.5, 1\}$,

beyond 300k steps; the isotropic run should rebound earlier ($d_{\text{latent}}^{\text{reb}} \in [88, 130]$) and more strongly.

2. *Conditional on Remark I.81, the rebound point scales linearly with n :* $d_{\text{latent}}^{\text{reb}} - d_{\text{intrinsic}} = n\psi^{\text{reb}}(\rho_{\perp})$, $\psi^{\text{reb}}(0.925) \in (0.19, 0.29)$. Test: rerun the scaled-width sweep at $n = 250$ and 1000 (predicted onset $d_{\text{latent}} \approx 55\text{--}80$ and $200\text{--}300$); the null-block signature needs generated-sample dumps (none exist).
3. *Four-bulk separation threshold:* the noise-dim/residual gap crosses one decade at $\sigma_{\perp}^* = 0.91$ ($d_{\text{latent}} = 10$), 0.70 (20), 0.50 (40), 0.40–0.45 (60–100), 0.52 (200); below $\sqrt{\Delta_t e^{2t}} = 0.14$ the block is diffusion-noise dominated and sits on the W -broadened floor. Test: RFNN spectra at $d_{\text{latent}} \in \{10, 40, 100\}$ over $\sigma_{\perp} \in \{0.05, \dots, 1.0\}$; expect $G = 1$ near 0.9 at $d_{\text{latent}} = 10$ and near 0.5 at $d_{\text{latent}} = 40$.
4. *Fixed-null-energy control* $\sigma_{\perp}^2 = c/D$: for $D \gg 50c$ the spectrum coincides with the $\sigma_{\perp} \rightarrow 0$ spectrum, τ_{gen} keeps growing linearly with $a_1(q)^2 \rightarrow 0.96$, the RFNN residual-bulk delay persists, and no rebound appears. Test: RFNN spectral sweep and scaled-width MLP sweep with $\sigma_{\perp}^2 = 1/D$.
5. $p = d_{\text{latent}} + n + 300$: block means unchanged, noise-dim width grows as $((1 + \sqrt{D/p})/(1 - \sqrt{D/p}))^2$ ($2.4\times$ at $d_{\text{latent}} = 40$, $6.8\times$ at 200), and four-bulk separation fails at $\sigma_{\perp} = 0.5$ for $d_{\text{latent}} \gtrsim 100$. Test: spectra at $d_{\text{latent}} \in \{40, 100, 200\}$ for $p \in \{d_{\text{latent}} + n + 300, 8d_{\text{latent}}, 64d_{\text{latent}}\}$.
6. *Feature-learning conjecture:* after one W -step following readout pre-training, signal-block eigenvalues rise by $O(\eta)$ while noise-dim edges move by $O(\eta\beta_t^2/\alpha_t^2)$; with continued training the residual-bulk top rises and $\tau_{\text{mem}}^{\text{onset}} - \tau_{\text{gen}}$ shrinks. Test: log the MLP first-layer conjugate-kernel spectrum during training.

Open gaps. (1) The measured residual-bulk top exceeds (75) by $4.6\times$ at $d_{\text{latent}} = 10$, decaying to $1.0\times$ at $d_{\text{latent}} = 200$; attributed to cluster-level residual spikes but not derived (Proposition I.27, Remark I.64); whether the small- d_{latent} top modes are cluster-level or sample-specific is unresolved and affects what “onset” means. This also covers the isotropic control’s reported $10.7\times$ gap growth, which the frozen-feature iso-residual model does not reproduce. (2) The rebound mechanism is a conjecture: the linear null model does not predict substantial replication in the stated high-dimensional approximation; $\varepsilon_{\perp}$ is a candidate diagnostic and ε_c is calibrated on the same two points previously claimed as a match; Prediction 2 is the only non-circular test. (3) Capacity confound: the fixed-width $\sigma_{\perp} = 0.5$ runs show 0.0000 throughout; the rebound exists only with hidden $= 8d_{\text{latent}}$ and is still rising at 300k steps; the main text must not say the theory “matches the MLP sweep”. (4) The cliff at index n holds for $U^{(1)}$; the saved 50-draw spectra show a soft shoulder ($\lambda_n/\lambda_{n+1} = 1.3\text{--}2.1$ for $d_{\text{latent}} \geq 40$) and have $N_{\text{nd}} = D$ for all d_{latent} ; the released block-summary script still partitions with the superseded cumulative count and only its noise-dim minimum and sample top are used here. (5) Block widths in (79) use a product-of-edges approximation that under-estimates the noise-dim lower edge by $\sim 25\%$; exact edges need the Silverstein fixed point. (6) Proposition I.79 neglects the signal–null cross covariance, of the same order as $\varepsilon_{\perp}^2$ at small ψ . (7) Feature learning is entirely conjectural; no MLP spectra exist; the one-step scaling is $O(\eta)$ (corrected from $\Theta(\eta^2)$) and whether the first-order term survives depends on alignment. (8) The isotropic control and $d_{\text{latent}} = 240$ do not exist as runs. (9) The real-data regime (continuous ρ_{Σ} , collapsed coordinates) is placed outside (S3) and handled in Appendix I.7. (10) Spectra are computed at the single time $t = 0.01$ while generation covers $t \in [0.01, 3]$; $\rho_{\perp}(t) \geq \frac{1}{2}$ only for $t \leq \frac{1}{2} \log(1 + \sigma_{\perp}^2)$ (0.11 at $\sigma_{\perp} = 0.5$), so the null block is data-dominated only on a short final segment of the reverse trajectory.

I.9 CONFIGURATIONS OUTSIDE THE HYPOTHESES

Remark I.86 (Out-of-scope configurations). 1. $d_{\text{latent}} = d_{\text{intrinsic}} = 5$ (both sweeps): $N_{\text{nd}} = 0$ and $p = 320 < n$, so (S2) fails twice; the residual bulk has $\leq p - d_{\text{latent}} = 315$ modes. A baseline, not a four-bulk spectrum.

2. $\sigma_{\perp} = 0.01$ at $t = 0.01$ (RFNN and MLP): $\rho_{\perp} = 4.9 \times 10^{-3}$, $\sigma_{\perp}^2 = 5 \times 10^{-3} \Delta_t$, measured noise-dim/residual gap 0.004–0.056 decades. (S3), (S5) fail: three-bulk regime. Counts and τ_{gen} still apply; Proposition I.82(i) is the operative statement and predicts no rebound

- (observed in the scaled sweep: memorization 0 for $20 \leq d_{\text{latent}} \leq 200$). Floor medians match $pa_1(q)^2 \Delta_t / d_{\text{latent}}$ for $d_{\text{latent}} \geq 30$ (0.67, 0.77 predicted vs 0.74, 0.81 measured at $d_{\text{latent}} = 30, 40$); at $d_{\text{latent}} = 10$ the floor block is merged with the residual bulk ($G = 0.03$) and the comparison is not meaningful.
3. $d_{\text{latent}} \geq 150$ at $n = 500$, $\sigma_{\perp} = 0.5$: $\psi_n \leq 3.3$, $\varepsilon_{\perp}^2 \geq 0.64$; regime of Remark I.81. (S5) also degrades (gap $1.26 \rightarrow 1.09$ decades from $d_{\text{latent}} = 150$ to 200). Memorization returns only in the scaled-width MLP sweep (0.0022, 0.0144 at $d_{\text{latent}} = 150, 200$, hidden $8d_{\text{latent}}$) and not in the fixed-width sweep (hidden 256: 0.0000).
 4. $d_{\text{intrinsic}}$ sweep at $d_{\text{latent}} = 20$: $d_{\text{intrinsic}} \in \{16, 20\}$ gives $N_{\text{nd}} \in \{4, 0\}$; the signal block is a finite-rank spike for every $d_{\text{intrinsic}}$, so “signal edges” are spike locations, not bulk edges.
 5. RFNN with $p = d_{\text{latent}} + n + 300$: run only at $d_{\text{latent}} = 20$; Remark I.83 predicts that width-side broadening breaks (S5) for $d_{\text{latent}} \gtrsim 100$ at $\sigma_{\perp} = 0.5$.
 6. Isotropic control $\sigma_{\perp} = \sigma_{\text{sig}} = 1$ (no released training run at $\sigma_{\perp} = 1$ exists; the spectral statements are checked by the simulation in Remark I.33): (S3) holds ($\rho_{\perp} = 0.98$) and the D noise-dim modes exist at $a_1(q)^2 \beta_t^2 / d_{\text{latent}}$ between signal and residual modes, but the signal/noise-dim ratio is $\alpha_t^2 / \beta_t^2 = 2.76$ (0.44 decades) for all d_{latent} : the two-atom *separation* of the signal block is absent, not the block (Remarks I.19 and I.33). The reported $10.7\times$ growth of the spectral gap under isotropy must then be a residual-bulk effect (fall of $a_*(q)^2$ and of the cluster-spike excess), not a noise-dim one. Remark I.81 with $\varepsilon_c^2 \in (0.30, 0.64)$ gives $\psi^{\text{reb}}(0.98) \in (0.166, 0.251)$, i.e. $d_{\text{latent}}^{\text{reb}} \in [88, 130]$ at $n = 500$, earlier than at $\sigma_{\perp} = 0.5$.
 7. All trainable-MLP runs (synthetic, hidden 256 fixed or $8d_{\text{latent}}$ scaled; real, depth 5, width 1024): (S1) fails; only Remark I.85 applies, and the clock statements of Remarks I.7 and I.40 do not cover Adam or momentum SGD.
 8. Real VAE latents ($t = 0.1$, $n = 1000$): ρ_{Σ} is continuous, effective rank $\approx 0.5 d_{\text{latent}}$, and for $d_{\text{latent}} \geq 140$ (CelebA) / 160 (CIFAR-10) a set of coordinates has $\lambda = \Delta_t$ to within 2×10^{-3} , exactly from $d_{\text{latent}} = 200$ (posterior collapse). (S3) fails as stated; the collapsed coordinates are the $\sigma_{\perp} = 0$ case of Proposition I.82(i): free modes at the floor with $\varepsilon_{\perp} = 0$, counted in N_{nd} but not contributing to $\rho_{\perp}$. The general- ρ_{Σ} analysis of Bonnaire et al. (2025, Thm. 3.1) motivates a surrogate, but its applicability to these non-Gaussian latents remains an assumption (Appendix I.7); the two-atom edges do not apply.
 9. Time mismatch: spectra are computed at the single time $t = 0.01$, while training and generation cover $t \in [0.01, 3]$ (the reverse SDE is integrated from $t_{\text{max}} = 3$ to $t_{\text{min}} = 0.01$; $t_{\text{eval}} = 0.1$ enters only the score-error metric). $\rho_{\perp}(t) \geq \frac{1}{2}$ only for $t \leq \frac{1}{2} \log(1 + \sigma_{\perp}^2)$ (0.11 at $\sigma_{\perp} = 0.5$, 5×10^{-5} at 0.01), so the null block is data-dominated only on a short final segment of the reverse trajectory; the bridge of Appendix I.5 places the attachment time t_c between 0.05 and 1.0 across the sweep, outside that segment for $d_{\text{latent}} \gtrsim 20$.
 10. Synthetic score-error curves: all saved values were computed with a transposed whitening matrix and are artifacts; the predictions of Appendix I.4 are for the corrected metric and are currently unanchored.